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APUL

NEET FLIX

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Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION TO BIOLOGY

1. Which of the following is a defining characteristic of living organisms?
(1) Growth
(2) Ability to make sound
(3) Reproduction
(4) Response to external stimuli

BLW089

2. In the given columns, column I contains the terms and column II contains its description. Select the correct match from the option given below.

Column I		Column II	
A.	Growth	I.	Production of offspring
B.	Reproduction	II.	Composed of one or more living cells
C.	Metabolism	III.	Increase in mass and increase in number of individuals
D.	Cellular organization	IV.	Sum total of all chemical reactions occurring in a living body

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-II, C-II, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-III, D-I

BLW094

3. Which of the followings is a unique feature of living organisms?
(1) Growth and reproduction.
(2) Reproduction and ability to sense environment.
(3) Metabolism and consciousness.
(4) All of the above

BLW095

4. Which of the followings is not true?
(1) Emergent properties is found in every level of biological organization
(2) Emergent properties do not present in a level that is beyond the organism
(3) Homeostasis indicates ability of self-regulation
(4) Mechanisms of homeostasis is different for different parameters.

BLW096

5. Adaptation is _____.
(1) Types of living beings present on earth
(2) Essential character of an organism that is necessary for it's survival.
(3) Inability to respond against environmental cue
(4) Types of flora present in an area.

BLW097

6. Which two points are known as the twin characteristics of growth?
(i) Increase in mass
(ii) Differentiation
(iii) Increase in number of individuals
(iv) Response to stimuli
(1) (i) and (ii)
(2) (i) and (iv)
(3) (ii) and (iii)
(4) (i) and (iii)

BLW100

7. Growth and reproduction are mutually exclusive events in _____.
(1) All organisms
(2) Only in unicellular organisms
(3) Only in multicellular organisms
(4) All animates and in animates

BLW111

8. Growth occurs _____ in plants and _____ in animals.
(1) continuously, only up to a certain age
(2) only up to a certain age, continuously
(3) continuously, never
(4) once, twice

BLW112

9. Defining characters of all livings is/are :-
(1) Metabolism
(2) Consciousness
(3) Cellular organization
(4) All of the above

BLW113

10. All livings show :-
(1) Growth (2) Reproduction
(3) Metabolism (4) Both (1) & (3)

BLW114

DIVERSITY IN THE LIVING WORLD, TAXONOMY

11. Artificial system of classification classifies plants on the basis of :-
(1) One or few characters
(2) Phylogenetic trends
(3) Many naturally existing characters
(4) None of the above

BLW002

12. The term systematics was introduced by :-
(1) Mayr (2) Bentham
(3) Hutchinson (4) Linnaeus

BLW003

13. ICBN was first published in :-
(1) 1961 (2) 1964 (3) 1975 (4) 1753

BLW005

14. The term taxon refers to :-
(1) Name of a species
(2) Name of genus
(3) Name of family
(4) A taxonomic group of any rank

BLW006

15. Taxonomy term was given by :-
(1) Linnaeus (2) Mayr
(3) Haeckel (4) A.P. de. Candolle

BLW009

16. Plant nomenclature means :-
(1) To give names to plants without any rules.
(2) Nomenclature of plants under the international rules.
(3) Nomenclature of plants in local language
(4) Nomenclature of plants in English language.

BLW010

17. Which of the following is a correct name ?
(1) *Solanum tuberosum*
(2) *Solanum Tuberosum*
(3) *Solanum tuberosum* Linn.
(4) All the above

BLW011

18. Systematics deals with :-
(1) Classification
(2) Nomenclature
(3) Identification
(4) All of these

BLW012

19. Scientific name of Mango plant is *Mangifera indica* Linn. in the above name Linn. refers to:
(1) Variety of Mango
(2) A taxonomist who proposed the present nomenclature in honour of Linnaeus
(3) A scientist who for the first time described Mango plant
(4) A scientist who changed the name proposed by Linnaeus and proposed present name

BLW013

20. Who wrote *Species Plantarum* ?
(1) Linnaeus (2) Mayr
(3) Bentham (4) Aristotle
BLW016

21. For higher plants, flowers are chiefly used as a basis of classification, because :–
(1) These show a great variety in colour
(2) It can be preserved easily
(3) Reproductive parts are more conservative than vegetative parts
(4) None of these
BLW017

22. Term systematics was derived from _____ word "Systema" which means _____.
(1) Greek, evolutionary relationship
(2) Latin, systematic arrangement of organism
(3) English, taxonomy of organisms
(4) Both (2) & (3)
BLW018

23. Which of the followings is not an advantage of taxonomy?
(1) Avoid confusion in scientific community
(2) Provides a valid name of an organism
(3) Provides help in studying them.
(4) Prevent loss of biodiversity
BLW019

24. Book '*Systema Naturae*' was written by :-
(1) Mayr
(2) Linnaeus
(3) John–Ray
(4) Theophrastus
BLW025

25. Most of the botanical names are derived from the following language :-
(1) German (2) Greek
(3) Latin (4) Spanish
BLW026

26. 'Taxa' differs from 'taxon' due to being: -
(1) A higher taxonomic category than taxon.
(2) Lower taxonomic category than taxon.
(3) The plural of taxon.
(4) The singular of taxon.
BLW029

27. Evolutionary history of an organism is known as :-
(1) Paleontology
(2) Ontogeny
(3) Phylogeny
(4) Ancestry
BLW031

28. ICBN stands for :
(1) Indian Code of Botanical Nomenclature
(2) Indian Congress of Biological Names
(3) International Code of Botanical Nomenclature
(4) International Congress of Biological Names
BLW033

TAXONOMIC CATEGORIES

29. Mayr proposed which type of concept of species :-
(1) Taxonomic concept
(2) Biological concept
(3) Taxonomic and Biological concept
(4) Genetic concept
BLW001

30. Group of organisms that closely resemble each other and freely interbreed in nature, constitute a:
(1) Species (2) Genus
(3) Family (4) Taxon
BLW004

31. The basic smallest unit of classifications is :-
(1) Genus (2) Species
(3) Order (4) All of the above
BLW008

32. Angiospermae is taxa at the rank of?

(1) Genus (2) Order

(3) Class (4) Division

BLW014

33. Which one of the following animals is correctly matched with its correct taxonomic category?

(1) Tiger – *Panthera*, the genus

(2) Solanum – china rose, the genus

(3) Humans – Primata, the family

(4) Housefly – *Musca*, an order

BLW020

34. Which of the following taxonomic category of housefly is incorrectly matched?

(1) Genus – *Musca* (2) Family – Muscidae

(3) Order – Primata (4) Class – Insecta

BLW021

35. Which of the following organisms is not correctly matched with its category?

(1) Human – *Sapiens*, specific epithet

(2) Lion – *Panthera*, genus

(3) Cats – Felidae, genus

(4) Datura – Solanaceae, family

BLW022

36. Who is regarded as 'Darwin of 20th Century'?

(1) John Ray (2) Lamarck

(3) Ernst Mayr (4) Darwin

BLW023

37. Two plants can be conclusively said to belong to the same species if they :

(1) Have same number of chromosomes

(2) Can reproduce freely with each other and form seeds

(3) Have more than 90 percent similar genes

(4) Look similar and possess identical secondary metabolites.

BLW032

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	4	2	2	4	3	1	4	4	1	4	1	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	4	3	1	3	2	4	2	3	3	3	3	2	1
Question	31	32	33	34	35	36	37								
Answer	2	4	1	3	3	3	2								

Exercise - II (Previous Year Questions) AIPMT/NEET

NEET-I 2016

1. Nomenclature is governed by certain universal rules. Which one of the following is contrary to the rules of nomenclature?

(1) Biological names can be written in any language

(2) The first word in a biological name represents the genus name, and the second is a specific epithet

(3) The names are written in Latin and are italicised

(4) When written by hand, the names are to be underlined

BLW035

NEET-II 2016

2. The label of a herbarium sheet does not carry information on :

(1) Local names (2) height of the plant

(3) date of collection (4) name of collector

BLW036

[PYQ from deleted syllabus as per NMC 2023 guidelines]

3. Match Column–I with Column–II for housefly classification and select the correct option using the codes given below :

Column–I		Column–II	
(a)	Family	(i)	Diptera
(b)	Order	(ii)	Arthropoda
(c)	Class	(iii)	Muscidae
(d)	Phylum	(iv)	Insecta

Codes :

	(a)	(b)	(c)	(d)
(1)	(iv)	(iii)	(ii)	(i)
(2)	(iv)	(ii)	(i)	(iii)
(3)	(iii)	(i)	(iv)	(ii)
(4)	(iii)	(ii)	(iv)	(i)

BLW037

4. Study the four statements (A–D) given below and select the two correct ones out of them :

(A) Definition of biological species was given by Ernst Mayr.

(B) Photoperiod does not affect reproduction in plants.

(C) Binomial nomenclature system was given by R.H. Whittaker.

(D) In unicellular organisms, reproduction is synonymous with growth.

The two correct statements are

(1) A and D (2) A and B

(3) B and C (4) C and D

BLW038

NEET-UG 2018

5. Match the items given in Column-I with those in Column-II and select the correct option given below:-

	Column-I		Column-II
(a)	Herbarium	(i)	It is a place having a collection of preserved plants and animals.
(b)	Key	(ii)	A list that enumerates methodically all the species found in an area with brief description aiding identification.
(c)	Museum	(iii)	Is a place where dried and pressed plant specimens mounted on sheets are kept.
(d)	Catalogue	(iv)	A booklet containing a list of characters and their alternates which are helpful in identification of various taxa.

(1) (a)-i, (b)-iv, (c)-iii, (d)-ii

(2) (a)-iii, (b)-ii, (c)-i, (d)-iv

(3) (a)-ii, (b)-iv, (c)-iii, (d)-i

(4) (a)-iii, (b)-iv, (c)-i, (d)-ii

BLW039

[PYQ from deleted syllabus as per NMC 2023 guidelines]

NEET-UG 2019

6. Select **correctly** written scientific name of Mango which was first described by Carolus Linnaeus:

(1) *Mangifera indica* Car. Linn.

(2) *Mangifera indica* Linn.

(3) *Mangifera indica*

(4) *Mangifera Indica*

BLW040

NEET-UG 2019 (Odisha)

7. Which of the following is against the rules of ICBN?

(1) Hand written scientific names should be underlined.

(2) Every species should have a generic name and a specific epithet.

(3) Scientific names are in Latin and should be italicized.

(4) Generic and specific names should be written starting with small letters.

BLW041

8. The contrasting characteristics generally in a pair used for identification of animals in taxonomic key are referred to as :

(1) Lead (2) Couplet

(3) Doublet (4) Alternate

BLW042

[PYQ from deleted syllabus as per NMC 2023 guidelines]

NEET-UG 2021

9. Which one of the following belongs to the family Muscidae?

(1) Fire fly (2) Grasshopper

(3) Cockroach (4) House fly

BLW043

NEET-UG 2022

10. In the taxonomic categories which hierarchical arrangement in ascending order is **correct** in case of animals ?

(1) Kingdom, Class, Phylum, Family, Order, Genus, Species

(2) Kingdom, Order, Class, Phylum, Family, Genus, Species

(3) Kingdom, Order, Phylum, Class, Family, Genus, Species

(4) Kingdom, Phylum, Class, Order, Family, Genus, Specie

BLW044

Re-NEET-UG 2022

11. Which of the following are true about the taxonomical aid 'key' ?

(a) Keys are based on the similarities and dissimilarities.

(b) Key is analytical in nature.

(c) Keys are based on the contrasting characters in pair called couplet.

(d) Same key can be used for all taxonomic categories.

(e) Each statement in the key is called Lead.

Choose the most appropriate answer from the options given below :

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (a), (b), (c) and (e) only

(4) (a), (c), (d) and (e) only

BLW045

[PYQ from deleted syllabus as per NMC 2023 guidelines]

NEET (UG) 2023 (Manipur)

12. 'X' and 'Y' are the components of Binomial nomenclature. This naming system was proposed by 'Z' :

(1) X-Generic name, Y-Specific epithet, Z-Carolus Linnaeus

(2) X-Specific epithet, Y-Generic name, Z-R.H. Whittaker

(3) X-Specific epithet, Y-Generic name, Z-Carolus Linnaeus

(4) X-Generic name, Y-Specific epithet, Z-R.H. Whittaker

BLW115

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	1	2	3	1	4	2	4	2	4	4	3	1

Solution :

[PYQ from deleted syllabus as per NMC 2023 guidelines]

Sol.2 Herbarium sheet carry a label providing information about date, place of collection, english/ local and botanical name, family, collector's name.

Sol.5 **Herbarium -** It is a place where dried and pressed plant specimens mounted on sheets are kept.

Key - A booklet containing a list of characters and their alternates which are helpful in identification of various taxa.

Museum - It is a place having a collection of preserved plants and animals.

Catalogue - A list that enumerates methodically all the species found in an area with brief description aiding identification.

Sol.8 Key is another taxonomical aid used for identification of plants and animals based on the similarities and dissimilarities. The keys are based on the contrasting characters generally in a pair called **couplet**. It represents the choice made between two opposite options.

Sol.11 Statement (d) is incorrect because separate taxonomic keys are required for each taxonomic category such as family, genus and species for identification purposes.

Exercise - III

Master Your Understanding

(A) NCERT BASED QUESTIONS

1. The number and types of organisms present on earth make :-

(1) Taxonomy (2) Plant diversity

(3) Animal diversity (4) Biodiversity

BLW046

2. The process by which anything is grouped into convenient categories based on some easily observable characters is called as:-

(1) Biodiversity (2) Classifications

(3) Identification (4) Nomenclature

BLW047

3. The process in which biologists follow universally accepted principles to provide name of any organism is called as :-

(1) Identification (2) Classification

(3) Nomenclature (4) Systematics

BLW048

4. What are the essential basis of modern taxonomic studies ?

(1) External and internal structure

(2) Structure of cell

(3) Developmental process and ecological information

(4) All of these

BLW049

5. Characterisation of families is done on the basis of:-

(1) Vegetative features

(2) Reproductive features

(3) Both (1) and (2)

(4) None of them

BLW050

6. In the names *Mangifera indica*, *Solanum tuberosum* and *Panthera leo*, the words *indica*, *tuberosum* and *leo* represent :-

(1) Generic name

(2) Generic epithet

(3) Name of species

(4) Specific epithet

BLW051

7. Why the determination of relationship becomes more complex in higher taxonomic categories ?

(1) Number of common characters goes on decreasing in lower taxa

(2) Number of common characters goes on decreasing in higher taxa

(3) Because classification itself is very difficult process

(4) Number of common characters goes on increasing in higher taxa

BLW052

8. Which of the following is **not** a correct statement?

(1) Order is the assemblage of families which exhibit a few similar characters in comparisation to families

(2) Convolvulaceae and Solanaceae families are included in order Polymoniales mainly on the basis of floral characters

(3) In hierarchy both broad categories and sub categories are used

(4) Class is a sub category which includes related order

BLW053

9. All the categories used in classification of organism constitute :-
(1) Taxonomy
(2) Systematics
(3) Taxonomic hierarchy
(4) Taxonomic affinity

BLW059

10. The number of species that are known and described ranges between :
(1) 1-2 million (2) 1.7 - 1.8 billion
(3) 1.7 - 1.8 million (4) 7 million

BLW060

11. Branch of biology, which deals with study of evolutionary relationship between organism:
(1) Plant anatomy (2) Systematics
(3) Ecology (4) Physiology

BLW061

12. In botanical gardens various plant species are grown for which purpose?
(1) Identification (2) Nomenclature
(3) Classification (4) Both (2) and (3)

BLW062

13. ICZN is used for
(1) Virus (2) Bacteria
(3) Cultivated plants (4) Animals

BLW066

(B) ANALYTICAL QUESTIONS

14. Practical purpose of taxonomy or classification :-
(1) Facilitate the identification of unknown species.
(2) Explain the origin of organisms.
(3) To know the evolutionary history
(4) Identification of medicinal plants

BLW067

15. The correct sequence of evolution is :-
(1) Bryophyta → Pteridophyta → Gymnosperms → Dicots → Monocots
(2) Pteridophyta → Gymnosperms → Dicots → Monocots → Bryophyta
(3) Bryophyta → Gymnosperms → Dicots → Monocots → Angiosperms
(4) Bryophyta → Pteridophyta → Gymnosperms → Monocots → Dicots

BLW068

16. Which of the following more similar in characters as compared to genus?
(1) Species (2) Division
(3) Class (4) Family

BLW069

17. Phenetic classification is based on :-
(1) The ancestral lineage of existing organisms
(2) Observable characteristics of existing organisms
(3) Dendograms based on DNA characteristics
(4) Sexual characteristics

BLW070

18. Species are considered as :-
(1) Real, basic and lowest units of classification.
(2) The units of classification of any rank.
(3) Artificial concept of human mind.
(4) Biggest units of classification devised by taxonomists.

BLW071

19. Which of the following is not true for a species?
(1) Variations occur among members of a species
(2) Gene flow does not occur between the populations of a species

(3) Each species is reproductively isolated from every other species

(4) Members of a species can interbreed

BLW072

20. As we go from species to kingdom in a taxonomic hierarchy, the number of common characteristics

(1) Will decrease

(2) Will increase

(3) Remain same

(4) May increase or decrease

BLW073

21. Which of the following is / are not correctly match?

	Common Name	Genus	Family
(i)	Man	*Homo*	Hominidae
(ii)	Housefly	*Musa*	Muscidae
(iii)	Mango	*Mangifera*	Anacardiaceae
(iv)	Wheat	*Triticum*	Poaceae

(1) i & ii (2) only i

(3) only ii (4) iii & iv

BLW075

22. How many of the followings are taxa ?

Dogs, Cats, Mammals, Wheat, Rice, Plants, Wood, Mango, Leaves, Jointed legs.

(1) Eight (2) Six

(3) Seven (4) One

BLW076

23. How many are the **incorrect** statements from the followings ?

(A) Lower the taxa, more are the characteristics that the members within the taxon share

(B) Higher the taxonomic category, greater is the difficulty of determining the relationships to other taxa at the same level

(C) In lower taxa the problems of classification becomes more complex

(D) As we go higher from species to kingdom, the number of common characters increases

(1) One (2) Two

(3) Three (4) Four

BLW078

24. Nomenclature is very must in taxonomy which of the following is **not correct** about scientific nomenclature ?

(1) They ensure that each organism has only one name

(2) They also ensure that such a name has not been used for any other known organism

(3) Scientific nomenclature is a standardised naming system

(4) Different countries of the world use different kinds of scientific nomenclature system

BLW080

25. The relation of Solanaceae and Convolvulaceae with Polymoniales is similar to the relation occurring in:-

(1) Felidae and Canidae with Carnivora

(2) Primata and Carnivora with Mammalia

(3) Amphibia and Reptilia with Chordata

(4) *Solanum* and *Petunia* with Solanaceae

BLW081

26. Which of the following 'suffixes' used for units of classification in plants indicates a taxonomic category of 'family'.

(1) Ales (2) Onae

(3) Aceae (4) Ae

BLW082

27. The term 'systematics' refers to:
(1) Identification and classification of plants and animals
(2) Nomenclature and identification of plants and animals
(3) Diversity of kinds of organisms and their relationship
(4) Different kinds of organisms and their classification

BLW083

28. Genus represents -
(1) An individual plant or animal
(2) A collection of plants or animals
(3) Group of closely related species of plants or animals
(4) None of these

BLW084

29. The taxonomic unit 'Phylum' in the classification of animals is equivalent to which hierarchical level in classification of plants -
(1) Class (2) Order
(3) Division (4) Family

BLW085

30. All living organisms are linked to one another because
(1) They have common genetic material of the same type
(2) They share common genetic material but to varying degrees
(3) All have common cellular organization
(4) All of above

BLW088

31. Match the following and choose the correct option:

A. Family	i. *Solanum tuberosum*
B. Kingdom	ii. Polymoniales
C. Order	iii. *Solanum*
D. Species	iv. Plantae
E. Genus	v. Solanaceae

Options:
(1) i-D, ii-C, iii-E, iv-B, v-A
(2) i-E, ii-D, iii-B, iv-A, v-C
(3) i-D, ii-E, iii-B, iv-A, v-C
(4) i-E, ii-C, iii-B, iv-A, v-D

BLW090

32. What is been deified by early man?
(1) All in animates
(2) All animates
(3) Some animates and some inanimates
(4) All living beings

BLW091

33. What was the biggest spin off (outcome) of systematic studies?
(1) Recognition of the sharing of similarities among living organisms only horizontally
(2) Recognition of the sharing of similarities among living organisms only vertically
(3) Recognition of the sharing of similarities among living organisms horizontally as well as vertically
(4) Recognition of importance of non-living.

BLW092

34. Revelation of which information humbled man and they started conservation of biodiversity?
(1) All the organisms have soul.
(2) All living organisms has RNA.
(3) All living organisms share common genetic material.
(4) All living organisms has life span.

BLW093

35. Match column I with column II and selected the correct option from the codes given below.

Column I		Column II	
(a)	Family	i.	*tuberosum*
(b)	Kingdom	ii.	Polymonials
(c)	Order	iii.	*Solanum*
(d)	Species	iv.	Plantae
(e)	Genus	v.	Solanaceae

(1) (a)-v, (b)-iv, (c)-ii, (d)-i, (e)-iii

(2) (a)-v, (b)-iv, (c)-i, (d)-iii, (e)-ii

(3) (a)-i, (b)-ii, (c)-iii, (d)-v, (e)-iv

(4) (a)-ii, (b)-i, (c)-iii, (d)-iv, (e)-v

BLW098

36. Genus represents aggregate of closely related species. Which of the following group of species does not belong to same genus?

(1) Potato, Brinjal

(2) Brinjal, Makoi

(3) Lion, Leopard

(4) Potato, Sweet potato

BLW099

37. Which of the following statements are not correct ?

(i) Lower the taxon, more are the characteristics that the members within the taxon share.

(ii) Order is the assemblage of genera which exhibit a few similar characters.

(iii) Cat and dog are included in the same family Felidae.

(iv) Binomial nomenclature was introduced by Carolus Linnaeus.

(1) (i), (ii) and (iii) (2) (ii), (iii) and (iv)

(3) (i) and (iv) (4) (ii) and (iii)

BLW101

38. Which of the following statements are correct?

(i) Genus comprises a group of related species.

(ii) Taxon represents a taxonomic group of individual organisms.

(iii) Family comprises a group of related genera.

(iv) Taxonomic category class includes related orders.

(1) (i), (ii), and (iv)

(2) (ii) and (iv)

(3) (i), (iii) and (iv)

(4) (ii), (iii) and (iv)

BLW102

39. Which of the following statement(s) is/are correct ?

(i) Classification is the providing of standardize names to the organisms such that a particular organism known by the same all over the world.

(ii) Taxonomy is the correct description of an organism and to recognition by its scientific name.

(iii) The system of naming with two components (binomial nomenclature) is proposed by Carolus Linneaus.

(iv) Zoological names are based on International Code of Zoological Nomenclature (ICZN).

(1) Only (i)

(2) Both (iii) and (iv)

(3) Only (iv)

(4) All of these

BLW103

40. Which of the following statement(s) is/are not correct ?

(i) Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(ii) Three different genera such as *Solanum*, *Datura* and *Petunia* are placed in the family Malvaceae.

(iii) In case of plants, classes with a few similar characters are assigned to a higher category called phylum.

(1) Both (i) and (ii)

(2) Only (iii)

(3) Both (ii) and (iii)

(4) All of these

BLW104

41. Select the correct statements (i-v) regarding taxonomic categories.

(i) Each step or rank in hierarchy is called taxonomic category.

(ii) Species is a group of individual organisms with fundamental similarities capable of breeding among themselves.

(iii) Taxonomic studies of all unknown organisms have led to the development of common categories like kingdom, phylum or division, class, order, family, genus and species.

(iv) Lower the category, greater is the difficulty of determining the relationship to other taxa at the same level.

(1) (i) & (ii) only

(2) (ii) & (iv) only

(3) (i), (ii) & (iii) only

(4) All of the above

BLW105

42. **Assertion :** Taxon and category are different things.

Reason : Category shows hierarchical classification.

(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) If Assertion is true but Reason is false.

(4) If both Assertion and Reason are false.

BLW106

43. **Assertion :** The species is reproductively isolated natural population.

Reason : Prokaryotes cannot be kept under different species on the basis of reproductive isolation.

(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) If Assertion is true but Reason is false.

(4) If both Assertion and Reason are false.

BLW107

44. **Assertion :** Binomial nomenclature given by Linnaeus is being practiced by biologists all over the world.

Reason : Each name of this system has two components, the generic name and the specific epithet.

(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) If Assertion is true but Reason is false.

(4) If both Assertion and Reason are false.

BLW116

45. **Assertion :** Scientific names for plants have been standardized through ICBN.

Reason : Naming system which used three word format was given by Linnaeus.

(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) If Assertion is true but Reason is false.

(4) If both Assertion and Reason are false.

BLW110

EXERCISE-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	4	3	4	2	4	3	3	2	1	4	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	1	2	1	3	3	2	4	1	3	3	3	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	3	3	1	4	4	3	2	3	1	2	2	1	3

Exercise - I (Conceptual Questions) Build Up Your Understanding

HISTORY OF TAXONOMY

1. According to Whittaker, BGA are included in:
(1) Mycota (2) Protista
(3) Plantae (4) Monera
BBC001

2. Kingdom Monera comprises the :-
(1) Plants of economic importance
(2) All the plants studied in botany
(3) Prokaryotic organisms
(4) Plants of Thallophyta group
BBC002

3. Whittaker is famous for :-
(1) Two kingdom classification
(2) Four kingdom classification
(3) Five kingdom classification
(4) Distinguishing in Bacteria & blue green Algae
BBC003

4. System of classification proposed by Linnaeus was:-
(1) Artificial (2) Natural
(3) Sexual (4) Both (1) and (3)
BBC004

5. The group "Plantae" proposed by Whittaker includes:-
(1) Pteridophytes (2) Gymnosperms
(3) Angiosperms (4) All the above
BBC005

6. In Whittaker's five kingdom classification, eukaryotes were assigned to :-
(1) All the five kingdom
(2) Only four of the five kingdoms
(3) Only three kingdom
(4) Only one kingdom
BBC006

7. Carolus Linnaeus classified plant kingdom on the basis of :-
(1) Floral morphology
(2) Overall morphology of plants
(3) Type of sexual reproduction
(4) Anatomical character
BBC007

8. First plant classification was given by :-
(1) Linnaeus (2) John–Ray
(3) Aristotle (4) Darwin
BBC008

9. "*Systema Naturae*" book was written by :-
(1) Angler and Prantle (2) Darwin
(3) Linnaeus (4) Oswald - Tippo
BBC009

10. According to Whittaker kingdom Protista includes :-
(1) Prokaryotes
(2) Unicellular eukaryotes
(3) Slime molds & protozoa
(4) Multicellular & eukaryotes
BBC010

KINGDOM – MONERA

11. Infoldings of plasma membrane in bacteria are called as :-
(1) Episomes (2) Plasmid
(3) Pili (4) Mesosomes
BBC011

12. The organisms participating most actively in nitrogen cycle in nature are :-
(1) Bacteria (2) Legumes
(3) Parasitic algae (4) Fungi
BBC012

13. Heterocyst is a structure which is associated with :-
(1) Reproduction
(2) Respiration
(3) Nitrogen fixation
(4) Locomotion

BBC013

14. *Trichodesmium erythraeum* which imparts red colour to sea water of red sea is a :-
(1) Cyanobacterium (2) Red Algae
(3) Diatom (4) Red Coral

BBC014

15. Archaebacterial cell lacks :-
(1) Peptidoglycan
(2) DNA
(3) Ribosomes
(4) Branched Chain Lipids

BBC015

16. Most common method of reproduction in prokaryotes :-
(1) Budding (2) Binary fission
(3) Transduction (4) Conjugation

BBC016

17. Harmful activity of Blue green algae is:-
(1) Denitrification
(2) Water - bloom
(3) Increase alkalinity of soil
(4) Decrease fertility of soil

BBC017

18. The function of mesosome in prokaryotes is:-
(1) Aerobic respiration
(2) Cell wall formation
(3) Both (1) and (2)
(4) N_2 - fixation

BBC018

19. During rainy season ground surface become slippery due to :-
(1) Fungi (2) Blue green algae
(3) Bryophytes (4) Gymnosperm

BBC019

20. Photosynthesis of Blue green algae is:-
(1) Oxygenic
(2) Non oxygenic
(3) Both oxygenic and non-oxygenic
(4) None

BBC020

21. Which of the following is the only group of organisms capable of using inorganic compounds as source of energy ?
(1) Fungi
(2) Chemo autotrophs
(3) Both the above
(4) None of the above

BBC021

22. Link between prokaryotes and multicellular eukaryotes :–
(1) Cyanobacteria (2) Protista
(3) Fungi (4) Plants

BBC022

23. Which structure of prokaryotes is analogous to lysosome ?
(1) Mesosome
(2) Genophore
(3) Periplasmic space
(4) Perinuclear space

BBC023

24. Which of the following performs respiration with the help of plasma membrane ?
(1) Bacteria (2) Algae
(3) Fungi (4) All the above

BBC024

25. Richest source of bacteria is : -
(1) Air (2) Soil (3) Water (4) Milk
BBC025

26. The most primitive monerans are :-
(1) Archaebacteria
(2) Eubacteria
(3) Filamentous bacteria
(4) Cyanobacteria
BBC026

27. Organisms which obtain energy by the oxidation of reduced inorganic compounds are called
(1) Photo autotrophs (2) Chemo autotrophs
(3) Saprozoic (4) Heterotrophs
BBC027

28. Which bacteria are utilized in Gobar gas plant?
(1) Methanogens
(2) Nitrifying bacteria
(3) Ammonifying bacteria
(4) Denitrifying bacteria
BBC028

29. Plasmid are :-
(1) Virus
(2) New types of micro organism
(3) Extra chromosomal genetic material of bacteria
(4) Essential bacterial genetic materials
BBC029

30. A free living aerobic bacteria capable of fixing nitrogen is :-
(1) *Azotobactor*
(2) *Rhizobium*
(3) *Clostridium botulinum*
(4) *Streptomyces*
BBC030

31. Which one of the following fixes CO_2 in to carbohydrates ?
(1) *Rhizobium*
(2) *E.coli*
(3) *Bacillus*
(4) *Rhodospirillum*
BBC032

32. Antibiotics are mostly obtained from :-
(1) Bacteria (2) Viruses
(3) Fungi (4) Angiosperm
BBC033

33. The main difference between gram +ve and gram –ve resides in the composition of :-
(1) Cilia (2) Cell-wall
(3) Cell-membrane (4) Cytoplasm
BBC034

34. Free living nitrogen-fixing bacteria are found in-
(1) Air (2) Soil
(3) Root nodules (4) None of above
BBC035

35. Cell membrane of bacteria is made up of -
(1) Cellulose and lipid
(2) Chitin
(3) Lipid + Protein
(4) Protein and Cellulose
BBC036

36. Which organism is most useful for soil fertility ?
(1) Bryophyta (2) Fungi
(3) Bacteria (4) Bacteriophage
BBC037

37. Bacterial flagella are made of-
(1) Carbohydrate (2) Lipid
(3) Protein (4) Amide
BBC038

38. The mode of the nutrition of bacteria is usually
(1) Photo autotrophic
(2) Chemo autotrophic
(3) Heterotrophic and autotrophic
(4) None
BBC039

39. Fertility of soil is increased by-
(1) Nitrogen - fixing bacteria
(2) Denitrifying bacteria
(3) Plasmalemma
(4) Cell membrane
BBC040

40. Plant pathogenic bacteria are :-
(1) Gram +ve (2) Gram –ve
(3) Both (1) & (2) (4) None
BBC041

41. Souring of milk is due to -
(1) Aerobic bacteria (2) Anaerobic bacteria
(3) Both (1) & (2) (4) None
BBC042

42. At which place bacteria are not found :
(1) Soil (2) Ice
(3) Sea (4) Distilled water
BBC043

43. Halophile are _____.
(1) Those can live at low pH
(2) Those can live at high salt concentration
(3) Those can live at high pressure
(4) Those can live at high light intensity
BBC044

44. The cyanobacteria are also referred to as?
(1) Slime moulds
(2) Blue green algae
(3) Decomposer protist
(4) Cell wall less prokaryote
BBC045

45. Besides paddy fields, cyanobacteria are also found inside vegetative part of:
(1) *Psilotum* (2) *Pinus*
(3) *Cycas* (4) *Equisetum*
BBC046

46. Cyanobacteria are _____.
(1) Chemosynthetic
(2) Oxygenic photosynthetic
(3) Anaerobic
(4) Facultative anaerobic
BBC047

KINGDOM-PROTISTA

47. "Golden Algae" is the common name of Algae belonging to :-
(1) Chrysophyta (2) Pyrrophyta
(3) Euglenophyta (4) Cyanophyta
BBC048

48. Armoured cell wall and biflagellated cells are characteristic of :-
(1) Chrysophyta (2) Pyrrophyta
(3) Euglenophyta (4) Cyanophyta
BBC049

49. Oils and Leucosine are characteristic stored food in :
(1) Dinoflagellates (2) Euglenoids
(3) Diatoms (4) None
BBC050

50. Armoured algae are :
(1) Dinoflagellates (2) Euglenoids
(3) Red algae (4) Cyanobacteria
BBC051

51. The diatoms do not easily decay like most of the other Algae because:
(1) They have water proof cells
(2) Their walls are mucilagenous
(3) They have highly siliceous wall
(4) They are non-living
BBC052

52. 'Keiselgurh' a heat resistant material is obtained from :-
(1) Red Algae (2) Brown Algae
(3) Diatoms (4) Fungi
BBC053

53. The diatomaceous earth is used to insulate boilers and steam pipes because:-
(1) The wall of diatoms is deposited with calcium.
(2) The diatomaceous earth is cheap.
(3) It is a good conductor of heat.
(4) The wall of diatoms is made of silica.
BBC054

54. Shell of diatoms is made up of :-
(1) Silica (2) Calcium carbonate
(3) Keratin (4) Calcium oxalate
BBC055

55. "Diatomite" (Keiselgurh) is obtained from:-
(1) Myxophyceae (2) Chrysophyta
(3) Phaeophyceae (4) Rhodophyceae
BBC056

56. Most characteristic feature of diatoms is :-
(1) Pigments
(2) Stored food
(3) Cell wall
(4) Non oxygenic photosynthesis
BBC057

57. Taxonomically the most controversial group is-
(1) Dinoflagellates (2) Diatoms
(3) Euglenoids (4) Prokryote
BBC058

58. Decomposer protists are :-
(1) Diatoms (2) Dinoflagellates
(3) Slime moulds (4) Euglenoid
BBC059

59. The dead remains of diatoms are known as:-
(1) Coenobium (2) Sporangia
(3) Keiselgurh (4) Sporocarp
BBC060

60. Taxonomists feel difficulty in classification of :-
(1) Procaryotes
(2) Unicellular eucaryotes
(3) Plants
(4) Animals
BBC061

61. Which of the following unicellular algae reproduce by auxospores, have silicified cell wall and store food in the form of fats, leucosine and chrysolaminarin ?
(1) Diatoms (2) Red algae
(3) Dinoflagellates (4) Euglenoids
BBC062

62. Paramylum is stored food of :-
(1) Dinoflagellate (2) Euglenoid
(3) Diatom (4) Slime mould
BBC063

63. The most efficient locomotion in protists is through :-
(1) Pseudopodia (2) Flagella
(3) Cilia (4) Tentacles
BBC064

64. Organism of which kingdom feed like animals and perform photosynthesis like plants :-
(1) Monera (2) Protista
(3) Mycota (4) Animalia
BBC065

65. "Fire algae" belongs to group :-
(1) Pyrrophyta (2) Chrysophyta
(3) Euglenophyta (4) Rhodophyta
BBC066

66. Slime mould is known as naked fungi.
(1) Cell membrane absent
(2) Cell wall absent
(3) Cell wall and cell membrane absent
(4) Never naked
BBC067

67. Dinoflagellates are called fire algae due to which character :-
(1) They appear like fire due to pigments
(2) They produce fire due to friction
(3) They occur on burnt places
(4) They show bioluminescence
BBC068

68. Toxins (Saxitoxins) secreted by some dinoflagellates enter the body of human beings through food chain and result in :-
(1) Madness (2) Paralysis
(3) Syphilis (4) Plague
BBC069

69. Stored food of Diatoms :-
(1) Leucosin (2) Starch
(3) Floridian starch (4) Glycogen
BBC070

70. Dead remains of diatoms at sea bed are called :-
(1) Keiselgurh (2) Peat
(3) Coral reefs (4) Sporopollenin
BBC071

71. Protists include :-
(1) Unicellular Eukaryotes
(2) Prokaryotes
(3) Multicellular eukaryotes
(4) Multicellular prokaryotes
BBC072

72. Which of the following eukaryotes are devoid of histone proteins ?
(1) Golden algae (2) Euglenoids
(3) Fire algae (4) Slime Mould
BBC073

73. Diatoms perform which type of movement in water :-
(1) Swimming (2) Amoeboid
(3) Floating (4) Ciliary
BBC074

74. Protist used for the construction of sound proof rooms, is :-
(1) Dinoflagellate (2) Diatoms
(3) Euglenoids (4) Zooflagellates
BBC075

75. Auxospores are formed by :-
(1) Diatoms (2) Euglenoids
(3) Dinoflagellates (4) bacteria
BBC076

76. Protists which are diploid reproduce sexually by the process of :-
(1) Zygotic meiosis (2) Cyst formation
(3) Binary fission (4) Gametic meiosis
BBC077

77. 'Red tides' are produced by :-
(1) Red algae (2) Dinoflagellates
(3) Diatoms (4) Brown algae
BBC078

78. In which of the following nuclear dimorphism is seen?
(1) *Entamoeba* (2) *Leishmania*
(3) *Trypanosoma* (4) *Paramoecium*
BBC461

79. Slipper animalcule is -
(1) *Entamoeba* (2) *Paramoecium*
(3) *Giardia* (4) *Euglena*
BBC462

80. Which group of organisms belongs to class sporozoa?
(1) *Plasmodium & Monocystis*
(2) *Entamoeba & Paramoecium*
(3) *Euglena & Trypanosoma*
(4) *Plasmodium & Vorticella*
BBC463

81. What is common about *Trypanosoma, Noctiluca, Monocystis* and *Giardia* ?
(1) These are all parasites.
(2) These are all unicellular protists.
(3) They have flagella.
(4) They produce spores.

BBC464

82. Which of the following unicellular organism has a macronucleus for trophic function and one or more micronuclei for reproduction?
(1) *Trypanosoma* (2) *Paramoecium*
(3) *Euglena* (4) *Amoeba*

BBC465

KINGDOM – FUNGI

83. Fungal hyphae penetrate hard cell walls of their hosts with the help of:-
(1) Enzymes (2) Hormones
(3) Sharp tips (4) Sugar Exudates

BBC079

84. A fungus completing its life cycle on a single host is known as:-
(1) Dikaryotic (2) Autoecious
(3) Heterocious (4) Heterothallic

BBC080

85. Which of the following secretes toxins during storage conditions of crop plants ?
(1) *Aspergillus* (2) *Penicillium*
(3) *Fusarium* (4) *Colletotrichum*

BBC081

86. Which of the following characters indicate similarity between fungi and animals ?
(1) Heterotrophic nutrition
(2) Type of stored food
(3) Presence of chitin
(4) All the above

BBC082

87. The sac fungi belongs to :-
(1) Ascomycetes (2) Basidiomycetes
(3) Phycomycetes (4) Deuteromycetes

BBC083

88. *Neurospora*, which is popularly known as *Drosophilla* of plant kingdom, belongs to :-
(1) Phycomycetes (2) Ascomycetes
(3) Basidiomycetes (4) Deuteromycetes

BBC084

89. The basidiomycetes includes :-
(1) Rusts (2) Smuts
(3) Mushrooms (4) All the above

BBC085

90. Which of the following causes wheat rust disease?
(1) A red alga (2) A green alga
(3) A fungus (4) Mycoplasma

BBC086

91. Sexual cycle is absent in :-
(1) Phycomycetes (2) Deuteromycetes
(3) Ascomycetes (4) Basidiomycetes

BBC087

92. *Penicillium* fungi belong to class :-
(1) Zygomycetes (2) Oomycetes
(3) Deuteromycetes (4) Ascomycetes

BBC088

93. The fungi are :-
(1) Autotrophic (2) Holotrophic
(3) Chemotrophic (4) Heterotrophic

BBC089

94. All fungi are :-
(1) With chlorophyll
(2) Without chlorophyll
(3) With carotene
(4) Wall less

BBC090

95. In class phycomycetes the mycelium is :-
(1) Coenocytic and aseptate.
(2) Coenocytic and septate.
(3) Uninucleate and aseptate.
(4) Multinucleate and septate.
BBC091

96. Coenocytic mycelium is found in :-
(1) *Rhizopus* (2) *Mucor*
(3) *Penicillium* (4) Both (1) and (2)
BBC092

97. Stored food material of fungi :-
(1) Cellulose
(2) Starch
(3) Glycogen and starch
(4) Glycogen and oil
BBC093

98. The cell wall of Fungi is composed of:-
(1) Chitin (2) Cellulose
(3) Mucopeptide (4) Pseudomurein
BBC094

99. The chief characteristic of class ascomycetes is:
(1) Formation of spores
(2) Hyphae
(3) Formation of ascospores
(4) Formation of zoospores
BBC095

100. Which is commonly called '*Drosophilla* of plant kingdom' ?
(1) *Morchella* (2) *Neurospora*
(3) *Rhizopus* (4) *Claviceps*
BBC096

101. Normally how many ascospores are formed in an ascus :-
(1) 4 - ascospores (2) 8 - ascospores
(3) 16 ascospores (4) 24 ascospores
BBC097

102. Edible part in mushrooms is :-
(1) Basidiospores
(2) Mycelium
(3) Pseudomycelium
(4) Complete basidiocarp
BBC098

103. Penicillin is obtained from:-
(1) *Chara* (2) *Penicillium*
(3) *Aspergillus* (4) *Albugo*
BBC100

104. Cell wall of Chitin is found in :-
(1) Fungi (2) Bryophyta
(3) Bacteria (4) Angiosperms
BBC101

105. The fungus without mycelium is :-
(1) *Phytophthora* (2) *Rhizopus*
(3) *Saccharomyces* (4) *Microsporum*
BBC102

106. Pseudomycelium occurs in :–
(1) Mushroom (2) *Mucor*
(3) Bread mould (4) Yeast
BBC104

107. Occurrence of dikaryotic mycelium mainly is the characteristic of :–
(1) Myxomycetes
(2) Phycomycetes
(3) Deuteromycetes
(4) Basidiomycetes
BBC105

108. Deuteromycetes are called 'Imperfect fungi' as:
(1) They have no cell wall
(2) No mycelium
(3) No sexual reproduction
(4) No asexual reproduction
BBC106

109. Absorptive mode of nutrition is found in :-
(1) Algae (2) Fungi
(3) Bryophytes (4) Euglenoids
BBC109

110. Which of the following is called 'toad stools'?
(1) All mushrooms
(2) Edible mushrooms
(3) Poisonous mushrooms
(4) None **BBC110**

111. Yeast grows more quickly in :–
(1) Salt water
(2) Sugar solution
(3) Double distilled water
(4) Marine water **BBC111**

112. Which of the following is a form class ?
(1) Deuteromycetes (2) Basidiomycetes
(3) Rhodophyceae (4) Euglenophyceae
BBC112

113. Ergot fungi belongs to :–
(1) Ascomycetes (2) Basidiomycetes
(3) Phycomycetes (4) Deuteromycetes
BBC113

114. Fungi are ecologically important because :-
(1) They yield antibiotics
(2) They are used in genetic studies
(3) They function as decomposers
(4) All the above **BBC115**

115. Citric acid is obtained from :-
(1) *Streptomyces* (2) *Aspergillus*
(3) *Penicillium* (4) *Albugo*
BBC116

116. In fungi lump of hyphae is referred to as :-
(1) Thallus (2) Haustorium
(3) Mycelium (4) Archegonia
BBC117

117. Plant group which shows Heterotrophic mode of nutrition is :-
(1) Algae (2) Fungi
(3) Bryophytes (4) Pteridophytes
BBC118

118. Sexual reproduction is absent in :-
(1) Phycomycetes (2) Deuteromycetes
(3) Zygomycetes (4) Basidiomycetes
BBC119

119. Non-septate mycelium occurs in :-
(1) Phycomycetes (2) Ascomycetes
(3) Basidiomycetes (4) Deuteromycetes
BBC120

120. Basidiocarp is present in:-
(1) Basidiomycetes (2) Ascomycetes
(3) Deuteromycetes (4) Phycomycetes
BBC121

121. All fungi are:-
(1) Symbionts (2) Parasites
(3) Saprophytes (4) Heterotrophs
BBC122

122. Aspergillosis is caused by :–
(1) Virus (2) Bacteria
(3) Fungi (4) Mycoplasma
BBC123

VIRUS, VIROIDS, LICHEN, MYCORRHIZA

123. Find the incorrect statements from the followings :-
(1) Growth, reproduction and consciousness are unique features of living organisms.
(2) Growth and reproduction are defining characteristics of living organisms.
(3) Metabolism, cellular organisation and consciousness are defining characters of livings.
(4) Living organisms are self replicating, evolving and self regulating.
BBC124

124. Which is a group of organisms in which reproduction is synonymous with growth :-
(1) *Amoeba*, bacteria and unicellular algae
(2) Algae, fungi and lichen
(3) Lower organisms
(4) Higher multicellular organisms

BBC125

125. Why viruses did not find a place in classification?
(1) They are obligate parasites, cannot survive without host.
(2) Viruses are unicellular organisms so can not be included in livings.
(3) They are not truly living because they don't have cell structure.
(4) They can pass through the bacteria proof filter

BBC126

126. Which of the followings is/are symbiotic associations between autotrophic and heterotrophic organisms only
(A) Mycorrhiza between fungi and roots of higher plants
(B) Lichen between phycobiont and mycobiont.
(C) Coralloid roots of *Cycas* having cyanobacteria.
(D) Obligate association between *Pinus* roots and fungi making mycorrhizae.
(1) A & B (2) B & C
(3) A, B & C (4) A, B & D

BBC127

127. Bacteriophages are made of both proteins and genetic material. Their genetic material is generally-
(1) ss DNA (2) ds DNA
(3) ss RNA (4) ds RNA

BBC128

128. Find the correct from the followings.
(1) T.O. Diener discovered a new infectious agent, smaller than virus and named them as prions.
(2) Viroids lack the protein coat and made of only free RNA.
(3) The RNA of viroids is of high molecular weight.
(4) Lichens are very good pollution indicators as they grow only in polluted areas.

BBC129

129. Match the column A with column B and find out the correct as shown ?

	Column-A	**Column-B**
a.	Pollution indicator	p. Fungal
b.	Mycobiont component	q. Viroid
c.	Potato spindle tuber disease	r. Lichen
d.	Obligate parasite	s. Virus

(1) a - r, b - p, c - s, d - q
(2) a - r, b - s, c - p, d - q
(3) a - s, b - r, c - p, d - q
(4) a - r, b - p, c - q, d - s

BBC136

130. In lichen fungal component is responsible to provide?
(1) Shelter to phycobiont
(2) Mineral absorption
(3) Water absorption
(4) All of the above

BBC137

131. The protein coat of viruses is called :-
(1) Capsomeres (2) Capsid
(3) Sheath (4) Envelope

BBC141

132. Lichens are :-

(1) Symbiotic association between algae and fungi

(2) Pollution indicators

(3) Not placed in any kingdom by R.H. Whittaker

(4) All of these are correct

BBC142

133. Phycobiont is a :

(1) algal, autotrophic component of mycorrhiza

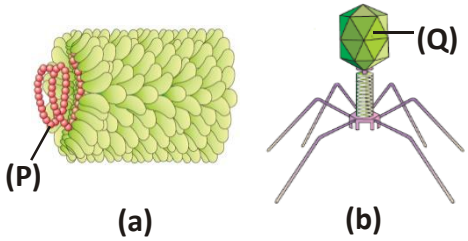
(2) fungal, heterotrophic component of lichen

(3) fungal component of mycorrhiza

(4) algal, autotrophic component of lichen

BBC143

134.



Identify the (a) & (b) figure and their (P) and (Q) part.

(1) a - Tobacco Mosaic virus P → RNA
b - Bacteriophage Q → Head

(2) a - Bacteriophage P → RNA
b - Tobacco Mosaic virus Q → Head

(3) a - Potato Mosaic virus P → Head
b - Cyanophage Q → RNA

(4) a - Smallpox virus P → RNA
b - Influenza virus Q → Head

BBC144

135. Choose the **incorrect** pair :-

(1) Virus – Nucleocapsid

(2) Viroids – Only RNA

(3) Lichen – Algae and fungi

(4) Mycorrhiza – BGA and higher plants

BBC145

136. Which of the given organism(s) belong to kingdom protista according to five kingdom system?

(A) Virus (B) *Euglena*

(C) Viroids (D) Lichen

(1) A and B (2) C and D

(3) Only B (4) A and C

BBC146

137. The transfer of genetic material from one bacterium to other bacterium from virus, is called:-

(1) Conjugation (2) Transduction

(3) Transformation (4) Crossing over

BBC147

138. Read the following statements :

(i) Mycorrhiza is the symbiotic association between roots of higher plants and fungus.

(ii) Lichens are biological indicator for water pollution.

(iii) Viroids contain only low molecular weight DNA and protein.

(iv) W.M. Stanley crystallised TMV

Which of the above statements are correct:

(1) (i), (ii) and (iv)

(2) (i) and (iv)

(3) (ii) and (iii)

(4) (i) and (ii)

BBC148

139. Which of the following benefit is not provided by fungi to symbiotic plant ?

(1) Protection from pathogen

(2) Availability of water

(3) Availability of phosphate

(4) Ability to fix nitrogen

BBC149

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	4	4	2	1	3	3	2	4	1	3	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	3	2	1	2	2	3	1	2	1	2	1	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	2	2	3	3	3	3	1	3	2	4	2	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	2	3	1	3	3	4	1	2	3	3	3	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	2	3	2	1	2	4	2	1	1	1	3	3	2	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	2	4	2	1	2	2	1	2	1	4	1	2	4	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	4	4	2	1	4	4	1	3	2	2	4	2	1	3
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	4	4	3	2	3	2	1	1	3	2	3	2	2	1	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	4	3	2	1	3	4	2	2	4	4	2	4	4	1	4
Question	136	137	138	139											
Answer	3	2	2	4											

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Five kingdom system of classification suggested by R.H. Whittaker is **not** based on
(1) Presence or absence of a well defined nucleus.
(2) Mode of reproduction.
(3) Mode of nutrition.
(4) Complexity of body organisation.

BBC202

2. Archaebacteria differ from eubacteria in :
(1) Cell membrane
(2) Mode of nutrition
(3) Cell shape
(4) Mode of reproduction

BBC204

3. Anoxygenic photosynthesis is characteristic of :
(1) *Rhodospirillum*
(2) *Spirogyra*
(3) *Chlamydomonas*
(4) *Ulva*

BBC205

4. Which of the following shows coiled RNA strand and capsomeres?
(1) Polio virus
(2) Tobacco masaic virus
(3) Measles virus
(4) Retrovirus

BBC206

5. Transformation was discovered by :-
(1) Meselson and Stahl
(2) Hershey and Chase
(3) Griffith
(4) Watson and Crick

BBC207

6. Viruses have :-
(1) DNA enclosed in a protein coat
(2) Prokaryotic nucleus
(3) Single chromosome
(4) Both DNA and RNA

BBC208

7. Which one of the following living organisms completely *lacks* a cell wall?
(1) Cyanobacteria
(2) Sea–fan (*Gorgonia*)
(3) *Saccharomyces*
(4) Blue–green algae

BBC209

AIPMT 2015

8. Which one of the following matches is **correct** ?

(1)	*Alternaria*	Sexual reproduction absent	Deuteromycetes
(2)	*Mucor*	Reproduction by Conjugation	Ascomycetes
(3)	*Agaricus*	Parasitic fungus	Basidiomycetes
(4)	*Phytophthora*	Aseptate mycelium	Basidiomycetes

BBC210

9. True nucleus is absent in :-
(1) *Mucor* (2) *Vaucheria*
(3) *Volvox* (4) *Anabaena*

BBC211

10. Which one of the following is **not** an inclusion body found in prokaryotes ?
(1) Cyanophycean granule
(2) Glycogen granule
(3) Polysome
(4) Phosphate granule

BBC212

11. The guts of cow and buffalo possess:
(1) *Chlorella* spp. (2) Methanogens
(3) Cyanobacteria (4) *Fucus* spp.
BBC213

Re-AIPMT 2015

12. Which of the following structures is **not** found in prokaryotic cells?
(1) Plasma membrane (2) Nuclear envelope
(3) Ribosome (4) Mesosome
BBC214

13. Choose the **wrong** statement :
(1) Yeast is unicellular and useful in fermentation.
(2) *Penicillium* is multicellular and produces antibiotics.
(3) *Neurospora* is used in the study of biochemical genetics.
(4) Morels and truffles are poisonous mushrooms.
BBC215

14. Which of the following are **not** membrane-bound?
(1) Mesosomes (2) Vacuoles
(3) Ribosomes (4) Lysosomes
BBC216

15. Cell wall is absent in :
(1) *Nostoc* (2) *Aspergillus*
(3) *Funaria* (4) Mycoplasma
BBC217

16. In which group of organisms the cells walls form two thin overlapping shells which fit together ?
(1) Slime moulds (2) Chrysophytes
(3) Euglenoids (4) Dinoflagellates
BBC218

17. Chromatophores take part in :
(1) Respiration (2) Photosynthesis
(3) Growth (4) Movement
BBC220

18. The structures that help some bacteria to attach to rocks and/or host tissues are :
(1) Holdfast (2) Rhizoids
(3) Fimbriae (4) Mesosomes
BBC221

19. The imperfect fungi which are decomposers of litter and help in mineral cycling belong to:
(1) Ascomycetes (2) Deuteromycetes
(3) Basidiomycetes (4) Phycomycetes
BBC222

20. Pick up the **wrong** statement :
(1) Nuclear membrane is present in Monera
(2) Cell wall is absent in Animalia.
(3) Protista have photosynthetic and heterotrophic modes of nutrition.
(4) Some fungi are edible.
BBC223

NEET-I 2016

21. Which one of the following statements is wrong ?
(1) Cyanobacteria are also called blue-green algae.
(2) Golden algae are also called desmids.
(3) Eubacteria are also called false bacteria.
(4) Phycomycetes are also called algal fungi
BBC224

22. Chrysophytes, Euglenoids, Dinoflagellates and Slime moulds are included in the kingdom:-
(1) Monera (2) Protista
(3) Fungi (4) Animalia
BBC225

23. The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals, include the :-

(1) Halophiles (2) Thermoacidophiles

(3) Methanogens (4) Eubacteria

BBC227

NEET-II 2016

24. Methanogens belong to :

(1) Dinoflagellates

(2) Slime moulds

(3) Eubacteria

(4) Archaebacteria

BBC228

25. Select the **wrong** statement :-

(1) Diatoms are chief producers in the oceans.

(2) Diatoms are microscopic and float passively in water.

(3) The walls of diatoms are easily destructible.

(4) 'Diatomaceous earth' is formed by the cell walls of diatoms.

BBC229

26. Select the **mismatch** :-

(1) Protists-Eukaryotes

(2) Methanogens-Prokaryotes

(3) Gas vacuoles-Green bacteria

(4) Large central vacuoles - Animal cells

BBC230

27. Select the **wrong** statement :-

(1) Cyanobacteria lack flagellated cells.

(2) *Mycoplasma* is a wall-less microorganism

(3) Bacterial cell wall is made up of peptidoglycan.

(4) Pilli and fimbriae are mainly involved in motility of bacterial cells

BBC231

28. Which of the following sets of diseases is caused by bacteria?

(1) Tetanus and mumps

(2) Herpes and influenza

(3) Cholera and tetanus

(4) Typhoid and smallpox

BBC233

NEET (UG) 2017

29. Which of the following are found in extreme saline conditions ?

(1) Eubacteria (2) Cyanobacteria

(3) Mycobacteria (4) Archaebacteria

BBC234

30. Select the mismatch :-

(1) *Rhodospirillum* — Mycorrhiza

(2) *Anabaena* — Nitrogen fixer

(3) *Rhizobium* — Alfalfa

(4) *Frankia* — *Alnus*

BBC235

31. Which of the following components provides sticky character to the bacterial cell ?

(1) Nuclear membrane

(2) Plasma membrane

(3) Glycocalyx

(4) Cell wall

BBC236

32. Viroids differ from viruses in having :-

(1) DNA molecules without protein coat

(2) RNA molecules with protein coat

(3) RNA molecules without protein coat

(4) DNA molecules with protein coat

BBC237

33. Which of the following is correctly matched for the product produced by them ?

(1) *Methanobacterium* : Lactic acid

(2) *Penicillium notatum* : Acetic acid

(3) *Saccharomyces cerevisiae* : Ethanol

(4) *Acetobacter aceti* : Antibiotics

BBC238

34. Which among the following are the smallest living cells, known without a definite cell wall, pathogenic to plants as well as animals and can survive without oxygen ?

(1) *Pseudomonas* (2) *Mycoplasma*

(3) *Nostoc* (4) *Bacillus*

BBC239

NEET (UG) 2018

35. Which among the following is ***not*** a prokaryote?

(1) *Saccharomyces*

(2) *Mycobacterium*

(3) *Nostoc*

(4) *Oscillatoria*

BBC240

36. Select the ***wrong*** statement :-

(1) Cell wall is present in members of fungi and plantae

(2) Mushrooms belong to basidiomycetes

(3) Pseudopodia are locomotory and feeding structures in sporozoans

(4) Mitochondria are the powerhouse of the cell in all kingdoms except Monera

BBC241

37. After karyogamy followed by meiosis, spores are produced exogenously in

(1) *Neurospora* (2) *Alternaria*

(3) *Agaricus* (4) *Saccharomyces*

BBC242

38. Oxygen is ***not*** produced during photosynthesis by :-

(1) Green sulphur bacteria

(2) *Nostoc*

(3) *Cycas*

(4) *Chara*

BBC243

39. Which of the following organisms are known as chief producers in the oceans ?

(1) Dinoflagellates

(2) Diatoms

(3) Cyanobacteria

(4) Euglenoids

BBC244

NEET (UG) 2019

40. Which of the following statements is **incorrect**?

(1) Morels and truffles are edible delicacies.

(2) *Claviceps* is a source of many alkaloids and LSD.

(3) Conidia are produced exogenously and ascospores endogenously.

(4) Yeasts have filamentous bodies with long thread-like hyphae.

BBC245

41. Which of the following statements is **incorrect** ?

(1) Viroids lack a protein coat.

(2) Viruses are obligate parasites.

(3) Infective constituent in viruses is the protein coat.

(4) Prions consist of abnormally folded proteins.

BBC246

42. *Thiobacillus* is a group of bacteria helpful in carrying out :

(1) Nitrogen fixation

(2) Chemoautotrophic fixation

(3) Nitrification

(4) Denitrification

BBC247

43. Match **Column-I** with **Column-II.**

Column-I	Column-II
(a) Saprophyte	(i) Symbiotic association of fungi with plant roots
(b) Parasite	(ii) Decomposition of dead organic materials
(c) Lichens	(iii) Living on living plants or animals
(d) Mycorrhiza	(iv) Symbiotic association of algae and fungi

Choose the **correct** answer from the options given below :-

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(ii)	(i)	(iii)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

BBC248

NEET (UG) 2019 (ODISHA)

44. Match the organisms in column-I with habitats in column-II.

Column-I		Column-II	
(a)	Halophiles	(i)	Hot springs
(b)	Thermoacidophiles	(ii)	Aquatic environment
(c)	Methanogens	(iii)	Guts of ruminants
(d)	Cyanobacteria	(iv)	Salty area

Select the **correct** answer from the options given below :-

(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

BBC249

45. Mad cow disease in cattle is caused by an organism which has :-

(1) inert crystalline structure

(2) abnormally folded protein

(3) free RNA without protein coat

(4) free DNA without protein coat

BBC250

46. Which of the following statements is correct ?

(1) Lichens do not grow in polluted areas.

(2) Algal component of lichens is called mycobiont.

(3) Fungal component of lichens is called phycobiont

(4) Lichens are not good pollution indicators.

BBC251

47. Which of the following bacteria reduce nitrate in soil into nitrogen ?

(1) *Nitrobacter* (2) *Nitrococcus*

(3) *Thiobacillus* (4) *Nitrosomonas*

BBC252

48. Among the following pairs of microbes, which pair has both the microbes that can be used as biofertilizers?

(1) *Aspergillus* and *Rhizopus*

(2) *Rhizobium* and *Rhizopus*

(3) *Cyanobacteria* and *Rhizobium*

(4) *Aspergillus* and *Cyanobacteria*

BBC253

49. Which of the following statements about methanogens is not correct?

(1) They can be used to produce biogas.

(2) They are found in the rumen of cattle and their excreta.

(3) They grow aerobically and breakdown cellulose-rich food.

(4) They produce methane gas.

BBC254

NEET (UG) 2020

50. Which of the following is correct about viroids?

(1) They have free DNA without protein coat.

(2) They have RNA with protein coat.

(3) They have free RNA without protein coat.

(4) They have DNA with protein coat.

BBC256

51. Which of the following statements about inclusion bodies is **incorrect** ?

(1) These represent reserve material in cytoplasm.

(2) They are not bound by any membrane.

(3) These are involved in ingestion of food particles.

(4) They lie free in the cytoplasm.

BBC257

NEET (UG) 2020 (COVID-19)

52. Inclusion bodies of blue- green, purple and green photosynthetic bacteria are :-

(1) Contractile vacuoles

(2) Gas vacuoles

(3) Centrioles

(4) Microtubules

BBC258

53. Which of the following is **incorrect** about Cyanobacteria ?

(1) They are photoautotrophs.

(2) They lack heterocyst.

(3) They often form blooms in polluted water bodies.

(4) They have chlorophyll A similar to green plants.

BBC259

NEET (UG) 2021

54. Which of the following statements is correct?

(1) Fusion of two cells is called Karyogamy.

(2) Fusion of protoplasm between two motile on non-motile gametes is called plasmogamy.

(3) Organisms that depend on living plants are called saprophytes.

(4) Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.

BBC262

NEET (UG) 2022

55. **Statement-I:** Mycoplasma can pass through less than 1 micron filter size.

Statement-II : Mycoplasma are bacteria with cell wall.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and II are correct.

BBC264

56. Which of the following is a **correct** statement?

(1) Bacteria are exclusively heterotrophic organisms.

(2) Slime moulds are saprophytic organisms classified under Kingdom Monera.

(3) Mycoplasma have DNA, Ribosome and cell wall.

(4) Cyanobacteria are a group of autotrophic organisms classified under Kingdom Monera.

BBC265

Re-NEET (UG) 2022

57. Mad cow disease in cattle and Cr Jacob disease in humans are due to infection by_____.

(1) Bacterium (2) Virus

(3) Viroid (4) Prion

BBC266

NEET (UG) 2023

58. Match **List-I** with **List-II.**

	List-I		List-II
A.	Ringworm	I.	*Haemophilus influenzae*
B.	Filariasis	II.	*Trichophyton*
C.	Malaria	III.	*Wuchereria bancrofti*
D.	Pneumonia	IV.	*Plasmodium vivax*

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-II, C-IV, D-I

(4) A-II, B-III, C-IV, D-I

BBC390

59. **Statement-I :** RNA mutates at a faster rate.

Statement-II : Viruses having RNA genome and shorter life span mutate and evolve faster.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) Both Statement-I and Statement-II are false.

(2) Statement I is true but Statement II is false.

(3) Statement I false but Statement II is true.

(4) Both Statement I and Statement II are true.

BBC391

NEET (UG) 2024

60. Which one of the following is not a criterion for classification of fungi?

(1) Morphology of mycelium

(2) Mode of nutrition

(3) Mode of spore formation

(4) Fruiting body

BBC398

61. Match **List-I** with **List-II**.

	List-I		List-II
A.	*Rhizopus*	I.	Mushroom
B.	*Ustilago*	II.	Smut fungus
C.	*Puccinia*	III.	Bread mould
D.	*Agaricus*	IV.	Rust fungus

Choose the **correct** answer from the options given below :-

(1) A-III, B-II, C-IV, D-I

(2) A-I, B-III, C-II, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-IV, B-III, C-II, D-I

BBC399

RE-NEET (UG) 2024

62. Match **List-I** with **List-II**.

	List-I (Organisms)		List-II (Mode of Nutrition)
A.	Euglenoid	I.	Parasitic
B.	Dinoflagellate	II.	Saprophytic
C.	Slime mould	III.	Photosynthetic
D.	*Plasmodium*	IV.	Switching between photosynthetic and heterotrophic mode

Choose the **correct** answer from the options given below :-

(1) A-III, B-IV, C-II, D-I

(2) A-IV, B-II, C-I, D-III

(3) A-IV, B-III, C-II, D-I

(4) A-IV, B-II, C-III, D-I

BBC400

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	2	3	1	2	1	4	3	2	2	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	3	2	1	3	2	3	4	3	4	4	3	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	3	2	1	3	3	1	2	4	3	4	4	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	3	3	3	3	3	2	2	2	2	4	4	4	4	2
Question	61	62													
Answer	1	3													

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Archaebacteria differ from other bacteria in having a:-
(1) Different cell wall and cell membrane structure
(2) Different cellular organization
(3) Parasitic nature
(4) Both (2) and (3)
BBC267

2. Nitrogen fixing cyanobacteria are :-
(1) *Rhizobium, Nostoc*
(2) *Rhizobium, Anabaena*
(3) *Nostoc, Anabaena*
(4) *Anabaena, Azolla*
BBC268

3. The vast majority of bacteria are :-
(1) Symbiotic (2) Autotrophic
(3) Heterotrophic (4) Nitrogen fixing
BBC269

4. Which character indicates that Cyanobacteria are similar to higher green plants ?
(1) Type of cell wall
(2) Nitrogen fixation ability
(3) Presence of chlorophyll 'a'
(4) Presence of gelatinous sheath
BBC270

5. Typhoid and tetanus are caused by :-
(1) Bacteria (2) Virus
(3) Viroids (4) Fungi
BBC271

6. Which one of the following prokaryotes lack cell wall?
(1) Virus (2) Cyanobacteria
(3) Mycoplasma (4) Protozoa
BBC272

7. According to five kingdom classification Protista includes :-
(1) Chrysophytes, Dinoflagellates, Protozoans
(2) Diatoms, Euglenoids, Virus
(3) Dinoflagellates, Protozoans, Red algae
(4) Chrysophytes, Bryophytes, Protozoans, Slime mould
BBC273

8. Which organisms have silica in their cell wall?
(1) Dinoflagellates (2) Slime mould
(3) Diatoms (4) Euglenoids
BBC274

9. Which organism is responsible for red surface of sea?
(1) *Euglena* (2) *Gonyaulax*
(3) *Amoeba* (4) Paramoecium
BBC275

10. Which unicellular, photosynthetic eukaryote lacks cell wall ?
(1) Mycoplasma (2) *Amoeba*
(3) *Euglena* (4) Slime mould
BBC276

11. Select incorrect statement about fungi ?
(1) The cell walls are composed of chitin and polysaccharides
(2) These are heterotrophic organisms
(3) They are autotrophic organism
(4) They are absorptive in nature
BBC278

12. In fungi, vegetative reproduction takes place by:-
(1) Fragmentation (2) Conidia
(3) Zygospore (4) Oospore
BBC279

13. In ascomycetes and basidiomycetes, the dikaryotic cell is :-
(1) Haploid (2) Diploid
(3) Triploid (4) Polyploid
BBC280

14. Which of the following is not a member of class phycomycetes ?
(1) *Albugo* (2) *Mucor*
(3) *Rhizopus* (4) *Neurospora*
BBC281

15. The mycelium is aseptate and coenocytic in:-
(1) Phycomycetes (2) Basidiomycetes
(3) Deuteromycetes (4) Ascomycetes
BBC282

16. Septate and branched mycelium is found in:-
(1) Basidiomycetes (2) Deuteromycetes
(3) Ascomycetes (4) All the above
BBC283

17. Select correct matched pair / pairs :-
(1) *Claviceps* – Ascomycetes
(2) Morels – Ascomycetes
(3) *Agaricus* – Basidiomycetes
(4) All the above
BBC284

18. Asexual reproduction takes place by conidia formation in :-
(1) *Aspergillus* (2) *Penicillium*
(3) *Colletotrichum* (4) All the above
BBC285

19. Which of the following are examples of unicellular algae ?
(1) *Chlamydomonas* (2) *Riccia*
(3) *Chlorella* (4) Both (1) & (3)
BBC286

20. Which of the following are not considered in five kingdom system of classification ?
(1) Lichen (2) Virus
(3) Viroid (4) All the above
BBC287

21. The name virus that means venom or poisonous fluid was given by :-
(1) Pasteur (2) Ivanowsky
(3) Stanley (4) Both (2) & (3)
BBC288

22. Viroid was discovered by :-
(1) Haeckel (2) Pasteur
(3) T.O. Diener (4) Theophrastus
BBC289

23. Bacteriophages are generally :-
(1) ss DNA virus (2) ds DNA virus
(3) ss RNA virus (4) ds RNA virus
BBC290

24. Select true statement :-
(1) Viruses are obligate parasites
(2) In Lichens, algal component is called phycobiont
(3) In Lichens, Fungal component is called mycobiont
(4) All the above
BBC291

25. Which group shows the most extensive metabolic diversity ?
(1) Plantae (2) Animalia
(3) Monera (4) Fungi
BBC292

26. Archaebacteria differ from other bacteria in:-
(1) Nucleoid
(2) Composition of cell wall
(3) Composition of cytoplasm
(4) Composition of genetic material
BBC293

27. Which group of organisms is responsible for the production of biogas from the dung of cows and buffaloes ?
(1) *Methanomonas* (2) Methanogens
(3) Cyanobacteria (4) *Mycoplasma*
BBC294

28. Choose the correct statement :-
(1) Most of the bacteria are autotrophs
(2) All bacteria are heterotrophs
(3) Some bacteria are heterotrophs
(4) Most of the bacteria are heterotrophs
BBC295

29. Which of the following character is common in both cyanobacteria and green plants ?
(1) Cell wall composition
(2) Presence of chlorophyll 'a'
(3) Presence of heterocyst
(4) Presence of 80s ribosome
BBC296

30. *Nostoc* can fix atmospheric nitrogen in specialised cell, called :-
(1) Chromatophore (2) Zoospore
(3) Heterocyst (4) Oospore
BBC297

31. Which process is not performed by bacterial activity?
(1) Production of antibiotics
(2) Fixing nitrogen in legumes
(3) Making curd from milk
(4) Production of pseudo mycelium
BBC298

32. Choose the incorrect statement regarding *Mycoplasma* :-
(1) They lack cell wall
(2) They are smallest living cell
(3) They can survive without oxygen
(4) They have mesosome for respiration
BBC299

33. Which are chief producers of oceans ?
(1) Diatoms (2) Dinoflagellates
(3) Euglenoid (4) Green algae
BBC300

34. Which organism is responsible for making the sea appear red by rapid multiplication?
(1) *Euglena* (2) *Paramoecium*
(3) *Gonyaulax* (4) *Trypanosoma*
BBC301

35. Euglenoids have a protein rich layer instead of a cell wall. This layer is called as :-
(1) Epidermis (2) Skin
(3) Pellicle (4) Middle layer
BBC302

36. Choose the incorrect statement from following:
(1) Dinoflagellates have stiff cellulosic plates on the outer surface.
(2) Euglenoids have two flagella, one lies longitudinally and the other transversely.
(3) Slime mould spores are dispersed by air currents.
(4) In diatoms, the cell walls form two thin overlapping shells, which fit together as in a soap box.
BBC303

37. Which group of organisms is used to make bread and beer?
(1) *Rhizopus* (2) *Yeast*
(3) *Albugo* (4) *Neurospora*
BBC304

38. Fungus prefers to grow in :-
(1) Warm and humid places
(2) Cold and humid places
(3) Warm and cold both
(4) Warm, cold and humid places
BBC305

39. In fungi when the hyphae are continuous and branched tubes and filled with multi-nucleated cytoplasm, these are called:-
(1) Unicellular hyphae
(2) Coenocytic hyphae
(3) Multicellular hyphae
(4) Both (1) and (3)

BBC306

40. In fungi sexual reproduction is by :-
(1) Fragmentation, Ascospores and Basidiospores
(2) Budding, Conidia, and Basidiospores
(3) Oospores, Ascospores and Basidiospores
(4) Fission, Zoospores, Oospores

BBC307

41. Mycelium of members of phycomycetes is:-
(1) Septate and coenocytic
(2) Aseptate and coenocytic
(3) Septate and multicellular
(4) Septate and unicellular

BBC308

42. Choose the incorrect statement about phycomycetes :-
(1) Members are found in aquatic habitats
(2) Spores are endogenously produced in sporangium
(3) Zygospore is formed by reduction division
(4) Its members reproduce sexually may be isogamous, anisogamous or oogamous.

BBC309

43. Which fungus is used extensively in biochemical and genetic work ?
(1) *Agaricus*
(2) *Aspergillus*
(3) *Claviceps*
(4) *Neurospora*

BBC310

44. Choose the incorrect statement about ascomycetes :-
(1) Mycelium is branched and septate.
(2) Sexual spores are produced endogenously .
(3) They have sac like structure in which karyogamy takes place.
(4) Reduction division occurs in fruiting bodies and form conidia.

BBC311

45. Which group of organism is related with basidiomycetes ?
(1) Mushroom, *Ustilago*, *Aspergillus*
(2) Puffballs, *Agaricus*, *Aspergillus*
(3) Mushroom, *Ustilago*, *Agaricus*
(4) *Ustilago*, *Aspergillus*, *Agaricus*

BBC312

46. Which set of diseases is caused by members of basidiomycetes ?
(1) Rust and smut
(2) Ergot and smut
(3) Ergot and rust
(4) Citrus canker and rust

BBC313

47. Fungi imperfecti are called :
(1) members of phycomycetes
(2) members of ascomycetes
(3) members of basidiomycetes
(4) members of deuteromycetes

BBC314

48. Which is not found in members of deuteromycetes ?
(1) Asexual reproduction
(2) Mode of nutrition
(3) Structure of mycelium
(4) Sexual reproduction

BBC315

49. Which of the following are noncellular organisms that are characterized by having an inert crystalline structure outside the living cell:-

(1) Bacteria (2) *Mycoplasma*

(3) Virus (4) Lichen

BBC316

50. Select false statement :-

(1) Lichens are symbiotic association between algae & fungi

(2) Viruses are smaller than bacteria

(3) Virus name are obligate parasites

(4) Viruses are facultative parasite

BBC317

51. Select incorrect statement about viroid :-

(1) Free infectious RNA

(2) It was discovered by T.O. Diener

(3) It causes potato spindle tuber disease

(4) It contains high molecular weight RNA

BBC318

52. Which is incorrect statement ?

(1) The *Mycoplasma* are organisms that completely lack a cell wall.

(2) *Mycoplasma* can survive without oxygen.

(3) *Mycoplasma* are the smallest living cells.

(4) *Mycoplasma* can cause disease only in plants.

BBC319

53. Generally plant viruses are :-

(1) Double stranded DNA viruses

(2) Single stranded RNA viruses

(3) Single stranded DNA viruses

(4) Double stranded RNA viruses

BBC320

(B) ANALYTICAL QUESTIONS

54. Modern farmer's can increase the yield of Paddy upto 50% by the use of :-

(1) Cyanobacteria

(2) *Rhizobium*

(3) Cyanobacteria in *Azolla pinnata*

(4) Farm yard manure

BBC322

55. Plant pathogenic bacteria are mostly :-

(1) Gram + Non spore forming

(2) Gram - Negative non spore forming

(3) Gram + spore forming

(4) Gram (-) spore forming

BBC323

56. Nitrogen fixing bacteria convert :-

(1) $N_2 \rightarrow NH_3$ (2) $NH_4^+ \rightarrow$ Nitrates

(3) $NO_2 \rightarrow NO_3$ (4) $NO_3 \rightarrow N_2$

BBC324

57. Main reason of water bloom in rivers, lakes, sea etc. is:-

(1) Brown algae and green algae

(2) Cyanobacteria and dinoflagellates

(3) *Eichornia*

(4) Fishes

BBC325

58. Genetic material of prokaryotic cell:-

(1) Non histonic double stranded DNA

(2) Histonic double stranded DNA

(3) Histone & DNA both are absent

(4) Histone without DNA

BBC326

59. Diatomaceous earth is used as heat insulator in boilers and steam pipes because the cell wall of diatom -

(1) Composed of Iron

(2) Composed of Silicon dioxide

(3) Is conductor of heat

(4) Is bad conductor of electricity

BBC327

60. Enzymes are not found in-
(1) Fungi (2) Algae
(3) Virus (4) Cyanobacteria
BBC328

61. Plant Decomposers belong to :-
(1) Monera and fungi
(2) Fungi and plants
(3) Protista and Animalia
(4) Animalia and Monera
BBC329

62. Sexual reproduction in fungus occurs in the following sequential event. Arrange them properly.
(A) Fusion of two nuclei called karyogamy.
(B) Fusion of protoplasm between two motile or non-motile gametes.
(C) Meiosis in zygote resulting in haploid spores.
(1) A $\rightarrow$ B $\rightarrow$ C (2) B $\rightarrow$ A $\rightarrow$ C
(3) C $\rightarrow$ B $\rightarrow$ A (4) C $\rightarrow$ A $\rightarrow$ B
BBC401

63. Which bacteria is utilized in Gober gas plant?
(1) Methanogens
(2) Nitrifying bacteria
(3) Ammonifying bacteria
(4) Denitrifying bacteria
BBC331

64. In bacteria, plasmid is :-
(1) Extra chromosomal material
(2) Main DNA
(3) Non functional DNA
(4) Repetitive gene
BBC332

65. Sequence of which of the following is used to know the phylogeny :-
(1) m-RNA (2) r-RNA
(3) t-RNA (4) DNA
BBC333

66. In five kingdom system, the main basis of classification :-
(1) Structure of nucleus
(2) Nutrition
(3) Structure of cell wall
(4) Asexual reproduction
BBC334

67. Organisms which obtain energy by the oxidation of reduced inorganic compounds are called
(1) Photo autotrophs
(2) Chemo autotrophs
(3) Saprozoic
(4) Coproheterotrophs
BBC335

68. Which one of the following bacteria has found extensive use in genetic engineering work in plants:-
(1) *Clostridium septicum*
(2) *Xanthomonas citri*
(3) *Bacillus coagulens*
(4) *Agrobacterium tumefaciens*
BBC336

69. In which kingdom would you classify the archaea and nitrogen-fixing organisms, if the five-kingdom system of classification is used:-
(1) Plantae (2) Fungi
(3) Protista (4) Monera
BBC337

70. In oogamy, fertilization involves :-
(1) A large motile female gamete and a small non–motile male gamete
(2) A small non–motile female gamete and a large motile male gamete
(3) A large non–motile female gamete and a small motile male gamete
(4) A large non–motile female gamete and a large non–motile male gamete
BBC340

71. Flagella of prokaryotic and eukaryotic cells differ in :-
(1) Microtubular organization and function
(2) Type of movement and placement in cell
(3) Location in cell and mode of functioning
(4) Microtubular organization and type of movement

BBC341

72. If by radiation all nitrogenase enzyme are inactivated, then there will be no :-
(1) Conversion from ammonium to nitrate in soil
(2) Fixation of nitrogen in legumes
(3) Fixation of atmospheric nitrogen
(4) Conversion from nitrate to nitrite in legumes

BBC342

73. A free living nitrogen–fixing cyanobacterium which can also form symbiotic association with the water fern *Azolla* is :-
(1) *Anabaena* (2) *Tolypothrix*
(3) *Chlorella* (4) *Nostoc*

BBC344

74. Carbohydrates, the most abundant biomolecules on earth, are produced by :
(1) some bacteria, algae and green plant cells
(2) all bacteria, fungi and algae
(3) fungi, algae and green plant cells
(4) viruses, fungi and bacteria

BBC345

75. Barophilic prokaryotes :
(1) occur in water containing high concentrations of barium hydroxide.
(2) readily grow and divide in sea water enriched in any soluble salt of barium
(3) grow and multiply in very deep marine sediments
(4) grow slowly in highly alkaline frozen lakes at high altitudes

BBC346

76. Auxospores and hormocysts are formed, respectively, by :
(1) Several diatoms and a few cyanobacteria
(2) Some diatoms and several cyanobacteria
(3) Several cyanobacteria and several diatoms
(4) Some cyanobacteria and many diatoms

BBC347

77. Which of the following is non-cellular organism?
(1) Bacteria
(2) Fungi
(3) Algae
(4) Virus

BBC348

78. Find the **incorrect** match :-

Column-I		Column-II
(1)	*Ustilago*	belongs to class basidiomycetes
(2)	*Agaricus*	belongs to class basidiomycetes
(3)	*Trichoderma*	sexual cycle is found
(4)	*Alternaria*	sexual cycle is not found

BBC349

79. Find the **incorrect** statement about fungi :-
(1) They show a great diversity in morphology and habitats
(2) Fungi are cosmopolitan and occur in air, water, soil and as parasites also
(3) They prefer to grow in cold and dry places
(4) With the exception of yeasts, fungi are filamentous

BBC350

80. Find the **correct** statements about fungi :-
(A) Most of the fungi are saprophytes
(B) Fungi can also survive as symbionts
(C) Sexual reproduction in fungi involves three steps
(D) Reduction division in fungi occurs in fruiting bodies which forms haploid spores
(1) A, B are correct
(2) C, D are correct
(3) A, B & C are correct
(4) A, B, C & D are correct

BBC351

81. Choose the **incorrect** statement regarding decomposer :-
(1) They may be prokaryotes or eukaryotes.
(2) They may be unicellular or multicellular
(3) They convert inorganic into organic compounds
(4) They play a great role in ecology

BBC352

82. With respect to fungal sexual cycle, what is the correct sequence ?
(1) Karyogamy, plasmogamy & meiosis
(2) Meiosis, plasmogamy & karyogamy
(3) Plasmogamy, karyogamy & meiosis
(4) Meiosis, karyogamy & plasmogamy

BBC353

83. In prokaryotes, chromatophores are –
(1) Specialized granules responsible for colouration of cells
(2) Structures responsible for organizing the shape of the organism
(3) Inclusion bodies lying free inside the cells for carrying out various metabolic activities
(4) Internal membrane systems that may become extensive and complex in photosynthetic bacteria

BBC354

84. All eukaryotic unicellular organisms belong to-
(1) Monera (2) Protista
(3) Fungi (4) Bacteria

BBC355

85. The five kingdom classification was proposed by-
(1) R.H. Whittaker (2) C.Linnaeus
(3) A. Roxberg (4) Virchow

BBC356

86. Organisms living in salty areas are called as
(1) Methanogens
(2) Halophiles
(3) Heliophytes
(4) Thermoacidophiles

BBC357

87. Naked cytoplasm, multinucleated and saprophytic are the characteristics of :
(1) Monera (2) Protista
(3) Fungi (4) Slime molds

BBC358

88. An association between roots of higher plants and fungi is called-
(1) Lichen (2) Fern
(3) Mycorrhiza (4) BGA

BBC359

89. A dikaryon is formed when
(1) Meiosis is arrested
(2) The two haploid nuclei do not fuse immediately
(3) Cytoplasm does not fuse
(4) None of the above

BBC360

90. *Contagium vivum fluidum* was proposed by
(1) D.J. Ivanowsky (2) M.W. Beijerinek
(3) Stanley (4) Robert Hook

BBC361

91. Mycobiont and Phycobiont are found in
(1) Mycorrhiza (2) Root
(3) Lichens (4) BGA
BBC362

92. Difference between Virus and Viroid is
(1) Absence of protein coat in viroid but present in virus
(2) Presence of low molecular weight RNA in virus but absent in viroid
(3) Both (1) and (2)
(4) None of the above
BBC363

93. The following features belong to class
(A) Asexual reproduction by zoospores or aplanospores.
(B) Fusion of gametes may be isogamous, anisogamous or oogamous.
(C) Mycelium is aseptate and coenocytic.
(D) Spores are endogenously produced in sporangium.
(1) Ascomycetes (2) Deuteromycetes
(3) Phycomycetes (4) Basidiomycetes
BBC402

94. Viruses are non-cellular organisms but replicate themselves once they infect the host cell. To which of the following kingdom do viruses belong to?
(1) Monera (2) Protista
(3) Fungi (4) None of the above
BBC365

95. Members of phycomycetes are found in
i. Aquatic habitats
ii. On decaying wood
iii. Moist and damp places
iv. As obligate parasites on plants
Choose from the following options
(1) None of the above (2) i and iv
(3) ii and iii (4) All of the above
BBC366

96. Cyanobacteria are classified under
(1) Protista (2) Plantae
(3) Monera (4) Algae
BBC367

97. Fusion of two gametes which are dissimilar in size is termed as
(1) Oogamy (2) Isogamy
(3) Anisogamy (4) Zoogamy
BBC368

98. Which of the following criterion was not used for classification of fungi?
(1) Structure of mycelium
(2) Structure of fruiting body
(3) Complexity of cell wall
(4) Mode of spore formation
BBC369

99. In which kingdom would you classify the archaebacteria and nitrogen-fixing organisms. If the five-kingdom system of classification is used ?
(1) Monera (2) Plantae
(3) Fungi (4) Protista
BBC370

100. How many kingdoms include prokaryotic organisms only, in three domain classification?
(1) One (2) Three
(3) Two (4) Four
BBC371

101. A bacterium is able of withstand extreme heat, dryness, and toxic chemical this indicates that it is probably able to form :
(1) Endospore
(2) Thick peptidoglycan wall
(3) Capsule
(4) Toxins
BBC372

102. Heterocysts as specialized cells to fix atmospheric nitrogen are produced by:
(1) *Rhodospirillum* and *Nostoc*
(2) *Anabaena* and *Azolla*
(3) *Nostoc* and *Anabaena*
(4) *Rhodopseudomonas* and *Oscillatoria*

BBC373

103. Which of the following are the characters of dinoflagellates?
(i) They are planktonic golden yellow algae with soap box like structure.
(ii) They are marine red biflagellated Protista.
(iii) They appear yellow, green, brown, blue and red in colour.
(iv) They are biflagellated organisms with pellicle.
(v) They are saprophytic (or) parasitic unicellular forms.
(1) (ii) and (iii) (2) (ii) and (v)
(3) (i), (ii) and (iii) (4) (ii), (iv) and (v)

BBC374

104. Which statement is incorrect regarding fungi :
(1) Stored food is glycogen and oil
(2) Fungi are haploid in nature
(3) Higher Fungi are more advanced than lower fungi in complexity of sex organs.
(4) Haplontic life cycle and zygotic meiosis occurs in most of the fungi

BBC376

105. Match column I with column II and choose the right option :

Column-I	**Column-II**
(A) Ascomycetes	(i) *Rhizopus*
(B) Basidiomycetes	(ii) *Penicillium*
(C) Deuteromycetes	(iii) *Ustilago*
(D) Phycomycetes	(iv) *Alternaria*

(1) A-(iv), B-(iii), C-(i), D-(ii)
(2) A-(ii), B-(iii), C-(iv), D-(i)
(3) A-(iv), B-(i), C-(ii), D-(iii)
(4) A-(iii), B-(iv), C-(ii), D-(i)

BBC377

106. Which of the following is not a sexual spore
(a) Sporangiospore
(b) Ascospore
(c) Basidiospore
(d) Oospore
(e) Zygospore
(1) (a), (b) and (c)
(2) only (a)
(3) (b), (c) and (d)
(4) (d) and (e)

BBC378

107. Which of the following group of kingdom Protista is being described in the statements given below ?
(i) This group includes diatoms and golden algae.
(ii) They are microscopic and float passively in water currents (plankton).
(iii) Most of them are photosynthetic.
(iv) They have deposits in their habitat; this accumulation over billion of years is referred to as 'diatomaceous earth'.
(1) Dinoflagellates
(2) Chrysophytes
(3) Euglenoids
(4) Slime moulds

BBC379

108. The given statements are some characters of a particular group of Kingdom Protista.
(i) Most of them are fresh water organisms found in standing water.

(ii) They have a protein rich layer (called pellicle) which makes their body flexible.

(iii) They have two flagella, a short and a long one.

(iv) Though they are photosynthetic in the presence of sunlight, but in the absence of sunlight they behave like heterotrophs by predating on other smaller organisms.

Identify the correct group on the basis of these characters.

(1) Protozoans

(2) Chrysophytes

(3) Slime moulds

(4) Euglenoids

BBC380

109. Which of the following statement(s) is/are correct about ascomycetes ?

(i) *Neurospora*, which is used in biochemical and genetic work is a member of this class.

(ii) They are mostly multicellular, *e.g.*, Yeast, or rarely unicellular, *e.g., Penicillium*.

(iii) They are saprophytic, decomposers, parasitic or coprophilous.

(iv) Some examples are *Aspergillus*, *Claviceps* and *Neurospora*.

(1) Both (i) and (ii)

(2) Only (ii)

(3) (i), (iii) and (iv)

(4) All of these

BBC381

110. Which of the following statement(s) is/are correct about class basidiomycetes ?

(i) They are commonly known as imperfect fungi because only the asexual or vegetative phases of these fungi are known.

(ii) They grow in soil, on logs and tree stumps and in living plant bodies as parasites, e.g., rusts and smuts.

(iii) The mycelium is branched and septate.

(iv) Some common members are *Agaricus*, *Ustilago* and *Puccinia*.

(1) Only (i)

(2) Both (ii) and (iii)

(3) (ii), (iii) and (iv)

(4) All of these

BBC382

111. **Assertion :** *Euglena* can be placed in the plant kingdom due to the presence of chlorophyll.

Reason : *Euglena* cannot be classified on the basis of two kingdom system of classification.

(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) If Assertion is true but Reason is false.

(4) If both Assertion and Reason are false.

BBC383

112. **Assertion :** Outside a living cell, viruses have must crystalline statements.
Reason : Viroids have a protein coat.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BBC384

113. **Assertion :** TMV is a virus which causes Tobacco mosaic disease.
Reason : TMV has RNA as genetic material.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BBC385

114. **Assertion :** Yeasts such as *Saccharomyces cerevisiae* are used in baking industry.
Reason : Carbon dioxide produced during fermentation causes bread dough to rise by thermal expansion.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BBC386

115. Match the terms given in column-I with their examples given in column-II and choose the correct option :

	Column-I (Terms)		Column-II (Examples)
A.	Ascus	I.	*Spirulina*
B.	Basidium	II.	*Penicillium*
C.	Protista	III.	*Agaricus*
D.	Cyanobacteria	IV.	*Euglena*
E.	Animalia	V.	Sponges

(1) A – II, B – III, C – IV, D – V, E – I
(2) A – I, B – II, C – III, D – V, E – IV
(3) A – II, B – V, C – III, D – I, E – IV
(4) A – II, B – III, C – IV, D – I, E – V
BBC387

116. Given below are two statements:
Statement-I: Fungi and members of the Monera and Protista having cell walls have now been excluded from Plantae, although according to earlier classifications they were placed in the same kingdom.
Statement-II: The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves etc.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.
BBC388

117. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Cyanobacteria that are referred to as blue green algae are not "algae" anymore.

Reason (R) : All unicellular prokaryotes are placed in kingdom Monera and algae are placed in kingdom Plantae

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBC389

118. The thalloid body of a slime mould (Myxomycetes) is known as: (kingdom Protista)

(1) Fruiting body

(2) Mycelium

(3) Protonema

(4) Plasmodium

BBC151

119. The bacterium (*Clostridium botulinum*) that causes botulism is :

(1) a facultative aerobe

(2) an obligate aerobe

(3) a facultative anaerobe

(4) an obligate anaerobe

BBC153

120. *Thermococcus*, *Methanococcus* and *Methanobacterium* exemplify :-

(1) Bacteria whose DNA is relaxed or positively supercoiled but which have a cytoskeleton as well as mitochondria

(2) Bacteria that contain a cytoskeleton and ribosomes

(3) Archaebacteria that contain protein homologous to eukaryotic core histones

(4) Archaebacteria that lack any histones resembling those found in eukaryotes but whose DNA is negatively supercoiled.

BBC161

121. Nitrogen fixation in root nodules of Alnus is brought about by :-

(1) *Frankia* (2) *Azorhizobium*

(3) *Bradyrhizobium* (4) *Clostridium*

BBC165

122. Single-celled eukaryotes are included in :-

(1) Monera (2) Protista

(3) Fungi (4) Archaea

BBC170

123. The common nitrogen-fixer in paddy fields is:

(1) *Frankia* (2) *Rhizobium*

(3) *Azospirillum* (4) *Oscillatoria*

BBC174

124. Select the correct combination of the statement (a-d) regarding the *characteristics* of certain organisms :

(a) Methanogens are Archaebacteria which produce methane in marshy areas

(b) *Nostoc* is filamentous blue-green alga which fixes atmospheric nitrogen

(c) Chemosynthetic autotrophic bacteria synthesize cellulose from glucose

(d) Mycoplasma lack a cell wall and can survive without oxygen

The correct statements are :

(1) (a), (b), (c)

(2) (b), (c), (d)

(3) (a), (b), (d)

(4) (b), (c)

BBC176

125. Pigment-containing membranous extensions in some cyanobacteria are :

(1) Chromatophores

(2) Heterocysts

(3) Basal bodies

(4) Pneumatophores

BBC201

126. Match the type of protozoans given in column-I with their examples given in column-II and choose the correct option.

	Column – I (Type of protozoans)		**Column – II (Examples)**
A.	Amoeboid protozoans	I.	*Paramoecium*
B.	Ciliated protozoans	II.	*Plasmodium*
C.	Flagellated protozoans	III.	*Amoeba*
D.	Sporozoans	IV.	*Trypanosoma*

(1) A – I, B – III, C – IV, D – II

(2) A – III, B – I, C – II, D – IV

(3) A – III, B – I, C – IV, D – II

(4) A – III, B – IV, C – I, D – II

BBC392

127. Which of the following organisms is/are correctly assigned its/their taxonomic group?

(1) *Paramoecium* and *Plasmodium* belong to the same kingdom as that of *Penicillium*.

(2) Lichen is a composite organism formed from the symbiotic association of an alga and a protozoan.

(3) Yeast used in making bread and beer is a fungus.

(4) *Nostoc* and *Anabaena* are examples of protista.

BBC393

128. Protozoans are not included in kingdom animalia because they are

(1) mostly asymmetrical.

(2) unicellular eukaryotes.

(3) heterotrophic in nature.

(4) multicellular prokaryotes.

BBC394

129. **Assertion :** Longitudinal binary fission in found in *Trypanosoma*.

Reason : Binary fission is a method of sexual reproduction.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BBC395

130. **Statement-1 :**

Amoeba has one and *Paramoecium* has two contractile vacuoles.

Statement-2 :

Fresh water protozoans get rid of excess water through contractile vacuole.

(1) Both Statements true

(2) Both Statements false

(3) Only statement-1 true

(4) Only statement-2 true

BBC396

131. Match the column-A and Column-B and select the correct option ?

Column-A		Column-B	
I.	*Trichomonas vaginalis*	a.	Kala-azar
II.	*Trypanosoma gambiense*	b.	Amoebic dysentery
III.	*Leishmania donovani*	c.	Leucorrhoea
IV.	*Entamoeba histolytica*	d.	Sleeping sickness disease

(1) I - c, II - b, III - a, IV - d

(2) I - a, II - b, III - c, IV - d

(3) I - c, II - d, III - a, IV - b

(4) I - a, II - c, III - d, IV – b

BBC397

EXERCISE-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	3	3	1	3	1	3	2	3	3	1	1	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	4	4	4	1	3	2	4	3	2	2	4	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	1	3	3	2	2	1	2	3	2	3	4	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	4	3	4	4	4	2	3	2	1	2	1	2	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	2	1	1	2	2	2	4	4	3	4	2	1	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	4	3	3	4	3	3	4	2	1	2	4	3	2	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	1	3	4	4	3	3	3	1	3	1	3	1	3	2
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	2	4	3	3	2	3	2	3	4	1	4	4	4	3
Question	121	122	123	124	125	126	127	128	129	130	131				
Answer	1	2	4	3	1	3	3	2	3	1	3				

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

KINGDOM – PLANTAE – ALGAE

1. Which algal groups have similarity in pigment composition ?
(1) Red algae and brown algae
(2) Green algae and blue green algae
(3) Kelps and diatoms
(4) Diatoms and euglenoids
BPK001

2. No zoospore formation has been observed in the algal members belonging to:-
(1) Chlorophyceae (2) Brown algae
(3) Phaeophyceae (4) Cyanophyceae
BPK004

3. Which pigment is found in Phaeophyceae?
(1) Chl. a, c and fucoxanthin
(2) Chl. a, d and violaxanthin
(3) β Carotene and phycocyanin
(4) None of these **BPK005**

4. Food reserve in Rhodophyta is :-
(1) Floridean starch (2) Mannitol
(3) Leucosin (4) All of the above
BPK006

5. Zygotic meiosis is characteristic of :-
(1) Prokaryotes (2) Thallophyta
(3) Bryophyta (4) Spermatophyta
BPK007

6. Photosynthetic pigments common to all algae:
(1) Chlorophyll 'b' and carotene
(2) Chlorophyll 'a' and 'b'
(3) Chlorophyll 'a' and carotene
(4) Chlorophyll and xanthophyll
BPK008

7. *Chlorella*, a very good source of protein, belongs to:-
(1) Chlorophyta (2) Rhodophyta
(3) Pyrrophyta (4) Phaeophyta
BPK009

8. Deepest algae in sea are :-
(1) Red Algae (2) Brown Algae
(3) Green Algae (4) Golden Algae
BPK010

9. Phycobilins are characteristic pigments of :-
(1) Rhodophyta and Phaeophyta
(2) Rhodophyta and Pyrophyta
(3) Pyrophyta and Cyanophyta
(4) Rhodophyta and Cyanophyta
BPK011

10. Which of the following plant groups have similar pigment composition ?
(1) Rhodophyta and Phaeophyta
(2) Chlorophyta and Phaeophyta
(3) Rhodophyta and Cyanophyta
(4) All of the above
BPK012

11. Globule and nucule are sex organs of :–
(1) *Chara* (2) *Chlorella*
(3) *Laminaria* (4) *Polysiphonia*
BPK013

12. Flagellated cells are absent in :-
(1) Red algae (2) Blue green algae
(3) Higher seed plants (4) All the above
BPK014

13. Green algae are considered as ancestors of higher plants due to their resemblance with higher plants in :-
(1) Pigments (2) Cell wall
(3) Stored food (4) All the above
BPK015

14. Pyrenoids are characteristically found in algae. A pyrenoid consists of :-
(1) Core of starch surrounded by protein
(2) Core of protein surrounded by starch
(3) Core of fatty acids covered by starch
(4) Nucleic acid and protein

BPK016

15. In Chlorophyta the mode of sexual reproduction is :-
(1) Isogamy
(2) Anisogamy
(3) Oogamy
(4) Isogamy, Anisogamy and oogamy

BPK017

16. Unique character of Thallophyta is :-
(1) Thalloid body
(2) Absence of vascular tissue
(3) Zygotic meiosis
(4) All the above

BPK018

17. In Thallophyta main plant body is:-
(1) Gametophyte (2) Sporophyte
(3) Diploid plant body (4) Leafy plant body

BPK019

18. Sexual reproduction in Thallophyta takes place by:-
(1) Isogamy (2) Anisogamy
(3) Oogamy (4) Any of the above

BPK020

19. Most advanced group of Algae is :-
(1) Myxophyta (2) Chlorophyta
(3) Brown algae (4) Phaeophyta

BPK021

20. "Agar-agar" is obtained from :-
(1) Green Algae (2) Red Algae
(3) Brown Algae (4) Yellow green Algae

BPK022

21. Motile stages are not found in life cycle of:-
(1) Red Algae & green Algae
(2) Red Algae & brown Algae
(3) Red Algae & blue green Algae
(4) Green Algae & brown Algae

BPK023

22. Embryo is not formed in Thallophyta due to:-
(1) Zygotic meiosis
(2) Zygotic mitosis
(3) Sporangial meiosis
(4) Gametic meiosis

BPK024

23. Oogonia of Thallophyta differs with archegonia of Bryophyta :-
(1) Being multicellular
(2) Being jacketed
(3) Being stalked
(4) Being unicellular and jacket less

BPK025

24. Which of the following best explains the evolution of sexual reproduction ?
(1) *Chlamydomonas* (2) *Ulothrix*
(3) *Puccinia* (4) *Albugo*

BPK026

25. *Volvox* belongs to :-
(1) Brown Algae (2) Red Algae
(3) Golden Algae (4) Green Algae

BPK028

26. Gametes are non-motile in :-
(1) Blue green Algae
(2) Red Algae
(3) Both (1) and (2)
(4) Green Algae

BPK029

27. Blue - green algae resembles more closely to:
(1) Green Algae
(2) Brown Algae
(3) Red Algae and bacteria
(4) Slime moulds

BPK030

28. Which of the following statement is true for algae ?
(1) Algae have root, stem and leaves
(2) Algae have true roots but lack leaves
(3) Algae have rhizoids and leaves
(4) Body of algae is thallus

BPK031

29. In which plant group reproductive organs are not enclosed in a layer of sterile cells?
(1) Pteridophyta (2) Thallophyta
(3) Angiosperm (4) Gymnosperm

BPK032

30. Classification of algae is mainly based on :-
(1) Reproductive organs
(2) Structure of spores
(3) Pigments
(4) Stored food

BPK033

31. "Carrageenin" is obtained from :-
(1) *Chondrus crispus* (2) *Laminaria*
(3) *Gelidium* (4) *Macrocystis*

BPK034

32. Female sex organ of algae is called :-
(1) Carpel (2) Oogonium
(3) Archegonia (4) Oosphere

BPK035

33. Which of the following is not correctly matched ?
(1) Heterocyst = N_2-fixation structure of B.G.A.
(2) Hormogonia = Reproductive structure of B.G.A
(3) Floridean starch = Stored food of brown algae
(4) Cyanophycean starch = Stored food of B.G.A.

BPK036

34. Cilia & flagella are absent in life cycle of :–
(1) Red algae
(2) Brown algae
(3) Green algae
(4) Red algae & B.G.A.

BPK037

35. Which algae best explains the evolution of sexual reproduction ?
(1) Green algae (2) Red algae
(3) Brown algae (4) B. G. Algae

BPK038

KINGDOM–PLANTAE – BRYOPHYTA

36. Embryo is present but true vasculature is absent in the group :-
(1) Cyanophyta (2) Tracheophyta
(3) Bryophyta (4) Chlorophyta

BPK039

37. The unique feature of bryophytes compared to other green plant groups is that:-
(1) They produce spores
(2) They lack vascular tissue
(3) They lack root
(4) Their sporophyte is attached to gametophyte

BPK040

38. In bryophytes diploid number of chromosomes occur in:-
(1) Gametes
(2) Spores
(3) Spore mother cells
(4) Nuclei of gametes

BPK041

39. The group Bryophyta includes :-
(1) Liverworts and ferns
(2) Liverworts and club moss
(3) Moss and ferns
(4) Liverworts and moss

BPK042

40. A leafy non-vascular plant with dependent sporophytic generation should properly be classified in
(1) Thallophyta (2) Bryophyta
(3) Pteridophyta (4) Spermatophyta
BPK043

41. Bryophyta includes :-
(1) Mosses (2) Psilopsida
(3) Horse tails (4) All the above
BPK044

42. A leafy gametophyte plant with multicellular rhizoids and sporophyte differentiated in foot, seta and capsule should belong to :-
(1) Psilopsida (2) Hepaticopsida
(3) Bryopsida (4) Lycopsida
BPK045

43. Bryophytes differ from thallophytes in having:-
(1) Embryo
(2) Rhizoids
(3) Sterile jacket around sex organs
(4) All the above **BPK046**

44. Spores do not form protonema but directly grow into flat branching thallus in :-
(1) Liverworts (2) Mosses
(3) Ferns (4) Gymnosperms
BPK047

45. In Bryophyta, simplest sporophyte occur in:-
(1) *Riccia* (2) *Marchantia*
(3) *Funaria* (4) *Anthoceros*
BPK048

46. In which of the following bryophytes there are gemmae, the means of vegetative reproduction?
(1) *Riccia* (2) *Marchantia*
(3) *Sphagnum* (4) *Anthoceros*
BPK049

47. In Bryophytes what is absent :-
(1) Embryo formation
(2) Fertilization
(3) Motile gametes
(4) True roots and vascular tissue
BPK050

48. In bryophytes fertilization takes place:-
(1) At low temp.
(2) In dry condition
(3) In presence of water
(4) In above all situation
BPK051

49. Which statement is true about bryophytes?
(1) They are non photosynthetic
(2) Zygote produces gametophyte on germination
(3) Spores form gametophyte plant on germination
(4) They have vascular tissues
BPK052

50. Which bryophyte is of economic importance ?
(1) *Funaria* (2) *Marchantia*
(3) *Riccia* (4) *Sphagnum*
BPK053

51. Mosses are gregarious because they :-
(1) Have vascular tissue
(2) Have indirect germination of spores
(3) Have direct germination of spores
(4) Have spore mother cells
BPK054

52. Aquatic ancestry of bryophytes is evidenced by:
(1) Their green colour
(2) Algae like protonema
(3) Many aquatic bryophytes
(4) Flagellated male gametes
BPK055

53. Moss sporophyte is differentiated in :-
(1) Stem & leaves
(2) Root, stem and leaves
(3) Rhizoids, stem & leaves
(4) None of these
BPK056

54. Oblique septa are found in which part of moss:
(1) Rhizoids of sporophyte
(2) Rhizoids of gametophyte
(3) Leaves
(4) Stem
BPK057

55. Mosses are ecologically important because
I. The first organism to colonize base rocks.
II. It decompose rocks making suitable for the growth of higher plant.
III. It forms dense mats on soil.
IV. It reduces the impact of falling rain and soil erosion.
(1) Only I (2) Only II
(3) III and IV (4) All of these
BPK366

56. Which of the following plants are similar in requirement of water for fertilisation ?
(1) Bryophyta
(2) Pteridophyta
(3) Angiosperm
(4) Both (1) and (2)
BPK059

57. The bryophyte which have great water holding capacity :-
(1) *Anthoceros*
(2) *Sphagnum*
(3) *Funaria*
(4) *Marchantia*
BPK060

58. In which of the following zygote further develops to form a diploid structure ?
(1) Thallophyta (2) Bryophyta
(3) Algae (4) Fungi
BPK061

59. Bryophytes are :–
(1) First successful land plant
(2) Vascular cryptogams
(3) Non vascular cryptogams
(4) Vascular embryophytes
BPK062

60. Sex organ in bryophytes are :–
(1) Unicellular and jacketed
(2) Unicellular and non-jacketed
(3) Multicellular and jacketed
(4) Multicellular and non jacketed
BPK063

61. Fossilised fuel obtained from bog is:-
(1) Tar (2) Peat
(3) Bio-gas (4) Petrol
BPK064

62. Which structure produces the gamete bearing plant of moss ?
(1) Spore (2) Bud
(3) Protonema (4) Zygote
BPK065

63. Bryophytes show an advancement over algae in:
(1) Having multicellular sporophytic generation
(2) Having parasitic sporophyte
(3) Having zygotic meiosis
(4) None of the above
BPK066

64. Which bryophyte is known as Peat moss ?
(1) *Riccia* (2) *Riella*
(3) *Sphagnum* (4) *Marchantia*
BPK067

65. Leafy gametophyte occurs in:-
(1) Liverworts (2) Hornworts
(3) Moss (4) Fern
BPK068

66. *Sphagnum* may be used as a substitute of :-
(1) Absorbent cotton
(2) Non-absorbent cotton
(3) Plastic
(4) Polythene
BPK070

67. Non vascular Embryophyta are :-
(1) Thallophyta (2) Bryophyta
(3) Pteridophyta (4) Both (1) and (2)
BPK071

68. The water conducting tissue in Bryophyta is:-
(1) Parenchyma (2) Sclerenchyma
(3) Tracheids (4) Sieve tubes
BPK072

69. Bryophyta are not tall plants due to:-
(1) Absence of meristem
(2) Absence of vascular tissues
(3) Presence of root system
(4) All the above
BPK073

70. The first cell of sporophytic generation in Bryophyta is:-
(1) Spore
(2) Spore mother cell
(3) Zygote
(4) Protonema
BPK074

71. Structures for dispersal of spores in Bryophyta are :-
(1) Elaters (2) Pseudo elaters
(3) Peristomial teeth (4) All the above
BPK075

72. Oblique septa in rhizoids are characteristic of:
(1) Liverworts (2) Hornworts
(3) Mosses (4) Ferns
BPK076

73. In which Bryophyta germination of spore is indirect:-
(1) *Riccia* (2) *Rhizopus*
(3) *Puccinia* (4) *Funaria*
BPK077

74. Male gametes of bryophytes are :-
(1) Uniflagellate (2) Multiflagellate
(3) Biflagellate (4) Triflagellate
BPK078

75. The vascular tissue is absent in :-
(1) Algae, fungi and pteridophytes
(2) Thallophytes and bryophytes
(3) Bryophytes and pteridophytes
(4) Angiosperm and gymnosperm
BPK080

76. The sporophyte of bryophyte is :-
(1) Parasitic
(2) Autotrophic
(3) Saprophytic
(4) Semiparasitic or parasitic
BPK081

77. Non vascular land plants are called:-
(1) Bryophytes (2) Pteridophytes
(3) Fungi (4) Algae
BPK082

KINGDOM–PLANTAE – PTERIDOPHYTA

78. Vascular cryptogams or seed less vascular plants belongs to:-
(1) Bryophyta (2) Pteridophyta
(3) Thallophyta (4) Spermatophyta
BPK083

79. Seed habit first established in:-
(1) Pteridophytes (2) Gymnosperms
(3) Angiosperms (4) None of the above
BPK084

80. Most conspicuous alternation of generation occurs is:-
(1) Thallophyta (2) Bryophyta
(3) Pteridophyta (4) Spermatophyta
BPK085

81. Rhizoids containing sporophytic plants are characteristic of:-
(1) Bryopsida (2) Sphenopsida
(3) Cycadophyta (4) Psilopsida
BPK086

82. Which group includes green leaf microphyllous plants:-
(1) Lycopsida (2) Sphenopsida
(3) Psilotopsida (4) Pteropsida
BPK087

83. Roots first originated in :-
(1) Algae (2) Fungi
(3) Bryophyta (4) Pteridophyta
BPK088

84. Pteridophyta differs from Bryophyta in having:-
(1) Vascular tissue
(2) Archegonia
(3) Alternation of generations
(4) Motile sperm
BPK089

85. In Pteridophyta, reduction division occurs when:-
(1) Prothallus is formed
(2) Spores are formed
(3) Sex organs are formed
(4) Gametes are formed
BPK090

86. The main plant body of Pteridophytes is:-
(1) Sporophyte
(2) Gametophyte
(3) Haploid
(4) None of the above
BPK091

87. Cryptogamic plants are:-
(1) Seedless (2) Embryo less
(3) Leafless (4) Rootless
BPK092

88. *Adiantum* is called "walking fern" due to :-
(1) Power of locomotion
(2) Vegetative reproduction
(3) Motile antherozoids
(4) All the above
BPK093

89. Plants having vascular tissues but lacking seeds are :-
(1) Bryophyta (2) Pteridophyta
(3) Gymnosperms (4) Angiosperms
BPK094

90. Heterospory occurs in :-
(1) *Selaginella* (2) *Pteridium*
(3) *Funaria* (4) *Riccia*
BPK095

91. Sporangia are found in fruiting structures called sporocarps in aquatic ferns, which of the following is aquatic fern :–
(1) *Salvinia* (2) *Selaginella*
(3) *Pteridium* (4) *Equisetum*
BPK096

92. The antherozoids of fern are :–
(1) Uniflagellate (2) Biflagellate
(3) Quadriflagellate (4) Multiflagellate
BPK097

93. In Pteridophytes the spores germinate to form:-
(1) Protonema (2) Prothallus
(3) Sporophyte (4) Archegonium
BPK098

94. Aquatic fern which supports the growth of blue green algae, *Anabaena*, and used to increase the yield of paddy crop is :-
(1) *Salvinia* (2) *Marsilea*
(3) *Selaginella* (4) *Azolla*
BPK099

95. Most distinct type of alternation of generations is demonstrated by :–
(1) Angiosperms (2) Ferns
(3) Gymnosperms (4) Bryophytes
BPK100

96. Presence of motile stage in life cycle & requirement of water as a medium to complete life cycle is diagnostic characters of:
(1) Thallophyta (2) Bryophyta
(3) Pteridophyta (4) Cryptogams
BPK101

97. Evolution of seed habit first started in :-
(1) *Selaginella* like ancestral pteridophytes
(2) *Psilotum* like ancestral pteridophytes
(3) Gymnosperms
(4) Mosses
BPK102

98. Which is an aquatic fern
(1) *Psilotum* (2) *Salvinia*
(3) *Chara* (4) *Lycopodium*
BPK103

99. In ferns, the permanent roots are :-
(1) Tap root
(2) Adventitious roots
(3) Tuberous roots
(4) Rhizome
BPK104

100. Independent alternation of generation is found in-
(1) Pteridophyta (2) Spermatophyta
(3) Thallophyta (4) Bryophyta
BPK105

101. Stem distinctly differentiated in to node and internode in :-
(1) Psilopsida (2) Lycopsida
(3) Sphenopsida (4) Pteropsida
BPK106

102. Spore producing part of pteridophytes is:-
(1) Sporangia of gametophytes
(2) Capsule of sporophytes
(3) Sporangia of sporophytes
(4) Capsule of gametophytes
BPK107

103. In pteridophytes, reduction division takes place in :-
(1) Zygote
(2) Spore mother cells
(3) Gametangia
(4) Prothallus
BPK108

KINGDOM–PLANTAE – GYMNOSPERM

104. Most advanced Gymnosperm belongs to:-
(1) Cycadales (2) Coniferales
(3) Gnetales (4) Cycadofillicales
BPK109

105. Which of the following is called father of forest?
(1) *Pinus* (2) *Banyan*
(3) *Sequoia* (4) *Cedrus*
BPK110

106. All Gymnosperms are:-
(1) Heterosporous
(2) Arborescent/Woody
(3) Seed plants
(4) All the above
BPK111

107. Gymnosperm plants lack :-
(1) Vessels (2) Fruits
(3) Companion cells (4) All the above
BPK112

108. Gymnosperm plants do not produce fruits because they do not have:-
(1) ovary (2) gametes
(3) fertilization (4) None of these
BPK113

109. Ovule is morphologically equivalent to:-
(1) Megaspore (2) Megasporangium
(3) Microspore (4) Megasporophyll
BPK114

110. Cones in gymnosperm plants are usually :-
(1) Bisexual (2) Unisexual
(3) Sterile (4) Any of the above
BPK115

111. Zygote of Pteridophyte :-
(1) Undergoes reduction division just often formation
(2) Produces multicellular gametophyte
(3) Produces multicellular sporophyte
(4) Remains dormant
BPK367

112. Majority of the pteridophytes are :-
I. Homosporous and produce only one kind of spores.
II. Aquatic in nature.
III. All are Heterosporous.
IV. Restricted to narrow geographical zone because of specific requirement of water.
(1) I and II (2) II and III
(3) III and IV (4) IV and I
BPK368

113. Heterosporous plants are :-
(1) Plants producing one kind of spores.
(2) Plants producing large and small spores.
(3) Condition found in *Funaria* and *Marcantia*.
(4) None of these
BPK369

114. Gametophyte depends on sporophyte in:-
(1) Bryophyta (2) Pteridophyta
(3) Cryptogams (4) Spermatophyta
BPK119

115. Pteridophytes have only
(1) Small microphyllous leaves
(2) Large macrophyllous leaves
(3) Both microphyllous as well as macrophyllous in some of the ferns
(4) All of the above
BPK370

116. Ephedrine is obtained by :-
(1) *Ephedra* (2) *Gnetum*
(3) *Pinus* (4) *Cycas*
BPK121

117. Resin turpentine is obtained from:-
(1) *Pinus* (2) *Adiantum*
(3) Club mosses (4) *Sequoia*
BPK122

118. Which one is false about pteridophyte?
(1) They flourish well moist and shady condition.
(2) They are found in cool damp and shady place.
(3) The xylem posses vessels.
(4) None of these
BPK371

119. Spore bearing tracheophytes (Vascular plants):
(1) Pteridophyta (2) Gymnosperms
(3) Angiosperms (4) All the above
BPK124

120. Which of the following Gymnospermic orders resembles with angiosperms ?
(1) Cycadales (2) Coniferales
(3) Gnetales (4) Ginkgoales
BPK125

121. Living fossil:-
(1) *Cycas* (2) *Ginkgo*
(3) *Psilotum* (4) All the above
BPK126

122. Gymnosperms :-
(1) Are naked seeds plants
(2) Have tap roots
(3) Are heterosporous
(4) All of these

BPK372

123. Which of the following are absent in group gymnosperm ?
(1) Trees (2) Shrubs
(3) Liana (4) Herbs

BPK128

124. Which plant group is exclusively perennial ?
(1) Dicots (2) Ferns
(3) Gymnosperms (4) Monocots

BPK129

125. In *Ginkgo*, male gametes are :-
(1) Motile (2) Non-motile
(3) Amoeboid (4) Absent

BPK130

126. Male gamete of *Cycas* is largest in plant kingdom, is :-
(1) Non motile (2) Biflagellate
(3) Multiciliate (4) Uniflagellate

BPK131

127. The mode of pollination in Gymnospermae is:-
(1) Anemophily (2) Entomophily
(3) Hydrophily (4) Any of the above

BPK132

128. Which of the following are correct for gymnosperms?
I. Have adventitious root system.
II. The leaves in gymnosperms can withstand extreme temperature, humidity and wind.
III. Microspores are produced in microsporangia.
IV. They include medium or tall sized trees and shrubs.
(1) I and III (2) I, II and II
(3) II, III and IV (4) I and IV

BPK373

129. Which of the following remained unchanged for last many million years ?
(1) *Pinus* (2) Rice
(3) *Cedrus* (4) *Ginkgo*

BPK134

130. Life cycle of gymnosperm is :-
(1) Haplontic (2) Haplodiplontic
(3) Diplontic (4) Diplohaplontic

BPK135

131. Which of the following is commonly known as "Chilgoza pine" ?
(1) *Cycas* (2) *Cedrus*
(3) *Pinus* (4) *Ginkgo*

BPK136

132. The gymnosperm resembles with angiosperm in having :-
(1) Ciliated sperms
(2) Presence of ovary
(3) Presence of seed
(4) Presence of fruit

BPK137

133. In pteridophytes and gymnosperm sporangia are born on which side of sporophyll :–
(1) Adaxial (2) Abaxial
(3) Lateral (4) Terminal

BPK138

134. Megaspore germinate into :-
(1) Female gametophyte.
(2) Gametophyte having both male and female sex organs.
(3) Male gametophyte.
(4) Male sporophyte.

BPK374

135. Fruits are not formed in gymnosperm because:
(1) Fertilization is absent
(2) Pollination is absent
(3) Seeds are not formed
(4) Ovary is absent

BPK140

136. The gametophyte of pteridophyte grows in damp, moist and shady places because :-
(1) They are limited and restricted to a narrow geographical region.
(2) They need water for fertilization of gametes formed on the gametophyte.
(3) Water is required for gametogenesis.
(4) Egg cell swims in water to reach to the antheridia.

BPK375

137. Gymnosperms differ from pteridophytes in having–
(1) Presence of tracheids
(2) Presence of embryo
(3) Presence of ovule
(4) Companian cell

BPK142

138. Most gymnosperms have :–
(1) Both archegonia and antheridia
(2) Antheridia but no archegonia
(3) Archegonia but no antheridia
(4) No antheridia or archegonia

BPK143

139. The "endosperm" of a gymnosperm represent:
(1) Gametophytic tissue
(2) Sporophytic tissue
(3) Tissue formed by double fertilization
(4) Polyploid tissue

BPK144

140. Vessels occur in the following gymnosperm plant :-
(1) *Ginkgo* (2) *Taxus*
(3) *Gnetum* (4) All the above

BPK145

141. In gymnosperm endosperm is :-
(1) Triploid (2) Diploid
(3) Haploid (4) Tetraploid

BPK146

142. Which of the following is not heterosporous ?
(1) *Selaginella* (2) *Pinus*
(3) *Pteridium* (4) *Cycas*

BPK147

143. Multiciliate male gametes are found in :–
(1) *Pinus* (2) *Cycas*
(3) *Gnetum* (4) *Mango*

BPK148

144. Which of the following plant form seed and have pollen tube ?
(1) Angiosperm
(2) Pteridophytes
(3) Gymnosperm
(4) Siphonogamous plants

BPK149

145. The plants frequently grown as ornamentals and are used for medicinal purpose belong to :-
(1) Angiosperms
(3) Pteridophytes
(3) Algae
(4) Some protist

BPK376

KINGDOM–PLANTAE – ANGIOSPERMS

146. Which of the following is not a part of flower?
(1) Androecium (2) Gynoecium
(3) Calyx (4) Trophophyll

BPK151

147. Smallest angiosperm is: -

(1) *Wolffia* (2) *Eucalyptus*

(3) *Mustard* (4) *Tomato*

BPK152

148. Pulses are: -

(1) Monocots plants (2) Dicot plants

(3) Both (1) and (2) (4) Conifers

BPK153

149. Which of the following products are provided by angiosperms?

(1) Fuel

(2) Fodder

(3) Food

(4) Fuel, Fodder and Food

BPK154

150. The seeds are enclosed in ovary in :-

(1) Angiosperms (2) Gymnosperms

(3) Bryophytes (4) Pteridophytes

BPK155

151. Male sex organ in flower is :-

(1) Antheridia (2) Stamens

(3) Calyx (d) Nucule

BPK377

152. Which of the following plant groups, produces flowers ?

(1) Bryophytes (2) Pteridophytes

(3) Algae (4) Angiosperms

BPK157

153. After fertilization ovule converts into: -

(1) Fruit (2) Ovary wall

(3) Seed (4) Nucellus

BPK158

154. Life cycle of gymnosperm is: -

(1) Haplontic

(2) Haplo-diplontic

(3) Diplontic

(4) Diplohaplontic

BPK159

155. Which kind of life-cycle pattern is exhibited by seed bearing plants?

(1) Haplontic

(2) Diplontic

(3) Haplo-diplontic

(4) All of these

BPK160

156. Haplontic life cycle is present in: -

(1) Wheat (2) Gymnosperm

(3) *Fucus* (4) *Volvox*

BPK161

157. Which of the following life cycle has only one celled zygote as sporophytic generation?

(1) Haplontic

(2) Diplontic

(3) Haplo-diplontic

(4) All of these

BPK162

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	1	2	3	1	1	4	3	1	4	4	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	4	2	2	3	1	4	1	4	2	3	4	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	3	4	1	3	4	3	4	2	1	3	4	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	4	3	3	4	2	4	4	2	4	4	2	2	3	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	1	3	3	1	2	1	2	3	4	3	4	3	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	1	2	2	3	4	1	4	1	2	1	1	2	2	1
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	4	2	4	2	4	1	2	2	1	3	3	2	3	3
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	4	4	1	2	2	3	4	2	4	3	1	1	3	4	3
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	4	4	4	3	1	3	1	3	4	3	3	3	2	1	4
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	2	3	3	1	3	3	3	2	4	2	4	1	2	4	1
Question	151	152	153	154	155	156	157								
Answer	2	4	3	3	2	4	1								

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which one of the following shows isogamy with non-flagellated gametes?

(1) *Sargassum* (2) *Ectocarpus*

(3) *Ulothrix* (4) *Spirogyra*

BPK209

2. Which one of the following is wrong about *Chara*?

(1) Upper oogonium and lower round antheridium.

(2) Globule and nucule present on the same plant.

(3) Upper antheridium and lower oogonium

(4) Globule is male reproductive structure

BPK210

3. Which of the following is responsible for peat formation ?

(1) *Marchantia* (2) *Riccia*

(3) *Funaria* (4) *Sphagnum*

BPK211

4. Male gametophyte with least number of cell is present in :

(1) *Pteris* (2) *Funaria*

(3) *Lilium* (4) *Pinus*

BPK212

5. An alga which can be employed as food for human being is :-

(1) *Ulothrix* (2) *Chlorella*

(3) *Spirogyra* (4) *Polysiphonia*

BPK213

AIPMT 2015

6. In which of the following gametophyte is **not** independent free living ?

(1) *Marchantia* (2) *Pteris*

(3) *Pinus* (4) *Funaria*

BPK214

7. Read the following five statements (A to E) and select the option with **all correct** statements :-

(A) Mosses and Lichens are the first organisms to colonise a bare rock.

(B) *Selaginella* is a homosporous pteridophyte

(C) Coralloid roots in *Cycas* have VAM

(D) Main plant body in bryophytes is gametophytic, whereas in pteridophytes it is sporophytic

(E) In gymnosperms, male and female gametophytes are present within sporangia located on sporophyte

(1) (B), (C) and (D) (2) (A), (D) and (E)

(3) (B), (C) and (E) (4) (A), (C) and (D)

BPK215

8. Male gametes are flagellated in :

(1) *Anabaena* (2) *Ectocarpus*

(3) *Spirogyra* (4) *Polysiphonia*

BPK216

9. Which one of the following statements is **wrong**?

(1) Agar - agar is obtained from *Gelidium* and *Gracilaria*

(2) *Chlorella* and *Spirulina* are used as space food

(3) Mannitol is stored food in Rhodophyceae

(4) Algin and carrageen are products of algae

BPK217

Re-AIPMT 2015

10. Which one is a **wrong** statement ?

(1) Brown algae have chlorophyll a and c, and fucoxanthin.

(2) Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms.

(3) *Mucor* has biflagellate zoospores.

(4) Haploid endosperm is typical feature of gymnosperms.

BPK218

NEET-I 2016

11. Select the **correct** statement :-

(1) Gymnosperms are both homosporous and heterosporous.

(2) *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms.

(3) *Sequoia* is one of the tallest trees.

(4) The leaves of gymnosperms are not well adapted to extremes of climate.

BPK219

12. In bryophytes and pteridophytes, transport of male gametes requires :-

(1) Wind (2) Insects (3) Birds (4) Water

BPK220

13. Which of the following would appear as the pioneer organisms on bare rocks?

(1) Lichens (2) Liverworts

(3) Mosses (4) Green algae

BPK221

NEET-II 2016

14. The ovule of an angiosperm is technically equivalent to :

(1) megaspore mother cell

(2) megaspore

(3) megasporangium

(4) megasporophyll **BPK222**

15. Conifers are adapted to tolerate extreme environmental conditions because of :

(1) thick cuticle

(2) presence of vessels

(3) broad hardy leaves

(4) superficial stomata

BPK223

16. Which one of the following statements is **wrong**?

(1) Agar-agar is obtained from *Gelidium* and *Gracilaria*.

(2) *Laminaria* and *Sargassum* are used as food.

(3) Algae increase the level of dissolved oxygen in the immediate environment.

(4) Algin is obtained from red algae, and carrageenan from brown algae.

BPK224

NEET (UG) 2017

17. An example of colonial alga is :

(1) *Volvox*

(2) *Ulothrix*

(3) *Spirogyra*

(4) *Chlorella*

BPK225

18. Zygotic meiosis is characteristic of;

(1) *Fucus*

(2) *Funaria*

(3) *Chlamydomonas*

(4) *Marchantia*

BPK226

19. Life cycle of *Ectocarpus* and *Fucus* respectively are:

(1) Diplontic, Haplodiplontic

(2) Haplodiplontic, Diplontic

(3) Haplodiplontic, Haplontic

(4) Haplontic, Diplontic

BPK227

20. Select the mismatch

(1) *Cycas* – Dioecious

(2) *Salvinia* – Heterosporous

(3) *Equisetum* – Homosporous

(4) *Pinus* – Dioecious

BPK228

NEET (UG) 2018

21. Which of the following statement is ***correct***?

(1) Ovules are not enclosed by ovary wall in gymnosperms.

(2) *Selaginella* is *heterosporous*, while *Salvinia* is homosporous.

(3) Horsetails are gymnosperms.

(4) Stems are usually unbranched in both *Cycas* and *Cedrus.*

BPK229

22. Which one is ***wrongly*** matched ?

(1) Uniflagellate gametes - *Polysiphonia*

(2) Biflagellate zoospores - Brown algae

(3) Gemma cups - *Marchantia*

(4) Unicellular organism - *Chlorella*

BPK230

23. Winged pollen grains are present in

(1) Mustard

(2) *Cycas*

(3) Mango

(4) *Pinus*

BPK231

NEET (UG) 2019

24. Phloem in gymnosperms lacks :

(1) Albuminous cells and sieve cells

(2) Sieve tubes only

(3) Companion cells only

(4) Both sieve tubes and companion cells

BPK232

25. From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in :

(1) Liverworts (2) Mosses

(3) Pteridophytes (4) Gymnosperms

BPK233

NEET (UG) 2020

26. Floridean starch has structure similar to:

(1) Laminarin and cellulose

(2) Starch and cellulose

(3) Amylopectin and glycogen

(4) Mannitol and align

BPK234

27. Which of the following pairs is of unicellular algae?

(1) *Chlorella* and *Spirulina*

(2) *Laminaria* and *Sargassum*

(3) *Gelidium* and *Gracilaria*

(4) *Anabaena* and *Volvox*

BPK235

28. Strobili or cones are found in :

(1) *Equisetum* (2) *Salvinia*

(3) *Pteris* (4) *Marchantia*

BPK236

NEET (UG) 2020 (COVID-19)

29. Phycoerythrin is the major pigment in :

(1) Red algae (2) Blue green algae

(3) Green algae (4) Brown algae

BPK237

30. Which of the following statements is incorrect about gymnosperms ?

(1) They are heterosporous

(2) Male and female gametophytes are free living

(3) Most of them have narrow leaves with thick cuticle

(4) Their seeds are not covered

BPK238

31. Male and female gametophytes do not have an independent free living existence in

(1) Pteridophytes (2) Algae

(3) Angiosperms (4) Bryophytes

BPK239

NEET (UG) 2021

32. Gemmae are present in :

(1) Mosses

(2) Pteridophytes

(3) Some Gymnosperms

(4) Some Liverworts

BPK240

33. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as:

(1) Homosorus (2) Heterosorus

(3) Homosporous (4) Heterosporous

BPK241

34. Which of the following plants is monoecious?

(1) *Carica papaya*

(2) *Chara*

(3) *Marchantia polymorpha*

(4) *Cycas circinalis*

BPK242

35. Which of the following algae contains mannitol as reserve food material ?

(1) *Ectocarpus* (2) *Gracilaria*

(3) *Volvox* (4) *Ulothrix*

BPK243

36. Which of the following algae produce Carrageen ?

(1) Green algae

(2) Brown algae

(3) Red algae

(4) Blue-green algae

BPK244

NEET (UG) 2022

37. Hydrocolloid carrageen is obtained from :

(1) Phaeophyceae and Rhodophyceae

(2) Rhodophyceae only

(3) Phaeophyceae only

(4) Chlorophyceae and Phaeophyceae

BPK245

38. Which of the following is **incorrectly** matched ?

(1) *Ulothrix* – Mannitol

(2) *Porphyra* - Floridian Starch

(3) *Volvox* – Starch

(4) *Ectocarpus* – Fucoxanthin

BPK246

39. Match the plant with the kind of life cycle it exhibits:

List-I		List-II	
(a)	*Spirogyra*	(i)	Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
(b)	Fern	(ii)	Dominant haploid free-living gametophyte
(c)	*Funaria*	(iii)	Dominant diploid sporophyte alternating with reduced gametophyte called prothallus
(d)	*Cycas*	(iv)	Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte

Choose the **correct answer** from the options given below:

(1) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)

(2) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

(3) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)

(4) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

BPK247

Re-NEET (UG) 2022

40. Match **List - I** with **List - II** :

List-I		List-II	
(a)	*Chlamydomonas*	(i)	Moss
(b)	*Cycas*	(ii)	Pteridophyte
(c)	*Selaginella*	(iii)	Alga
(d)	*Sphagnum*	(iv)	Gymnosperm

Choose the **correct answer** from the options given below :

(1) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)

(2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(3) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(4) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)

BPK248

41. Read the following statements and identify the characters related to the alga shown in the diagram:



(a) It is a member of Chlorophyceae

(b) Food is stored in the form of starch

(c) It is a monoecious plant showing oogonium and antheridium

(d) Food is stored in the form of laminarin or mannitol

(e) It shows dominance of pigments chlorophyll a, c and Fucoxanthin.

Choose the **correct answer** from the options given below :

(1) (a) and (b) only

(2) (a), (b) and (c) only

(3) (a), (c) and (d) only

(4) (c), (d) and (e) only

BPK249

NEET (UG) 2023

42. Identify the pair of heterosporous pteridophytes among the following :

(1) *Selaginella* and *Salvinia*

(2) *Psilotum* and *Salvinia*

(3) *Equisetum* and *Salvinia*

(4) *Lycopodium* and *Selaginella*

BPK361

43. **Assertion:** The first stage of gametophyte in the life cycle of moss is protonema stage.

Reason: Protonema develops directly from spores produced in capsule.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(2) Assertion is correct but Reason is not correct.

(3) Assertion is not correct but Reason is correct.

(4) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

BPK362

44. **Assertion:** In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.

Reason: Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

(2) **A** is true but **R** is false.

(3) **A** is false but **R** is true

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

BPK363

NEET (UG) 2023 (Manipur)

45. Which classes of algae possess pigment fucoxanthin and pigment phycoerythrin, respectively ?

(1) Phaeophyceae and Chlorophyceae

(2) Phaeophyceae and Rhodophyceae

(3) Chlorophyceae and Rhodophyceae

(4) Rhodophyceae and Phaeophyceae

BPK364

46. Which of the following statements is true ?

(1) All pteridophytes exhibit haplo-diplontic pattern.

(2) Seed bearing plants follow haplontic pattern.

(3) Most algal genera are diplontic

(4) Most bryophytes do not have haplo-diplontic life cycle.

BPK365

NEET (UG) 2024

47. Read the following statements and choose the set of correct statements :

In the members of Phaeophyceae,

A. Asexual reproduction occurs usually by biflagellate zoospores.

B. Sexual reproduction is by oogamous method only.

C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.

D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.

E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below :

(1) A, B, C and D only

(2) B, C, D and E only

(3) A, C, D and E only

(4) A, B, C and E only

BPK378

RE-NEET (UG) 2024

48. Identify the **incorrect** pair :

(1) Sphenopsida — *Adiantum*

(2) Pteropsida — *Dryopteris*

(3) Psilopsida — *Psilotum*

(4) Lycopsida — *Selaginella*

BPK379

49. Which of the following is the **correct** match?

(1) Gymnosperms : *Cedrus, Pinus, Sequoia*

(2) Angiosperms : *Wolffia, Eucalyptus, Sequoia*

(3) Bryophytes : *Polytrichum, Polysiphonia, Sphagnum*

(4) Pteridophytes : *Equisetum, Ginkgo, Adiantum*

BPK380

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	4	3	2	3	2	2	3	3	3	4	1	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	3	2	4	1	1	4	4	3	3	1	1	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	4	2	1	3	2	1	1	2	2	1	4	2	2
Question	46	47	48	49											
Answer	1	3	1	1											

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Algae reproduce by :-
(1) Asexual (2) Sexual
(3) Vegetative (4) All the above
BPK250

2. Stored food in Phaeophyceae is :-
(1) Laminarin or mannitol
(2) Starch
(3) Cellulose
(4) Algin **BPK251**

3. Asexual reproduction in maximum brown algae is by :-
(1) Biflagellated zoospore
(2) Single flagellated zoospore
(3) Aplanospore
(4) None of these
BPK252

4. Pigments in Phaeophycae :-
(1) Chlorophyll a (2) Chlorophyll c
(3) Fucoxanthin (4) All the above
BPK253

5. Generally red algae do not form :-
(1) Zoospore
(2) Non-motile spore
(3) Non-motile gamete
(4) None of these
BPK254

6. The main plant body of bryophyte is :-
(1) Haploid (2) Diploid
(3) Haplo-diploid (4) Diplo-haploid
BPK255

7. The plant body of liverworts is :-
(1) Sporophyte (2) Thalloid
(3) With roots (4) Xerophytic
BPK256

8. Vegetative reproduction in mosses is by :-
(1) Fragmentation (2) Budding
(3) Both (1) & (2) (4) By spore
BPK257

9. *Marchantia* is an example of :-
(1) Mosses (2) Liverworts
(3) Sphenopsida (4) Lycopsida
BPK258

10. In Mosses creeping, green, branched and frequently filamentous stage is called :-
(1) Protonema (2) Rhizome
(3) Rhizophore (4) All of these
BPK259

11. The leaves of Pteridophyta are :-
(1) Microphylls (2) Macrophylls
(3) Both (1) & (2) (4) None of these
BPK260

12. In majority of the Pteridophytes all the spores are of similar kind such plants are called :-
(1) Homosporous (2) Heterosporous
(3) Prothallus (4) Protonema
BPK261

13. Which is the dominant phase of Pteridophytes?
(1) Gametophyte (2) Sporophyte
(3) Spores (4) Gametes
BPK262

14. In Pteridophytes, sporophylls may form distinct compact structure, called :-
(1) Strobili or cones (2) Microphyll
(3) Macrophyll (4) Tropophyll
BPK263

15. Pteridophytes are classified in to :-
(1) Three classes (2) Two classes
(3) Four classes (4) Five classes
BPK264

16. Gymnospermic plants are :-
(1) Homosporous (2) Heterosporous
(3) Both (1) & (2) (4) Without spores
BPK265

17. The nucellus is protected by envelopes and the composite structure is called :-
(1) Megaspore (2) Microspore
(3) Ovule (4) Cone
BPK266

18. In *Cycas* specialised roots are associated with N_2 fixing cyanobacteria, called :-
(1) Tap root (2) Coralloid root
(3) Adventitious root (4) All the above
BPK267

19. Gymnosperms include :-
(1) Medium sized trees (2) Tall tree
(3) Shrubs (4) All the above
BPK268

20. In Gymnosperm which is not found :-
(1) Antheridia (2) Archegonia
(3) Both (4) None of these
BPK269

21. Which group does not produce embryo ?
(1) Algae (2) Moss
(3) Liverworts (4) Club moss
BPK270

22. In which group the male and female sex organs are called antheridia and archegonia, respectively ?
(1) Blue green algae (2) Eubacteria
(3) Protista (4) Bryophyta
BPK271

23. The group of plants, in which body is differentiated into root, stem and leaf :-
(1) Chlorophyceae (2) Phaeophyceae
(3) Lycopsida (4) Hepaticopsida
BPK272

24. Which is not an example of moss plants ?
(1) *Funaria* (2) *Polytrichum*
(3) *Sphagnum* (4) *Colletotrichum*
BPK273

25. Strobili or cones are not formed in :-
(1) Lycopods (2) Sphenopsids
(3) Conifers (4) Ferns
BPK274

26. In which group of plants both male and female gametophytes do not have an independent free living existence ?
(1) Bryophytes (2) Pteridophytes
(3) Gymnosperms (4) Both (1) and (2)
BPK275

27. The gametophyte of pteridophytes require to grow :-
(1) Warm, damp, and shady place
(2) Cool, damp, and shady place
(3) Warm, dry, and shady place
(4) Cool, dry, and place of well sunshine
BPK276

28. Zygotic meiosis occurs in :-
(1) *Pinus* (2) *Funaria*
(3) *Pteridium* (4) *Chara*
BPK277

29. The megaspore mother cell in gymnosperms is differentiated from :-
(1) Integument (2) Embryo sac
(3) Nucellus (4) Endosperm
BPK278

30. A group of algae, having following characters:

(a) Chlorophyll a and b are present

(b) Chloroplast has one or many pyrenoids

(c) Cell wall made of cellulose and pectose

(d) Reserve food material is starch

Which one of the following is also a very important character including above characters to call it as a member of Chlorophyceae?

(1) Plant body colonial only

(2) Plant body is made of haploid cells called as gametophyte

(3) Its flagellar character may be 2–8 in number, equal and apical

(4) It may occur in fresh water, brackish water or salt water

BPK279

31. Integumented megasporangia are formed in:-

(1) Phanerogames

(2) Cryptogames only

(3) Gymnosperms only

(4) Pteridophytes only

BPK280

32. Needle like leaves to reduce the surface area, thick cuticle, and sunken stomata to reduce water loss etc. are the xerophytic characters present in :-

(1) Pteridophytes

(2) Gymnosperms

(3) Angiosperms

(4) Sphenopsida

BPK281

33. Consider the following characters :-

(A) Formation of only one functional megaspore in a megasporangium

(B) Formation of hard covering around megasporangium

(C) Development of embryo from zygote within the female gametophyte

(D) Retention of megaspore inside the megasporangium

These were very essential events occurred during the course of evolution for the phenomenon of:-

(1) Heterospory

(2) Seed habit

(3) Fruit formation

(4) Covered seed formation

BPK282

34. In gymnosperms, the megaspore forms :-

(1) Endosperm (2) Female gamete

(3) Male gamete (4) Male cone

BPK283

35. Find the odd one with respect to ploidy?

(1) Spore (2) Sporophyte

(3) Archegonium (4) Antheridium

BPK284

36. Which of the following is odd one from the following about liverworts ?

(1) Foot, Seta, Capsule

(2) Spore, Archegonium, Antheridium

(3) Multicellular rhizoids, Protonema

(4) Rhizoids, Gemmae

BPK285

37. Which is not true about fern plants ?

(1) They produce sori on sporophylls

(2) They produce male and female cone by spiral arrangement of sporophylls

(3) Some fern plants are heterosporous also

(4) Some ferns are aquatic also

BPK286

38. Which group of plants is of less economic importance but have great ecological importance?
(1) Gymnosperms (2) Angiosperms
(3) Bryophyta (4) Pteridophyta
BPK287

39. Normally in gymnosperms how many archegonia are produced in an ovule ?
(1) One (2) Two
(3) Two or more (4) Always one
BPK288

40. Fusion between a large, nonmotile (static) female gamete and a smaller, motile male gamete is termed as :-
(1) Isogamy (2) Oogamy
(3) Anisogamy (4) Autogamy
BPK289

41. Vascular plants producing integumented megasporangia but not ovary are :-
(1) Pteridophytes (2) Gymnosperms
(3) Angiosperms (4) Bryophytes
BPK290

(B) ANALYTICAL QUESTIONS

42. Water is essential in Bryophyta :-
(1) For fertilization and homosporous nature
(2) Water should be filled in archegonium for fertilization
(3) Water is necessary for movement of sperm
(4) For dissemination spores
BPK291

43. Which statement is wrong for *Cycas* ?
(1) Xylem have vessels
(2) Male cones are well developed
(3) It has coralloid roots
(4) Circinate ptyxis/Circinate vernation
BPK292

44. Walking fern name of *Adiantum* is due to :-
(1) Dispersal by animals
(2) Reproduction by spores
(3) Vegetative reproduction
(4) Power of locomotion
BPK293

45. Which of the following statement is true for Bryophyta ?
(1) Along with water absorption roots also provide anchorage to plants
(2) Sporophyte is dominant
(3) Gametophyte is dominant and sporophyte is mostly parasitic
(4) Gametophyte is parasitic
BPK294

46. *Cycas* have two cotyledons but not included in angiosperms because of :-
(1) Naked ovules
(2) Seems like monocot
(3) Circinate ptyxis
(4) Compound leaves
BPK295

47. What is true for cyanobacteria ?
(1) Oxygenic with nitrogenase
(2) Oxygenic without nitrogenase
(3) Non oxygenic with nitrogenase
(4) Non oxygenic without nitrogenase
BPK296

48. Vessels are found in :–
(1) All Angiosperms and some Gymnosperm
(2) Most of the Angiosperms and few Gymnosperms
(3) All Angiosperms, all Gymnosperms and some Pteridophyta
(4) All Pteridophyte
BPK297

49. Plants reproducing by spores such as mosses and ferns are grouped under the general term

(1) Cryptogams (2) Bryophytes

(3) Sporophytes (4) Thallophytes

BPK298

50. Chlorenchyma is known to develop in the :-

(1) Cytoplasm of *Chlorella*

(2) Mycelium of a green mould such as *Aspergillus*

(3) Spore capsule of a moss

(4) Pollen tube of *Pinus*

BPK299

51. Angiosperms have dominated the land flora primarily because of their :-

(1) Domestication by man

(2) Power of adaptability in diverse habitat

(3) Property of producing large number of seeds

(4) Nature of self pollination

BPK300

52. Top-shaped multiciliate male gametes, and the mature seed which bears only one embryo with two cotyledons, are characteristic features of :

(1) Cycads

(2) Gamopetalous angiosperms

(3) Conifers

(4) Polypetalous angiosperms

BPK301

53. Holdfast, stipe and frond constitutes the plant body in case of

(1) Rhodophyceae

(2) Chlorophyceae

(3) Phaeophyceae

(4) All of the above

BPK302

54. A plant shows thallus level of organization. It shows rhizoids and is haploid. It needs water to complete its life cycle because the male gametes are motile. Identify the group to which it belongs to

(1) Pteridophytes (2) Gymnosperms

(3) Monocots (4) Bryophytes

BPK303

55. A Prothallus is :-

(1) A structure in pteridophytes formed before the thallus develops

(2) A sporophytic free living structure formed in pteridophytes

(3) A gametophyte free living structure formed in pteridophytes

(4) A primitive structure formed after fertilization in pteridophytes

BPK304

56. Plants of this group are diploid and well adapted to extreme conditions. They grow bearing sporophylls in compact structures called cones. The group in reference is

(1) Monocots (2) Dicots

(3) Pteridophytes (4) Gymnosperms

BPK305

57. Protonema is

(1) Haploid and is found in mosses

(2) Diploid and is found in liverworts

(3) Diploid and is found in pteridophytes

(4) Haploid and is found in pteridophytes

BPK306

58. The giant Redwood tree (*Sequoia sempervirens*) is a/an

(1) Angiosperm (2) Free fern

(3) Pteridophyte (4) Gymnosperm

BPK307

59. Zoospores (motile) are formed in :-
(1) Sexual reproduction
(2) Asexual reproduction
(3) Vegetative reproduction
(4) Sexual and asexual both

BPK308

60. Binary fission is found in :-
(1) Multicellular algae
(2) Unicellular algae
(3) Multicellular algae and unicellular algae
(4) None of these

BPK309

61. Hypnospores are formed in :-
(1) Favourable condition
(2) Unfavourable condition
(3) Favourable condition and Unfavourable condition
(4) None of these

BPK310

62. At least how much amount of carbon dioxide is fixed by algae on earth?
(1) ¼ of total carbon dioxide
(2) ½ of total carbon dioxide
(3) ¾ of total carbon dioxide
(4) Small quantity of total carbon di oxide

BPK311

63. Which of the following is not a hydrocolloid?
(1) Carrageen (2) Algin
(3) Agar-agar (4) *Chlorella*

BPK312

64. Iodine and bromine are obtained from :-
(1) *Laminaria* (2) *Chlorella*
(3) *Porphyra* (4) *Sargassum*

BPK313

65. Alginate is obtained from :-
(1) *Fucus* (2) *Laminaria*
(3) *Macrocyctis* (4) All of these

BPK314

66. Algae are of paramount importance :-
(1) As producers
(2) As consumers
(3) Both of these
(4) As microconsumers

BPK315

67. Protonema is a :-
(1) Stage of gametophytic generation.
(2) Green branched and develops directly from a spore.
(3) Produces lateral bud which forms leafy plant body.
(4) All of the above

BPK316

68. Gymnosperm plants do not produce fruits because they do not have :-
(1) Ovary (2) Gametes
(3) Fertilization (4) None of these

BPK317

69. Life cycle of *Ectocarpus* and *Fucus* are ____ and _____ respectively: -
(1) Diplontic, Haplo-diplontic
(2) Haplo-diplontic, Diplontic
(3) Haplo-diplontic, Haplontic
(4) Haplontic, Diplontic

BPK164

70. Select the correct statement :-
(1) *Sequoia* is one of the tallest trees
(2) The leaves of gymnosperms are not well adapted to extremes of climate
(3) Gymnosperms are both homosporous and heterosporous.
(4) *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms.

BPK319

71. Which of the following is a fruit bearing group?
(1) Pteridophytes (2) Bryophytes
(3) Gymnosperms (4) Angiosperms
BPK320

72. Essential parts of the flower are :-
(1) Androecium (2) Gynoecium
(3) Both (4) Calyx
BPK321

73. In angiosperm the seed are enclosed in :-
(1) Thalamus (2) Petiole
(3) Stamen (4) Fruit
BPK322

74. Which of the followings are monocotyledons:-
(1) Pulses (2) Grasses
(3) Cereals (4) Grasses and cereals
BPK323

75. Which of the following are the characters present in dicot angiosperms?
(1) Tetramerous flower
(2) Two cotyledons in seeds
(3) Reticulated venation
(4) All of them
BPK324

76. The grasses and cereals are respectively : -
(1) Monocotyledon, Dicotyledon
(2) Dicotyledon, Monocotyledon
(3) Dicotyledon, Dicotyledon
(4) Monocotyledon, Monocotyledon
BPK325

77. Leaves of Mosses and Ferns are :-
(1) Analogous and homologous both
(2) Analogous but not homologous
(3) Homologous but not analogous
(4) None of the above
BPK058

78. Antheridia and archegonia are absent in :-
(1) Bryophyta (2) Pteridophyta
(3) Gymnosperms (4) Angiosperms
BPK120

79. Vessels and companion cells are found in :–
(1) Pteridophyta (2) *Gnetum*
(3) *Ephedra* (4) Angiosperm
BPK139

80. Which one have maximum power of adaptation ?
(1) Bryophyta (2) Pteridophyta
(3) Gymnosperm (4) Angiosperm
BPK141

81. Modern day (Advanced) plants are :-
(1) Monocots (2) Dicots
(3) Gnetales (4) Ferns
BPK150

82. In haploid life cycle the dominant phase is :-
(1) Diploid sporophyte
(2) Haploid sporophyte
(3) Diploid gametophyte
(4) Haploid gametophyte
BPK331

83. Choose the **incorrect** statement :-
(1) Haploid life cycle is present in *Spirogyra*
(2) Diplontic life cycle is present in *Fucus.*
(3) Haplontic diplontic life cycle is present in Pteridophyte.
(4) Haplontic life cycle is present in Gymnosperm.
BPK332

84. Which one of the following has haplontic life cycle :-
(1) Wheat (2) *Funaria*
(3) *Polytrichum* (4) *Ustilago*
BPK333

85. Life cycle patterns of all seed-bearing plants and maximum algae respectively are :-
(1) Haplontic, Diplontic
(2) Haplontic, Haplontic
(3) Diplontic, Diplontic
(4) Diplontic, Haplontic

BPK334

86. In haplontic life cycle the haploid spore divides by A division and forms the B, identify the A and B :-
(1) A-Meiotic, B-Sporophyte
(2) A-Mitotic, B-Sporophyte
(3) A-Meiotic, B-Gametophyte
(4) A-Mitotic, B-Gametophyte

BPK335

87. Life cycle of *Polysiphonia* and *Fucus* is :-
(1) Diplontic, haplodiplontic
(2) Haplodiplontic, diplontic
(3) Haplodiplontic, haplontic
(4) Haplontic, diplontic

BPK336

88. Choose the incorrect statement :-
(1) Haplontic life cycle is present in Spirogyra.
(2) Diplontic life cycle is present in focus.
(3) Haplo-diplontic life cycle is present in pteridophytes.
(4) Haplontic life cycle is present in gymnosperm.

BPK337

89. Most algal genera are __A__ some of them such as *Ectocarpus*, *Polysiphonia*, kelps are __B__ Identify the A and B :-
(1) Diplontic and Diplontic
(2) Haplontic and haplontic
(3) Haplontic and diplontic
(4) Haplontic and haplo-diplontic

BPK338

90. Sporophyte $\rightarrow$ Spore $\rightarrow$ "A''
Identify the A :-
(1) Zygote
(2) Syngamy
(3) Gametophyte
(4) Gametogenesis

BPK339

91. Sporophyte is developed from :-
(1) Spore
(2) Gametophyte
(3) Zygote
(4) Meiosis

BPK340

92. Given below are two statements:

Statement-I : Natural classification systems also based on evolutionary relationships between the various organisms.

Statement-II : In numerical taxonomy each character is given equal importance and at the same time hundreds of characters can be considered.

In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BPK341

93. Given below are two statements:

Statement-I : The form and size of algae is highly variable, ranging from colonial forms like *Ulothrix* and *Spirogyra* and to the filamentous forms like *Volvox*.

Statement-II : Anisogamous type of reproduction is found in *Eudorina*.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK342

94. Given below are two statements:

Statement-I : Agar is one of the commercial products obtained from members of Phaeophyceae.

Statement-II : Major pigments chlorophyll a, d and phycoerythrin is found in the members of Phaeopyceae.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK343

95. Given below are two statements:

Statement-I : Most of the members of Chlorophyceae have one or more storage bodies called pyrenoids located in the chloroplasts.

Statement-II : Cell wall of members of Rhodophyceae is made up of cellulose, pectin and polysulphate esters.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK344

96. Given below are two statements:

Statement-I : Algae are non Embryophytes, Atracheophytes, Spermatophytes and homosporous.

Statement-II : Sex organs of algae are multicellular and jacketed.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK345

97. Given below are two statements:

Statement-I : *Selaginella* and *Salvinia* are homosporous pteridophytes.

Statement-II: Gymnosperms are Tracheophytes, Embryophytes, heterosporous, Spermatophytes, Phanerogams.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

BPK346

98. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Artificial system of classification separated closely related species.

Reason (R) : Artificial system of classification was based on a few characteristics.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK347

99. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Most algal genera are haplontic.

Reason (R) : In algae, mitosis in zygote results in the formation of diploid spores, the diploid spores divide meiotically to form Gametophyte.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK348

100. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Mosses are of great ecological importance.

Reason (R) : Mosses along with lichens are the first organisms to colonise rocks. They decompose rocks making the substrate suitable for the growth of higher plants.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK349

101. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Majority of Pteriodophytes are heterosporous.

Reason (R) : Genera like *Selaginella* and *Salvinia* produce two kinds of spores, macro (large) and micro (small).

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK350

102. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In heterosporous Pteridophytes, gametophyte is dioecious.

Reason (R) : The female gametophytes in heterosporous Pteridophytes, are retained on parent sporophyte for variable periods.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BPK351

103. Identify the correct set of statements :

(a) Natural classification systems were not based on natural affinities among the organisms and evolutionary relationship between various organisms.

(b) In *Volvox*, *Fucus* sexual reproduction is oogamous.

(c) In numerical taxonomy, each character is given equal importance and at the same time hundreds of characters can be considered.

(d) In algae, asexual reproduction is by production of different types of spores, the most common being the conidia.

(e) At least half of the total carbon di-oxide fixation on earth is carried out by algae through photosynthesis.

Choose the correct answer from the options given below :

(1) (a) and (b) only

(2) (b), (c), (d) and (e) only

(3) (a), (b), (d) and (e) only

(4) (b), (c) and (e) only

BPK352

104. Identify the correct set of statements :

(a) Red colour of red algae is due to predominance of the red pigment fucoxanthin.

(b) *Chlorella* is a multicellular alga rich in proteins is used as food supplement even by space travellers.

(c) Green algae usually have a rigid cell wall made of an outer layer of cellulose and an inner layer of pectose.

(d) In red algae food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.

(e) *Fucus* is a brown alga shows haplo-diplontic life cycle.

Choose the correct answer from options given below :

(1) (a), (b) and (c)

(2) (a), (b), (c) and (e)

(3) only (d)

(4) (c) and (d) only

BPK353

105. Identify the correct set of statements :

(a) *Polysiphonia*, *Porphyra* are red algae, and *Laminaria*, *Sargassum* are brown algae.

(b) Asexual reproduction in most brown algae is by biflagellated zoospores that are pear shaped and have two equal laterally attached flagella.

(c) The plant body of bryophytes is less differentiated than Pteriodophytes.

(d) Species of *Gelidium*, a moss provide peat that have long been used as fuel.

(e) *Marchantia*, is the leafy member of liverworts has tiny leaf like appendages in two rows on stem like structure.

Choose the correct answer from options given below :

(1) (a) and (c) only

(2) (b), (d) and (e) only

(3) (a), (b) and (c) only

(4) (c) only

BPK354

106. Identify the correct set of statements :

(a) The main plant body of the bryophytes is haploid.

(b) In bryophytes, zygotes do not undergo reduction division immediately and they produce a multicellular body called sporophyte.

(c) In *Selaginella*, *Equisetum* sporophylls may form distinct compact structures called strobili or cones.

(d) *Funaria*, *Polytrichum* and *Sphagnum* are liverworts.

(e) In mosses rhizoids are unbranched and unicellular.

Choose the correct answer from options given below :-

(1) (a), (b) and (c) only

(2) (a), (b), (c) and (d) only

(3) (d) and (e) only

(4) (a), (b) and (d) only

BPK355

107. Identify the correct set of statements :-

(a) Pteridophytes are first spermatophytes evolutionarily.

(b) In homosporous pteridophytes, spores germinate to give rise to inconspicuous, small but multicellular, free living mostly photosynthetic thalloid gameteophytes called prothallus.

(c) Majority of Pteriodophytes are homosporous.

(d) Homosporous Pteridophyes show event precursor to the seed habit.

(e) *Equisetum* belongs to class Lycopsida.

Choose the correct answer from options given below :

(1) (a), (d) and (e) only

(2) (b) and (c) only

(3) (a), (b), (c) and (d) only

(4) (b), (c) and (d) only

BPK356

108. Identify the correct set of statements :

(a) One of the gymnosperms, the giant red wood tree *sequoia* is one of the tallest tree species.

(b) Gymnosperms are heterosporous cryptogams.

(c) Gymnosperms are evolutionarily last archegoniate plant group.

(d) Pteridophytes are evolutionarily last antheridial plant group.

(e) The stems are unbranched in *Cycas* and branched in *Pinus* and *Cedrus*.

Choose the correct answer from options given below :

(1) (a), (b) and (d) only

(2) (c), (d) and (e) only

(3) (a), (b), (c) and (d) only

(4) (a), (c), (d) and (e) only

BPK357

109. Identify the correct set of statements :-

(a) Roots in some genera have fungal association in the form of mycorrhiza (like *Cycas*), while in some others (like *Pinus*) small specialised roots called coralloid roots are associated with N_2-fixing cyanobacteria.

(b) In gymnosperms the multicellular female gametophytes is also retained within megasporangium.

(c) In gymnosperm's endosperm represents female gametophyte.

(d) Gymnosperms show anemophilous pollination.

(e) In *Cycas* the pinnate leaves persist for few year.

Choose the **correct** answer from options given below :-

(1) (a), (b), (c) and (e) only

(2) (a), (c), (d) and (e) only

(3) (b), (c), (d) and (e) only

(4) (a), (b), (c), (d) and (e)

BPK358

110. Identify the correct set of statements :

(a) Angiosperms range in size form the smallest *Wolffia* to tall tree like Eucalyptus. (Over 100 meter)

(b) In angiosperms endosperm is haploid and it represents the female gametophyte.

Choose the **correct** answer from options given below :-

(1) (b) only (2) (a) only

(3) Both (a) and (b) (4) None of these

BPK359

111. Identify the correct set of statements :-

(a) In *Spirogyra* sporophytic generation is represented only by one called zygote.

(b) *Fucus* is a brown alga shows diplontic life cycle.

(c) Bryophytes and pteridophytes show diplontic life cycle.

(d) *Ectocarpus*, *Polysiphonia*, kelps are haplodiplontic.

(e) Gymnosperms and angiosperms show haplontic life cycle.

Choose the correct answer from options given below :

(1) (a), (b) and (e) only

(2) (a), (b), (d) and (e) only

(3) (a), (b) and (d) only

(4) (b) and (d) only

BPK360

112. Conifers differ from grasses in the
(1) absence of pollen tubes
(2) formation of endosperm before fertilization
(3) production of seeds from ovules
(4) lack of xylem tracheids

BPK169

113. Moss peat is used as a packing material for sending flowers and live plants to distant places because :
(1) it reduces transpiration
(2) it serves as a disinfectant
(3) it is easily available
(4) it is hygroscopic

BPK170

114. In a moss the sporophyte :
(1) arises from a spore produced from the gametophyte
(2) manufactures food for itself, as well as for the gametophyte
(3) is partially parasitic on the gametophyte
(4) produces gametes that give rise to the gametophyte

BPK171

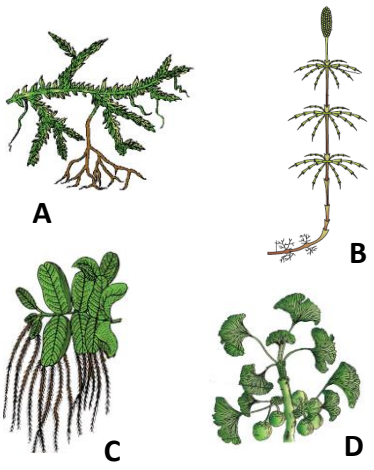
115. If you are asked to classify the various algae into distinct groups, which of the following characters you should choose ?
(1) Chemical composition of the cell wall
(2) Types of pigments present in the cell
(3) Nature of stored food materials in the cell
(4) Structural organization of thallus

BPK173

116. Algae have cell wall made up of :
(1) Cellulose, hemicellulose and pectins
(2) Cellulose, galactans and mannans
(3) Hemicellulose, pectins and proteins
(4) Pectins, cellulose and proteins

BPK188

117. Examine the figure A, B, C and D. In which one of the four options all the items A, B, C and D are correct ?



Options :

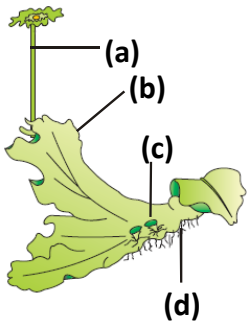
	A	**B**	**C**	**D**
(1)	*Equisetum*	*Ginkgo*	*Selaginella*	*Lycopodium*
(2)	*Selaginella*	*Equisetum*	*Salvinia*	*Ginkgo*
(3)	*Funaria*	*Adiantum*	*Salvinia*	*Riccia*
(4)	*Chara*	*Marchantia*	*Fucus*	*Pinus*

BPK190

118. Archegoniophore is present in :-
(1) *Marchantia*
(2) *Chara*
(3) *Adiantum*
(4) *Funaria*

BPK193

119. Examine the figure given below and select the right option giving all the four parts (a, b, c, d) correctly identified.



	(a)	(b)	(c)	(d)
(1)	Antheridiophore	Male thallus	Globule thallus	Roots
(2)	Antheridiophore	Male thallus	Gemma-cup	Rhizoids
(3)	Archegoniophore	Female thallus	Bud	Foot
(4)	Seta	Sporophyte	Protonema	Rhizoids

BPK196

120. *Selaginella* and *Salvinia* are considered to represent a significant step toward evolution of seed habit because :-

(1) Embryo develops in female gametophyte which is retained on parent sporophyte

(2) Female gametophyte is free and gets dispersed like seeds.

(3) Female gametophyte lacks archegonia

(4) Megaspores possess endosperm and embryo surrounded by seed coat.

BPK197

121. Consider the following four statements whether they are correct or wrong :

(a) The sporophyte in liverworts is more elaborate than that is mosses.

(b) *Salvinia* is heterosporous

(c) The life-cycle in all seed-bearing plants is diplontic.

(d) In Pinus male and female cones are borne on different trees.

The two wrong statements together are :

(1) Statements (a) and (b)

(2) Statements (a) and (c)

(3) Statements (a) and (d)

(4) Statements (b) and (c)

BPK198

122. Which one of the following is a correct statement ?

(1) Antheridiophores and archegoniophores are present in pteridophytes.

(2) Origin of seed habit can be traced in pteridophytes.

(3) Pteridophyte gametophyte has a protonemal and leafy stage.

(4) In gymnosperms female gametophyte is free living.

BPK200

123. A nitrogen fixing microbe associated with *Azolla* in rice-fields is :-

(1) *Frankia* (2) *Rhizobium*

(3) *Spirulina* (4) *Anabaena*

BPK201

124. Read the following five statements (A-E) and answer as asked next to them :-

(A) In *Equisetum* the female gametophyte is retained on the parent sporophyte

(B) In *Ginkgo* male gametophyte is not independent

(C) The sporophyte in *Riccia* is more developed than that in *Polytrichum*

(D) Sexual reproduction in - Volvox is isogamous

(E) The spores of slime molds lack cell walls

How many of the above statements are correct ?

(1) Four (2) One

(3) Two (4) Three

BPK202

125. Which one of the following pairs is wrongly matched?

(1) Viroids – RNA

(2) Mustard – Synergids

(3) *Ginkgo* – Archegonia

(4) *Salvinia* – Prothallus

BPK203

126. Megasporangium is equivalent to :

(1) Ovule (2) Embryo sac

(3) Fruit (4) Nucellus

BPK204

127. Isogamous condition with non-flagallated gametes is found in :

(1) *Fucus*

(2) *Chlamydomonas*

(3) *Spirogyra*

(4) *Volvox*

BPK205

128. Which of the following is **not** correctly matched for the organism and its cell wall degrading enzyme?

(1) Fungi – Chitinase

(2) Bacteria – Lysozyme

(3) Plant cells – Cellulase

(4) Algae – Methylase

BPK207

129. Read the following statements (A–E) and answer the question which follows them.

(a) In liverworts, mosses and ferns gametophytes are free living.

(b) Gymnosperms and some ferns are heterosporous.

(c) Sexual reproduction in *Fucus*, *Volvox* and *Albugo* is oogamous.

(d) The sporophyte in liverworts is more elaborate than that in mosses.

(E) Both, *Pinus* and *Marchantia* are dioecious.

How many of the above statements are correct?

(1) Four (2) One

(3) Two (4) Three

BPK208

130. The life cycle pattern of *Fucus* is: -

(1) Haplontic (2) Diplontic

(3) Haplo-diplontic (4) All of these

BPK163

131. The pattern of life cycle in Bryophyte is :-

(1) Haplontic

(2) Diplontic

(3) Haplo-diplontic

(4) Diploid gametophyte

BPK165

132. Life cycle of kelps is :-

(1) Haplontic

(2) Diplontic

(3) Haplo-diplontic

(4) Diplontic- haplontic

BPK166

133. Life cycle of the Pteridophytes is :-

(1) Haplontic (2) Diplontic

(3) Haplo-diplontic (4) All of them

BPK167

134. Which pattern of life cycle is followed by moss and fern respectively :-

(1) Haplontic, Diplontic

(2) Diplontic, Haplontic

(3) Haplodiplontic, Diplohaplontic

(4) Haplodiplontic, Haplodiplontic

BPK168

EXERCISE-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	1	4	1	1	2	3	2	1	3	1	2	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	2	4	1	1	4	3	4	4	3	2	4	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	2	1	2	3	2	3	3	2	2	3	1	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	2	1	3	2	1	3	4	3	4	1	4	2	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	4	1	4	1	4	1	2	1	4	3	4	4	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	2	4	4	4	1	4	4	4	4	4	2	4	4	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	1	4	2	1	2	4	4	2	4	3	1	4	3	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	1	2	4	3	2	3	2	4	3	2	2	2	1	2	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	
Answer	3	2	4	2	4	1	3	4	4	2	3	3	3	4	

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

ROOT, STEM

1. Roots associated with nitrogen fixing bacteria are :-
(1) Fusiform roots (2) Napiform roots
(3) Nodulated roots (4) Conical roots
BMP002

2. Shoot/Stem develops from :-
(1) Plumule (2) Radicle
(3) Both (1) & (2) (4) Endosperm
BMP005

3. Which is an example of offset?
(1) *Cynodon dactylon* (2) *Eichhornia*
(3) *Fragaria* (4) *Mentha*
BMP006

4. Rhizome of ginger is a modification of stem because :-
(1) It bears Adventitious roots
(2) It bears nodes and internodes
(3) It is underground
(4) It stores food material
BMP007

5. Phylloclade is found in :-
(1) *Opuntia* (2) Cactus
(3) *Acacia* (4) Both (1) & (2)
BMP008

6. Nodulated roots occurs in:-
(1) *Liliaceae* (2) *Solanaceae*
(3) *Malvaceae* (4) *Fabaceae*
BMP009

7. Thorns and spines are :-
(1) Defensive organs
(2) Respiratory organs
(3) Both (1) & (2)
(4) Storage organs
BMP010

8. In onion the reduced disc like underground stem is :-
(1) Root (2) Rhizome
(3) Bulb (4) Tuber
BMP011

9. Stem modified into leaf like structure and leaves are changed into spines in :-
(1) Phyllode (2) Tuber
(3) Phylloclade (4) All the above
BMP012

10. What is the eye of potato ?
(1) Axillary bud
(2) Accessory bud
(3) Adventitious bud
(4) Apical bud
BMP020

11. In *Ruscus,* the modification is :-
(1) Phyllode (2) Cladode
(3) Offset (4) Sucker
BMP021

12. In turmeric, stem is a :-
(1) Tuber (2) Bulb
(3) Rhizome (4) Corm
BMP022

13. Edible part of potato is:-
(1) Inflorescence (2) Leaves
(3) Roots (4) Stem
BMP060

LEAF, INFLORESCENCE

14. Which is not a stem modification :-
(1) Rhizome of ginger
(2) Corm of *Colocasia*
(3) Pitcher of *Nepenthes*
(4) Tuber of potato
BMP016

15. A modification of petiole is :-
(1) Phyllode (2) Phylloclade
(3) Cladode (4) Corm
BMP017

16. Catkin inflorescence is found in :-
(1) Wheat (2) Oat
(3) Mulberry (4) Fig
BMP023

17. Spathe is present in inflorescence of :-
(1) Maize (2) Rice
(3) Marigold (4) Sunflower
BMP024

18. Parallel venation occurs in :-
(1) Liliaceae (2) Solanaceae
(3) Brassicaceae (4) Fabaceae
BMP090

19. Disc florets are
(1) Bisexual
(2) in centre
(3) small flower in Asteraceae
(4) All of the above
BMP323

20. The most advanced type of inflorescence is:
(1) Spike of spikelets (2) Capitulum
(3) Verticillaster (4) Catkin
BMP325

21. Modification of petiole in to leaf-like structure is called :-
(1) Cladode (2) Phylloclade
(3) Phyllode (4) Pistillode
BMP326

22. How many plants in the list given below have opposite phyllotaxy ?
China rose, *Ocimum*, Sunflower, Mustard, *Alstonia*, Guava, *Calotropis* and *Nerium*.
(1) Four (2) Three
(3) Five (4) Two
BMP327

23. Identify the type of leaf modification in the given diagram (A-C) and choose correct option in which modified structure is correctly identified with their related function :-


(A) (B) (C)

(1) A-spines (Support), B-tendril (Protection), C- fleshy leaves (Storage)
(2) A- tendril (Support), B- spine (Protection), C- fleshy leaves (Storage)
(3) A- tendril (Protection), B- spine (Support), C-fleshy leaves (Storage)
(4) A- spine (Protection), B-tendril (Support), C-fleshy leaves (Storage)
BMP328

24. Match the column-I with column-II and choose correct option :-

Column-I		Column-II	
(a)	Alternate phyllotaxy	(i)	*Alstonia*
(b)	Opposite phyllotaxy	(ii)	Mustard
(c)	Whorled phyllotaxy	(iii)	Guava

(1) a-ii, b-iii, c-I (2) a-iii, b-i, c-ii
(3) a-i, b-ii, c-iii (4) a-ii, b-i, c-iii
BMP329

25. Which one of the following statement is **incorrect**?
(1) Pulvinus leaf base is found in some legume plants.
(2) Whorled phyllotaxy is present in *Alstonia*
(3) Thorn in *citrus* are modified leaves.
(4) Pinnately compound leaf is found in Neem.
BMP330

26. Opposite simple leaves with reticulate venation are present in :-

(1) Neem (2) *Calotropis*

(3) Sunflower (4) *Alstonia*

BMP331

27. Select correct option in which compound leaf is found :-

(1) Neem (2) Mango

(3) Peepal (4) Radish

BMP332

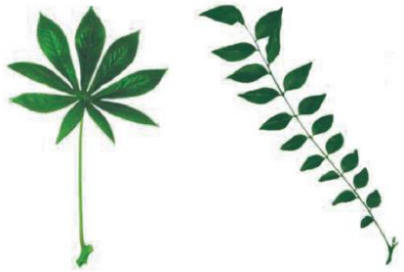
28. In which of following type of phyllotaxy a pair of leaves arise at each node?

(1) Guava (2) Mustard

(3) *Alstonia* (4) Sunflower

BMP333

29. The two sketches A and B given below represents two different types of leaves. Select the option in which both are correctly identified :-



(A) (B)

	A	B
(1)	Simple leaf	Compound leaf
(2)	Compound leaf	Simple leaf
(3)	Pinnately compound leaf	Palmately compound leaf
(4)	Palmately compound leaf	Pinnately compound leaf

BMP334

30. The green and expanded part of the leaf with vein and veinlets is known as :-

(1) Leaf base (2) Petiole

(3) Lamina (4) Stipule

BMP335

31. The pattern of arrangement of leaves on the stem or branch is known as :-

(1) Venation (2) Aestivation

(3) Phyllotaxy (4) Inflorescence

BMP336

32. Function of petiole :-

(1) Hold the leaf blade

(2) Flutter the leaf blade in wind

(3) Bringing fresh air to leaf surface

(4) All of the above

BMP337

33. In *Nepenthes* (pitcher plant), the pitcher is formed due to modification of :-

(1) Leaf petiole (2) Leaf lamina

(3) Tendril (4) Leaflet

BMP338

34. Pulvinus is found in :-

(1) Grasses

(2) Some crucifers

(3) Some cucurbits

(4) Some leguminous plants

BMP339

35. Identify the order where plants show alternate, opposite and whorled phyllotaxy.

(1) China rose, *Calotropis* and *Alstonia*

(2) China rose, *Nerium* and *Calotropis*

(3) *Nerium*, *Calotropis* and China rose

(4) *Calotropis*, China rose and *Nerium*

BMP340

36. Sheathing leaf base is found in :-

(1) Leguminous plants

(2) Monocot plants

(3) Gymnosperm plants

(4) Pteridophytes

BMP341

37. Winged petiole is commonly found in :-
(1) Leguminous plants (2) Crucifer plants
(3) Monocots (4) Citrus

BMP342

38.



Choose correct example for above given figure.
(1) China rose, mustard
(2) Gauva, *calotropis*
(3) *Alstonia*
(4) Neem, Mango

BMP343

39. Find the incorrect statement.
(1) In pinnately compound leaf rachis is present.
(2) Rachis, actually represents the mid-rib of the leaf in a pinnately compound leaf.
(3) In palmately compound leaf rachis is present.
(4) Silk cotton has got palmately compound leaf.

BMP344

40. Swollen leaf base is called _____.
(1) Pulvinus
(2) Sheathing leaf base
(3) Stipule
(4) Pulvinate

BMP345

41. In ___A___ type of inflorescence, main axis terminates in a flower, hence is limited in growth and flowers are borne in ___B___ succession. Choose the correct option to replace A and B.
(1) 1 (2) 2 (3) 3 (4) 4

BMP346

42. When a shoot tip transforms into a flower ?
(1) Pollination can not be normal
(2) It is always solitary
(3) Flower opening is centripetal
(4) Growth is indefinite

BMP347

43. In Cymose inflorescence the main axis :-
(1) has unlimited growth
(2) bear a solitary flower
(3) has unlimited growth but lateral branches end in flower
(4) terminates in a flower

BMP348

44. **Assertion :** In Racemose inflorescence flowers are arranged in acropetal succession.
Reason : In this type of inflorescence shoot apex transforms into flower.
(1) Both **A** and **R** true but **R** is not correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) Both **A** and **R** are true and **R** correct explanation of **A**.
(4) **A** is false but **R** is true.

BMP349

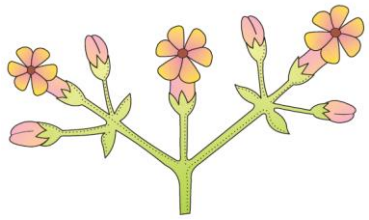
45. Identify this inflorescence :-



(1) Cymose (2) Cyathium
(3) Racemose (4) Hypanthodium

BMP350

46. Which of the following is correct for the type of inflorescence given below ?



(1) The main axis terminates in a flower.
(2) The main axis is of limited growth.
(3) The flowers are borne in a basipetal order.
(4) All are correct

BMP351

47. The arrangement of veins & the veinlets in the lamina of leaf is termed as :-

(1) Phyllotaxy (2) Placentation
(3) Venation (4) Vernation

BMP352

FLOWER

48. The floral leaves arise from :-

(1) Mother axis (2) Thalamus
(3) Root (4) Pedicel

BMP019

49. If the filaments are fused in a single group the condition is :-

(1) Monoadelphous (2) Polyadelphous
(3) Both (1) & (2) (4) Diadelphous

BMP034

50. Pappus is modification of :-

(1) Bracts (2) Corolla
(3) Calyx (4) All

BMP035

51. Most important characteristic of angiosperm is :-

(1) Flowers (2) Roots
(3) Seed (4) Stem

BMP036

52. Adhesion in a flower is :-

(1) Union of dissimilar parts
(2) Union of similar parts
(3) Both (1) and (2)
(4) Fusion of carpels

BMP037

53. The fourth whorl of flower is of :-

(1) Petals (2) Stamens
(3) Carpels (4) Sepals

BMP038

54. The first whorl of flower is of :-

(1) Petals (2) Stamens
(3) Carpels (4) Sepals

BMP043

55. Epigynous flowers are present in :-

(1) Mustard (2) Brinjal
(3) China rose (4) Cucumber

BMP050

56. In *Dianthus,* placentation is :-

(1) Basal (2) Free central
(3) Axile (4) Marginal

BMP051

57. Ovary is half-inferior in the flower of :-

(1) Apple (2) Guava
(3) Peach (4) Garlic

BMP052

58. The term "Keel" is used for special type of:-

(1) Sepals (2) Petals
(3) Stamens (4) Carpels

BMP053

59. Polyadelphous stamens are found in :-

(1) Cotton (2) China rose
(3) Pea (4) Lemon

BMP054

60. Zygomorphic flowers are found in:-

(1) Solanaceae (2) Malvaceae
(3) Fabaceae (4) Liliaceae

BMP056

61. Which type of Placentation is found in mustard :-
(1) Parietal (2) Axile
(3) Basal (4) Marginal
BMP059

62. Which of the following statements is correct:-
(1) Flower is a modified root
(2) Flower is a modified shoot
(3) Flower is a modified leaf
(4) Flower is a modified inflorescence
BMP084

FRUIT & SEED

63. In which of the following fruits the edible part is the aril ?
(1) Custard apple (2) Pomegranate
(3) Orange (4) Litchi
BMP109

64. Cotyledons and testa respectively are edible parts in :-
(1) Cashew nut and litchi
(2) Groundnut and pomegranate
(3) Walnut and tamarind
(4) French bean and coconut
BMP117

65. A fruit developed from hypanthodium inflorescence is called :-
(1) Caryopsis (2) Hesperidium
(3) Sorosis (4) Syconus
BMP118

66. Caryopsis fruit is found in :-
(1) Wheat (2) Pea
(3) Gram (4) Lentil
BMP033

67. Siliqua is the fruit of :-
(1) Cruciferae (2) Fabaceae
(3) Liliaceae (3) Solanaceae
BMP305

68. Fruit of brinjal is :-
(1) Berry (2) Hesperidium
(3) Drupe (4) Pome
BMP039

69. Which of the following is false fruit ?
(1) Pome (2) Pepo
(3) Hesperidium (4) Drupe
BMP040

70. A true fruit develops from :-
(1) Ovary (2) Thalamus
(3) Petals (4) Receptacle
BMP041

71. Fruit of 'Ground nut" is :-
(1) Nut (2) Legume
(3) Lomentum (4) Capsule
BMP042

72. Geocarpic fruit is :-
(1) Carrot (2) Radish
(3) Ground nut (4) Turnip
BMP045

73. Edible part in pomegranate is :-
(1) Testa (2) Epicarp
(3) Endocarp (4) Epidermis
BMP047

74. Edible Juicy hair-like structures are found in:
(1) Apple (2) Pear
(3) Lemon (4) Strawberry
BMP306

75. Which of the following represents the edible part of the fruit of Litchi:–
(1) Endocarp (2) Pericarp
(3) Juicy aril (4) Mesocarp
BMP049

76. What is the major edible part of brinjal :-
(1) Calyx (2) Thalamus
(3) Placenta (4) Style
BMP078

77. Fruit legume is characteristic feature of :-
(1) Solanaceae (2) Liliaceae
(3) Fabaceae (4) Brassicaceae
BMP088

78. Grain of Maize is :–
(1) Seed
(2) Fruit
(3) Neither seed nor fruit
(4) Bud of flower
BMP102

79. The fruit type cypsela is found in :-
(1) Poaceae (2) Asteracea
(3) Malvaceae (4) Fabaceae
BMP317

BRASSICACEAE, MALVACEAE, FABACEAE

80. Tetradyanamous condition of stamens occur in members of :–
(1) Cruciferae (2) Malvaceae
(3) Solanaceae (4) Lilliaceae
BMP046

81. Point out the correct example of Cruciferae:-
(1) Mustard (2) Pea
(3) Onion (4) Brinjal
BMP058

82. Ground nut belongs to family :-
(1) Fabaceae (2) Brassicaceae
(3) Liliaceae (4) Solanaceae
BMP061

83. Bicarpellary ovary with parietal placentation and false septum is found in:-
(1) Cruciferae
(2) Leguminosae
(3) Solanaceae
(4) Liliaceae
BMP062

84. Diadelphous condition is common in :-
(1) Solanaceae (2) Cruciferae
(3) Liliaceae (4) Fabaceae
BMP080

85. Pulses yielding main family of plants is :-
(1) Solanaceae (2) Brassicaceae
(3) Liliaceae (4) Fabaceae
BMP082

86. The floral formula of pea is represented by:-
(1) Br% ♀ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G_1}$
(2) Br ⊕ ♀ $K_{(5)}\ C_5\ A_{1+(9)}\ \underline{G_1}$
(3) Br% ♀ $K_{(5)}\ C_{1+2+(2)}\ A_{9+1}\ \underline{G_1}$
(4) Ebr or Br % $K_{(5)}\ C_{1+2+(2)}\ A_{10}\ \underline{G_1}$
BMP083

87. Bean and gram belong to the family :-
(1) Liliaceae (2) Fabaceae
(3) Solanaceae (4) Cruciferae
BMP087

88. Tetradynamous stamens and cruciform corolla are characteristic features of :–
(1) ***Solanum tuberosum*** (Potato)
(2) ***Solanum nigrum*** (Makoi)
(3) ***Allium cepa*** (Onion)
(4) ***Brassica campestris*** (Mustard)
BMP100

89. Fibre yielding plants belongs to which of the following plant families.
(1) Liliaceae, Poaceae
(2) Malvaceae, Fabaceae
(3) Asteraceae, Brassicaceae
(4) Solanaceae, Graminae
BMP324

SOLANACEAE, ASTERACEAE, LILIACEAE, POACEAE

90. Pentamerous, actinomorphic flowers, bicarpellary ovary with oblique septa, and fruit a capsule or berry, are characteristic features of :-
(1) Asteraceae (2) Brassicaceae
(3) Solanaceae (4) Liliaceae
BMP112

91. *Aloe* and Garlic are the plants of:–
(1) Leguminosae (2) Asteraceae
(3) Liliaceae (4) Poaceae
BMP055

92. Replum is:-
(1) False placenta (2) False septum
(3) False ovule (4) False thalamus
BMP057

93. Inflorescence in Liliaceae is generally: -
(1) Racemose
(2) Solitary/Cymose
(3) Catkin
(4) Hypanthodium
BMP063

94. Inflorescence in Solanaceae is :-
(1) Racemose (2) Cymose
(3) Capitulum (4) Hypanthodium
BMP072

95. ***Lycopersicum esculentum*** (Tomato) belongs to family :-
(1) Solanaceae (2) Malvaceae
(3) Cruciferae (4) Cucurbitaceae
BMP073

96. ***Atropa belladona***, an important medicinal plant is of the family :-
(1) Liliaceae (2) Cucurbitaceae
(3) Cruciferae (4) Solanaceae
BMP074

97. 'Simla mirch', chillies and potato belong to family :-
(1) Solanaceae (2) Compositae
(3) Gramineae (4) Cruciferae
BMP075

98. ***Nicotiana,*** Petunia belong to :-
(1) Malvaceae (2) Liliaceae
(3) Solanaceae (4) Cruciferae
BMP076

99. The drug 'Belladona' is obtained from :-
(1) *Atropa* (2) *Rauwolfia*
(3) *Solanum* (4) *Capsicum*
BMP077

100. Floral formula $\oplus\,\male\,\female\; K_{(5)}\overset{\frown}{C_{(5)}A_5}\underline{G}_{(2)}$ is shown by :-
(1) Onion, Garlic (2) Pea, Bean
(3) Tomato, Brinjal (4) All of the above
BMP089

101. In which of the families the stamens are in two whorls and epiphyllous :-
(1) Brassicaceae (2) Solanaceae
(3) Liliaceae (4) Fabaceae
BMP091

102. A distinct monocot character shown by the flowers of Liliaceae is :-
(1) Hypogynous flowers
(2) Actinomorphic flowers
(3) Trimerous flowers
(4) Bisexual flowers
BMP092

103. ***Allium cepa*** (onion) belongs to the family :-
(1) Solanaceae (2) Liliaceae
(3) Cruciferae (4) Brassicaceae
BMP093

104. Which of the family possess perianth of six tepals:-
(1) Fabaceae (2) Solanaceae
(3) Liliaceae (4) Brassicaceae
BMP094

105. ***Colchicum autumnale*** belongs to :-

(1) Leguminosae (2) Cruciferae

(3) Liliaceae (4) Malvaceae

BMP095

106. ***Smilax*** and ***Gloriosa*** belong to :-

(1) Liliaceae (2) Solanaceae

(3) Fabaceae (4) Cruciferae

BMP096

107. In which of the following family, perianth and trimerous flowers are found :–

(1) Solanaceae (2) Crucifereae

(3) Liliaceae (4) Fabaceae

BMP101

108. $Br\oplus\ \varnothing\ P_{(3+3)}A_{3+3}\ \underline{G}_{(3)}$ floral formula belongs:- to which family :-

(1) Leguminosae (2) Liliaceae

(3) Labiatae (4) Graminae

BMP103

109. The new name for family compositae is

(1) Asteraceae (2) Brassicaceae

(3) Cruciferae (4) Poaceae

BMP315

110. The following diagram is :-



(1) Ray floret (2) Disc floret

(3) Neuter floret (4) All of the above

BMP316

111. The odd one among following members of family Asteraceae is

(1) *Helianthus annus* (2) *Chryserthemum*

(3) *Allium cepa* (4) *Tagetus*

BMP318

112. Other than Liliaceae, which of the following family shows perianth :-

(1) Poaceae (2) Fabaceae

(3) Asteraceae (4) Malvaceae

BMP319

113. Lodicules are reduced perianth in

(1) Green (2) Sunflower

(3) Rice (4) Tulip

BMP320

114. Source of common sugar is :-

(1) Maize (2) Wheat

(3) Sugar cane (4) Rice

BMP321

115. Feathery stigma is found in

(1) Poaceae – Wheat

(2) Brassicaceae – Mustard

(3) Asteraceae – Sunflower

(4) Malvaceae – Chinarose

BMP322

AIM: FREE EDUCATION TO ALL APUL

Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	2	2	4	4	1	3	3	1	2	3	4	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	1	4	2	3	2	2	1	3	2	1	1	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	2	4	1	2	4	3	2	1	4	2	4	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	3	2	1	3	1	1	3	4	4	2	3	2	4	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	2	4	2	4	1	1	1	1	1	3	3	1	3	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	3	2	2	1	1	1	1	4	4	1	2	4	2	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	2	2	2	1	4	1	3	1	3	3	3	2	3	3
Question	106	107	108	109	110	111	112	113	114	115					
Answer	1	3	2	1	1	3	1	3	3	1					

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Placenta and pericarp are both edible portions in:-

(1) Apple (2) Banana

(3) Tomato (4) Potato

BMP151

2. When the margins of sepals or petals overlap one another without any particular direction, the condition is termed as :-

(1) Vexillary (2) Imbricate

(3) Twisted (4) Valvate

BMP152

3. An example of edible underground stem is :

(1) Carrot (2) Groundnut

(3) Sweet potato (4) Potato

BMP153

4. An aggregate fruit is one which develops from:

(1) Multicarpellary syncarpous gynoecium

(2) Multicarpellary apocarpous gynoecium

(3) Complete inflorescence

(4) Multicarpellary superior ovary

BMP154

AIPMT 2015

5. Leaves become modified into spines in :-

(1) Pea (2) Onion

(3) Silk Cotton (4) *Opuntia*

BMP155

6. Perigynous flowers are found in :-

(1) Cucumber (2) China rose

(3) Rose (4) Guava

BMP156

7. In ginger vegetative propagation occurs through:

(1) Offsets (2) Bulbils

(3) Runners (4) Rhizome

BMP157

8. $\oplus \male\!\!\female K_{(5)}\overset{\curvearrowleft}{C}_{(5)}A_5\underline{G}_{(2)}$ is the floral formula of :-

(1) *Sesbania* (2) *Petunia*

(3) *Brassica* (4) *Allium*

BMP158

9. Keel is the characteristic feature of flower of:-

(1) *Indigofera* (2) *Aloe*

(3) Tomato (4) Tulip

BMP159

Re-AIPMT 2015

10. Axile placentation is present in :

(1) *Argemone* (2) *Dianthus*

(3) Lemon (4) Pea

BMP160

11. Among china rose, mustard, brinjal, potato, guava, cucumber, onion and tulip, how many plants have superior ovary?

(1) Four (2) Five

(3) Six (4) Three

BMP161

12. Which one of the following fruits is parthenocarpic?

(1) Banana (2) Brinjal

(3) Apple (4) Jackfruit

BMP162

13. Roots play insignificant role in absorption of water in :-

(1) Wheat (2) Sunflower

(3) *Pistia* (4) Pea

BMP163

NEET-I 2016

14. The standard petal of a papilionaceous corolla is also called :-

(1) Carina (2) Pappus

(3) Vexillum (4) Corona

BMP164

15. Tricarpellary syncarpous gynoecium is found in flowers of :-

(1) Liliaceae (2) Solanaceae

(3) Fabaceae (4) Poaceae

BMP165

16. Which of the following is **not** a stem modification?

(1) Pitcher of *Nepenthes*

(2) Thorns of citrus

(3) Tendrils of cucumber

(4) Flattened structures of *Opuntia*

BMP166

17. Cotyledon of maize grain is called :-

(1) plumule (2) coleorhiza

(3) coleoptile (4) scutellum

BMP167

18. Stems modified into flat green organs performing the functions of leaves are known as :-

(1) Cladodes

(2) Phyllodes

(3) Phylloclades

(4) Scales

BMP168

NEET-II 2016

19. The term 'polyadelphous' is related to :-

(1) Corolla (2) Calyx

(3) Gynoecium (4) Androecium

BMP169

20. How many plants among *Indigofera, Sesbania, Salvia, Allium, Aloe*, mustard, groundnut, radish, gram and turnip have stamens with different lengths in their flowers ?

(1) Five (2) Six

(3) Three (4) Four

BMP170

21. Radial symmetry is found in the flowers of:-

(1) *Pisum* (2) *Cassia*

(3) *Brassica* (4) *Trifolium*

BMP171

22. Free-central placentation is found in :-

(1) *Brassica* (2) *Citrus*

(3) *Dianthus* (4) *Argemone*

BMP172

NEET (UG) 2017

23. Coconut fruit is a :-

(1) Berry (2) Nut

(3) Capsule (4) Drupe

BMP173

24. In *Bougainvillea* thorns are the modifications of :-

(1) Adventitious root (2) Stem

(3) Leaf (4) Stipules

BMP174

25. Root hairs develop from the region of :-

(1) Elongation

(2) root cap

(3) Meristematic activity

(4) Maturation

BMP175

26. The morphological nature of the edible part of coconut is:-

(1) Cotyledon (2) Endosperm

(3) Pericarp (4) Perisperm

BMP176

NEET (UG) 2018

27. Offsets are produced by:-

(1) Meiotic divisions (2) Mitotic divisions

(3) Parthenocarpy (4) Parthenogenesis

BMP177

28. Pneumatophores occur in :-

(1) Halophytes

(2) Free-floating hydrophytes

(3) Carnivorous plants

(4) Submerged hydrophytes

BMP178

29. Sweet potato is a modified :-

(1) Stem

(2) Adventitious root

(3) Tap root

(4) Rhizome

BMP179

NEET (UG) 2019

30. Placentation, in which ovules develop on the inner wall of the ovary or in peripheral part, is:

(1) Basal (2) Axile

(3) Parietal (4) Free central

BMP180

NEET (UG) 2019 (Odisha)

31. Which of the following shows whorled phyllotaxy ?

(1) Mustard (2) China rose

(3) *Alstonia* (4) *Calotropis*

BMP181

32. Bicarpellary ovary with obliquely placed septum is seen in :-

(1) *Brassica* (2) *Aloe*

(3) *Solanum* (4) *Sesbania*

BMP182

33. Match the placental types **(column-I)** with their examples **(column-II):-**

Column-I	**Column-II**
(a) Basal	(i) Mustard
(b) Axile	(ii) China rose
(c) Parietal	(iii) *Dianthus*
(d) Free central	(iv) Sunflower

Choose the correct answer from the following: **options:**

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

BMP183

NEET (UG) 2020

34. Ray florets have :-

(1) Half inferior ovary

(2) Inferior ovary

(3) Superior ovary

(4) Hypogynous ovary

BMP184

35. The roots that originate from the base of the stem are :-

(1) Lateral roots (2) Fibrous roots

(3) Primary roots (4) Prop roots

BMP185

36. The ovary is half inferior in :-

(1) Plum (2) Brinjal

(3) Mustard (4) Sunflower

BMP186

NEET (UG) 2020 (COVID-19)

37. In some plants thalamus contributes to fruit formation. Such fruits are termed as :-

(1) False fruits

(2) Aggregate fruits

(3) True fruits

(4) Parthenocarpic fruit

BMP187

38. Correct position of floral parts over thalamus in mustard plant is :-

(1) Gynoecium occupies the highest position, while the other parts are situated below it.

(2) Margin of the thalamus grows upward, enclosing the ovary completely, and other parts arise below the ovary.

(3) Gynoecium is present in the centre and other parts cover it partially.

(4) Gynoecium is situated in the centre, and other parts of the flower are located at the rim of the thalamus, at the same level.

BMP188

39. Which of the following is the correct floral formula of Liliaceae ?

(1) $\% \, ♀ \, C_{1+2+(2)} A_{(9)+1} \underline{G}_1$ (2) $\oplus \, ♂♀ \, K_{(5)} \overset{\frown}{C_{(5)}} A_5 \, \underline{G}_{(2)}$

(3) $Br \oplus \, ♀ \, \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G}_{(3)}$ (4) $\oplus \, ♀ \, K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$

BMP189

40. Vegetative propagule in *Agave* is termed as:-

(1) Rhizome (2) Bulbil

(3) Offset (4) Eye

BMP190

41. Identify the correct features of Mango and Coconut fruits:-

(i) In both fruit is a drupe

(ii) Endocarp is edible in both

(iii) Mesocarp in Coconut is fibrous, and in Mango it is fleshy

(iv) In both, fruit develops from monocarpellary ovary

Select the correct option from below :

(1) (i), (iii) and (iv) only

(2) (i), (ii) and (iii) only

(3) (i) and (iv) only

(4) (i) and (ii) only

BMP191

NEET (UG) 2021

42. Diadelphous stamens are found in:-

(1) China rose

(2) *Citrus*

(3) Pea

(4) China rose and *Citrus*

BMP192

43. Match **Column - I** with **Column - II**:-

Column - I	**Column - II**
(a) $\% \, ♂♀ \, K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$	(i) Brassicaceae
(b) $\oplus \, ♂♀ \, K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_2$	(ii) Liliaceae
(c) $\oplus \, ♂♀ \, \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G}_{(3)}$	(iii) Fabaceae
(d) $\oplus \, ♂♀ \, K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(iv)	(ii)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(iv)	(ii)	(i)	(iii)

BMP193

Re-NEET (UG) 2021

44. Vegetative propagation in mint and jasmine occurs by

(1) Runner (2) Sucker

(3) Stolon (4) Offset

BMP293

45. The following figure represents the root system of



(1) *Triticum* (2) Mango
(3) Banyan (4) *Brassica*

BMP294

46. Among mustard, mulaithi, watermelon, soyabean, pumpkin, ashwagandha, *Asparagus*, brinjal, *Allium*, *Trifolium*, Rose, *Indigofera*, bitter gourd, plum, cucumber, arhar, moong and peach, how many plants have hypogynous flowers ?

(1) Three (2) Ten
(3) Four (4) Eleven

BMP295

47. Pinnate, compound leaves with reticulate venation are present in

(1) Neem (2) Mango
(3) *Alstonia* (4) Silk cotton

BMP296

NEET (UG) 2022

48. Identify the **correct** set of statements:-

(a) The leaflets are modified into pointed hard thorns in *Citrus* and *Bougainvillea*

(b) Axillary buds form slender and spirally coiled tendrils in cucumber and pumpkin

(c) Stem is flattened and fleshy in *Opuntia* and modified to perform the function of leaves

(d) *Rhizophora* shows vertically upward growing roots that help to get oxygen for respiration

(e) Subaerially growing stems in grasses and strawberry help in vegetative propagation

Choose the **correct answer** from the options given below:

(1) (a) and (d) Only

(2) (b), (c), (d) and (e) Only

(3) (a), (b), (d) and (e) Only

(4) (b) and (c) Only

BMP194

49. The flowers are Zygomorphic in:-

(a) Mustard (b) Gulmohaur
(c) *Cassia* (d) *Datura*
(e) Chilly

Choose the **correct answer** from the options given below:

(1) (b), (c) Only (2) (d), (e) Only
(3) (c), (d), (e) Only (4) (a), (b), (c) Only

BMP195

50. Which one of the following plants shows vexillary aestivation and diadelphous stamens ?

(1) *Pisum sativum*

(2) *Allium cepa*

(3) *Solanum nigrum*

(4) *Colchicum autumnale*

BMP196

NEET (UG) 2022 (OVERSEAS)

51. Identify the **correct** statements related to the androecium in the flower.

(a) The sterile stamens are called staminodes.

(b) When stamens are attached to petals they are called epipetalous.

(c) Monoadelphy is seen in China-rose.

(d) Polyadelphy is seen in Pea.

(e) Variation in the length of anther filaments is seen in Mustard.

Choose the **correct** answer from the options given below :
(1) (a), (b), (c) and (e) only
(2) (a), (b) and (c) only
(3) (b), (c) and (d) only
(4) (a), (c), (d) and (e) only

BMP297

52. Which of the following plants possesses the placentation of ovules borne on central axis with no septa?
(1) Pea (2) China-rose
(3) Primrose (4) Lemon

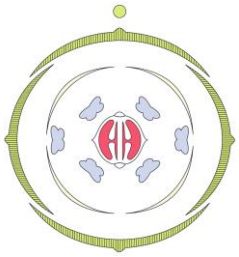
BMP298

53. Choose the mismatched pair of leaf character with its example :
(1) Alternate phyllotaxy - China-rose
(2) Leaf tendril - Pea
(3) Opposite phyllotaxy - *Calotropis*
(4) Palmately compound - *Alstonia*

BMP299

Re-NEET (UG) 2022

54. The Floral Diagram represents which one of the following families ?



(1) Fabaceae (2) Brassicaceae
(3) Solanaceae (4) Liliaceae

BMP197

55. Match **List - I** with **List - II** :-

	List I		List II
a.	Imbricate	i.	*Calotropis*
b.	Valvate	ii.	*Cassia*
c.	Vexillary	iii.	Cotton
d.	Twisted	iv.	Bean

Choose the **correct answer** from the options given below :-
(1) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)
(2) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
(3) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)
(4) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

BMP198

NEET (UG) 2023

56. Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens, pick out the characteristics specific to family. Fabaceae but not found in Solanaceae or Liliaceae.
(1) Polyadelphous and epipetalous stamens
(2) Monoadelphous and Monothecous anthers
(3) Epiphyllous and Dithecous anthers
(4) Diadelphous and Dithecous anthers

BMP300

57. Axile placentation is observed in
(1) China rose, Beans and Lupin
(2) Tomato, Dianthus and Pea
(3) China rose, Petunia and Lemon
(4) Mustard, Cucumber and Primrose

BMP301

NEET (UG) 2023 (Manipur)

58. In *Calotropis*, aestivation is :
(1) Valvate (2) Vexillary
(3) Imbricate (4) Twisted

BMP302

59. In a pea flower, five petals are arranged in a specialized manner with one posterior, two lateral and two anterior. These are named as _____, _____ and _____ respectively.
(1) Keel, Wings and Standard
(2) Vexillum, Keel and Standard
(3) Keel, Standard and Carina
(4) Standard, Wings and Keel

BMP303

60. Match the following :-

Type of flower		Example	
(A)	Zygomorphic	(I)	Mustard
(B)	Hypogynous	(II)	Plum
(C)	Perigynous	(III)	*Cassia*
(D)	Epigynous	(IV)	Cucumber

Select the **correct** option :-

(1) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(2) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

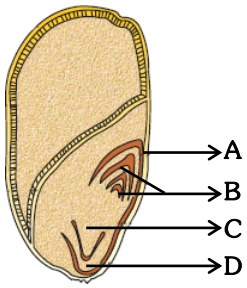
(3) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

BMP304

NEET (UG) 2024

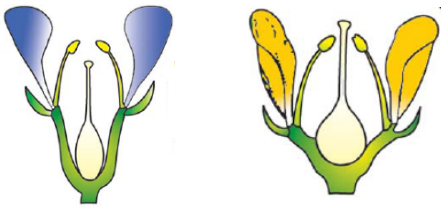
61. Identify the part of the seed from the given figure which is destined to form root when the seed germinates.



(1) A (2) B (3) C (4) D

BMP353

62. Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (a) and (b).



(a) **(b)**

(1) (a)–Epigynous; (b)–Hypogynous

(2) (a)–Hypogynous; (b)–Epigynous

(3) (a)–Perigynous; (b)–Epigynous

(4) (a)–Perigynous; (b)–Perigynous

BMP354

63. Which of the following is an example of actinomorphic flower ?

(1) *Datura*

(2) *Cassia*

(3) *Pisum*

(4) *Sesbania*

BMP355

64. Match List-I with List-II.

List I (Types of Stamens)		List II (Example)	
A.	Monoadelphous	I.	Citrus
B.	Diadelphous	II.	Pea
C.	Polyadelphous	III.	Lily
D.	Epiphyllous	IV.	China-rose

Choose the correct answer from the options given below :-

(1) A–IV, B–II, C–I, D–III

(2) A–IV, B–I, C–II, D–III

(3) A–I, B–II, C–IV, D–III

(4) A–III, B–I, C–IV, D–II

BMP356

65. Match **List-I** with **List-II**.

	List I		List II
A.	Rose	I.	Twisted aestivation
B.	Pea	II.	Perigynous flower
C.	Cotton	III.	Drupe
D.	Mango	IV.	Marginal placentation

Choose the **correct** answer from the options given below :-

(1) A-II, B-IV, C-I, D-III

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-III, C-IV, D-I

BMP357

Re-NEET (UG) 2024

66. Match **List-I** with **List-II**.

List I		List II	
A.	Vexillary aestivation	I.	Brinjal
B.	Epipetalous stamens	II.	Peach
C.	Epiphyllous stamens	III.	Pea
D.	Perigynous flower	IV.	Lily

Choose the **correct** answer from the options given below :-

(1) A-III, B-I, C-IV, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-II, C-I, D-IV

(4) A-II, B-I, C-IV, D-III

BMP358

67. Match **List-I** with **List-II**.

List I		List II	
A.	China rose	I.	Free central
B.	Mustard	II.	Basal
C.	Primrose	III.	Axile
D.	Marigold	IV.	Parietal

Choose the **correct** answer from the options given below

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-IV, C-II, D-I

BMP359

68. Which of the following examples show monocarpellary, unilocular ovary with many ovules ?

A. *Sesbania*

B. Brinjal

C. *Indigofera*

D. Tobacco

E. *Asparagus*

Choose the **correct** answer from the options given below :-

(1) B and E only (2) C, D and E only

(3) A, B and D only (4) A and C only

BMP360

Exercise-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	2	4	3	4	2	1	3	3	1	3	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	3	4	4	3	3	4	2	4	2	2	1	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	3	2	2	1	1	1	3	2	1	3	1	3	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	1	2	1	1	1	3	4	2	2	4	3	1	4	4
Question	61	62	63	64	65	66	67	68							
Answer	3	4	1	1	1	1	3	4							

Exercise - III

Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Roots developing from plant parts other than radicle are :-

(1) Tap roots (2) Adventitious roots

(3) Both (1) & (2) (4) Conical roots

BMP199

2. Prop roots are :-

(1) Tap roots (2) Adventitious roots

(3) Secondary roots (4) All

BMP200

3. A monocot can be distinguished from a dicot by:

(1) Aestivation (2) Venation

(3) Both (1) & (2) (4) Placentation

BMP201

4. In Racemose, flowers are arranged in :-

(1) Acropetal order (2) Centrifugal order

(3) Both (1) & (2) (4) Basipetal order

BMP202

5. Axis of inflorescence is :-

(1) Pedicel (2) Peduncle

(3) Petiole (4) All

BMP203

6. Stilt roots occur in :-

(1) Groundnut (2) Rice

(3) Sugarcane (4) Wheat

BMP204

7. Function of stem is to :-

(1) Bear leaves and branches

(2) Conduction of water and minerals

(3) Conduction and storage of food

(4) All

BMP205

8. Arrangement of leaves on a stem or branch is:

(1) Venation (2) Vernation

(3) Inflorescence (4) Phyllotaxy

BMP206

9. A slender lateral branch arises from the base of the main axis and after growing aerially for some time arch downwards to touch the ground. Such type of modification is :-

(1) Runner (2) Sucker

(3) Stolon (4) Offset

BMP207

10. A lateral branch with short internodes and each node bearing a rosette of leaves and a tuft of roots in aquatic plants, such type of modification is :-

(1) Runner (2) Stolon

(3) Sucker (4) Offset

BMP208

11. Lateral branches originate from the basal and underground portion of the main stem, grow horizontally beneath the soil and then comes out obliquely upward giving rise to leafy shoots, such type of modification is

(1) Runner (2) Stolon

(3) Sucker (4) Offset

BMP209

12. In some leguminous plants the leaf base may become swollen, it is called :-

(1) Pulvinus (2) Lamina

(3) Leaf margin (4) Stipule

BMP210

13. When the veins run parallel to each other within a lamina, the venation is termed as:-
(1) Parallel (2) Reticulate
(3) Both (1) and (2) (4) Pinnate
BMP211

14. If the leaflets are present on a common axis, the rachis, leaf is called :-
(1) Palmately compound leaf
(2) Pinnately compound leaf
(3) Simple leaf
(4) Trifoliate leaf
BMP212

15. If the leaf lets are attached at the tip of petiole, leaf is called :-
(1) Pinnately compound leaf
(2) Palmately compound leaf
(3) Simple leaf
(4) Unipinnate leaf
BMP213

16. When single leaf arises at each node then phyllotaxy is called :-
(1) Alternate (2) Opposite
(3) Whorled (4) Pinnate
BMP214

17. Opposite phyllotaxy is present in :-
(1) Mustard (2) Guava
(3) China rose (4) *Alstonia*
BMP215

18. Arrangement of sepals and petals in floral bud with respect of other members of same whorl is known as :-
(1) Venation (2) Aestivation
(3) Cohesion (4) Adhesion
BMP216

19. Didynamous condition is related to :-
(1) Androecium (2) Inflorescence
(3) Gynoecium (4) All
BMP217

20. Stamens attached to petals are :-
(1) Epipetalous (2) Epiphyllous
(3) Episepalous (4) All
BMP218

21. Androecium is the whorl of :-
(1) Anthers (2) Stamens
(3) Filaments (4) Tepals
BMP219

22. Syncarpous gynoecium has two or more :-
(1) Free carpels (2) Fused carpels
(3) Free Ovaries (4) All
BMP220

23. A typical flower with superior ovary and other floral parts inferior is called :-
(1) Polygamous (2) Hypogynous
(3) Perigynous (4) Epigynous
BMP221

24. Arrangement of ovules within the ovary is known as:-
(1) Aestivation (2) Placentation
(3) Both (1) & (2) (4) Cohesion
BMP222

25. When gynoecium is present in the topmost position of thalamus, the ovary is known as:-
(1) Inferior (2) Half inferior
(3) Half superior (4) Superior
BMP223

26. Drupe contains :-
(1) Stony endocarp (2) Stony mesocarp
(3) Edible epicarp (4) Edible endocarp
BMP224

27. Plants which yield pulses belong to family :-
(1) Fabaceae (2) Liliaceae
(3) Brassicaceae (4) Solanaceae
BMP226

28. Edible portion of Mango is :-
(1) Epicarp (2) Endocarp
(3) Mesocarp (4) Placenta

BMP227

29. Fibrous coir is obtained from which part of the Coconut ?
(1) Epicarp (2) Mesocarp
(3) Endocarp (4) Seed coat

BMP228

30. A characteristic feature of Ovary of Mustard is:
(1) Presence of false septum
(2) Axile placentation
(3) Epigynous
(4) All of the above

BMP229

31. In Cacti the spines are modification of :-
(1) Leaf (2) Branch
(3) Epidermis (4) Axillary bud

BMP230

32. Placentaion in Solanaceae is :-
(1) Parietal (2) Marginal
(3) Axile (4) Basal

BMP231

33. Coconut is which type of fruit ?
(1) Drupe (2) Cypsela
(3) Berry (4) Pome

BMP232

34. Arrangement of flowers on the floral axis is termed as inflorescence. Regarding to inflorescence which of the following statement is not correct :-
(1) In racemose–main axis is continues to grow
(2) In cymose–main axis terminates into flower
(3) In racemose – flowers are in basipetal succession
(4) In cymose – growth of main axis is limited

BMP233

35. Regarding to symmetry of flower which of the following plant is odd:-
(1) Pea (2) Mustard
(3) *Datura* (4) Chilli

BMP234

36. Find out the wrong match:-
(1) Actinomorphic – *Datura*
(2) Radial symmetry – Mustard
(3) Zygomorphic – Bean
(4) Bilateral Symmetry – Chilli

BMP235

37. In which of the following plant flower can not be divided into two similar halves by any vertical plane:-
(1) Mustard (2) *Cassia*
(3) *Canna* (4) *Datura*

BMP236

38. In which of the following plant gynoecium occupies the highest position while the other parts situated below it?
(1) Brinjal (2) Plum
(3) Rose (4) Guava

BMP237

39. Match the following and select correct option:-

(a) Hypogynous	(i) Lily, Onion
(b) Perigynous	(ii) Cucumber, Ray florets of sunflower
(c) Epigynous	(iii) Plum, Peach
(d) Perianth	(iv) China rose, Brinjal

(1) a-(iv), b-(iii), c-(ii), d-(i)
(2) a-(iv), b-(ii), c-(iii), d-(i)
(3) a-(iii), b-(ii), c-(iv), d-(i)
(4) a-(iii), b-(iv), c-(ii), d-(i)

BMP238

40. Calyx is the outermost accessory whorl of flower. What is the function of calyx?
(1) Help in pollination
(2) Help in protection of flower during bud condition
(3) Help in fertilization
(4) Help in seed germination

BMP239

41. Match the following with respect to aestivation in petals and select correct option:-

(a) Valvate	(i) China rose
(b) Twisted	(ii) *Calotropis*
(c) Imbricate	(iii) Pea
(d) Vexillary	(iv) *Cassia*

(1) a-(ii), b-(i), c-(iv), d-(iii)
(2) a-(ii), b-(iii), c-(iv), d-(i)
(3) a-(i), b-(ii), c-(iii), d-(iv)
(4) a-(iv), b(-iii), c-(ii), d-(i)

BMP240

42. Match the following:-

(a) Epiphyllous stamen	(i) *Citrus*
(b) Monoadelphous stamen	(ii) Pea
(c) Diadelphous stamen	(iii) China rose
(d) Polyadelphous stamen	(iv) Lily

(1) a-(ii), b-(i), c-(iii), d-(iv)
(2) a-(i), b-(ii), c-(iii), d-(iv)
(3) a-(iv), b-(iii), c-(i), d-(ii)
(4) a-(iv), b-(iii), c-(ii), d-(i)

BMP241

43. In which type of placentation, ovules are present on central axis:-
(1) Axile (2) Parietal
(3) Free central (4) Both (1) & (3)

BMP242

44. The floral formula, $Br\oplus\,\female\,\overset{\frown}{P_{(3+3)}}\,A_{3+3}\,\underline{G}_{(3)}$ belongs to :-
(1) Onion (2) Makoi
(3) Pea (4) Mustard

BMP243

45. Diadelphous stamens are found in :-
(1) Liliaceae (Onion)
(2) Fabaceae (Pea)
(3) Poaceae (Wheat)
(4) Malvaceae (China rose)

BMP244

46. Pneumatophores take part in :-
(1) Excretion (2) Feeding
(3) Respiration (4) Reproduction

BMP245

47. Gram belongs to the family :-
(1) Gramineae (2) Fabaceae
(3) Liliaceae (4) Solanaceae

BMP246

48. Ginger is :-
(1) Stem (2) Root
(3) Leaf (4) Fruit

BMP247

49. Inflorescence in members of papilionideae is:-
(1) Racemose
(2) Cymose
(3) Cyathium
(4) Hypanthodium

BMP248

50. *Sesbania* belongs to :-
(1) Liliaceae (2) Fabaceae
(3) Solanaceae (4) Cruciferae

BMP249

51. Swollen placenta with oblique septum can be seen in:-
(1) Brassicaceae (2) Fabaceae
(3) Liliaceae (4) Solanaceae

BMP251

52. Colchicine a mitotic poison can be obtained from a plant of:-
(1) Brassicaceae (2) Fabaceae
(3) Solanaceae (4) Liliaceae
BMP252

53. Perianth condition is characteristic of:-
(1) Brassicaceae (2) Fabaceae
(3) Solanaceae (4) Liliaceae
BMP253

54. Match the following and select correct option :-

(a) Mustard	(i) Liliaceae
(b) Mulaithi	(ii) Solanaceae
(c) Ashwagandha	(iii) Fabaceae
(d) *Tulip*	(iv) Brassicaceae

(1) a–(iv), b–(iii), c–(ii), d–(i)
(2) a–(iv), b–(iii), c–(i), d–(ii)
(3) a–(iii), b–(iv), c–(ii), d–(i)
(4) a–(i), b–(ii), c–(iii), d–(iv)
BMP254

(B) ANALYTICAL QUESTIONS

55. Which is correct pair for edible part :–
(1) Tomato – Thalamus
(2) Mango – Cotyledons
(3) Guava – Endosperm
(4) Date palm – Pericarp
BMP256

56. Edible part of Banana :–
(1) Epicarp
(2) Mesocarp and less developed endocarp
(3) Endocarp and less developed mesocarp
(4) Epicarp & mesocarp
BMP257

57. A small, dry, one-seeded fruit with its pericarp (fruit wall) fused with the seed-coat, developing from a monocarpellary gynoecium is called :-
(1) Cypsela (2) Siliqua
(3) Caryopsis (4) Samara
BMP258

58. Flowers are zygomorphic in :-
(1) Mustard & Chilli (2) Radish & Datura
(3) Lily & Canna (4) Candytuft & Pea
BMP259

59. Dye 'Neel' is obtained from:-
(1) *Indigofera tinctoria*
(2) *Brassica oleracea*
(3) *Brassica rapa*
(4) *Capsella bursa pestoris*
BMP260

60. A floral formula represents :-
(1) Plant symmetry
(2) Floral position
(3) Floral characters
(4) Floral functions
BMP261

61. Bicarpellary, syncarpous gynoecium with bilocular ovary, axile swollen placenta and oblique septum occurs in:-
(1) Cruciferae (2) Solanaceae
(3) Cucurbitaceae (4) Liliaceae
BMP262

62. Colchicine is obtained from :-
(1) *Atropa belladona*
(2) *Colchicum autumnale*
(3) *Withania somnifera*
(4) *Nicotiana tabacum*
BMP263

63. The leguminous plants are important in agriculture because :-
(1) They are disease resistant
(2) They require very little irrigation
(3) They help in nitrogen fixation
(4) Crops of legumes can be produced in a year
BMP266

64. In family fabaceae, 5 petals form a unique association, In which 3 different elements participate, these are standard (vexillum), Wings (alae) & Keel (carina). What is the number of these elements :-
(1) 1, 2, 2 respectively
(2) 2, 1, 2 respectively
(3) 1, 1, 3 respectively
(4) 2, 2, 1 respectively
BMP267

65. The ovules after fertilisation, develop into:-
(1) Seeds (2) Fruits
(3) Endosperm (4) Pericarp
BMP270

66. Seeds are endospermous in :–
(1) Bean (2) Gram
(3) Pea (4) Castor
BMP271

67. In which of the following membranous seed coat is fused with the Fruit wall ?
(1) Pea (2) Gram
(3) Maize (4) Castor
BMP272

68. Replum is present in the ovary of flower of:
(1) Sun flower (2) Pea
(3) Lemon (4) Mustard
BMP113

69. The fruit is chambered, developed from inferior ovary and has seeds with succulent testa in :-
(1) Guava (2) Cucumber
(3) Pomegranate (4) Orange
BMP114

70. Dry indehiscent single-seeded fruit formed from bicarpellary syncarpous inferior ovary is:
(1) Berry (2) Cremocarp
(3) Caryopsis (4) Cypsela
BMP116

71. An example of axile placentation is :-
(1) Marigold (2) *Argemone*
(3) *Dianthus* (4) Lemon
BMP119

72. Vegetative propagation in mint occurs by :-
(1) Sucker (2) Runner
(3) Offset (4) Rhizome
BMP120

73. The floral formula $\oplus \, ♀ \, K_{(5)} \overset{\frown}{C_{(5)}} A_5 \, \underline{G}_{(2)}$ is that of :-
(1) Tobacco (2) Tulip
(3) Soyabean (4) Sunnhemp
BMP121

74. In unilocular ovary with a single ovule the placentation is :-
(1) Axile (2) Marginal
(3) Basal (4) Free Central
BMP122

75. The technical term used for the androecium in a flower of China rose *(Hibiscus rosa-sinensis)* is :
(1) Polyadelphous (2) Monoadelphous
(3) Diadelphous (4) Polyandrous
BMP123

76. Ovary is half-inferior in the flowers of:-
(1) Cucumber (2) Guava
(3) Plum (4) Brinjal
BMP124

77. Keel is characteristic of the flowers of:-
(1) Bean (2) Gulmohur
(3) *Cassia* (4) *Calotropis*
BMP125

78. Vegetative propagation in *Pistia* occurs by:-
(1) Offset (2) Runner
(3) Sucker (4) Stolon
BMP126

79. Which one of the following is a xerophytic plant in which the stem is modified into a flat, green and succulent structure ?

(1) Casuarina (2) *Hydrilla*

(3) *Acacia* (4) *Opuntia*

BMP127

80. Consider the following four statements A, B, C and D and select the right option for two correct statements :-

Statements :

(A) In vexillary aestivation, the large posterior petal is called - *standard*, two lateral ones are *wings* and two small anterior petals are termed *keel*.

(B) The floral formula for Liliaceae is

$\oplus \male \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G_{(3)}}$

(C) In pea flower the stamens are monoadelphous

(D) The floral formula for Solanaceae is

$\oplus \male K_{(3)} C_{(3)} A_{(4)} \underline{A_{(2)}}$

The correct statements are :-

(1) (A) and (B) (2) (B) and (C)

(3) (C) and (D) (4) (A) and (C)

BMP128

81. The correct floral formula of soyabean is :-

(1) % $\male K_5 C_{1+(2)+2} A_{(9)+1} \underline{G_1}$

(2) % $\male K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G_1}$

(3) % $\male K_{(5)} C_{1+2+(2)} A_{1+(9)} \underline{G_1}$

(4) % $\male K_{(5)} C_{1+(2)+2} A_{(9)+1} G_{\bar{1}}$

BMP129

82. Aestivation of petals in the flower of cotton is correctly shown in :-

(1)  (2) 

(3)  (4) 

BMP130

83. The "Eyes" of the potato tuber are :-

(1) Root buds

(2) Flower buds

(3) Shoot buds

(4) Axillary buds

BMP131

84. Flowers are Zygomorphic in :-

(1) Mustard (2) Gulmohur

(3) Tomato (4) *Datura*

BMP132

85. The ovary is half inferior in flowers of :-

(1) Peach (2) Cucumber

(3) Cotton (4) Guava

BMP133

86. A drupe develops in :-

(1) Mango (2) Wheat

(3) Pea (4) Tomato

BMP134

87. Which one of the following statements is correct?

(1) In tomato, fruit is a capsule

(2) Seeds of orchids have oil-rich endosperm

(3) Placentation in *Primrose* is basal

(4) Flower of tulip is a modified shoot

BMP135

88. The correct floral formula of chilli is :-

(1) $\oplus \male K_{(5)} C_5 A_5 \underline{G_{(2)}}$

(2) $\oplus \male K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G_{(2)}}$

(3) $\oplus \male K_{(5)} C_{(5)} A_{(5)} G_2$

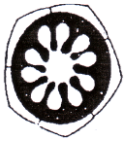
(4) $\oplus \male K_5 \overset{\frown}{C_5} A_{(5)} G_2$

BMP136

89. Which one of the following diagrams represents the placentation in *Dianthus* ?

(1) 

(2) 

(3) 

(4) 

BMP137

90. Sweet potato is homologous to :-

(1) Turnip (2) Potato

(3) *Colocasia* (4) Ginger

BMP138

91. Whorled, simple leaves with reticulate venation are present in :

(1) *Alstonia* (2) *Calotropis*

(3) Mustard (4) China Rose

BMP139

92. Which one of the following pairs is wrongly matched while the remaining three are correct?

(1) *Agave* - Bulbils

(2) Grass - Runner

(3) Water hyacinth - Runner

(4) *Bryophyllum* - Leaf buds

BMP140

93. Placentation in tomato and lemon is :-

(1) Marginal (2) Axile

(3) Parietal (4) Free central

BMP141

94. Vexillary aestivation is characteristic of the family:

(1) Solanaceae (2) Brassicaceae

(3) Fabaceae (4) Asteraceae

BMP142

95. Phyllode is present in :-

(1) *Australian Acacia*

(2) *Opuntia*

(3) *Asparagus*

(4) *Euphorbia*

BMP143

96. How many plants in the list given below have composite fruits that develop from an inflorescence? Walnut, poppy, radish, fig, pineapple, apple, tomato, mulberry:-

(1) Two (2) Three

(3) Four (4) Five

BMP144

97. Cymose inflorescence is present in :-

(1) *Trifolium* (2) *Brassica*

(3) *Solanum* (4) *Sesbania*

BMP145

98. The coconut water and the edible part of coconut are equivalent to :-

(1) Mesocarp

(2) Embryo

(3) Endosperm

(4) Endocarp

BMP146

99. The gynoecium consists of many free pistils in flowers of :-

(1) *Papaver* (2) *Michelia*

(3) *Aloe* (4) Tomato

BMP147

100. How many plants in the list given below have marginal placentation ?

Mustard, Gram, Tulip, *Asparagus*, Arhar, Sun hemp, Chilli, Colchicine, Onion, Moong, Pea, Tobacco, Lupin

(1) Six (2) Three

(3) Four (4) Five

BMP148

101. In china rose the flowers are :-

(1) Zygomorphic, epigynous with twisted aestivation

(2) Actinomorphic, hypogynous with twisted aestivation

(3) Actinomorphic, epigynous with valvate aestivation

(4) Zygomorphic, hypogynous with imbricate aestivation

BMP149

102. Among bitter gourd, mustard, brinjal, pumpkin china rose, lupin, cucumber, sunnhemp, gram, guava, bean, chilli, plum, *Petunia*, tomato, rose, *Withania*, potato, onion, *Aloe* and tulip how many plants have hypogynous flower?

(1) Eighteen (2) Six

(3) Ten (4) Fifteen

BMP150

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	2	1	2	3	4	4	3	4	3	1	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	1	1	2	2	2	2	4	1	1	3	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	1	3	1	4	3	1	1	2	1	4	4	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	1	1	2	4	4	4	1	4	2	3	4	1	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	3	1	1	4	3	4	3	4	4	1	1	3	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	1	1	4	1	2	3	4	2	1	1	4	2	3	1
Question	91	92	93	94	95	96	97	98	99	100	101	102			
Answer	1	3	2	3	1	2	3	3	2	1	2	4			

Exercise - I (Conceptual Questions) Build Up Your Understanding

THE TISSUE (MERISTEMATIC TISSUE TO SIMPLE PERMANENT TISSUE)

1. A meristem may be defined as the group of cells which -
(1) Does not divide
(2) Conserve food
(3) Divide continuously to give rise to new cells.
(4) Elongate, mature and add to the group of cells.

BAP001

2. The length of different internodes in a culm of sugarcane is variable because of :-
(1) Position of axillary buds.
(2) Size of leaf lamina at the node below each internode.
(3) Intercalary meristem.
(4) Shoot apical meristem.

BAP118

3. The secondary meristem originates from-
(1) Promeristem
(2) Primary meristem
(3) Primary permanent tissue
(4) Secretory tissue

BAP006

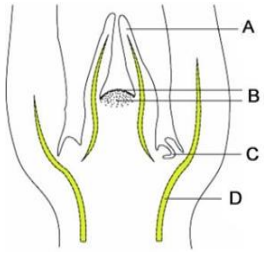
4. The function of root cap is-
(1) Provide protection to root apex
(2) Storage of food products
(3) Absorption of nutrients
(4) None of the above

BAP007

5. Aerenchyma is helpful to plants by :-
(1) Providing buoyancy to hydrophytes
(2) Promoting photosynthesis
(3) Give mechanical strength to plants
(4) Giving flexibility to plants

BAP009

6. Label the figure correctly.



(1) A= Shoot apical meristematic zone,
B= Leaf primordium,
C= Axillary bud,
D= Differentiating vascular tissue
(2) A= Leaf primordium,
B= Shoot apical meristematic zone,
C= Axillary bud,
D= Differentiating vascular tissue
(3) A= Axillary bud,
B= Leaf primordium,
C= Shoot apical meristematic zone,
D= Differentiating vascular tissue
(4) A= Axillary bud,
B= Shoot apical meristematic zone,
C= Leaf primordium,
D= Differentiating vascular tissue

BAP272

7. In plants, which of the following cells are living :-
(1) Xylem vessels (2) Meristem
(3) Cork (4) Fibres

BAP011

8. Which of the following tissues form the main bulk of storage organ-
(1) Parenchyma (2) Collenchyma
(3) Sclerenchyma (4) Aerenchyma

BAP012

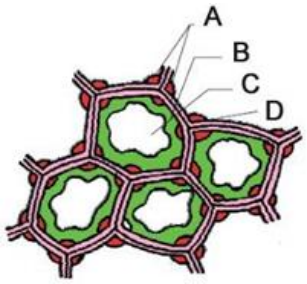
9. Mechanical tissue consisting of living cells is
(1) Sclerenchyma (2) Collenchyma
(3) Chlorenchyma (4) Parenchyma

BAP014

10. **Statement-1 :** Often collenchyma contain chloroplast.
Statement-2: Collenchyma with chloroplast assimilate food.
(1) Only statement 1 is correct.
(2) Only statement 2 is correct.
(3) Both 1 and 2 statements are incorrect.
(4) Both 1 and 2 statements are correct.

BAP273

11. Identify the following figure and label A, B, C and D.



(1) Collenchyma. A= Thickened corner B= Protoplasm, C= Vacuole, D= Cell wall
(2) Parenchyma. A= Thickened corners, B= Protoplasm, C= Vacuole, D= Cell wall
(3) Collenchyma. A= Thickened corner B= Vacuole, C= Protoplasm, D= Cell wall
(4) Sclerenchyma. A= Thickened corners, B= Protoplasm, C= Vacuole, D= Cell wall

BAP274

12. Read the following statements carefully and choose the correct option accordingly :-
(a) During the formation of leaves and elongation of stem, some cells "left behind" from shoot apical meristem.
(b) Meristem which occurs between mature tissue is known as intercalary meristem.
(c) Cylindrical meristems such as fascicular vascular cambium, interfascicular cambium and cork cambium are examples of apical meristem.
(d) During the formation of the primary plant body, specific regions of the lateral meristem produce dermal tissues, ground tissues and vascular tissues.
(1) Only a and c are correct
(2) Only a and d are correct
(3) Only a and b are correct
(4) Only a, b and c are correct

BAP275

13. Which one of the following is *not* a lateral meristem?
(1) Intercalary meristem
(2) Intrafascicular cambium
(3) Interfascicular cambium
(4) Phellogen

BAP123

COMPLEX PERMANENT TISSUE

14. Transport of food material in higher plants takes place through :
(1) Transfusion tissue (2) Tracheids
(3) Sieve elements (4) Companion cells

BAP126

15. Which of following plant cells are without vacuoles without nuclei and are dead -
(1) Cambium cells (2) Xylem vessels
(3) Root hairs (4) Companion cells

BAP018

16. The cell functionally associated with sieve tube element is -
(1) Phloem fibres
(2) Phloem Parenchyma
(3) Companion cell
(4) Collenchyma

BAP020

17. Vessels and companion cells are respectively present in the xylem and phloem of :-
(1) Gymnosperm (2) Pteridophyte
(3) Angiosperm (4) Bryophyta

BAP022

18. Phloem parenchyma is absent in :-
(1) Dicot stem (2) Dicot leaf
(3) Monocot stem (4) Dicot root
BAP023

19. Palisade parenchyma is *absent* in leaves of:-
(1) Gram (2) *Sorghum*
(3) Mustard (4) Soybean
BAP120

20. The chief function of a xylem vessel in a plant body is to -
(1) Conduct sap
(2) Conduct mineral salts only
(3) Eliminate excess of water at night
(4) Translocate organic nutrients
BAP025

21. Long and pointed sclerenchyma cells are
(1) Fibres (2) Tracheae
(3) Wood parenchyma (4) Sclereids
BAP027

22. The chief water conducting elements of xylem in gymnosperms are :
(1) Tracheids (2) Vessels
(3) Fibres (4) Transfusion tissue
BAP124

23. Which xylem element is living :-
(1) Vessels (2) Tracheids
(3) Fibre (4) Parenchyma
BAP035

24. Sugar transport elements of gymnosperms & pteridophytes are :-
(1) Sieve cells
(2) Sieve elements
(3) Sieve tubes
(4) Sieve tube elements
BAP060

25. Gymnosperms are also called soft wood spermatophytes because they lack :-
(1) Thick-walled tracheids
(2) Xylem fibres
(3) Cambium
(4) Phloem fibres
BAP131

TISSUE SYSTEM, VASCULAR BUNDLES

26. Ground tissue includes :-
(1) All tissues external to endodermis
(2) All tissues except epidermis and vascular bundles
(3) Epidermis and cortex
(4) All tissues internal to endodermis
BAP127

27. Bast fibres are mostly found in :-
(1) Secondary xylem
(2) Secondary phloem
(3) Primary phloem
(4) Primary xylem
BAP021

28. When xylem and phloem are on same radius, the vascular bundles are said to be -
(1) Radial
(2) Conjoint
(3) Concentric
(4) Exarch
BAP030

29. Conjoint collateral vascular bundles are found in :-
(1) Stem
(2) Leaf
(3) Root
(4) Both (1) and (2)
BAP276

30. The basic difference between stem and root is that xylem in stem is :-
(1) Endarch (2) Exarch
(3) Mesarch (4) Polyarch
BAP034

31. In which of the following order, an exarch xylem develops :-
(1) Centripetal
(2) Centrifugal
(3) Both centripetal & centrifugal
(4) Irregular
BAP051

32. **Statement-I :** The epidermal tissue systems are made of epidermal cells, stomata and the epidermal appendages.
Statement-II : The ground tissue system form the main bulk of the plant. It is divided into three zones-cortex pericycle and pith.
(1) Statement I and II both are correct.
(2) Statement I and II both are incorrect.
(3) Only statement I is correct
(4) Only statement II is correct.
BAP277

33. Pith is produced by :-
(1) Ground meristem
(2) Procambium
(3) Periblem
(4) Dermatogen
BAP059

34. When protoxylem faces pericycle, it is called :-
(1) Endarch (2) Mesarch
(3) Exarch (4) Polyarch
BAP061

35. External protective tissues are :-
(1) Cortex and epidermis
(2) Cork and pericycle
(3) Cortex and pericycle
(4) Cork and epidermis
BAP071

36. Gymnosperm wood is non-porous because it
(1) Lacks vessels
(2) Contains tracheae
(3) Has abundant fibres
(4) Contains no fibres
BAP090

37. Porous wood is characterised by :-
(1) Absence of tracheids
(2) Presence of vessels
(3) Absence of vessels
(4) Presence of sieve-tubes
BAP091

38. Intrafascicular cambium is situated :-
(1) In between the vascular bundles
(2) Inside the vascular bundles
(3) Outside the vascular bundles
(4) In pith
BAP095

39. Radial vascular bundles are found in :-
(1) Only dicot root
(2) Only monocot root
(3) Only Pteridophyta
(4) Roots of vascular plants
BAP103

ANATOMY OF DICOT AND MONOCOT (ROOT)

40. In dicot root :-
(1) Vascular bundles are scattered with cambium.
(2) Vascular bundles are open and arranged in a ring.
(3) Xylem and pholem are radial.
(4) Xylem is always endarch.
BAP047

41. A dicot root differs from a monocot root in which of the following-
(1) Presence of piliferous layer.
(2) Presence of exodermis.
(3) Presence of ill-developed (Poorly developed) pith.
(4) Separate radial vascular bundle.
BAP048

42. Polyarch and exarch xylem is the characteristic of :-
(1) Dicot stem (2) Dicot root
(3) Monocot stem (4) Monocot root
BAP049

43. Passage cells are found in endodermis of-
(1) Dicot stem (2) Monocot stem
(3) Orchid stem (4) Monocot root
BAP058

44. Passage cells are thin-walled cells found in:-
(1) Central region of style through which the pollen tube grows towards the ovary.
(2) Endodermis of roots facilitating rapid transport of water from cortex to pericycle
(3) Phloem elements that serve as entry points for substances for transport to other plant parts
(4) Testa of seeds to enable emergence of growing embryonic axis during seed germination.

BAP116

45. Monocot root is differ from dicot root in having :
(1) Open vascular bundle
(2) Scattered vascular bundle
(3) Large pith
(4) Radial vascular bundle

BAP098

46. Casparian strip is found in :-
(1) Epidermis (2) Endodermis
(3) Endothecium (4) Endothelium

BAP104

ANATOMY OF DICOT AND MONOCOT (STEM)

47. In barley stem vascular bundles are :-
(1) Closed and radial
(2) Open and scattered
(3) Closed and scattered
(4) Open and in a ring

BAP119

48. Collenchymatous hypodermis is characteristic feature of :-
(1) Dicot stem
(2) Monocot stem
(3) Monocot as well as dicot stem
(4) Hydrophytes

BAP043

49. Innumerable (many) vascular bundles, lack of cambium and lack of a well demarcated pith is found in :-
(1) Sugarcane, Grass
(2) Sunflower, Neem
(3) Radish, Neem
(4) Pea, Peepal

BAP044

50. **Cortex** and **pith** are not distinguished in :-
(1) Monocot stem (2) Monocot root
(3) Dicot stem (4) Dicot root

BAP045

51. What is the characteristics of a vascular bundle of monocot stem -
(1) Open and surrounded by a sclerenchymatous bundle sheath.
(2) Closed and not surrounded by bundle sheath.
(3) Closed and surrounded by bundle sheath.
(4) Open and not surrounded by a bundle sheath.

BAP046

52. Water cavity & V or Y-shaped xylem occurs in
(1) Dicot stem
(2) Monocot root
(3) Monocot stem
(4) Dicot root

BAP050

53. Vascular bundles are scattered in ground tissue in :-
(1) Maize stem (2) Sunflower stem
(3) Gram root (4) Isobilateral leaf

BAP056

54. The hypodermis present in maize stem is -
(1) Parenchymatous
(2) Collenchymatous
(3) Sclerenchymatous
(4) Meristematic

BAP057

55. How many types of cells are present in vascular cambium of dicot stem
(1) Two types, fusiform & ray initial
(2) Only fusiform initial
(3) Only ray initial
(4) Three types fusiform, ray and medullary rays.

BAP079

56. In monocot stems, secondary growth cannot occur because vascular bundles are
(1) Scattered (2) Open
(3) Closed (4) Radial

BAP089

57. There is no result of 'Girdling Experiment' in monocot plants, due to :-
(1) Presence of wax layer on the surface of its stem.
(2) Stem is comparatively thin.
(3) Phloem is inside xylem.
(4) Vascular bundles are not in specific position.

BAP101

58. Conjoint, Collateral, open vascular bundle and vascular bundles are arranged in a ring in :-
(1) Dicot stem (2) Monocot stem
(3) Monocot root (4) Dicot Root

BAP102

ANATOMY OF DICOT AND MONOCOT (LEAF)

59. Position of xylem & phloem in leaf respectively -
(1) Abaxial & Adaxial (2) Adaxial & Abaxial
(3) Both Adaxial (4) Both abaxial

BAP053

60. In leaves, the vascular bundles are :
(1) Bicollateral & open
(2) Collateral & open
(3) Collateral & closed
(4) Radial & exarch

BAP055

SECONDARY GROWTH

61. For a critical study of secondary growth in plants, which one of the following pairs is suitable?
(1) Wheat and maiden hair fern
(2) Sugarcane and sunflower
(3) Teak and pine
(4) Deodar and fern

BAP115

62. In dicot stem, hypodermis is
(1) Parenchymatous
(2) Sclerenchymatous
(3) Collenchymatous
(4) Chlorenchymatous

BAP054

63. Which wood conduct sap :-
(1) Heart wood
(2) Sap wood
(3) Wood with lots of fibres and tyloses
(4) All of the above

BAP062

64. Phelloderm is formed by :-
(1) Vascular cambium
(2) Phellogen
(3) Fascicular cambium
(4) Interfascicular cambium

BAP063

65. Dendrochronology is the study of determination of :-
(1) Height of a tree
(2) Diameter of a tree
(3) Age of a tree with the help of annual rings
(4) Counting of the number of branches

BAP064

66. A timber merchant told his customer that log of wood which he was purchasing comes from a 20 years old tree, he told so by inspecting the :-

(1) Diameter of log
(2) Thickness of the heart wood
(3) Number of cork layers
(4) Annual rings

BAP065

67. In trees, the annual rings represent :-
(1) Primary xylem (2) Secondary xylem
(3) Secondary phloem (4) Cambium

BAP066

68. Annual rings are formed due to the activity of:
(1) Intrastelar cambium
(2) Intercalary cambium
(3) Extrastelar cambium
(4) Primary cambium

BAP068

69. When a tree grows older which of the following increased rapidly :-
(1) Heart wood (2) Sap wood
(3) Pith (4) Cortex

BAP069

70. Which of the following provide maximum mechanical strength to a tree trunk.
(1) Heart wood (2) Sap wood
(3) Cork (4) Late wood

BAP072

71. Extra stelar secondary growth in dicot stem occurs due to the activity of
(1) Intrafascicular cambium
(2) Interfascicular cambium
(3) Vascular cambium
(4) Cork cambium

BAP073

72. Normally secondary growth takes place in -
(1) Dicots & Monocots
(2) Gymnosperms & Monocots
(3) Dicots & Gymnosperms
(4) Only in dicots

BAP074

73. Formation of which tissue is example of dedifferentiation
(1) Interfascicular cambium
(2) Apical meristem
(3) Intrafascicular cambium
(4) Intercalary meristem

BAP075

74. What is the fate of primary phloem in dicot stem after sec growth
(1) Compresses outside and degenerates
(2) Compresses inside and degenerates
(3) Becomes part of sec phloem
(4) Modifies in sclerenchyma

BAP076

75. Which tissue remains more active during spring
(1) Cork cambium
(2) Vascular cambium
(3) Parenchyma
(4) Sclerenchyma

BAP077

76. Cork cambium is
(1) Always primary meristem
(2) Always secondary meristem
(3) May be secondary or primary meristem
(4) Partly primary & partly secondary meristem

BAP080

77. Normally in dicot stems, phellogen develops from :-
(1) Hypodermis
(2) Phellem
(3) Endodermal cells
(4) Epidermal & pericycle cells

BAP081

78. Suberin in chiefly deposited in the cells of :-
(1) Sclerenchyma
(2) Collenchyma
(3) Cork
(4) Phelloderm

BAP082

79. Which of the following is a meristematic tissue:
(1) Phellem (2) Phellogen
(3) Phelloderm (4) Periderm
BAP083

80. The common bottle cork is a product of :-
(1) Xylem (2) Vascular Cambium
(3) Dermatogen (4) Phellogen
BAP135

81. Secondary growth in dicots and gymnosperms occurs by :-
(1) Formation of vascular rays.
(2) Thickening of tracheary elements.
(3) Formation of meristematic cells in vascular region.
(4) Development of meristematic cells in vascular & cortical regions.
BAP085

82. The balloon like outgrowths of parenchyma in the lumen of a vessel are known as
(1) Histogen (2) Tyloses
(3) Phellogen (4) Tunica
BAP086

83. Which of the following tissues originate from ray initials of cambium
(1) Tracheids & vessels
(2) Sieve tubes & companion cells
(3) Xylem & phloem fibres
(4) Radial rows of parenchyma
BAP087

84. Cork is an excellent material for making bottle stopper because it is
(1) Cheap
(2) Easily available
(3) Air tight
(4) Light
BAP088

85. Secondary growth is the production of
(1) New tissues from intercalary meristem
(2) New conduction cells
(3) New tissues from lateral meristem
(4) New ground cells
BAP093

86. Each annual ring consists of two strips of
(1) Autumn & spring wood
(2) Heart wood & sap wood
(3) Xylem and phloem
(4) Cork & cortex
BAP094

87. An example of monocots showing secondary growth in stem is
(1) *Lilium* (2) *Pea*
(3) *Asparagus* (4) *Yucca*
BAP096

88. Heartwood differs from sapwood in :
(1) Being susceptible to pests and pathogens
(2) Presence of rays and fibres
(3) Absence of vessels and parenchyma
(4) Having dead and non-conducting elements
BAP125

89. Autumn wood can be differentiated from spring wood by :-
(1) Broad vessels and tracheids
(2) Narrow vessels and tracheids
(3) Red colour of xylem
(4) Cambium
BAP099

90. Vascular cambium forms :-
(1) Secondary xylem & Secondary phloem
(2) Primary xylem & Primary Phloem
(3) Only Secondary phloem
(4) Only Primary xylem
BAP100

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Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	1	1	2	2	1	2	4	1	3	1	3	2
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Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	3	2	7	4	2	4	3	3	1	4	4	2	1

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. You are given a fairly old piece of dicot stem and a dicot root. Which of the following anatomical structures will you use to distinguish between the two?
(1) Secondary xylem
(2) Secondary phloem
(3) Protoxylem
(4) Cortical cells

BAP139

2. Tracheids differ from other tracheary elements in:
(1) Having casparian strips
(2) Being imperforate
(3) Lacking nucleus
(4) Being lignified

BAP140

AIPMT 2015

3. A major characteristic of the monocot root is the presence of :
(1) Scattered vascular bundles
(2) Vasculature without cambium
(3) Cambium sandwiched between phloem and xylem along the radius
(4) Open vascular bundles

BAP141

4. In a ring girdled plant:
(1) The root dies first
(2) The shoot and root die together
(3) Neither root nor shoot will die
(4) The shoot dies first

BAP142

5. Vascular bundles in monocotyledons are considered closed because:
(1) Cambium is absent
(2) There are no vessels with perforations
(3) Xylem is surrounded all around by phloem
(4) A bundle sheath surrounds each bundle

BAP143

Re-AIPMT 2015

6. Read the different components from (a) to (d) in the list given below and tell the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem:
(a) Secondary cortex (b) Wood
(c) Secondary phloem (d) Phellem
The correct order is :
(1) (d), (c), (a), (b) (2) (c), (d), (b), (a)
(3) (a), (b), (d), (c) (4) (d), (a), (c), (b)

BAP144

NEET-I 2016

7. Specialised epidermal cells surrounding the guard cells are called :-
(1) Complementary cells
(2) Subsidiary cells
(3) Bulliform cells
(4) Lenticels

BAP145

NEET-II 2016

8. Cortex is the region found between :-
(1) Endodermis and pith
(2) Endodermis and vascular bundle
(3) Epidermis and stele
(4) Pericycle and endodermis

BAP146

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9. The balloon-shaped structures called tyloses :-
(1) Are extensions of xylem parenchyma cells into vessels
(2) Are linked to the ascent of sap through xylem vessels
(3) Originate in the lumen of vessels
(4) Characterize the sapwood

BAP147

NEET(UG) 2017

10. The vascular cambium normally gives rise to
(1) Primary phloem (2) Secondary xylem
(3) Periderm (4) Phelloderm

BAP148

11. Which of the following is made up of dead cells?
(1) Collenchyma (2) Phellem
(3) Phloem (4)Xylem parenchyma

BAP149

12. Identify the wrong statement in context of heartwood:
(1) It is highly durable
(2) It conducts water and minerals efficiently
(3) It comprises dead elements with highly lignified walls
(4) Organic compounds are deposited in it

BAP150

NEET (UG) 2018

13. Stomata in grass leaf are
(1) Dumb-bell shaped (2) Kidney shaped
(3) Rectangular (4) Barrel shaped

BAP151

14. Secondary xylem and phloem in dicot stem are produced by
(1) Apical meristems
(2) Vascular cambium
(3) Phellogen
(4) Axillary meristems

BAP152

15. Casparian strips occur in
(1) Epidermis
(2) Pericycle
(3) Cortex
(4) Endodermis

BAP153

16. Plants having little or no secondary growth are:
(1) Grasses
(2) Deciduous angiosperms
(3) Conifers
(4) Cycads

BAP154

NEET(UG) 2019

17. Xylem translocates :-
(1) Water only
(2) Water and mineral salts only
(3) Water, mineral salts and some organic nitrogen only
(4) Water, mineral salts, some organic nitrogen and hormones

BAP155

18. Grass leaves curl inwards during very dry weather. Select the most appropriate reason from the following :
(1) Closure of stomata
(2) Flaccidity of bulliform cells
(3) Shrinkage of air spaces in spongy mesophyll
(4) Tyloses in vessels

BAP156

19. Phloem in gymnosperms lacks :
(1) Albuminous cells and sieve cells
(2) Sieve tubes only
(3) Companion cells only
(4) Both sieve tubes and companion cells

BAP157

20. Which of the statements given below is **not** true about formation of Annual Rings in trees?

(1) Annual ring is a combination of spring wood and autumn wood produced in a year.

(2) Differential activity of cambium causes light and dark bands of tissue - early and late wood respectively

(3) Activity of cambium depends upon variation in climate.

(4) Annual rings are not prominent in trees of temperate region.

BAP158

NEET(UG) 2019 (Odisha)

21. In the dicot root the vascular cambium originates from :-

(1) Tissue located below the phloem bundles and a portion of pericycle tissue above protoxylem.

(2) Cortical region.

(3) Parenchyma between endodermis and pericycle.

(4) Intrafascicular and interfascicular tissue in a ring.

BAP159

22. Regeneration of damaged growing grass following grazing is largely due to :-

(1) Lateral meristem

(2) Apical meristem

(3) Intercalary meristem

(4) Secondary meristem

BAP160

NEET (UG) 2020

23. The transverse section of a plant shows following anatomical features:

(a) Large number of scattered vascular bundles surrounded by bundle sheath.

(b) Large conspicuous parenchymatous ground tissue.

(c) Vascular bundles conjoint and closed.

(d) Phloem parenchyma absent.

Identify the category of plant and its part :-

(1) Dicotyledonous root

(2) Monocotyledonous stem

(3) Monocotyledonous root

(4) Dicotyledonous stem

BAP161

24. Identify the **incorrect** statement.

(1) Due to deposition of tannins, resins, oils etc., heart wood is dark in colour

(2) Heart wood does not conduct water but gives mechanical support

(3) Sapwood is involved in conduction of water and minerals from root to leaf

(4) Sapwood is the innermost secondary xylem and is lighter in colour

BAP162

NEET (UG) 2020 (COVID-19)

25. Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are

(1) Lenticels

(2) Guard cells

(3) Bundle sheath cells

(4) Bulliform cells

BAP163

26. Which of the following statements about cork cambium is **incorrect**?

(1) It forms secondary cortex on its outer side

(2) It forms a part of periderm

(3) It is responsible for the formation of lenticels

(4) It is a couple of layers thick

BAP164

NEET (UG) 2021

27. Match **List-I** with **List-II.**

List-I		List-II	
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iv)	(i)	(iii)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(i)	(ii)	(iii)	(iv)
(4)	(iii)	(ii)	(iv)	(i)

BAP165

28. Match **List -I** with **List -II.**

List-I		List-II	
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(i)	(iii)	(ii)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(iv)	(ii)	(i)	(iii)

BAP166

29. Select the **correct** pair.

(1) Large colorless empty cells in the epidermis of grass leaves – Subsidiary cells

(2) In dicot leaves, vascular bundles are surrounded by large thick-walled cells – Conjunctive tissue

(3) Cells of medullary rays that form part of cambial ring – Interfascicular cambium

(4) Loose parenchyma cells rupturing the epidermis and forming a lens-shaped opening in bark – Spongy parenchyma

BAP167

Re-NEET(UG) 2021

30. Read A to D and tell the correct order of components from outer side to inner side in a young dicot stem.

(A) Cortex (B) Pericycle

(C) Epidermis (D) Pith

(1) C, D, B, A (2) A, B, C, D

(3) C, A, B, D (4) C, B, A, D

BAP259

31. During secondary growth in dicot stem, cork cambium or phellogen usually develops in

(1) Epidermal region

(2) Cortex region

(3) Pericycle region

(4) Pith region

BAP260

32. Transport of water and mineral in higher plants takes place through

(1) Sieve elements

(2) Companion cells

(3) Tracheids

(4) Transfusion tissue

BAP261

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NEET (UG) 2022

33. Read the following statements about the vascular bundles :-

(a) In roots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii.

(b) Conjoint closed vascular bundles do not process cambium.

(c) In open vascular bundles, cambium is present in between xylem and phloem.

(d) The vascular bundles of dicotyledonous stem possess endarch protoxylem.

(e) In monocotyledonous root, usually there are more than six xylem bundles present.

Choose the **correct** answer from the options given below :-

(1) (b), (c), (d) and (e) only

(2) (a), (b), (c) and (d) only

(3) (a), (c), (d) and (e) only

(4) (a), (b) and (d) only

BAP262

34. The anatomy of springwood shows some peculiar features. Identify the correct set of statements about springwood.

(a) It is also called as the earlywood

(b) In spring season cambium produces xylem elements with narrow vessels

(c) It is lighter in colour.

(d) The springwood along with autumn-wood shows alternate concentric rings forming annual rings.

(e) It has lower density

Choose the **correct** answer from the options given below :-

(1) (a), (c), (d) and (e) only

(2) (a), (b) and (d) only

(3) (c), (d) and (e) only

(4) (a), (b), (d) and (e) only

BAP170

NEET (UG) 2022 (Overseas)

35. Select the correct statement related to the activity of cork cambium.

(1) The cork differentiated from cork cambium, is impervious to water due to deposition of tannins and resins

(2) Cuts the cells only on the outer side

(3) Cuts the cells on inner as well as outer side.

(4) The outer cells differentiate into phelloderm

BAP263

36. Axillary buds are derived from the activity of:

(1) Secondary meristem

(2) Apical meristem

(3) Intercalary meristem

(4) Lateral meristem

BAP264

37. Which of the following is not a character of collenchyma tissue?

(1) They occur in layers below epidermis in dicotyledonous plants.

(2) They consists of cells with thick corners due to cellulose deposition

(3) They are usually dead and without protoplasts.

(4) They provide mechanical support to the growing part of the plant

BAP265

Re-NEET (UG) 2022

38. Initiation of lateral roots and vascular cambium during secondary growth takes place in cells of :

(1) Epiblema (2) Cortex

(3) Endodermis (4) Pericycle

BAP171

39. The type of tissue commonly found in the fruit wall of nuts is :

(1) Parenchyma (2) Collenchyma

(3) Sclerenchyma (4) Sclereid

BAP172

40. Interfascicular cambium is present between

(1) Primary xylem and primary phloem

(2) Pericycle and endodermis

(3) Two vascular bundles

(4) Secondary xylem and secondary phloem

BAP266

NEET (UG) 2023

41. Given below are two statements :-

Assertion (A) : Late wood has fewer xylary elements with narrow vessels.

Reason (R) : Cambium is less active in winters.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

(2) **A** is true but **R** is false.

(3) **A** is false but **R** is true.

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

BAP267

42. Given below are two statements :-

Statement-I : Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

Statement-II : Exarch condition is the most common feature of the root system.

In the light of the above statements, choose the correct answer from the options given below :-

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

(1) Both Statement I and Statement II are true

BAP268

43. Identify the correct statements:

A. Lenticels are the lens-shaped openings permitting the exchange of gases.

B. Bark formed early in the season is called hard bark.

C. Bark is a technical term that refers to all tissues exterior to vascular cambium.

D. Bark refers to periderm and secondary phloem.

E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below:

(1) A and D only

(2) A, B and D only

(3) B and C only

(4) B, C and E only

BAP269

NEET (UG) 2023 (Manipur)

44. The transverse section of a plant part showed polyarch, radial and exarch xylem, with endodermis and pericycle. The plant part is identified as :

(1) Monocot root

(2) Dicot root

(3) Dicot stem

(4) Monocot stem

BAP270

45. Consider the following plant tissues :

(A) Axillary buds

(B) Fascicular vascular cambium

(C) Interfascicular cambium

(D) Cork cambium

(E) Intercalary meristem

Identify the lateral meristems among the above.

(1) (A), (C) and (D) only

(2) (B), (C) and (D) only

(3) (A), (B), (C) and (E) only

(4) (A), (B), (D) and (E) only

BAP271

NEET (UG) 2024

46. Bulliform cells are responsible for

(1) Inward curling of leaves in monocots.

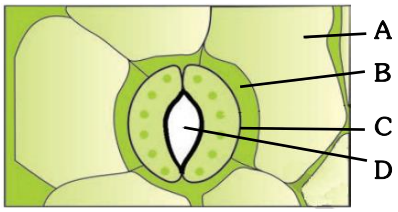
(2) Protecting the plant from salt stress.

(3) Increased photosynthesis in monocots.

(4) Providing large spaces for storage of sugars.

BAP278

47. In the given figure, which component has thin outer walls and highly thickened inner walls?



(1) C (2) D (3) A (4) B

BAP279

48. Formation of interfascicular cambium from fully developed parenchyma cells is an example for :-

(1) Differentiation

(2) Redifferentiation

(3) Dedifferentiation

(4) Maturation

BAP280

Re-NEET (UG) 2024

49. Which of the following simple tissues are commonly found in the fruit walls of nuts and pulp of pear ?

(1) Sclereids (2) Fibres

(3) Parenchyma (4) Collenchyma

BAP281

50. Which one of the following is **not** found in Gymnosperms?

(1) Sieve cells (2) Albuminous cells

(3) Tracheids (4) Vessels

BAP282

51. Which of the following helps in maintenance of the pressure gradient in sieve tubes ?

(1) Albuminous cells

(2) Sieve cells

(3) Phloem parenchyma

(4) Companion cells

BAP283

52. Given below are two statements :-

Statement-I : In a dicotyledonous leaf, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

Statement-II : In a dicotyledonous leaf, adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

BAP284

Exercise-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	2	1	1	4	2	3	1	2	2	2	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	2	4	4	1	3	2	4	4	1	1	2	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	1	1	3	2	3	4	4	3	4	3	1	1	2
Question	46	47	48	49	50	51	52								
Answer	1	1	3	1	4	4	2								

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Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Branch of botany related with study of internal structure of plants.
(1) Cytology (2) Anatomy
(3) Physiology (4) Ecology

BAP173

2. A tissue is a group of cells having a -
(1) Common origin and dissimilar in function.
(2) Dissimilar origin and common in function.
(3) Dissimilar origin and common in function
(4) Common origin and usually preforming a common function.

BAP174

3. In embryonic stage, cells are
(1) Collenchymatous
(2) Sclerenchymatous
(3) Meristematic
(4) Parenchymatous

BAP175

4. Apical meristem is present
(1) At the tips of roots
(2) At the tips of shoots
(3) In buds which are present in axils of leaves
(4) All of the above

BAP176

5. Which of the following is an example of primary meristem
(1) Phellogen
(2) Vascular cambium of root
(3) Inter fascicular cambium
(4) Intra fascicular cambium

BAP177

6. Which of the following are examples of lateral meristems
(1) Fascicular vascular cambium
(2) Inter-fascicular cambium
(3) Cork cambium
(4) All of the above

BAP178

7. Find out the **correct match** pairs from the given table:-

	Column-I	Column-II	Column-III
i	Dicot stems	Pericycle	only collenchyma
ii.	Monocot stems	Closed vascular bundle	cambium absent
iii.	Dicot roots	Casparian strips	suberised
iv.	Monocot roots	Hypodermis	collenchyma

(1) i, iv (2) i, ii, iii
(3) ii, iii (4) i, iii, iv

BAP285

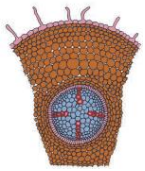
8. Which of the following statement(s) is/are true about parenchyma?
(1) The cells are generally isodiametric
(2) The cells may either be closely packed or have intercellular spaces.
(3) It performs various functions like photosynthesis, storage & secretion.
(4) All of the above

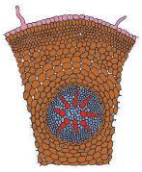
BAP180

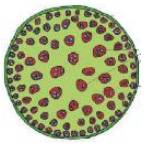
9. Which of the following tissue provides mechanical support to the growing parts of the plant such as young stem and petiole of a leaf.
(1) Parenchyma (2) Collenchyma
(3) Sclereids (4) Fibres

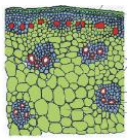
BAP181

10. Which of the following is observed in sunflower plant ?

(1) 

(2) 

(3) 

(4) 

BAP286

11. Companion cells are present in the phloem of

(1) Pteridophytes (2) Gymnosperms

(3) Angiosperms (4) Both (1) and (2)

BAP183

12. Vessels differ from tracheids in having

(1) In being living

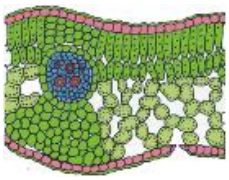
(2) Presence of perforation plates at both ends

(3) Enucleated condition

(4) All of the above

BAP184

13. Identify the given diagramatic sketch and select the **correct** option in respect of diagram?



(1) Hypostomatic leaf

(2) Amphistomatic leaf

(3) Found in monocot plant

(4) Presence of bulliform cells towards abaxial surface.

BAP287

14. When a meristematic tissue "Intra fascicular cambium" is present inside a vascular bundle, the vascular bundle is called

(1) Conjoint (2) Open

(3) Closed (4) Radial

BAP186

15. Which of the following is not included in stomatal apparatus

(1) Stomatal aperture

(2) Guard cells

(3) Subsidiary cells

(4) Sclerenchymatous cells

BAP187

16. Consider the following statements.

(a) Phloem fibres are made up of sclerenchymatous cells.

(b) Phloem parenchyma are generally present in monocots.

(c) The first formed primary phloem consists of narrow sieve tube and is referred to as protophloem and the later formed phloem has bigger sieve tubes and is referred to as metaphloem.

Which statement(s) is/are false ?

(1) Only a (2) Only b

(3) Only c (4) a, b, and c

BAP188

17. The outside of the epidermis is often covered with a waxy thick layer, called

(1) Hypodermis

(2) Cuticle

(3) Root hair

(4) Stem hair

BAP189

18. Cuticle is absent in

(1) Roots (2) Dicot Stem

(3) Leaves (4) Monocot stem

BAP190

19. In which plant vascular bundles are scattered in ground tissue and each vascular bundle is surrounded by sclerenchymatous bundle sheath

(1) Maize (2) Sunflower

(3) Gram (4) *Ficus*

BAP191

20. In below mentioned structure, how many structures are having 'Exarch' condition of vascular bundles.
'Dicot stem, Dicot root, Monocot stem, monocot root, Isobilateral leaf, Dorsivendral leaf'.
(1) 4 (2) 3
(3) 2 (4) All of these

BAP288

21. The dicot root is identified from the monocot root by the presence of
(1) Exarch xylem
(2) 2-4 xylem bundles
(3) > 6 xylem bundles
(4) Large and well developed pith

BAP193

22. Parenchymatous cells with large intercellular spaces which occupy the central portion of the stem constitute
(1) Cortex (2) Pith
(3) Hypodermis (4) Epidermis

BAP194

23. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These cells are called
(1) Bulliform cells
(2) Starch sheath cells
(3) Companion cells
(4) Complimentary cells

BAP195

24. In dicot stem, the cells of cambium present between primary xylem and primary phloem is the
(1) Interfascicular cambium
(2) Vascular cambium
(3) Intra fascicular cambium
(4) Cork cambium

BAP196

25. The innermost layer of cortex of dicot root is characterised by presence of suberin thickening. This suberin thickening occurs on
(1) Radial walls (2) Transverse wall
(3) Tangential wall (4) Both (1) & (3)

BAP197

26. Identify correct match from the column-I, II and III.

	Column-I	Column-II		Column-III	
(1)	Epidermis	(a)	Trichomes	(i)	Dicot root
(2)	Hypodermis	(b)	Casparian strips	(ii)	Monocot stem
(3)	Endodermis	(c)	Incons-picuous	(iii)	Dicot stem
(4)	Pith	(d)	Scleren-chymatous	(iv)	Monocot root

(1) 2-a-iii, 1-d-ii, 3-b-i, 4-c-i
(2) 1-a-iii, 2-b-ii, 3-d-iv, 4-c-i
(3) 2-d-ii, 1-a-iii, 3-b-iv, 4-c-
(4) 1-a-iii, 2-a-iv, 3-c-i, 4-b-ii

BAP289

27. Regarding to stele which of the following statement is correct ?
(1) All the tissues lies inner to pericycle
(2) All the tissues lies inner to endodermis
(3) All the tissues lies inner to hypodermis
(4) All the tissues lies inner to epidermis

BAP199

28. Complimentary cells are formed by the activity of
(1) Vascular cambium
(2) Inter fascicular cambium
(3) Phellogen
(4) Intra fascicular cambium

BAP200

29. Vascular cambium cuts off cells more
(1) Towards centre
(2) Towards periphery
(3) Both (1) and (2)
(4) None of these

BAP201

30. In a dicot root, vascular cambium originates from.
(1) Parenchymatous conjunctive tissue located just below phloem bundles
(2) Medullary rays
(3) A portion of pericycle tissue, lying opposite to protoxylem
(4) Both (1) and (3)

BAP202

31. In a dicot root, the pericycle gives rise to
(1) Lateral roots
(2) Cork cambium
(3) A portion of vascular cambium
(4) All of the above

BAP203

(B) ANALYTICAL QUESTIONS

32. Select true statement
(1) Lenticels occur in woody trees
(2) Bark is a non-technical term
(3) Vascular cambium and cork cambium are examples of lateral meristems
(4) All of the above

BAP204

33. During secondary growth in dicot stem, At some places, the cambium forms a narrow band of parenchyma, which passes through the secondary xylem and secondary phloem in radial directions. These are called :-
(1) Sap wood
(2) Heart wood
(3) Secondary medullary rays
(4) Primary medullary rays

BAP205

34. **Statement-I :** In dicot stem, cork cambium develop usually in steler region.
Statement-II : Cork cambium cuts off phelloderm on outside and phellem on inner side.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

BAP290

35. At certain regions, the phellogen cuts off closely arranged parenchymatous cells on the outer side instead of cork cells. These are called :-
(1) Phellem
(2) Periderm
(3) Bark
(4) Complimentary cells

BAP207

36. The cells of secondary cortex are :-
(1) Sclerenchymatous
(2) Parenchymatous
(3) Collenchymatous
(4) Meristematic

BAP208

37. In dicot root, the vascular cambium is.
(1) Primary in origin
(2) Completely secondary in origin
(3) Both primary and secondary in origin
(4) Neither primary nor secondary in origin

BAP209

38. Due to continuous growth of secondary xylem which of the following get crushed gradually
(1) Primary phloem
(2) Earlier formed secondary phloem
(3) Either (1) or (2)
(4) Both (1) and (2)

BAP210

39. After secondary growth what is the actual future of primary xylem ?
(1) Converts into secondary xylem
(2) Remains more or less intact in or around the centre
(3) Converts into secondary phloem
(4) Gets crushed

BAP211

40. In plants, during early embryonic condition-
(1) All cells of the embryo divide
(2) Meristematic activity is confined to single apical cell
(3) Meristematic activity is confined to a group of apical cells.
(4) Apical & lateral cells only divide

BAP212

41. **Statement-I :** In old trees, the greater part of secondary Xylem is dark brown due to deposition of organic compounds like tannins, resins, oils, gum, aromatic substances and essential oils.
Statement-II : The peripheral region of secondary phloem is lighter in colour and is also known as sapwood.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

BAP291

42. Identify the **correct** match from the column I and II :-

Column-I		Column-II	
(a)	Dicot stem	i.	Tetrach
(b)	Monocot stem	ii.	Polyarch
(c)	Monocot root	iii.	Starch sheath
(d)	Dicot root	iv.	Scattered vascular bundle

(1) a → iii; b → iv; c → ii; d → i
(2) a → iii; b → iv; c → i; d → ii
(3) a → iv; b → iii; c → ii; d → i
(4) a → ii; b → i; c → iii; d → iv

BAP292

43. Sieve plates in angiosperms are -
(1) Oblique & in lateral walls
(2) Pitted & in end wall
(3) Oblique & in end wall
(4) Straight & in lateral wall

BAP215

44. **Assertion :-** Amount of secondary xylem is produced more than secondary phloem
Reason :- The cambium is generally more active on the inner side than on the outer
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

BAP293

45. T.S. of a material exhibits conjoint, collateral endarch and closed bundles scattered in a ground tissue what should be the material -
(1) Monocot root
(2) Dicot root
(3) Monocot stem
(4) Dicot stem

BAP217

46. In respect of many grasses, the presence of bulliform or motor cells in the upper epidermis of leaves is to:-
(1) Increase the surface area of the leaf
(2) Store large amount of water
(3) Check transpiration by reducing the surface area of the leaf
(4) Bear unicellular trichomes

BAP218

47. Heart wood of most of the woody trees, fails to conduct sap due to the obstruction by ?
(1) Excess secondary wall deposition
(2) Storing a variety of material by tyloses
(3) Deposition of reserved food material
(4) Ray parenchyma

BAP219

48. The trees growing in deserts will -
(1) Show alternate rings of xylem and sclerenchyma
(2) Show distinct annual rings
(3) Not show distinct annual rings
(4) Have only conjunctive tissue and phloem formed by the activity of cambium

BAP220

49. Sap wood differ from heart wood in being -
(1) Darker and non-conducting
(2) Softer and non-conducting
(3) Lighter and conducting
(4) Harder, darker and less conducting

BAP221

50. If a stem is girdled-
(1) Root dies first
(2) Shoot dies first
(3) Both die together
(4) None of the above die

BAP222

51. Spring wood (early wood) differs with autumn wood (late wood) in -
(1) Size of vessels and tracheids
(2) Thickness of cell wall
(3) Amount of wood
(4) All the above

BAP223

52. Most conspicuous annual rings are formed in
(1) Temperate evergreen plants
(2) Tropical deciduous
(3) Temperate deciduous plants
(4) Tropical evergreen

BAP224

53. In dicot root which tissue becomes dead due to activity of phellogen
(1) All tissue outside cortex
(2) Tissue outside hypodermis
(3) Tissue outside endodermis
(4) Tissue outside pericycle

BAP225

54. Physiologically functional part of wood is
(1) Heart wood only
(2) Sap wood only
(3) Only primary wood
(4) Both heart wood & sap wood

BAP226

55. Find out the **correct** match.

	Spring wood	**Autumn wood**	**Heart wood**
(1)	Wood formed during Autumn	Wood formed during spring	Lighter in colour
(2)	Wood formed during spring season	Wood formed during autumn season	Dead-elements with highly lignified wall
(3)	Late wood	Early wood	Teloses
(4)	Dark in colour	Light in colour	Phellem

BAP294

56. Which is not correct about heartwood

(1) It is formed of living cells

(2) It contains resins, tannin and other organic contents

(3) It is of dark coloured

(4) It lies in the central region of secondary xylem

BAP228

57. Which of the following statements are correct :-

(a) In the spring season vascular cambium produce large number of xylary elements.

(b) Spring wood is lighter in colour with high density.

(c) In the winter season vascular cambium produce vessel with narrow lumen.

(d) Autumn wood is darker in colour with high density.

(1) a, b, c

(2) b, c, d

(3) a, c, d

(4) a, b, c, d

BAP295

58. After two or three years of secondary growth the cortex in dicot roots

(1) Remains intact

(2) Is completely sloughed away

(3) Is largely lost

(4) Is converted into cork

BAP230

59. When secondary growth in thickness is initiated in a dicot root, which of the following happens first ?

(1) Cambial initials between xylem and phloem will divide

(2) Pericycle strands, outside the primary xylem will divide

(3) Periclinal division takes place, so that the cambium becomes circular

(4) Parenchymatous cells between xylem and phloem become meristematic

BAP231

60. Complementary cells of lenticels are

(1) Compact and suberised

(2) Closely arranged and non-suberised

(3) Compact and lignified

(4) Loose and lignified

BAP232

61. In a woody dicotyledonous tree, which of the following parts will mainly consist of primary permanent tissues ?

(1) Stem and root

(2) All parts

(3) Shoot tips and root tips

(4) Flowers, fruits and leaves

BAP233

62. Anatomically fairly old dicotyledonous root is distinguished from the dicotyledonous stem by :-

(1) Position of protoxylem

(2) Absence of secondary xylem

(3) Absence of secondary phloem

(4) Presence of cortex

BAP122

63. The cork cambium, cork and secondary cortex are collectively called :-
(1) Phelloderm (2) Phellogen
(3) Periderm (4) Phellem

BAP128

64. Some vascular bundles are described as open because these :
(1) Are not surrounded by pericycle
(2) Are surrounded by pericycle but no endodermis
(3) Are capable of producing secondary xylem and phloem.
(4) Possess conjunctive tissue between xylem and phloem.

BAP130

65. Water containing cavities in vascular bundles are found in :-
(1) *Cycas* (2) *Pinus*
(3) Sunflower (4) Maize

BAP132

66. Closed vascular bundles lack :-
(1) Cambium
(2) Pith
(3) Ground tissue
(4) Conjunctive tissues

BAP133

67. Companion cells are closely associated with:
(1) Trichomes (2) Guard cells
(3) Sieve elements (4) Vessel elements

BAP134

68. Age of a tree can be estimated by :
(1) Diameter of its heartwood
(2) Its height and girth
(3) Biomass
(4) Number of annual rings

BAP136

69. Interfascicular cambium develops from the cells of:
(1) Pericycle
(2) Medullary rays
(3) Xylem parenchyma
(4) Endodermis

BAP137

70. Lenticels are involved in :
(1) Photosynthesis
(2) Transpiration
(3) Gaseous exchange
(4) Food transport

BAP138

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	3	4	4	4	3	4	2	1	3	2	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	1	1	3	2	2	1	3	4	3	2	3	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	3	4	4	2	2	4	2	1	2	1	3	1	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	3	3	1	4	3	4	2	2	1	3	2	4	2
Question	61	62	63	64	65	66	67	68	69	70					
Answer	4	1	3	3	4	1	3	4	2	3					

Exercise - I (Conceptual Questions) Build Up Your Understanding

CELL THEORY, CELL MEMBRANE AND CELL WALL

1. Which of the following is not correct?
(1) Robert Brown discovered the cell.
(2) Schleiden and Schwann formulated the cell theory.
(3) Virchow explained that cells are formed from pre-existing cells.
(4) A unicellular organism carries out its life activities within a single cell.

BCL009

2. The main arena of various types of activities of a cell is:
(1) Nucleus
(2) Plasma membrane
(3) Mitochondria
(4) Cytoplasm

BCL089

3. Cell membrane is composed of :-
(1) Proteins and cellulose
(2) Proteins and phospholipids
(3) Proteins and carbohydrates
(4) Proteins, phospholipids and some carbohydrates

BCL001

4. Carbohydrates are present in the plasma membrane in the form of :–
(1) Hemicellulose (2) Cellulose
(3) Starch (4) Glycoprotein

BCL002

5. Carbohydrates which are present in the cell membrane take part in :–
(1) Transport of substance
(2) Cell recognition
(3) Attachment to microfilament
(4) Attachment to microtubules

BCL003

6. Plasma membrane is fluid structure due to presence of :–
(1) Carbohydrates (2) Lipid
(3) Glycoprotein (4) Polysaccharide

BCL004

7. The fluid nature of the membrane is helpful in function of :-
(1) Cell growth (2) Cell division
(3) Endocytosis (4) All the above

BCL007

8. Plasma membrane is :–
(1) Selectively permeable
(2) Permeable
(3) Impermeable
(4) Semipermeable

BCL008

9. In fluid mosaic model of plasma membrane:
(1) Upper layer is non-polar and hydrophilic
(2) Polar layer is hydrophobic
(3) Phospholipids form a bimolecular layer in middle part
(4) Proteins form a middle layer

BCL011

10. Which of the following is correct:
(1) Cells of all living organisms have a nucleus.
(2) Both animal and plant cells have a well defined cell wall.
(3) In prokaryotes, there are no membrane bound organelles.
(4) Cells are formed *de novo* from abiotic materials.

BCL208

11. The plasma membrane consists mainly of:
(1) proteins embedded in a carbohydrate bilayer
(2) phospholipids embedded in a protein bilayer
(3) proteins embedded in a phospholipid bilayer
(4) proteins embedded in a polymer of glucose molecules

BCL086

12. Select the correct statement from the following regarding cell membrane: -
(1) Lipids are arranged in a bilayer with polar heads towards the inner part
(2) Fluid mosaic model of cell membrane was proposed by Singer and Nicolson
(3) Na^+ and K^+ ions move across cell membrane by passive transport
(4) Proteins make up 60 to 70% of the cell membrane

BCL096

13. The chemical substance abundantly present in middle lamella is :–
(1) Cutin (2) Chitin
(3) Lignin (4) Pectin

BCL005

14. Which of the following is capable of growth and gradually diminishes as the cell matures?
(1) Primary cell wall
(2) Secondary cell wall
(3) Middle lamella
(4) Cell membrane

BCL006

15. Plasmodesmata are: -
(1) Connections between adjacent cells
(2) Lignified cemented layers between cells
(3) Locomotary structures
(4) Membranes connecting the nucleus with plasmalemma

BCL083

16. Middle lamella is composed mainly of: -
(1) Phosphoglycerides
(2) Hemicellulose
(3) Muramic acid
(4) Calcium pectate

BCL084

17. Which one of the following structures between two adjacent cells is an effective transport pathway?
(1) Plasmalemma
(2) Plasmodesmata
(3) Plastoquinone
(4) Endoplasmic reticulum

BCL087

18. Algae have cell wall made up of:
(1) Cellulose, hemicellulose and pectins
(2) Cellulose, galactans and mannans
(3) Hemicellulose, pectins and proteins
(4) Pectins, cellulose and proteins

BCL090

19. Ingestion of large molecules by a cell is called -
(1) Diffusion (2) Osmosis
(3) Exocytosis (4) Endocytosis

BCL010

20. The main lipid components of the plant cell membrane are :-
(1) Phosphodiester (2) Glycocalyx
(3) Peptidoglycan (4) Phosphoglyceride

BCL012

ER, GOLGI COMPLEX AND LYSOSOME

21. Hydrolytic enzymes are abundantly found in which cell organelles ?
(1) Ribosome
(2) Lysosome
(3) Golgi body
(4) Endoplasmic reticulum

BCL018

22. At which pH lysosomal enzymes become active ?
(1) pH – 5 (2) pH – 7
(3) pH – 8 (4) pH – 10
BCL022

23. Which cell organelle(s) take(s) part in the formation of lysosomes ?
(1) Endoplasmic reticulum
(2) Golgi bodies
(3) Both (1) and (2)
(4) Mitochondria
BCL024

24. A single unit membrane organelle is :-
(1) Ribosome (2) Mitochondria
(3) Chloroplast (4) Lysosome
BCL025

25. Rough E.R. mainly responsible for:-
(1) Protein synthesis
(2) Cell wall formation
(3) Lipid synthesis
(4) Cholesterol synthesis
BCL013

26. Which cell organelle synthesises steroids ?
(1) E. R. (2) Golgi body
(3) Peroxisomes (4) Lysosomes
BCL016

27. Which of the following provides mechanical support and shape to the cell ?
(1) Golgi complex (2) Centrioles
(3) Lysosomes (4) E.R.
BCL017

28. The smooth E.R. is mainly made up of :–
(1) Cisternae (2) Tubules
(3) Vesicle (4) All the above
BCL019

29. Ribophorins occur on the surface of :–
(1) Rough E.R. (2) Smooth E.R.
(3) Golgi body (4) Lysosome
BCL021

30. Which of the following is the site of lipid synthesis?
(1) Rough ER (2) Smooth ER
(3) Golgi bodies (4) Ribosome
BCL027

31. A major site for synthesis of lipids is:
(1) Nucleoplasm (2) RER
(3) SER (4) Symplast
BCL102

32. Golgi body originates from :-
(1) E. R. (2) Mitochondria
(3) Nucleus (4) Proplastid
BCL015

33. Main function of golgi–complex is :–
(1) Fermentation
(2) Phosphorylation
(3) Respiration
(4) Packaging of materials for secretion
BCL023

34. Golgi body is concerned with :-
(1) Respiration (2) Secretion
(3) Excretion (4) Digestion
BCL030

35. Important site for formation of glycoproteins and glycolipids is :-
(1) Vacuole (2) Golgi apparatus
(3) Plastid (4) Lysosome
BCL093

36. The Golgi complex plays a major role:
(1) In post translational modification of proteins and glycosidation of lipids
(2) In trapping the light and transforming it into chemical energy
(3) In digesting proteins and carbohydrates
(4) As energy transferring organelles
BCL101

MITOCHONDRIA AND PLASTID

37. Mitochondrial DNA is :-
(1) Naked (2) Circular
(3) Double stranded (4) All the above
BCL014

38. Power house of cell is :-
(1) Nucleus (2) DNA
(3) Mitochondria (4) ATP
BCL031

39. Semiautonomous cell organelle is :–
(1) Mitochondria (2) Ribosome
(3) Golgi body (4) Peroxisome
BCL033

40. The cell organelles having abundance of oxidizing enzymes is :–
(1) Golgi body
(2) Endoplasmic reticulum
(3) Centrioles
(4) Mitochondria
BCL035

41. Aerobic respiration is performed by :-
(1) Mitochondria (2) Chloroplast
(3) Ribosome (4) Golgi body
BCL036

42. Ground substance present inside the mitochondria is called :-
(1) Stroma (2) Matrix
(3) Cell sap (4) Cytoplasm
BCL037

43. ATP factories of cells are –
(1) Chloroplast
(2) Mitochondria
(3) Ribosome
(4) Nucleus
BCL038

44. Cristae are found in :-
(1) Surface of grana
(2) Surface of plasma membrane.
(3) Inner Membrane of Mitochondria
(4) Nuclear Membrane
BCL039

45. Double layered organelles are -
(1) Ribosomes (2) Mitochondria
(3) Lysosomes (4) Centrioles
BCL040

46. Which one of the following has its own DNA?
(1) Peroxisome (2) Mitochondria
(3) Dictyosome (4) Lysosome
BCL088

47. Anthocyanin pigment is found in :–
(1) Chromoplasts (2) Amyloplasts
(3) Cytoplasm (4) Cell sap
BCL029

48. Which cell organelle releases oxygen ?
(1) Mitochondria (2) Golgi–body
(3) Chloroplast (4) Ribosome
BCL034

49. Chlorophyll in chloroplasts is located in –
(1) Grana and stroma thylakoids
(2) Only in stroma thylakoids
(3) Stroma
(4) Both grana and stroma
BCL041

50. Elaioplasts are absent in :-
(1) ***Solanum tuberosum*** (Potato)
(2) ***Cocos nucifera*** (Coconut)
(3) ***Arachis hypogea*** (Ground nut)
(4) ***Helianthus annuus*** (Sunflower)
BCL042

51. In higher plants, the chloroplasts are :-
(1) Discoidal or oval (2) Spiral
(3) Cup shaped (4) Reticulate
BCL043

52. Non-pigmented part of chloroplast is called:-
(1) Thylakoids (2) Grana
(3) Stroma (4) Lamellae
BCL045

INTRA-ENDOMEMBRANE SYSTEM

53. Which of the following is known as ***"System of membranes"*** ?

(1) Lysosome (2) E.R.

(3) Mitochondria (4) Chloroplast

BCL020

54. Cisternae are found -

(1) Only in mitochondria

(2) Only in Endoplasmic Reticulum

(3) In Endoplasmic Reticulum and Golgi body

(4) Only in Golgi body

BCL026

55. Vacuole in a plant cell: -

(1) Lacks membrane and contains air

(2) Lacks membrane and contains water and excretory substances

(3) Is membrane-bound and contains storage proteins and lipids

(4) is membrane-bound and contains water and excretory substances

BCL082

56. Which one of the following is not considered as a part of the endomembrane system?

(1) Lysosome (2) Golgi complex

(3) Peroxisome (4) Vacuole

BCL094

RIBOSOME

57. DNA is not found in :–

(1) Nucleus (2) Mitochondria

(3) Chloroplast (4) Ribosome

BCL048

58. Self duplication does not occur in :-

(1) Mitochondria (2) Centrioles

(3) Chloroplast (4) Ribosome

BCL049

59. "Palade particles" are :–

(1) Ribosomes (2) Golgi vesicles

(3) Lysosomes (4) Peroxisomes

BCL053

60. Polysome is a chain of :–

(1) Pinosomes (2) Phagosomes

(3) Microsomes (4) Ribosomes

BCL054

61. Prokaryotic ribosomes are 70 S, S refers to :

(1) Svedberg unit (2) Smallest unit

(3) Smooth (4) Speed

BCL056

62. Smallest cell organelle is :-

(1) Lysosome (2) Centrosome

(3) Ribosome (4) Golgibody

BCL059

63. 70 S type of ribosomes are found in :-

(1) Prokaryotic cells

(2) Prokaryotic cells, chloroplasts and mitochondria

(3) Mitochondria

(4) Nucleus, mitochondria

BCL060

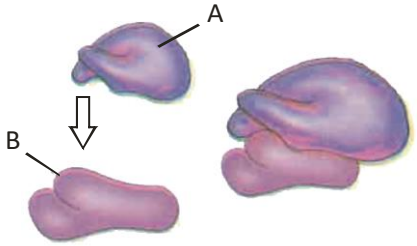
64. The Ribosomes are made up of -

(1) DNA + Protein (2) RNA + Protein

(3) DNA + RNA (4) Only protein

BCL061

65. Here 'S' (Svedberg's unit) stands for sedimentation coefficient then what are [A] & [B] for eukaryotic cell.



(1) A – 40S, B – 60S (2) A – 60S, B – 40S

(3) A – 80S, B – 60S (4) A – 40S, B – 80S

BCL064

66. Polysome is formed by: -
(1) A ribosome with several subunits
(2) Ribosomes attached to each other in a linear arrangement
(3) Several ribosomes attached to a single mRNA
(4) Many ribosomes attached to a strand of endoplasmic reticulum

BCL081

67. What is true about ribosomes?
(1) These are found only in eukaryotic cells
(2) These are self-splicing introns of some RNAs
(3) The prokaryotic ribosomes are 80S where "S" stands for sedimentation coefficient
(4) These are composed of ribonucleic acid and proteins

BCL097

68. Which one of the following structures is an organelle within an organelle?
(1) ER (2) Mesosome
(3) Ribosome (4) Peroxisome

BCL100

INTRA-MITOCHONDRIA, PLASTID AND RIBOSOMES

69. Match the following

	Column I		Column II
(a)	Cristae	(i)	Flat membranous sacs in stroma
(b)	Cisternae	(ii)	Infoldings in mitochondria
(c)	Thylakoids	(iii)	Disc-shaped sacs in Golgi apparatus

(1) (a)-(ii), (b)-(iii), (c)-(i)
(2) (a)-(ii), (b)-(i), (c)-(iii)
(3) (a)-(iii), (b)-(ii), (c)-(i)
(4) (a)-(i), (b)-(iii), (c)-(ii)

BCL028

70. Which of the following sets of cell organelles contain DNA ?
(1) Mitochondria, peroxisome
(2) Plasma membrane, ribosome
(3) Mitochondria, chloroplast
(4) Chloroplast, dictyosome

BCL032

71. Mitochondria and chloroplast are considered to be endosymbionts of cell because they :-
(1) Possess their own nucleic acid
(2) Have capacity of ATP synthesis
(3) Do not reproduce
(4) All of the above

BCL044

72. Which of following is not common in chloroplasts & mitochondria ?
(1) Both are present in animal cells
(2) Both contain their own genetic material
(3) Both are present in eukaryotic cells
(4) Both are present in plant cells

BCL046

MICROBODIES, CYTOSKELETON AND CENTROSOMES CENTRIOLE

73. During germination of seeds which cell organelle converts fatty acid into soluble carbohydrate ?
(1) Peroxisome (2) Glyoxysome
(3) Golgi body (4) Lysosome

BCL047

74. The peroxisomes are associated with the phenomenon of :-
(1) Krebs cycle
(2) Degradation of H_2O
(3) Anaerobic respiration
(4) Photorespiration and degradation of H_2O_2

BCL051

75. Which of the following termed as highly specialised peroxisomes ?
(1) Glyoxysomes (2) Mitochondria
(3) Golgi body (4) Lysosomes

BCL052

76. In which one of the following would you expect to find glyoxysomes ?
(1) Endosperm of wheat
(2) Endosperm of castor
(3) Palisade cells in leaf
(4) Root hairs

BCL063

77. In which of the following tubulin protein is not present?
(1) Plasma membrane (2) Cilia
(3) Flagella (4) Microtubules

BCL050

78. Cilia and flagella both -
(1) have 9 + 2 arrangement of microtubules
(2) are protective structure of cells
(3) are only present in protozoa animals
(4) are outgrowth structures of cytoplasm

BCL062

79. Cytoskeleton is made up of: -
(1) Proteinaceous filaments
(2) Calcium carbonate granules
(3) Callose deposits
(4) Cellulosic microfibrils

BCL085

80. An elaborate network of filamentous proteinaceous structures present in the cytoplasm which helps in the maintenance of cell shape is called:
(1) Endosplasmic Reticulum
(2) Plasmalemma
(3) Cytoskeleton
(4) Thylakoid

BCL091

81. Basal body is :-
(1) Centriole like
(2) Plastid like
(3) Ribosome like
(4) Mitochondria like

BCL055

82. Microtubules are composed of :-
(1) Actin protein (2) Myosin protein
(3) Tubulin protein (4) Dynein protein

BCL057

83. Arrangement of microtubules in centriole is:-
(1) 9 + 2 (2) 2 + 9
(3) 11 + 0 (4) 9 + 0

BCL058

CHROMOSOME AND NUCLEUS

84. Genome is :-
(1) Part of chromosome
(2) Half part of a chromosome
(3) Total DNA in cell
(4) Total chromosomes in a gamete

BCL065

85. Nucleolar organizer region is a :-
(1) Primary constriction
(2) Secondary constriction
(3) Tertiary constriction
(4) Centriole

BCL066

86. Kinetochore is present on :-
(1) Mitochondria (2) Peroxisome
(3) Chromosome (4) Flagella

BCL067

87. Chromosomes are composed of :-
(1) DNA, RNA, Histones, Non histones
(2) DNA and Histones
(3) DNA and RNA
(4) DNA, RNA and Histones

BCL068

88. Which part of chromosome is concern with ageing of organism?
(1) Centromere (2) Telomere
(3) Kinetochore (4) Satellite

BCL069

89. The non–sticky chromosomal ends are known as–
(1) Chromatids (2) Centromeres
(3) Satellites (4) Telomeres
BCL070

90. Salivary gland chromosome is concerned with :-
(1) vitellogenesis
(2) formation of ribosomes
(3) lipid synthesis
(4) metamorphosis in some insects
BCL072

91. Part of chromosome after secondary constriction is called :–
(1) Centromere
(2) Telomere
(3) Satellite
(4) Nucleolar organiser
BCL073

92. If the centromere is close to chromosome's end and the two arms are unequal then the chromosome is called as :-
(1) Metacentric (2) Submetacentric
(3) Acrocentric (4) Telocentric
BCL074

93. Hetero-chromatin is :-
(1) Darkly stained part of chromatin
(2) Lightly stained part of cristae
(3) Lightly stained part of grana
(4) Scattered Lobes in cytoplasm
BCL075

94. Chromosome with centromere at one end, is:
(1) Metacentric (2) Submetacentric
(3) Telocentric (4) Acrocentric
BCL076

95. Nucleus is absent in :-
(1) Cell of vascular cambium
(2) Root hair cell
(3) Companion cell
(4) Members of mature sieve tube
BCL077

96. The telomeres of eukaryotic chromosomes consist of short sequences of –
(1) Cytosine rich repeats
(2) Adenine rich repeats
(3) Guanine rich repeats
(4) Thymine rich repeats
BCL078

97. The protein nucleoplasmin occurs on :-
(1) Nuclear pore
(2) Sieve cells
(3) Nucleolus
(4) Hetero chromatin
BCL071

98. Ribosomal RNA is actively synthesized in:-
(1) Nucleoplasm
(2) Ribosomes
(3) Lysosomes
(4) Nucleolus
BCL098

99. Protein synthesis in an animal cell occurs –
(1) On ribosomes present in cytoplasm as well as in mitochondria
(2) On ribosomes present in the nucleolus as well as in cytoplasm
(3) Only on ribosomes attached to the nuclear envelope and endoplasmic reticulum
(4) Only on the ribosomes present in cytosol
BCL079

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	4	4	2	2	4	1	3	3	3	2	4	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	2	4	4	2	1	3	4	1	1	4	2	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	4	2	2	1	4	3	1	4	1	2	2	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	4	3	1	1	1	3	2	3	4	3	4	4	1	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	2	2	2	3	4	3	1	3	1	1	2	4	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	1	1	1	3	1	3	4	4	2	3	1	2	4	4
Question	91	92	93	94	95	96	97	98	99						
Answer	3	3	1	3	4	3	1	4	1						

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. The solid linear cytoskeletal elements having a diameter of 6 nm and made up of a single type of monomer are known as:
(1) Microtubules
(2) Microfilaments
(3) Intermediate filaments
(4) Lamins

BCL104

2. Match the following and select the correct answer :

Column-I		Column-II
(a) Centriole	(i)	Infoldings in mitochondria
(b) Chlorophyll	(ii)	Thylakoids
(c) Cristae	(iii)	Nucleic acids
(d) Ribozymes	(iv)	Basal body cilia or flagella

(1) (a)–(iv), (b)–(ii), (c)–(i), (d)–(iii)
(2) (a)–(i), (b)–(ii), (c)–(iv), (d)–(iii)
(3) (a)–(i), (b)–(iii), (c)–(ii), (d)–(iv)
(4) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)

BCL105

AIPMT 2015

3. DNA is not present in :-
(1) Ribosomes (2) Nucleus
(3) Mitochondria (4) Chloroplast

BCL106

4. The structures that are formed by stacking of organized flattened membranous sacs in the chloroplasts are:-
(1) Grana (2) Stroma lamellae
(3) Stroma (4) Cristae

BCL107

5. The chromosomes in which centromere is situated close to one end are:
(1) Acrocentric (2) Telocentric
(3) Sub-metacentric (4) Metacentric

BCL108

6. Which one of the following is not an inclusion body found in prokaryotes?
(1) Cyanophycean granule
(2) Glycogen granule
(3) Polysome
(4) Phosphate granule

BCL109

7. Select the correct matching in the following pairs:
(1) Smooth ER – Synthesis of lipids
(2) Rough ER – Synthesis of glycogen
(3) Rough ER – Oxidation of fatty acids
(4) Smooth ER – Oxidation of phospholipids

BCL110

RE-AIPMT 2015

8. Which of the following structures is not found in prokaryotic cells?
(1) Plasma membrane
(2) Nuclear envelope
(3) Ribosome
(4) Mesosome

BCL111

9. Which of the following are not membrane-bound?
(1) Mesosomes (2) Vacuoles
(3) Ribosomes (4) Lysosomes

BCL112

10. Cellular organelles with membranes are :
(1) Lysosomes, Golgi apparatus and mitochondria
(2) Nuclei, ribosomes and mitochondria
(3) Chromosomes, ribosomes and endoplasmic reticulum
(4) Endoplasmic reticulum, ribosomes and nuclei

BCL113

11. Match the columns and identify the correct option:

Column-I	Column-II
(a) Thylakoids	(i) Disc-shaped sacs in Golgi apparatus
(b) Cristae	(ii) Condensed structure of DNA
(c) Cisternae	(iii) Flat membranous sacs in stroma
(d) Chromatin	(iv) Infoldings in mitochondria

(1) (a)–(iii), (b)–(iv), (c)–(ii), (d)–(i)
(2) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)
(3) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)
(4) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

BCL114

NEET-I 2016

12. Mitochondria and chloroplast are :-
(a) semi-autonomous organelles
(b) formed by division of pre-existing organelles and they contain DNA but lack protein synthesizing machinery
Which one of the following options is correct?
(1) Both (a) and (b) are correct
(2) (b) is true but (a) is false
(3) (a) is true but (b) is false
(4) Both (a) and (b) are false

BCL115

13. Microtubules are the constituents of :-
(1) Cilia, Flagella and Peroxisomes
(2) Spindle fibres, Centrioles and Cilia
(3) Centrioles, Spindle fibres and Chromatin
(4) Centrosome, Nucleosome and Centrioles

BCL116

14. A complex of ribosomes attached to a single strand of RNA is known as :-
(1) Polysome (2) Polymer
(3) Polypeptide (4) Okazaki fragment

BCL117

15. Which one of the following cell organelles is enclosed by a single membrane?
(1) Mitochondria (2) Chloroplasts
(3) Lysosomes (4) Nuclei

BCL118

16. Water soluble pigments found in plant cell vacuoles are :-
(1) Xanthophylls (2) Chlorophylls
(3) Carotenoids (4) Anthocyanins

BCL119

NEET-II 2016

17. A cell organelle containing hydrolytic enzymes is:-
(1) Ribosome (2) Mesosome
(3) Lysosome (4) Microsome

BCL120

NEET 2017

18. Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?
(1) Ribosome (2) Chloroplast
(3) Mitochondrion (4) Lysosome

BCL121

NEET 2018

19. Which of the following is true for nucleolus?
(1) Larger nucleoli are present in dividing cells.
(2) It is a membrane-bound structure.
(3) It takes part in spindle formation.
(4) It is a site for active ribosomal RNA synthesis.

BCL122

20. The Golgi complex participates in :-
(1) Fatty acid breakdown
(2) Formation of secretory vesicles
(3) Respiration in bacteria
(4) Activation of amino acid

BCL123

21. Which of the following events does *not* occur in rough endoplasmic reticulum?
(1) Protein folding
(2) Protein glycosylation
(3) Cleavage of signal peptide
(4) Phospholipid synthesis

BCL124

22. Select the *incorrect* match :

(1) Lampbrush chromosomes	–	Diplotene bivalents
(2) Allosomes	–	Sex chromosomes
(3) Submetacentric chromosomes	–	L-shaped chromosomes
(4) Polytene chromosomes	–	Oocytes of amphibians

BCL125

NEET-UG 2019

23. The shorter and longer arms of a submetacentric chromosome are referred to as :-
(1) s-arm and l-arm respectively
(2) p-arm and q-arm respectively
(3) q-arm and p-arm respectively
(4) m-arm and n-arm respectively

BCL126

24. Which of the following pair of organelles does not contain DNA?
(1) Mitochondria and Lysosomes
(2) Chloroplast and Vacuoles
(3) Lysosomes and Vacuoles
(4) Nuclear envelope and Mitochondria

BCL127

25. Which of the following statements is not correct?
(1) Lysosomes have numerous hydrolytic enzymes.
(2) The hydrolytic enzymes of lysosomes are active under acidic pH.
(3) Lysosomes are membrane bound structures.
(4) Lysosomes are formed by the process of packaging in the endoplasmic reticulum.

BCL128

26. The concept of "*Omnis cellula-e cellula*" regarding cell division was first proposed by:
(1) Rudolf Virchow
(2) Theodore Schwann
(3) Schleiden
(4) Aristotle

BCL129

27. Which of the following statements regarding mitochondria is incorrect?
(1) Outer membrane is permeable to monomers of carbohydrates, fats and proteins.
(2) Enzymes of electron transport are embedded in outer membrane.
(3) Inner membrane is convoluted with infoldings.
(4) Mitochondrial matrix contains single circular DNA molecule and ribosomes.

BCL130

NEET-UG 2019 (Odisha)

28. Which of the following cell organelles is present in the highest number in secretory cells?
(1) Mitochondria
(2) Golgi complex
(3) Endoplasmic reticulum
(4) Lysosomes

BCL131

29. Non-membranous nucleoplasmic structures in nucleus are the site for active synthesis of:-
(1) Protein synthesis (2) mRNA
(3) rRNA (4) tRNA

BCL132

30. Match the column-I with column-II :-

Column-I	Column-II
(a) Golgi apparatus	(i) Synthesis of protein
(b) Lysosomes	(ii) Trap waste and excretory products
(c) Vacuoles	(iii) Formation of glycoproteins and glycolipids
(d) Ribosomes	(iv) Digesting biomolecules

Choose the right match from options given below :-

(1) (a)–(iii), (b)–(iv), (c)–(ii), (d)–(i)
(2) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)
(3) (a)–(iii), (b)–(ii), (c)–(iv), (d)–(i)
(4) (a)–(i), (b)–(ii), (c)–(iv), (d)–(iii)

BCL133

NEET-UG 2020

31. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells :

(1) Polysomes
(2) Endoplasmic reticulum
(3) Peroxisomes
(4) Golgi bodies

BCL134

NEET-UG 2021

32. The organelles that are included in the endomembrane system are:

(1) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
(2) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles
(3) Golgi complex, Mitochondria, Ribosomes and Lysosomes
(4) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes

BCL135

33. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as:

(1) Metacentric (2) Telocentric
(3) Sub-metacentric (4) Acrocentric

BCL136

34. Match List -I with List - II.

List -I	List -II
(a) Cristae	(i) Primary constriction in chromosome
(b) Thylakoids	(ii) Disc-shaped sacs in Golgi apparatus
(c) Centromere	(iii) Infoldings in mitochondria
(d) Cisternae	(iv) Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below.

(1) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)
(2) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii)
(3) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)
(4) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)

BCL137

35. Which of the following is an incorrect statement?

(1) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
(2) Microbodies are present both in plant and animal cells.
(3) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
(4) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.

BCL138

NEET-UG 2022

36. Match **List-I** with **List-II**.

	List-I		List-II
(a)	Metacentric	(i)	Centromere situated close to the end forming one extremely short and one very long arms
(b)	Acrocentric chromosome	(ii)	Centromere at the terminal end
(c)	Sub-metacentric	(iii)	Centromere in the middle forming two equal arms of chromosomes
(d)	Telocentric chromosome	(iv)	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the **correct answer** from the options given below:

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

BCL139

37. Which of the following statements with respect to Endoplasmic Reticulum is **incorrect?**

(1) SER is devoid of ribosomes

(2) In prokaryotes only RER are present

(3) SER are the sites for lipid synthesis

(4) RER has ribosomes attached to ER

BCL140

Re-NEET (UG) 2022

38. Which type of substance would face difficulty to pass through the cell membrane?

(1) Substance with hydrophobic moiety

(2) Substance with hydrophilic moiety

(3) All substance irrespective of hydrophobic and hydrophilic moiety

(4) Substance soluble in lipids

BCL141

39. If the pH in lysosomes is increased to alkaline, what will be the outcome?

(1) Hydrolytic enzymes will function more efficiently

(2) Hydrolytic enzymes will become inactive

(3) Lysosomal enzymes will be released into the cytoplasm

(4) Lysosomal enzymes will be more active

BCL142

NEET-UG 2023

40. Which of the following are NOT considered as the part of endomembrane system ?

A. Mitochondria

B. Endoplasmic Reticulum

C. Chloroplasts

D. Golgi complex

E. Peroxisomes

Choose the **most appropriate** answer from the options given below :

(1) A, C and E only (2) A and D only

(3) A, D and E only (4) B and D only

BCL204

41. Which of the following functions is carried out by cytoskeleton in a cell ?

(1) Protein synthesis (2) Motility

(3) Transportation (4) Nuclear division

BCL205

NEET-UG (MANIPUR) 2023

42. Given below are two statements:

Statement I : In bacteria, the mesosomes are formed by the extensions of plasma membrane.

Statement II : The mesosomes, in bacteria, help in DNA replication and cell wall formation.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

BCL206

43. Which of the following statements are **correct** with respect of Golgi apparatus ?

(A) It is the important site of formation of glycoprotein and glycolipids.

(B) It produces cellular energy in the form of ATP.

(C) It modifies the protein synthesized by ribosomes on ER.

(D) It facilitates the transport of ions.

(E) It provides mechanical support.

Choose the **most appropriate** answer from the options given below :

(1) (B) and (C) only

(2) (A) and (C) only

(3) (A) and (D) only

(4) (D) and (E) only

BCL207

NEET-UG 2024

44. Match List I with List II.

	List I		List II
A.	Nucleolus	I.	Site of formation of glycolipid
B.	Centriole	II.	Organization like the cartwheel
C.	Leucoplasts	III.	Site for active ribosomal RNA synthesis
D.	Golgi apparatus	IV.	For storing nutrients

Choose the correct answer from the options given below :

(1) A-III, B-II, C-IV, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-III, B-IV, C-II, D-I

(4) A-I, B-II, C-III, D-IV

BCL210

45. The DNA present in chloroplast is :

(1) Linear, double stranded

(2) Circular, double stranded

(3) Linear, single stranded

(4) Circular, single stranded

BCL211

46. Match List I with List-II :

List-I		List-II	
A.	Axoneme	I.	Centriole
B.	Cartwheel pattern	II.	Cilia and flagella
C.	Crista	III.	Chromosome
D.	Satellite	IV.	Mitochondria

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-II, D-I

(2) A-IV, B-II, C-III, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-I, C-IV, D-III

BCL212

47. Given below are two statements:

Statement-I : Mitochondria and chloroplasts are both double membrane bound organelles.

Statement-II : Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.

In the light of the above statement, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

BCL213

RE-NEET-UG 2024

48. Match **List-I** with **List-II** :

	List-I		**List-II**
A.	Flemming	I.	Disc shaped sac or cisternae near cell nucleus
B.	Robert Brown	II.	Chromatin
C.	George Palade	III.	Ribosomes
D.	Camillo Golgi	IV.	Nucleus

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-III, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-II, C-III, D-I

BCL214

49. Match **List-I** with **List-II** :

	List I		**List II**
A.	Metacentric chromosome	I.	Chromosome has a terminal centromere
B.	Sub-metacentric chromosome	II.	Middle centromere forming two equal arms of chromosome
C.	Acrocentric chromosome	III.	Centromere is slightly away from the middle of chromosome resulting into two unequal arms
D.	Telocentric chromosome	IV.	Centromere is situated close to its end forming one extremely short and one very long arm

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-IV, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

BCL215

50. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	F_1 Particles	I.	Chromosomes
B.	Histones	II.	Cilia
C.	Axoneme	III.	Golgi apparatus
D.	Cisternae	IV.	Mitochondria

Choose the **correct answer** from the options given below :

(1) A-II, B-I, C-IV, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-IV, B-III, C-I, D-II

BCL216

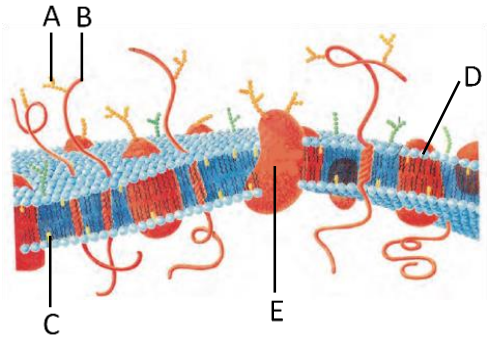
EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	1	1	3	1	2	3	1	3	3	2	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	3	4	2	4	4	2	3	4	1	2	2	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	1	3	1	4	2	2	2	1	2	3	2	1	2
Question	46	47	48	49	50										
Answer	4	3	1	4	2										

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. The figure given below shows the structure of plasma membrane, with its parts labelled from A to E, Identify the correct :-



(1) A → 52 percent of erythrocyte membrane

(2) B and E → Movement of these can be measured as fluidity of membrane

(3) C → Help in facilitated transport of polar molecules

(4) D → Its polar tail contains saturated hydrocarbons

BCL143

2. Contractile vacuole :-

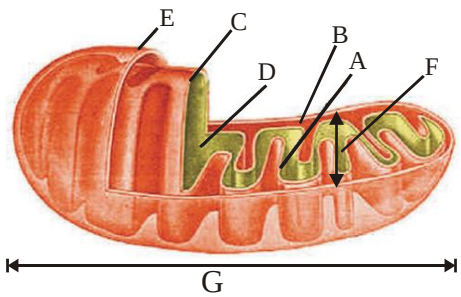
(1) is a membrane less structure

(2) is formed by engulfing the food

(3) is important for osmoregulation

(4) is called suicidal bag of cell

Q. No. 3 to 7 are based on given diagram :-



BCL144

3. Which represents aqueous compartments?

(1) A, B (2) D, C

(3) A, D (4) B, D

BCL145

4. What is the average value of labelled F?

(1) 0.2 µm (2) 0.5 µm

(3) 1.0 µm (4) 4.1 µm

BCL146

5. Cristae are infoldings of ....., present toward the .....

(1) E and B (2) B and C

(3) C and D (4) F and D

BCL147

6. Single, circular DNA molecule is found in :-

(1) B (2) D

(3) E (4) C

BCL148

7. D contains :-

(i) few RNA molecules (ii) 70s ribosome

(iii) Enzymes (iv) Circular DNA

Options :-

(1) ii, iv (2) i, ii, iv

(3) ii, iii, iv (4) i, ii, iii, iv

BCL149

8. (a) granular structure

(b) first observed under the electron microscope as dense particles by George Palade

(c) composed of RNA and proteins

(d) not surrounded by any membrane

Above given statements are true for which cell organelle?

Options :-

(1) Nucleolus (2) Ribosome

(3) Lysosome (4) Chloroplast

BCL150

9. Find **incorrect** statement with regard to centrosome and centrioles :-

(a) Centrosome is surrounded by amorphous pericentriolar material

(b) In centrosome, both centrioles lie parallel to each other in which each has an organisation like the cartwheel

(c) Centrioles are made up of nine unevenly spaced peripheral fibrils of tubulin

(d) Hub is the central proteinaceous part of centriole

(e) Proteinaceous radial spokes connect hub to peripheral triplets

Options :-

(1) a, b, e (2) only b

(3) b, c (4) All are correct

BCL151

10. .....observed that all plant tissues are made up of cells. At the same time....., studied different type of animal cells.

(1) Rudolf Virchow and Robert Brown respectively

(2) Mathias Schleiden and Theodore Schwann respectively

(3) Theodore Schwann and Mathias Schleiden respectively

(4) Robert Hooke and Schleiden respectively

BCL152

11. According to fluid-mosaic model, the quasi-fluid nature of.....enables lateral movement of.....within the overall bilayer. This ability to move within the membrane is measured as its...

(i) Carbohydrates (ii) Lipids

(iii) Proteins (iv) Fluidity

(v) Selective permeability

Correct sequence is :-

(1) ii, iii, iv (2) iii, i, iv

(3) iii, ii, v (4) i, ii, iv

BCL153

12. One of the most important functions of the plasma membrane is :-

(1) Formation of nuclear membrane

(2) Transport of molecules across it

(3) Exocytosis

(4) Detoxification

BCL154

13. .....can not pass through the lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the membrane

(1) Nonpolar molecules

(2) Polar molecules

(3) Hydrophobic molecules

(4) Both (2) and (3)

BCL155

14. Na^+/K^+ pump is an example of :-

(1) Passive transport (2) Osmosis

(3) Active transport (4) Simple diffusion

BCL156

15. Mark the incorrect match for transport of molecules across the membrane :-

(1) Neutral solute – simple diffusion

(2) Water – osmosis

(3) Non-polar molecules – facilitate diffusion

(4) ATP utilized – active transport

BCL157

16. Mitochondria :-

(a) are easily visible under the microscope (without specifically stained)

(b) are typically sausage-shaped or cylindrical

(c) are double membrane bound structures

(d) have two aqueous compartments

Options :-

(1) a, d correct and b, c incorrect

(2) a, b correct and c, d incorrect

(3) a incorrect and b, c, d correct

(4) a, d incorrect and b, c correct

BCL158

17. Inner mitochondrial membrane forms infoldings called :-
(1) Thylakoid (2) Cisternae
(3) Oxysomes (4) Cristae
BCL159

18. The number of mitochondria per cell is variable, depending on :-
(1) Size of cells
(2) Shape of cells
(3) Physiological activity of cells
(4) Type of genes present in mitochondrial DNA
BCL160

19. Which one of the following is not a component of endomembrane system ?
(a) Endoplasmic reticulum
(b) Golgi body (c) Lysosome
(d) Vacuole (e) Nucleus
Options :-
(1) Both a and c (2) Only c
(3) d and e both (4) Only e
BCL161

20. Membrane bound vesicular structures formed by the process of packaging in the Golgi apparatus and filled with hydrolytic enzymes, are called :-
(1) Contractile vacuoles (2) Food vacuoles
(3) Lysosomes (4) Centrosome
BCL162

21. Eukaryotes have 80S, while prokaryotes have 70S ribosomes in cytoplasm. Here "S":
(a) denotes sedimentation coefficient
(b) is an indirect measure of density
(c) is an indirect measure of size
Options :-
(1) a only (2) a and b only
(3) b and c only (4) a, b and c
BCL163

22. In r-RNA, ''r'' stands for :-
(1) Reversible (2) Ribozyme
(3) Ribosomal (4) Recognition
BCL164

23. Interphase nucleus has a loose and indistinct network of nucleoprotein fibres called chromatin, but during different stages of cell division, cells show "*structured chromosomes*" in place of the:-
(1) Nucleus (2) Plastids
(3) Mitochondria (4) Vacuole
BCL165

24. Nucleus as a "cell organelle" was first described by ..... as early as.....
(1) Robert Hooke, 1665
(2) Robert Brown, 1831
(3) Flemming, 1931
(4) Strasburger, 1831
BCL166

25. Space between parallel nuclear membranes is called perinuclear space which is :-
(1) 10–50 nm (2) 0.1 – 0.4 μm
(3) 10 – 50 Å (4) 1 – 4 nm
BCL167

26. A single human cell has approximately ..... long thread of DNA distributed among its ..... chromosomes:-
(1) 2 cm, 46 (2) 2 metre, 46
(3) 2 cm, 23 (4) 2 metre, 23
BCL168

27. Regarding to cell membrane find out the odd one
(1) Fluid mosaic model is widely accepted model
(2) Quasi fluid nature of lipids enables the lateral movement of proteins
(3) All types of molecules can easily pass through membrane
(4) Fluid nature of membrane is also important for cell growth & formation of intercellular junctions
BCL170

28. Which type of solutes may move across plasma membrane from higher to lower concentration along concentration gradient without help of transmembrane proteins?
(1) Positively charged solutes
(2) Negatively charged solutes
(3) Neutral solutes
(4) Any of the above

BCL171

29. Select out the wrong statement -
(1) Neutral solute can move according to concentration gradient across the nonpolar lipid bilayer
(2) Water can also move according to concentration gradient across the plasma membrane.
(3) Non polar molecules can not pass through non polar lipid bilayer
(4) Na^+ & K^+ can move across membrane through active transport

BCL172

30. Which of the following is not a function of cell wall ?
(1) Protection from mechanical damage and infection
(2) Cell to cell interaction
(3) Barrier to undesirable macromolecules
(4) Secretion

BCL173

31. Which of the following components is not a constituent of algal cell wall ?
(1) Cellulose (2) Galactans
(3) Mannans (4) Chitin

BCL174

32. Endoplasmic reticulum helps in :-
(1) The transport of substances
(2) Synthesis of lipoproteins
(3) Synthesis of glycogen
(4) All of the above

BCL175

33. Regarding to endoplasmic reticulum which of the following statements are wrong ?
(1) ER divides the intra cellular space into two distinct compartments
(2) RER frequently observed in cells actively involved in secretion
(3) In animals steroidal hormones are synthesized in RER
(4) SER is the major site of lipid synthesis

BCL176

34. Golgi complex receives proteins for modification from RER at which face ?
(1) Cis face (2) Trans face
(3) Concave face (4) Maturing face

BCL177

35. Classification of plastids into chloroplast, chromoplast and leucoplast is based on -
(1) Stored food (2) Pigments
(3) Structure (4) Size

BCL178

36. Chloroplast of higher plants contains -
(1) Only chlorophyll
(2) Only carotenoids
(3) Both chlorophyll and carotenoids
(4) Phycobilin

BCL179

37. Regarding to cilia and flagella which of the following statements is incorrect ?
(1) Cilia is small and flagella is long
(2) Cilia can move either cell or surrounding fluid
(3) Flagella is responsible for movement of surrounding fluid
(4) Cilia work like oars

BCL180

38. Plasma membrane covers the central core of the flagella and cilia this central core is known as -
(1) Bridge (2) Axoneme
(3) Radial spoke (4) Arms

BCL181

39. Radial spokes of flagella helps in connection between-

(1) Peripheral triplets
(2) Central singlet microtubules
(3) Peripheral doublet and central sheath
(4) Two successive peripheral doublets

BCL182

40. Match the following -

	Column-I		Column-II
(A)	Robert brown	(I)	Ribonucleo protein
(B)	Flemming	(II)	Nucleus as cell organelle
(C)	Palade	(III)	Packaging of materials
(D)	Camillo	(IV)	Staining of nucleus material

(1) A - (II) B - (IV) C - (I) D - (III)
(2) A - (II) B - (IV) C - (III) D - (I)
(3) A - (I) B - (II) C - (III) D - (IV)
(4) A - (IV) B - (III) C - (II) D - (I)

BCL183

41. Classification of chromosomes is based on -

(1) the size of satellite
(2) Number of telomeres
(3) Position of centromere
(4) Position of secondary constriction

BCL184

42. Chromosome with centromere slightly away from center is known as -

(1) Metacentric (2) Submetacentric
(3) Acrocentric (4) Telocentric

BCL185

43. Find out the incorrect about secondary constriction -

(1) Non staining
(2) Constant position
(3) Known as satellite
(4) Present in some chromosomes

BCL186

44. Match the column I and II and choose correct option :-

Column-I		Column-II	
(A)	Plasma membrane	(I)	Helps in cell division of animal cells
(B)	Centriole	(II)	Protein synthesis
(C)	Ribosomes	(III)	Not present in animal cell
(D)	Plastids	(IV)	Rich in hydrolytic enzymes
(E)	Lysosome	(V)	Barrier between cytoplasm and outer environment, in an animal cell.

Options :-

(1) A – V , B – I, C – II, D – III, E – IV
(2) A – V , B – III, C – II, D – I, E – IV
(3) A – IV , B – III, C – II, D – I, E – V
(4) A – II , B – IV, C – III, D – I, E – V

BCL187

45. Which of the following is not the function of cell wall ?

(i) Provides shape to the cell.
(ii) Protects the cell from mechanical damage and infection.
(iii) Helps in cell to cell connection.
(iv) Provides barrier to undesirable macromolecules.
(v) Helps in cell recognition

Options :-

(1) only (iii)
(2) only (iv)
(3) only (ii), (iii) & (v)
(4) only (v)

BCL188

46. (A) Unicellular organisms are capable of independent existence.
(B) Cell is the fundamental structural and functional unit of all unicellular organisms only.
(C) All cells arise from pre-existing cells.
(D) The cytoplasm is the main arena of cellular activities of cells.

Options :-

(1) Statements A, C and D are correct
(2) Statements A, B and D are not correct
(3) Statements A, B and C are correct
(4) Statements B, C and D are not correct

BCL189

47. Which one of the following statements is not correct for the vacuoles ?

(1) Contractile vacuoles are helpful in excretion
(2) Food vacuoles are formed by engulfing the food particles
(3) Sap vacuole is bound by tonoplast
(4) Tonoplast facilitates the transport of ions against the concentration gradient into cytoplasm

BCL190

48. Which of the following is true for mitochondria?

(A) Mitochondrial ribosome is smaller than prokaryotic ribosome
(B) Single stranded circular DNA is present in matrix
(C) Protein synthesis occurs in Mitochondrial matrix
(D) Outer membrane is highly folded and form cristae

Options :-

(1) A, C and D only
(2) A, B and D only
(3) Only C
(4) A and D only

BCL191

(B) ANALYTICAL QUESTIONS

49. Which of the following are **correct** for the cell which has naked ds circular DNA?

(a) Both 70S and 80S ribosomes
(b) Histone absent
(c) Compartmentalisation of cytoplasm absent
(d) Always diploid

Options :-

(1) a and d (2) b and c
(3) a and c (4) b and d

BCL192

50. Which of the following is/are function(s) of Golgi complex ?

(I) Modification of lipids and proteins
(II) Detoxification
(III) Formation of acrosome
(IV) Glycogen synthesis & breakdown

Choose the correct option –

(1) I and II are incorrect
(2) I and III are correct
(3) II is incorrect and remaining are correct
(4) II and III are incorrect

BCL193

51. If living cells, similar to those found on earth, were found on another planet, where there was no oxygen, then which cell organelle would most probably be absent ?

(1) Ribosomes
(2) Golgi apparatus
(3) Mitochondria
(4) Endoplasmic Reticulum

BCL194

52. Plant and prokaryotic cells are similar with each other but differ from animal cells :-

(1) In possessing 70 S ribosomes
(2) In possessing cell wall
(3) In possessing mitochondria
(4) In possessing chloroplasts

BCL195

53. Ratio of protein and lipid in the membrane of erythrocytes of human being is :-

(1) 1.8 (2) 1.3 (3) 0.80 (4) 0.60

BCL196

54. A student did the cell fractionation of a tissue but forgot to label his tubes. The content of one tube when studied showed organelles bounded by membrane with activity of catalase enzyme. These organelles could be :-

(1) SER (2) Chloroplast

(3) Lysosome (4) Peroxisome

BCL197

55. Given below are two statement -

Statement-I :- In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradient into the vacuole.

Statement-II :- Golgi apparatus is the important site of formation of glycoprotein and glycolipids.

In the light of the above statements, choose the correct answer from the options given below -

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

BCL198

56. **Assertion (A) :** Cell wall is a living rigid structure which form outer covering for the plasma membrane of fungi and plants.

Reason (R) : Secondary cell wall is formed on the outer (towards middle lamella) side of the cell.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

BCL199

57. **Assertion (A) :** Chloroplast & Mitochondria are considered as "Semi-autonomous organelle" of the cell.

Reason (R) : Both the organelle possess their own genetic material and have their own protein synthesising machinery.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

BCL200

58. Given below are two statement -

Statement-I :- At number of places the nuclear envelop is interrupted by minute pores.

Statement-II :- Material of the nucleus stained by the acidic dye was given the name chromatin by Flemming.

In the light of the above statements, choose the correct answer from the options given below -

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

BCL201

59. Read the following statements carefully and select the option which includes incorrect statements -

(a) RER is frequently observed in the cells actively involved in lipid synthesis & secretion.

(b) The SER is the major site for protein synthesis

(c) The golgi cisternae are concentrically arranged near the nucleus.

(d) Lysosomes are rich in alkaline hydrolases.

(1) Statement (a), (b) and (c)

(2) Statement (a), (b) and (d)

(3) Statement (b), (c) and (d)

(4) Statement (a), (c) and (d)

BCL202

60. New cells generate from-

(1) bacterial fermentation

(2) regeneration of old cells

(3) pre-existing cells

(4) abiotic materials

BCL209

61. Select the wrong statement from the following:

(1) The chloroplasts are generally much larger than mitochondria

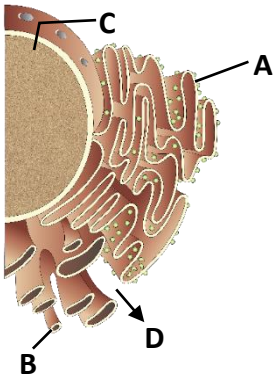
(2) Both chloroplasts and mitochondria contain an inner and an outer membrane

(3) Both chloroplasts and mitochondria have an internal compartment, the thylakoid space bounded by the thylakoid membrane

(4) Both chloroplasts and mitochondria contain DNA

BCL080

62. Identify the components labelled A, B, C and D in the diagram below from the list (i) to (viii) given with -



Components:

(i) Cristae of mitochondria

(ii) Inner membrane of mitochondria

(iii) Cytoplasm

(iv) Smooth endoplasmic reticulum

(v) Rough endoplasmic reticulum

(vi) Mitochondrial matrix

(vii) Cell vacuole

(viii) Nucleus

The correct component are:

(1) A–(i), B–(iv), C–(viii), D–(vi)

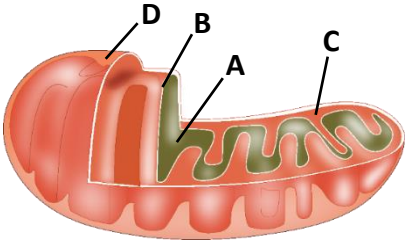
(2) A–(vi) B–(v), C–(iv), D–(vii)

(3) A–(v), B–(i), C–(iii), D–(ii)

(4) A–(v), B–(iv), C–(viii), D–(iii)

BCL092

63. The figure below shows the structure of a mitochondrion with its four parts labelled (A), (B), (C) and (D). Select the part correctly matched with its function.



(1) Part (A): Matrix - major site for respiratory chain enzymes

(2) Part (D): Outer membrane - gives rise to inner membrane by splitting

(3) Part (B): Inner membrane - forms infoldings called cristae

(4) Part (C): Cristae - possess single circular DNA molecule and ribosomes

BCL095

64. Which one of the following cellular parts is correctly described?

(1) Ribosomes - those on chloroplasts are larger (80 S) while those in the cytoplasm are smaller (70 S)

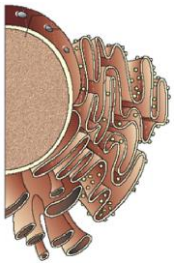
(2) Lysosomes-optimally active at a pH of about 8.5

(3) Thylakoids-flattened membranous sacs forming the grana of chloroplasts

(4) Centrioles - sites for active RNA synthesis

BCL099

65. Which one of the following organelle in the figure correctly matches with its function?



(1) Rough endoplasmic reticulum, protein synthesis

(2) Rough endoplasmic reticulum, formation of glycoproteins

(3) Golgi apparatus, protein synthesis

(4) Golgi apparatus, formation of glycolipids

BCL103

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	4	2	3	2	4	2	3	2	1	2	2	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	3	4	3	4	3	1	2	1	2	3	3	3	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	3	1	2	3	3	2	3	1	3	2	3	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	3	2	2	3	2	2	4	1	4	1	3	2	3
Question	61	62	63	64	65										
Answer	3	4	3	3	1										

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION TO MODIFICATIONS OF MITOSIS

1. During G_2 - phase a diploid cell contains the amount of DNA equal to a :-
(1) Diploid cell
(2) Tetraploid cell
(3) Haploid cell
(4) Hexaploid cell

BCD001

2. In which order, cytokinesis occurs in plants:
(1) Centripetal (2) Centrifugal
(3) Oblique (4) Equatorial

BCD002

3. Nuclear envelope reappears at :-
(1) Metaphase (2) Prophase
(3) Anaphase (4) Telophase

BCD003

4. Which does not occurs in prophase ?
(1) Decondensation of chromatin
(2) Condensation of chromatin
(3) Appearance of chromosome
(4) Disappearance of nuclear membrane and nucleolus

BCD004

5. In which stage of cell division, number of chromosomes can be best counted?
(1) Prophase (2) Metaphase
(3) Telophase (4) Interphase

BCD005

6. How many chromosomes shall be present in a diploid cell at mitotic anaphase if ovum of same species has ten chromosomes?
(1) 10 (Ten) (2) 20 (Twenty)
(3) 30 (Thirty) (4) 40 (Forty)

BCD006

7. In which stage of mitosis, the chromosomes are composed of two chromatids ?
(1) Prophase & metaphase
(2) Anaphase and telophase
(3) Prophase and telophase
(4) Metaphase and anaphase

BCD007

8. Gap between division phase and start of DNA-replication is called :-
(1) G_1 - phase (2) G_2 - phase
(3) M - phase (4) Interkinesis

BCD008

9. During cell division, spindle fibres attach to which part of chromosome ?
(1) Primary constriction
(2) Secondary constriction
(3) Telomere
(4) Satellite

BCD009

10. During which stage a diploid cell becomes tetraploid in mitosis?
(1) G_2 (2) Prophase
(3) Metaphase (4) Anaphase

BCD010

11. Division of centromere occurs in:-
(1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase

BCD011

12. What happens in synthesis phase during cell cycle?
(1) DNA replication
(2) Chromosome number becomes double
(3) Formation of two nuclei
(4) Synthesis of tubulin proteins

BCD012

13. Reappearance of nucleolus along with thinning & elongation in chromosomes are diagnostic characters for the :-
(1) Anaphase (2) Metaphase
(3) Interphase (4) Telophase
BCD013

14. Condensation of chromosomes and appearance of astral rays occur during :-
(1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase
BCD014

15. Chromosomal morphology (Structure) is best observed at :-
(1) Prophase (2) Metaphase
(3) Interphase (4) Anaphase
BCD015

16. M–phase of cell cycle consist of :–
(1) G_1, S and G_2 phase
(2) Prophase, Metaphase, Anaphase, Telophase
(3) Interphase, Prophase, Metaphase, Anaphase, Telophase
(4) Only prophase
BCD016

17. If the cell is diploid in G_1 than after the S phase cell remain/become :-
(1) n (2) 4n
(3) 8n (4) 2n
BCD017

18. Nuclear membrane disappears in :–
(1) Late prophase (2) Early prophase
(3) Metaphase (4) Telophase
BCD018

19. Pre - DNA synthesis phase is:-
(1) G_1 - phase (2) G_2 - phase
(3) S-phase (4) Prophase
BCD019

20. How many steps of division will occur in an isolated tip cell to form 128 cells ?
(1) 128 (2) 127 (3) 32 (4) 7
BCD020

21. The number of chromatids in a chromosome at G_2 state of cell cycle will be:-
(1) One (2) Two (3) Four (4) Eight
BCD021

22. Many cells function properly and divide mitotically even though they do not have :-
(1) Plasma membrane
(2) Cytoskeleton
(3) Mitochondria
(4) Plastids
BCD022

23. Centromere is required for –
(1) Movement of chromosomes towards poles
(2) Cytoplasmic cleavage
(3) Crossing over
(4) Transcription
BCD023

24. At which stage of the cell cycle are histone proteins synthesized in a eukaryotic cell ?
(1) During telophase
(2) During S–phase
(3) During G_2–phase
(4) During entire prophase
BCD024

25. Prophase which follows the S and G_2 phases of interphase, is the first stage of :-
(1) Meiosis-II (2) Karyokinesis
(3) Interphase (4) G_1 phase
BCD025

26. The two asters together with spindle fibres form:-
(1) Mitotic apparatus (2) Centromere
(3) Astral fibres (4) Centrosome
BCD026

27. In prophase centrosome, which had undergone duplication during interphase, begins to move towards :
(1) Same pole of the cell
(2) Opposite poles of the cell
(3) One towards centre while another towards pole
(4) Both towards centre

BCD027

MEIOSIS, SIGNIFICANCE OF MEIOSIS, DIFFERENCES BETWEEN MITOSIS AND MEIOSIS

28. Which of the following not occurs in Anaphase-I but occurs in Anaphase-II :–
(1) Condensation of chromosomes
(2) Poleward movement of chromosomes
(3) Shortening of spindle fibres
(4) Splitting of centromere

BCD028

29. Crossing over takes place in :-
(1) Zygotene (2) Pachytene
(3) Diplotene (4) Diakinesis

BCD029

30. Which of the two events maintain the constant number of chromosomes across generations?
(1) Mitosis and Meiosis
(2) Meiosis and fertilisation
(3) Fertilisation and mitosis
(4) Only meiosis

BCD030

31. Match the column–I with column–II and select the correct answer :-

Column-I	Column-II
(A) Pachytene	(i) Bouquet stage
(B) Zygotene	(ii) Chiasmata visible
(C) Diplotene	(iii) Terminalisation
(D) Leptotene	(iv) Gene exchange
(E) Diakinesis	(v) Synapsis

Options :-
(1) A–i, B–ii, C–iii, D–iv, E–v
(2) A–iv, B–v, C–ii, D–i, E–iii
(3) A–iii, B–iv, C–v, D–ii, E–i
(4) A–ii, B–iii, C–iv, D–i, E–v

BCD031

32. Which part of plant is suitable for the study of meiosis?
(1) Root apex (2) Ovary
(3) Anther (4) Shoot apex

BCD032

33. Slipping of chiasmata towards the ends of bivalent is called :-
(1) Terminalisation (2) Diakinesis
(3) Interkinesis (4) Congression

BCD033

34. "Bouquet-stage" occur in which sub stages of prophase-I ?
(1) Leptotene (2) Zygotene
(3) Pachytene (4) Diplotene

BCD034

35. At anaphase-II of meiosis each chromosome contains:-
(1) 4 chromatids (2) 3 chromatids
(3) 2 chromatids (4) 1 chromatid

BCD035

36. In Anaphase-I each chromosome composed of :-
(1) One chromatid
(2) Two chromatids
(3) Four chromatids
(4) Many chromatids

BCD036

37. In meiosis, division of centromere occurs during:-
(1) Interphase (2) Anaphase - I
(3) Anaphase - II (4) Metaphase - I

BCD037

38. In meiosis, nuclear membrane and nucleolus disappear during :-
(1) Zygotene (2) Pachytene
(3) Diakinesis (4) Metaphase - I
BCD038

39. Separation of homologous chromosomes during anaphase-I is called :-
(1) Synapsis (2) Disjunction
(3) Nondisjunction (4) Crossing over
BCD039

40. Diakinesis represents :-
(1) transition to prophase
(2) transition to metaphase
(3) transition to anaphase
(4) transition to telophase
BCD040

41. Synaptonemal complex is characteristic of :-
(1) Mitotic chromosomes
(2) Leptotene chromosomes
(3) Paired meiotic chromosomes
(4) Metaphase
BCD041

42. Each chromosome composed of one chromatid in:-
(1) Anaphase – I (2) Anaphase – II
(3) Metaphase – I (4) Metaphase – II
BCD042

43. If the number of bivalents are 8 in metaphase–I, what shall be the number of chromosomes in daughter cells after meiosis–I and meiosis–II respectively:-
(1) 8 and 4 (2) 4 and 4
(3) 8 and 8 (4) 16 and 8
BCD043

44. Which of the following not occurs in anaphase-I?
(1) Segregation of homologous chromosomes
(2) Shortening in spindle fibres
(3) Poleward movement of chromosomes
(4) Division of centromere
BCD044

45. In meiosis :-
(1) Division of nucleus twice but replication of DNA only once
(2) Division of nucleus twice and replication of DNA twice
(3) Division of nucleus once and replication of DNA is also once
(4) Division of nucleus once and DNA replication is twice
BCD045

46. After meiosis–I :-
(1) the two chromatids of a chromosome are genetically similar
(2) the two chromatids of a chromosome are genetically different
(3) There occurs only one chromatid in each chromosome
(4) Four similar daughter cells are formed
BCD046

47. The correct sequence of prophase–I of meiosis is :–
(1) Leptotene, pachytene, zygotene, diplotene, diakinesis
(2) Leptotene, diplotene, pachytene, zygotene, diakinesis
(3) Leptotene, zygotene, pachytene, diplotene, diakinesis
(4) Leptotene, zygotene, diakinesis, diplotene, pachytene
BCD047

48. Which of the following is called heterotypic division ?
(1) Meiosis–I (2) Meiosis–II
(3) Mitosis (4) Amitosis
BCD048

49. DNA replication occurs before :–
(1) Mitosis and meiosis–I
(2) Mitosis, meiosis–I and meiosis–II
(3) Meiosis-I only
(4) Mitosis only
BCD049

50. Thick–thread stage is :–
(1) Leptotene (2) Zygotene
(3) Pachytene (4) Diplotene
BCD050

51. Crossing over that results in genetic recombination in higher organisms occurs between –
(1) Non-sister chromatids of a bivalent
(2) Two daughter nuclei
(3) Two different bivalents
(4) Sister chromatids of a bivalent
BCD051

52. When synapsis is complete all along the chromosome, the cell is said to have entered a stage called :-
(1) Zygotene (2) Pachytene
(3) Diplotene (4) Diakinesis
BCD052

53. If in a diploid plant cell the 'n' value is 16 then what is possible in metaphase - I of meiosis ?
(1) 32 Bivalents (2) 64 Bivalents
(3) 16 Bivalents (4) 8 Bivalents
BCD053

54. Which one of the following statements is incorrect for interkinesis ?
(1) It is the stage between the two subphases (I & II) of a meiotic division
(2) There is no replication of DNA
(3) DNA replicate but chromosome number remains same
(4) It is generally short lived.
BCD054

55. At anaphase-II, sister chromatids move towards opposite poles of the cell by :
(1) Contraction in spindle fibre attached to kinetochores
(2) Shortening of microtubules attached to kinetochores
(3) Lengthening of microtubules attached to kinetochores
(4) Relaxation in spindle fibre attached to kinetochores
BCD055

56. Synapsis occurs between :-
(1) Two homologous chromosomes
(2) A male and a female gamete
(3) mRNA and ribosomes
(4) Spindle fibres and centromere
BCD056

57. During gamete formation, the enzyme recombinase participates during :-
(1) Prophase-I
(2) Prophase-II
(3) Metaphase-I
(4) Anaphase-II
BCD062

58. Identify the meiotic stage in which the homologous chromosomes separate while the sister chromatids remain associated at their centromeres :-
(1) Anaphase I
(2) Anaphase II
(3) Metaphase I
(4) Metaphase II
BCD064

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	4	1	2	4	1	1	1	4	3	1	4	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	1	1	4	2	4	1	2	2	1	2	4	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	1	1	4	2	3	3	2	2	3	2	3	4	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58		
Answer	2	3	1	1	3	1	2	3	3	2	1	1	1		

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. During which phase(s) of cell cycle, amount of DNA in a cell remains at 4C level if the initial amount is denoted as 2C ?
(1) G_0 and G_1 (2) G_1 and S
(3) Only G_2 (4) G_2 and M

BCD067

2. In 'S' phase of the cell cycle :-
(1) Amount of DNA doubles in each cell.
(2) Amount of DNA remains same in each cell.
(3) Chromosome number is increased.
(4) Amount of DNA is reduced to half in each cell.

BCD068

3. The enzyme recombinase is required at which stage of meiosis :
(1) Pachytene
(2) Zygotene
(3) Diplotene
(4) Diakinesis

BCD069

AIPMT 2015

4. A somatic cell that has just completed the S phase of its cell cycle, as compared to gamete of the same species, has :
(1) same number of chromosomes but twice the amount of DNA
(2) twice the number of chromosomes and four times the amount of DNA
(3) four times the number of chromosomes and twice the amount of DNA
(4) twice the number of chromosomes and twice the amount of DNA

BCD070

Re-AIPMT 2015

5. Arrange the following events of meiosis in correct sequence :
(a) Crossing over
(b) Synapsis
(c) Terminalisation of chiasmata
(d) Disappearance of nucleolus
(1) (b), (c), (d), (a)
(2) (b), (a), (d), (c)
(3) (b), (a), (c), (d)
(4) (a), (b), (c), (d)

BCD071

NEET-I 2016

6. Spindle fibres attach on to :-
(1) Telomere of the chromosome
(2) Kinetochore of the chromosome
(3) Centromere of the chromosome
(4) Kinetosome of the chromosome

BCD072

7. In meiosis crossing over is initiated at :
(1) Pachytene (2) Leptotene
(3) Zygotene (4) Diplotene

BCD073

NEET-II 2016

8. During cell growth, DNA synthesis takes place in:-
(1) G_2 phase (2) M phase
(3) S phase (4) G_1 phase

BCD074

9. When cell has stalled DNA replication fork, which checkpoint should be predominantly activated?
(1) M
(2) Both G_2/M and M
(3) G_1/S
(4) G_2/M

BCD075

10. Match the stages of meiosis in **Column–I** to their characteristic features in **Column–II** and select the correct option using the codes given below :

Column-I		Column-II	
a	Pachytene	i	Pairing of homologous chromosomes
b	Metaphase-I	ii	Terminalization of chiasmata
c	Diakinesis	iii	Crossing over takes place
d	Zygotene	iv	Chromosomes align at equatorial plate

Codes :

	a	b	c	d
(1)	ii	iv	iii	i
(2)	iv	iii	ii	i
(3)	iii	iv	ii	i
(4)	i	iv	ii	iii

BCD076

NEET (UG) 2017

11. Which of the following options gives the correct sequence of events during mitosis ?

(1) Condensation → nuclear membrane disassembly → arrangement at equator → centromere division → segregation → telophase

(2) Condensation→ crossing over → nuclear membrane disassembly → segregation → telophase

(3) Condensation → arrangement at equator→ centromere division → segregation → telophase

(4) Condensation → nuclear membrane disassembly → crossing over → segregation → telophase

BCD077

12. Anaphase Promoting Complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cells. If APC is defective in a human cell, which of the following is expected to occur ?

(1) Chromosomes will be fragmented

(2) Chromosomes will not segregate

(3) Recombination of chromosome arms will occur

(4) Chromosomes will not condense

BCD078

NEET (UG) 2018

13. The stage during which separation of the paired homologous chromosomes begins is

(1) Pachytene (2) Diplotene

(3) Diakinesis (4) Zygotene

BCD079

NEET (UG) 2019

14. The **correct** sequence of phases of cell cycle is:

(1) M→ G_1 → G_2 → S

(2) G_1→ G_2 → S → M

(3) S→ G_1 → G_2 → M

(4) G_1→ S → G_2 → M

BCD080

15. Cells in G_0 phase:

(1) exit the cell cycle

(2) enter the cell cycle

(3) suspend the cell cycle

(4) terminate the cell cycle

BCD081

NEET (UG) 2019 (Odisha)

16. After meiosis-I, the resultant daughter cells have:

(1) Same amount of DNA as in the parent cell in S phase

(2) Twice the amount of DNA in comparison to haploid gamete.

(3) Same amount of DNA in comparison to haploid gamete

(4) Four times the amount of DNA in comparison to haploid gamete

BCD082

17. Crossing over takes place between which chromatids and in which stage of the cell cycle?

(1) Non-sister chromatids of non-homologous chromosomes at Zygotene stage of prophase I.

(2) Non-sister chromatids of homologous chromosomes at Pachytene stage of prophase I.

(3) Non-sister chromatids of homologous chromosomes at Zygotene stage of prophase I.

(4) Non-sister chromatids of non-homologous chromosomes at Pachytene stage of prophase I.

BCD083

NEET (UG) 2020

18. Match the following with respect to meiosis:

(a) Zygotene	(i) Terminalization
(b) Pachytene	(ii) Chiasmata
(c) Diplotene	(iii) Crossing over
(d) Diakinesis	(iv) Synapsis

Select the **correct** option from the following:

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iv)	(iii)	(i)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(i)	(ii)	(iv)	(iii)

BCD084

19. Identify the **correct** statement with regard to G_1 phase (Gap 1) of interphase.

(1) Nuclear Division takes place.

(2) DNA synthesis or replication takes place.

(3) Reorganisation of all cell components takes place.

(4) Cell is metabolically active, grows but does not replicate its DNA.

BCD085

20. Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G_0). This process occurs at the end of :

(1) G_2 phase (2) M phase

(3) G_1 phase (4) S phase

BCD086

21. Dissolution of the synaptonemal complex occurs during :

(1) Leptotene (2) Pachytene

(3) Zygotene (4) Diplotene

BCD087

NEET (UG) 2020 (Covid-19)

22. Attachment of spindle fibres to kinetochores of chromosomes becomes evident in :

(1) Anaphase (2) Telophase

(3) Prophase (4) Metaphase

BCD088

23. In a mitotic cycle, the correct sequence of phases is

(1) S, G_1, G_2, M (2) G_1, S, G_2, M

(3) M, G_1, G_2, S (4) G_1, G_2, S, M

BCD089

24. During Meiosis-I, in which stage synapsis takes place ?

(1) Pachytene (2) Zygotene

(3) Diplotene (4) Leptotene

BCD090

25. Match the following events that occur in their respective phases of cell cycle and select the correct option :

(a)	G_1 phase	(i)	Cell grows and organelle duplication
(b)	S phase	(ii)	DNA replication and chromosome duplication
(c)	G_2 phase	(iii)	Cytoplasmic growth
(d)	Metaphase in M-phase	(iv)	Alignment of chromosomes

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

BCD091

NEET (UG) 2021

26. Which of the following stages of meiosis involves division of centromere ?
(1) Metaphase I (2) Metaphase II
(3) Anaphase II (4) Telophase II

BCD092

27. Match **List -I** with **List - II.**

List -I		List -II	
(a)	S phase	(i)	Proteins are synthesized
(b)	G_2 phase	(ii)	Inactive phase
(c)	Quiescent stage	(iii)	Interval between mitosis and initiation of DNA replication
(d)	G_1 phase	(iv)	DNA replication

Choose the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iv)	(ii)	(iii)	(i)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(ii)	(iv)	(iii)	(i)

BCD093

28. The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G_1 phase is 8, what would be the number of chromosomes after S-phase ?
(1) 8 (2) 16 (3) 4 (4) 32

BCD094

29. The centriole undergoes duplication during:
(1) S-phase (2) Prophase
(3) Metaphase (4) G_2 phase

BCD095

30. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature?
(1) Leptotene (2) Zygotene
(3) Diakinesis (4) Pachytene

BCD096

NEET (UG) 2022

31. The appearance of recombination nodules on homologous chromosomes during meiosis characterizes:
(1) Bivalent
(2) Sites at which crossing over occurs
(3) Terminalization
(4) Synaptonemal complex

BCD097

32. Which one of the following never occurs during mitotic cell division ?
(1) Movement of centrioles towards opposite poles
(2) Pairing of homologous chromosomes
(3) Coiling and condensation of the chromatids
(4) Spindle fibres attach to kinetochores of chromosomes

BCD098

33. Select the **incorrect** statement with reference to mitosis:
(1) Spindle fibres attach to centromere of chromosomes.
(2) Chromosomes decondense at telophase.
(3) Splitting of centromere occurs at anaphase.
(4) All the chromosomes lie at the equator at metaphase.

BCD099

34. Regarding Meiosis, which of the statements is **incorrect** ?
(1) DNA replication occurs in S phase of Meiosis-II
(2) Pairing of homologous chromosomes and recombination occurs in Meiosis-I
(3) Four haploid cells are formed at the end of Meiosis-II
(4) There are two stages in Meiosis, Meiosis-I and II

BCD100

Re-NEET (UG) 2022

35. Which stage of meiosis can last for months or years in the oocytes of some verterbrates?

(1) Leptotene (2) Pachytene

(3) Diplotene (4) Diakinesis

BCD101

36. In meiosis, crossing over and exchange of genetic material between homologous chromosomes are catalyzed by the enzyme.

(1) Phosphorylase (2) Recombinase

(3) Transferase (4) Polymerase

BCD102

37. Identify the **correct** sequence of events during Prophase I of meiosis :

(a) Synapsis of homologous chromosomes

(b) Chromosomes become gradually visible under microscope

(c) Crossing over between non-sister chromatids of homologous chromosomes

(d) Terminalisation of chiasmata

(e) Dissolution of synaptonemal complex

Choose the **correct answer** from the options given below :

(1) (a), (b), (c), (d), (e)

(2) (b), (c), (d), (e), (a)

(3) (b), (a), (c), (e), (d)

(4) (a), (c), (d), (e), (b)

BCD103

38. Bivalent or Tetrad formation is a characteristic feature observed during :

(1) Synaptonemal complex in zygotene stage

(2) Chiasmata in Diplotene stage

(3) Synaptonemal complex in Pachytene stage

(4) Chiasmata in zygotene stage

BCD104

39. With respect to metaphase, which of the following statements is incorrect ?

(1) Complete disintegration of nuclear envelope takes place

(2) Chromosomes are highly condensed

(3) Metaphase chromosomes are made up of four sister chromatids held together by centromere

(4) Chromosomes lie at the equator of the cell

BCD105

NEET (UG) 2023

40. The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis ?

(1) Pachytene (2) Diplotene

(3) Diakinesis (4) Zygotene

BCD146

41. Which of the following stages of meiosis involves division of centromere ?

(1) Metaphase II (2) Anaphase II

(3) Telophase (4) Metaphase I

BCD147

42. Among eukaryotes, replication of DNA takes place in -

(1) S phase (2) G_1 phase

(3) G_2 phase (4) M phase

BCD148

43. Match **List I** with **List II**:

	List-I		List-II
(A)	M Phase	(I)	Proteins are synthesized
(B)	G_2 Phase	(II)	Inactive phase
(C)	Quiescent stage	(III)	Interval between mitosis and initiation of DNA replication
(D)	G_1 Phase	(IV)	Equational division

Choose the correct answer from the options given below:

(1) A-IV, B-II, C-I, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-II, B-IV, C-I, D-III

(4) A-III, B-II, C-IV, D-I

BCD149

44. Select the correct statements.

A. Tetrad formation is seen during Leptotene.

B. During Anaphase, the centromeres split and chromatids separate.

C. Terminalization takes place during Pachytene.

D. Nucleolus, Golgi complex and ER are reformed during Telophase.

E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the **correct** answer from the options given below:

(1) B and D only (2) A, C and E only

(3) B and E only (4) A and C only

BCD150

NEET (UG) 2023 (Manipur)

45. The dissolution of synaptonemal complex occurs during :

(1) Pachytene (2) Diplotene

(3) Diakinesis (4) Leptotene

BCD151

46. Doubling of the number of chromosomes can be achieved by disrupting mitotic cell division soon after :

(1) Anaphase (2) Telophase

(3) Prophase (4) Metaphase

BCD152

47. During which stages of mitosis and meiosis, respectively does the centromere of each chromosome split ?

(1) Mataphase, Metaphase II

(2) Prophase, Telophase I

(3) Telophase, Anaphase I

(4) Anaphase, Anaphase II

BCD153

48. Which one of the following is the quiescent stage of cell cycle ?

(1) M (2) G_2 (3) G_1 (4) G_0

BCD154

49. Select **correct** sequence of substages of Prophase-I of Meiotic division :

(A) Zygotene (B) Pachytene

(C) Diakinesis (D) Leptotene

(E) Diplotene

Choose the **correct** answer from the options given below :

(1) (D), (B), (A), (E), (C)

(2) (A), (B), (D), (E), (C)

(3) (D), (A), (B), (E), (C)

(4) (A), (D), (B), (C), (E)

BCD155

NEET-UG 2024

50. Spindle fibers attach to kinetochores of chromosomes during

(1) Prophase (2) Metaphase

(3) Anaphase (4) Telophase

BCD156

51. Given below are two statements :

Statement-I : Chromosomes become gradually visible under light microscope during leptotene stage.

Statement-II : The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement-I and Statement-II are true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is true but Statement-II is false

(4) Statement-I is false but Statement-II is true

BCD157

52. Match List I with List-II :

List-I (Sub Phases of Prophase-I)		List-II (Specific characters)	
A.	Diakinesis	I.	Synaptonemal complex formation
B.	Pachytene	II.	Completion of terminalisation of chiasmata
C.	Zygotene	III.	Chromosomes look like thin threads
D.	Leptotene	IV.	Appearance of recombination nodules

Choose the correct answer from the options given below:

(1) A-IV, B-II, C-III, D-I

(2) A-I, B-II, C-IV, D-III

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-III, C-II, D-I

BCD158

53. Following are the stages of cell division :

A. Gap 2 phase

B. Cytokinesis

C. Synthesis phase

D. Karyokinesis

E. Gap 1 phase

Choose the correct sequence of stages from the options given below :

(1) C-E-D-A-B (2) E-B-D-A-C

(3) B-D-E-A-C (4) E-C-A-D-B

BCD159

RE-NEET-UG 2024

54. Recombination between homologous chromosomes is completed by the end of :

(1) Diakinesis (2) Zygotene

(3) Diplotene (4) Pachytene

BCD160

55. Match **List-I** with **List-II**.

List-I Event		List-II Stage of Prophase-I (Meiosis-I)	
A.	Chiasmata formation	I.	Pachytene
B.	Crossing over	II.	Diakinesis
C.	Synaptonemal complex formation	III.	Diplotene
D.	Terminalisation of chiasmata	IV.	Zygotene

Choose the **correct** answer from the options given below :

(1) A-III, B-I, C-IV, D-II

(2) A-II, B-I, C-III, D -IV

(3) A-III, B-I, C-II, D-IV

(4) A-II, B-III, C-IV, D-I

BCD161

56. Match **List-I** with **List-II**.

List-I		List-II	
A.	Cell are metabolically active and proliferate	I.	G_2 phase
B.	DNA replication takes place	II.	G_1 phase
C.	Proteins are synthesised	III.	G_0 phase
D.	Quiescent state with metabolically active cells	IV.	S phase

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I

(2) A-I, B-III, C-IV, D-II

(3) A-II, B-I, C-III, D-IV

(4) A-II, B-IV, C-I, D-III

BCD162

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	1	2	3	2	1	3	4	3	1	2	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	3	4	3	4	4	2	2	4	3	3	1	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	1	1	3	2	3	1	3	1	2	1	2	1	2
Question	46	47	48	49	50	51	52	53	54	55	56				
Answer	4	4	4	3	2	1	3	4	4	1	4				

Exercise - III

Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which one is correct for G_0 stage ?

I. It is a quiescent stage.

II. In this phase cell cycle is stopped

III. G_0 cells do not proliferate but metabolically active

IV. G_0 cells can divide in response to some stimulus

Options :-

(1) All are correct

(2) Only I, II, III are correct

(3) Only I, II are correct

(4) Only I and IV are correct

BCD106

2. Which of the following is **not** correctly matches a phase of the cell cycle with its function ?

(1) Second gap phase	Period of cytoplasmic growth
(2) First gap phase	Most of the organelle duplication
(3) Interphase	Phase of preparation for cell division
(4) DNA synthesis phase	Doubling of number of chromosomes in cell

BCD107

3. Mark **incorrect** statements :-

(A) Meiosis involves only a single cycle of DNA replication

(B) Four haploid cells are formed at the end of meiosis–I

(C) Meiosis occurs in diploid cells

(D) In yeast, cell cycle takes about 90 minutes.

Options :-

(1) A and B (2) A and C

(3) Only B (4) All are correct

BCD108

4. Prophase–I of meiotic division is typically longer and more complex, it is subdivided into five phases, on the basis of :-

(1) Staining

(2) Behaviour of chromosomes

(3) Duration

(4) Number of chromosomes

BCD109

5. Arrange the following events of meiosis in the correct sequence -

A. Terminalisation of chiasmata

B. Crossing over

C. Synapsis

D. Disjunction of chromosomes

E. Dissolution of synaptonemal complex

The correct sequence is-

(1) A → B → C → D → E

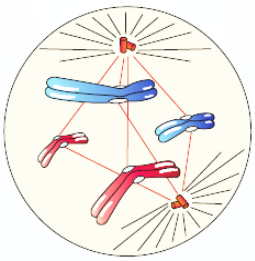
(2) E → D → C → B → A

(3) C → B → D → E → A

(4) C → B → E → A → D

BCD110

6. Identify the following diagram-



(1) Transition to Metaphase

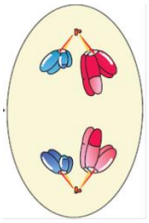
(2) Transition to Anaphase-II

(3) Transition to Metaphase-I

(4) Transition to Anaphase

BCD111

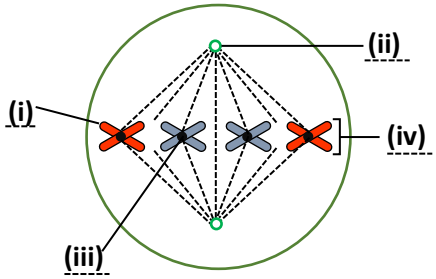
7. The drawing below shows a cell whose diploid chromosome number is four. This cell is in:-



(1) Metaphase of mitosis
(2) Anaphase of mitosis
(3) First anaphase of meiosis
(4) Second anaphase of meiosis

BCD112

8. Label the structures indicated by (i), (ii), (iii) and (iv) in below given diagram –



(1) (i)-Chromatid, (ii)-Centriole, (iii)-Centromere, (iv)-Chromosome
(2) (i)-Chromosome, (ii)-Centriole, (iii)-Centromere, (iv)-Chromatid
(3) (i)-Chromatid, (ii)-Centromere, (iii)-Centriole, (iv)-Chromosome
(4) (i)-Chromosome, (ii)-Centromere, (iii) Centriole, (iv)-Chromatid

BCD113

9. Interkinesis is stage between :-

(1) Two mitotic divisions
(2) Two phases of meiotic divisions
(3) Anaphase and telophase
(4) Leptotene and zygotene

BCD114

10. It is significant to note that in the 24 hour average duration of cell cycle of human cell, cell division proper lasts for only about:-

(1) Four hours (2) 90 minutes
(3) An hour (4) 10 hours

BCD115

11. In which stage of mitotic division, cells do not show Golgi complex, ER, nucleolus and nuclear envelope ?

(1) Metaphase (2) Late prophase
(3) Anaphase (4) All of these

BCD116

12. Condensation of chromosomes is completed in:-

(1) Prophase (2) Interphase
(3) Metaphase (4) Anaphase

BCD117

13. Regarding to cell cycle which of the following statements is wrong ?

(1) Cytoplasm increase is a continuous process
(2) DNA synthesis occurs only during one specific stage
(3) Replicated chromosomes distributed to daughter nuclei by complex series of events
(4) Events for replicated chromosomes distribution are not under genetic control

BCD118

14. About cell cycle, which of the following statements is correct ?

(1) In G_0 phase cells are metabolically inactive
(2) In G_0 phase cells are metabolically active
(3) Diploid somatic cells of animals divide only by meiotic division
(4) In plants only haploid cells can show mitotic divisions

BCD119

15. Chromatin condensation and movement of duplicated centrioles towards opposite poles can be observed during-

(1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase

BCD120

16. Regarding arrangement of chromosomes on equator during metaphase, which of the following statements is incorrect ?

(1) Each chromatid remains connected by one spindle fibres from both poles

(2) Each chromosome remains connected by spindle fibres from both poles

(3) Spindle fibres remain attached on kinetochores of both the chromatids

(4) Each chromatid remains connected at one pole by spindle fibres

BCD121

17. Match the following -

(a) Prophase	(I) Decondensation of chromosome
(b) Metaphase	(II) Division of centromere
(c) Anaphase	(III) Attachment of spindle fibres on kinetochores of chromosomes
(d) Telophase	(IV) Initiation of assembly of mitotic spindle

Options :-

(1) a–(IV) b–(III) c–(I) d–(II)

(2) a–(IV) b–(III) c–(II) d–(I)

(3) a–(III) b–(IV) c–(II) d–(I)

(4) a–(III) b–(IV) c–(I) d–(II)

BCD122

18. Which of the following is not a significance of mitosis ?

(1) Maintenance of identical genetic complement

(2) Cell repair

(3) Restore nucleo cytoplasmic ratio

(4) Genetic variability

BCD123

19. Meiosis ensures the production of ..... phase in life cycle of sexually reproducing organism, where as fertilisation restores ..... phase.

(1) diploid, haploid (2) haploid, triploid

(3) diploid, triploid (4) haploid, diploid

BCD124

20. Regarding key features of meiosis select out the wrong one -

(1) Meiosis involves two sequential cycles of nuclear and cell division called meiosis- I & meiosis- II

(2) Meiosis is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S - Phase

(3) Meiosis involves pairing of homologous chromosomes and recombination between non homologous chromosome

(4) Four haploid cells are formed at the end of meiosis-II

BCD125

21. Select the odd one -

(1) Zygotene	–	Synaptonemal complex appearance
(2) Pachytene	–	Appearance of recombination nodule
(3) Diplotene	–	Terminalisation of chiasmata
(4) Diakinesis	–	Assembly of meiotic spindle

BCD126

22. A cell has 46 chromosomes at each pole in mitotic telophase. In this division the number of chromatids at the metaphase was :-

(1) 23 (2) 46 (3) 92 (4) 69

BCD127

23. Match the following :-

(a)	Prophase-II	(I)	Enclosure of chromosomes in nuclear envelope
(b)	Metaphase-II	(II)	Separation of sister chromatids
(c)	Anaphase-II	(III)	Chromosome alignment on equator
(d)	Telophase-II	(IV)	Disappearance of nuclear membrane

Options :-

(1) a–(IV) b –(III) c–(I) d–(II)
(2) a–(IV) b–(III) c–(II) d–(I)
(3) a–(IV) b–(II) c–(III) d–(I)
(4) a–(IV) b–(I) c–(II) d–(III)

BCD128

24. Which of the following is not a significance of meiosis ?

(1) Helps in conservation of specific chromosome number in each species
(2) Increase in genetic variability
(3) Helps in evolution
(4) Helps in growth of organism

BCD129

25. Number of chromatids in each chromosome at anaphase is-

(1) One in mitosis, one in meiosis-I and two in meiosis-II
(2) One in mitosis, two in meiosis-I and one in meiosis-II
(3) Two in mitosis, one in meiosis-I and two in meiosis-II
(4) Two in mitosis, two in meiosis-I and two in meiosis-II

BCD130

26. During meiosis how many cycles of nucleus division, DNA replication and division of centromere take place respectively ?

(1) Three, Two, One (2) One, Two, Two
(3) Two, Two, One (4) Two, One, One

BCD131

27. Cell cycle includes :-

(1) Cell division (2) DNA replication
(3) Cell growth (4) All of the above

BCD132

28. Interphase of an animal cell is differ from interphase of a plant cell :-

(1) in having DNA replication
(2) in having centrioles duplication
(3) in having growth of cell
(4) in having metabolically active cell

BCD133

29. Multinucleated condition in the liquid endosperm of coconut arises because :-

(1) DNA replication not occurs during cell cycle
(2) Separation of sister chromatids not occurs in anaphase
(3) Condensation of chromosomal material not occurs in prophase
(4) Karyokinesis is not followed by cytokinesis

BCD134

30. Which of the following phases can last for months or years in oocytes of some vertebrates ?

(1) Leptotene (2) Zygotene
(3) Pachytene (4) Diplotene

BCD135

(B) ANALYTICAL QUESTIONS

31. Which of the following organelles can be observed in a cell even after completion of prophase?

(1) Golgi complex
(2) Mitochondria
(3) Nucleus
(4) Endoplasmic reticulum

BCD136

32. A mouse cell is treated with a chemical that interferes with the activity of actin protein. Which of the following will probably be affected the most?

(1) Formation of spindle fibres

(2) Division of cytoplasm

(3) Chromosome duplication

(4) Pairing of homologous chromosomes

BCD137

33. Number of chromosome pairs at equator in metaphase-I of a diploid plant cell (for which n = 25) shall be-

(1) 50 (2) 100 (3) 75 (4) 25

BCD138

34. In a diploid cell before S-phase if quantity of DNA is 20 pico gram (pg) then after meiosis I what will be the quantity of DNA in each daughter cell ?

(1) 10 pg (2) 20 pg (3) 5 pg (4) 40 pg

BCD139

35. Diploid cell in human, where cell division does not occur :-

(a) Heart cell (b) Muscle cell

(c) Nerve cell

Options :-

(1) Only c (2) b and c

(3) a and c (4) a, b and c

BCD140

36. Given below are two statement -

Statement-I : M-phase is the most dramatic phase of cell cycle, involving a major reorganisation of virtually all component of cell.

Statement-II : Cells in G_0-stage remain metabolically inactive but no longer proliferate unless called on to do so depending on the requirement of the organism.

In the light of the above statements, choose the correct answer from the options given below :-

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

BCD141

37. **Assertion (A) :** Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organism.

Reason (R) : Meiosis involve reduction in chromosome number to half as that of parent cell.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

BCD142

38. **Assertion (A) :** Cytokinesis in plant cell occur by cell plate formation method.

Reason (R) : Cytokinesis in animal cells occur by the formation of cell furrow.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

BCD143

39. **Assertion (A) :** No replication of DNA is needed between meiosis-I and meiosis-II.
Reason (R) : Chromosomes are already duplicated and they already have two sister chromatids.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is incorrect but (R) is correct.
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A).

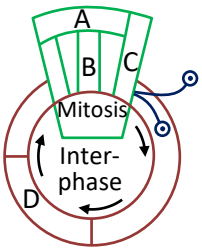
BCD144

40. What will be the ratio of number of chromosomes and amount of DNA in G_1, S and G_2 phases for a somatic cell?

	G_1	S	G_2
(1)	1 : 1	1 : 1	1 : 1
(2)	1 : 1	1 : 2	1 : 2
(3)	1 : 2	1 : 2	1 : 2
(4)	1 : 2	1 : 1	1 : 2

BCD145

41. Given below is a schematic break-up of the phases/stages of cell cycle :-



Which one of the following is the correct indication of the stage/phase in the cell cycle?

(1) A – Cytokinesis
(2) B – Metaphase
(3) C – Karyokinesis
(4) D – Synthetic phase

BCD057

42. During mitosis ER and nucleolus begin to disappear at :

(1) Early prophase (2) Late prophase
(3) Early metaphase (4) Late metaphase

BCD058

43. Which stages of cell division do the following figures A and B represent respectively ?

(A)  **(B)** 

(1) Prophase – Anaphase
(2) Metaphase – Telophase
(3) Telophase – Metaphase
(4) Late Anaphase – Prophase

BCD059

44. Select the correct option with respect to mitosis

(1) Chromatids separate but remain in the centre of the cell in anaphase.
(2) Chromatids start moving towards opposite poles in telophase.
(3) Golgi complex and endoplasmic reticulum are still visible at the end of prophase.
(4) Chromosomes move to the spindle equator and get aligned along equatorial plate in metaphase

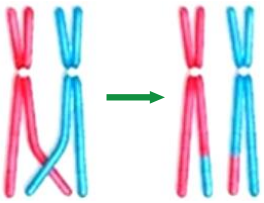
BCD060

45. At metaphase, chromosomes are attached to the spindle fibres by their :-

(1) Centromere
(2) Satellites
(3) Secondary constrictions
(4) Kinetochores

BCD061

46. Given below is the representation of a certain event at a particular stage of a type of cell division. Which is this stage ?



(1) Prophase of Mitosis

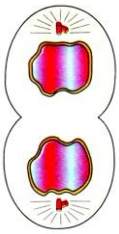
(2) Both prophase and metaphase of mitosis

(3) Prophase I during meiosis

(4) Prophase II during meiosis

BCD063

47. A stage in cell division is shown in the figure. Select the answer which gives correct identification of the stage with its characteristics.



(1)	Telophase	Endoplasmic reticulum and nucleolus not reformed yet.
(2)	Telophase	Nuclear envelop reforms, Golgi complex reforms.
(3)	Late anaphase	Chromosomes move away from equatorial plate, Golgi complex not present.
(4)	Cytokinesis	Cell plate formed, mitochondria distributed between two daughter cells.

BCD065

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	2	4	1	3	1	2	3	4	3	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	4	4	3	3	3	2	4	2	4	4	2	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	4	2	4	3	1	2	4	2	4	2	4	4	4
Question	46	47													
Answer	3	2													

Important Notes

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION TO DEVELOPMENT

1. Auxanometer is meant for measuring -
(1) Respiratory activity
(2) Photosynthetic activity
(3) Growth activity
(4) Osmotic pressure

BPD001

2. In plants growth is -
(1) Restricted to certain regions or structure
(2) Irreversible
(3) Change in size
(4) All the above

BPD002

3. Plant growth is primarily affected by two climatic factors which are :-
(1) Light and temperature
(2) Temperature and relative humidity
(3) Light and wind
(4) Rainfall and temperature

BPD003

4. In a growing plant, the first phase during the process of growth is -
(1) Cell division (2) Cell enlargement
(3) Cell differentiation (4) Cell maturation

BPD004

5. The maximum growth rate occurs in :-
(1) Exponential phase
(2) Stationary phase
(3) Senescent phase
(4) Lag phase

BPD005

PLANT GROWTH REGULATORS TO SEED DORMANCY

6. Primary precursor of I.A.A is :-
(1) Phenyl alanine (2) Tyrosine
(3) Tryptophan (4) Leucin

BPD006

7. The biological activity of I.A.A is tested by :-
(1) α-amylase test
(2) Avena curvature test
(3) Endosperm activity test
(4) Chlorophyll preservation test

BPD007

8. Indole, 3 acetic acid, called as auxin was first isolated from :-
(1) Human urine (2) Corn germ oil
(3) *Fusarium* (4) *Rhizopus*

BPD008

9. Stem elongation is affected by :-
(1) Gibberellin and florigen
(2) Auxin and gibberellin
(3) Florigen and kinin
(4) Kinin and auxin

BPD009

10. Apical dominance means :-
(1) Suppression of growth of apical bud by axillary buds
(2) Suppression of growth of axillary buds by the presence of apical bud.
(3) Stimulation of growth of axillary buds by removal of apical bud
(4) Inhibition of growth of axillary buds by removal of apical bud.

BPD010

11. Which of the following is not a natural occuring plant hormone ?
(1) 2, 4 - D (2) GA_2
(3) Gibberellin (4) I.A.A

BPD011

12. Auxin is mainly produced by -
(1) Apical root meristem
(2) Root cambium
(3) Apical shoot meristem
(4) Phloem in shoot tip

BPD012

13. Plants bend toward the light because -
(1) They need light for photosynthesis
(2) They need light for respiration
(3) Light attracts them
(4) Cells on the shaded side elongate more

BPD013

14. Growth hormone acts -
(1) Always as growth promoters
(2) Always as growth inhibitors
(3) Some as promoters and some as inhibitors
(4) Rarely as growth inducers

BPD014

15. The movement of auxin is -
(1) Acropetal (2) Basipetal
(3) Lateral (4) Centripetal

BPD015

16. Which growth hormone is responsible for apical dominance ?
(1) Auxin (2) Cytokinin
(3) Gibberellin (4) Ethylene

BPD016

17. Which of the following induces femaleness in plants ?
(1) Ethylene (2) Ethanol
(3) ABA (4) Gibberellin

BPD017

18. Agent orange is-
(1) Biodegradable insecticide
(2) Dioxin weedicide (2,4-D + 2,4,5-T)
(3) Biofertilizer
(4) Biopesticide

BPD018

19. Richmond - Lang effect is due to :-
(1) Cytokinin (2) Auxin
(3) ABA (4) Ethylene

BPD019

20. What causes a green plant exposed to the light on only one side, to bend toward the source of light as it grows?
(1) Green plants need light to perform photosynthesis.
(2) Green plants seek light because they are phototropic.
(3) Light stimulates plant cells on the lighted side to grow faster.
(4) Auxin accumulates on the shaded side, stimulating greater cell elongation there.

BPD170

21. Which of the following plant hormone substitutes for long photoperiod in flowering plant ?
(1) Auxin
(2) Gibberellin
(3) Cytokinin
(4) Ethylene

BPD021

22. Cytokinin -
(1) Is a hormone whose main function is to induce the cell division
(2) Induce bolting
(3) Induce senescence
(4) Causes dormancy

BPD022

23. Bolting hormone is -
(1) Auxin (2) Gibberellin
(3) ABA (4) Ethylene

BPD023

24. Gibberellins do not cause -
(1) Shortening of genetically tall plants
(2) Stimulation of seed germination
(3) Promotion of parthenocarpy
(4) Induction of α-amylase synthesis in barley

BPD024

25. Gibberellins can promote seed germination because of their influence on :
(1) Rate of cell division
(2) Production of hydrolysing enzymes
(3) Synthesis of abscisic acid
(4) Absorption of water through hard seed coat.
BPD025

26. Which of the following is a coconut milk factor ?
(1) Auxin (2) ABA
(3) Morphactin (4) Cytokinin
BPD026

27. In germinating seeds Amylase, Proteases, Lipases are stimulated by :-
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Ethylene
BPD027

28. Which one of the following is a gaseous plant hormone ?
(1) Auxin (2) Gibberellin
(3) Ethylene (4) Cytokinin
BPD028

29. Apical dominance can be overcome by application of :-
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Florigen
BPD029

30. Dormancy of seed is broken by :-
(1) Auxin (2) Gibberellins
(3) Ethylene (4) Cytokinin
BPD030

31. In tissue culture, differentiation of shoot is controlled by :-
(1) Light Intensity
(2) Temperature shock
(3) Low Auxin to high CK ratio
(4) High auxin to low CK ratio
BPD031

32. Among the following which helps in ripening of fruits?
(1) Methane (2) Ethylene
(3) CO_2 (4) CO
BPD032

33. Abscisic acid induces :-
(1) Shoot elongation
(2) Cell elongation and cell wall formation
(3) Cell division
(4) Leaf fall and dormancy
BPD033

34. Which of the following is growth inhibitor ?
(1) IAA (2) ABA
(3) NAA (4) GA_3
BPD034

35. Abscisic acid treatment results in -
(1) Leaf expansion
(2) Stem elongation
(3) Stomatal closure
(4) Root elongation
BPD035

36. Natural cell division inducing factor occurs in-
(1) Coconut milk
(2) Immature maize seeds
(3) Both (1) and (2)
(4) Heated t – RNA
BPD036

37. Which is a stress hormone ?
(1) Benzyl aminopurine (BAP)
(2) Dichlorophenoxy acetic acid
(3) Ethylene
(4) Abscisic acid
BPD037

38. Seed dormancy is due to the :-
(1) Ethylene (2) Abscisic acid
(3) IAA (4) Starch
BPD038

39. Hormone responsible to induce senescence :-
(1) ABA (2) Auxin
(3) GA (4) Cytokinin
BPD039

40. Decapitation i.e. removal of shoot tips in a plant usually results in :-
(1) Inactivation of lateral buds
(2) Growth of lateral buds
(3) Cessation of plant growth
(4) Yellowing of leaves
BPD171

41. Which hormone is used to initiate flowering and synchronizing fruit set in pineapple ?
(1) Ethylene (2) Kinetin
(3) GA_3 (4) ABA
BPD172

42. Cell elongation in internodal regions of the green plants takes place due to :-
(1) Ethylene (2) Indole acetic acid
(3) Cytokinin (4) Gibberellins
BPD045

43. The most widely used compound as sources of C_2H_4 is-
(1) Kinetin (2) Zeatin
(3) IBA (4) Ethephon
BPD152

44. Which of the following plants hormone show triple response on stem-
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Ethylene
BPD153

45. When a plant faces scarcity of water, which of the following hormone synthesized in mesophyll tissue-
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Abscisic acid
BPD154

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	1	1	3	2	1	2	2	1	3	4	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	2	1	4	2	1	2	1	2	4	2	3	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	4	2	3	3	4	2	1	2	1	4	4	4	4

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Dr. F. Went noted that if coleoptile tips were removed and placed on agar for one hour, the agar would produce a bending when placed on one side of freshly-cut coleoptile stumps. Of what significance is this experiment?

(1) It made possible the isolation and exact identification of auxin.

(2) It is the basis for quantitative determination of small amounts of growth-promoting substances.

(3) It supports the hypothesis that IAA is auxin.

(4) It demonstrated polar movement of auxins.

BPD063

2. A few normal seedlings of tomato were kept in a dark room. After a few days they were found to have become white-coloured like albinos. Which of the following terms will you use to describe them?

(1) Mutated

(2) Embolised

(3) Etiolated

(4) Defoliated

BPD064

3. Which one of the following growth regulators is known as stress hormone?

(1) Abscisic acid

(2) Ethylene

(3) GA_3

(4) Indole acetic acid

BPD065

AIPMT 2015

4. Typical growth curve in plants is: -

(1) Linear

(2) Stair-steps shaped

(3) Parabolic

(4) Sigmoid

BPD066

5. What causes a green plant exposed to the light on only one side, to bend toward the source of light as it grows?

(1) Green plants seek light because they are phototropic

(2) Light stimulates plant cells on the lighted side to grow faster

(3) Auxin accumulates on the shaded side, stimulating greater cell elongation there.

(4) Green plants need light to perform photosynthesis

BPD067

Re-AIPMT 2015

6. Auxin can be bio assayed by:

(1) Lettuce hypocotyl elongation

(2) Avena coleoptile curvature

(3) Hydroponics

(4) Potometer

BPD068

NEET-I 2016

7. The Avena curvature is used for bioassay of:

(1) ABA (2) GA_3

(3) IAA (4) Ethylene

BPD069

NEET-II 2016

8. You are given a tissue with its potential for differentiation in an artificial culture. Which of the following pairs of hormones would you add to the medium to secure shoots as well as roots?

(1) Auxin and abscisic acid

(2) Gibberellin and abscisic acid

(3) IAA and gibberellin

(4) Auxin and cytokinin

BPD070

9. Phytochrome is a:

(1) Lipoprotein (2) Chromoprotein

(3) Flavoprotein (4) Glycoprotein

BPD071

NEET-UG 2017

10. Fruit and leaf drop at early stages can be prevented by the application of:

(1) Ethylene (2) Auxins

(3) Gibberellic acid (4) Cytokinins

BPD072

NEET-UG 2019

11. What is the site of perception of photoperiod necessary for induction of flowering in plants?

(1) Lateral buds (2) Pulvinus

(3) Shoot apex (4) Leaves

BPD073

12. It takes very long time for pineapple plants to produce flowers. Which combination of hormones can be applied to artificially induce flowering in pineapple plants throughout the year to increase yield?

(1) Auxin and Ethylene

(2) Gibberellin and Cytokinin

(3) Gibberellin and Abscisic acid

(4) Cytokinin and Abscisic acid

BPD074

NEET-UG [Odisha] 2019

13. Removal of shoot tips is a very useful technique to boost the production of tea-leaves. This is because: -

(1) Gibberellins prevent bolting and are inactivated

(2) Auxins prevent leaf drop at early stages

(3) Effect of auxins is removed and growth of lateral buds is enhanced.

(4) Gibberellins delay senescence of leaves.

BPD075

14. In order to increase the yield of sugarcane crop, which of the following plant growth regulators should be sprayed?

(1) Ethylene (2) Auxins

(3) Gibberellins (4) Cytokinins

BPD076

NEET-UG 2020

15. The process of growth is maximum during :

(1) Dormancy (2) Log phase

(3) Lag phase (4) Senescence

BPD077

16. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop.

(1) Abscisic acid (2) Cytokinin

(3) Gibberellin (4) Ethylene

BPD078

NEET-UG [COVID-19] 2020

17. Inhibitory substances in dormant seeds cannot be removed by subjecting seeds to :

(1) Gibberellic acid

(2) Nitrate

(3) Ascorbic acid

(4) Chilling conditions

BPD079

18. Match the following concerning the activity/function and the phytohormone involved :-

(a) Fruit ripener	(i) Abscisic acid
(b) Herbicide	(ii) GA_3
(c) Bolting agent	(iii) 2, 4-D
(d) Stress hormone	(iv) Ethephon

Select the correct option from following:

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

BPD080

NEET-UG 2021

19. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called :

(1) Elasticity (2) Flexibility

(3) Plasticity (4) Maturity

BPD081

20. The site of perception of light in plants during photoperiodism is :

(1) Shoot apex (2) Stem

(3) Axillary bud (4) Leaf

BPD082

21. The plant hormone used to destroy weeds in a field is :

(1) IAA (2) NAA (3) 2,4-D (4) IBA

BPD083

NEET-UG [Paper-2] 2021

22. Which of the following is incorrectly matched pair?

(1) Ethylene – Cousins

(2) Kinetin – Skoog and Miller

(3) Gibberellic acid – F.W. Went

(4) Auxin – Darwin and Darwin **BPD084**

23. Match the columns I and II, and choose the correct combination from the options given.

Column-I	Column II
a. Zeatin	i. Tracheary element
b. Differentiation	ii. Secondary xylem
c. Auxin	iii. Cytokinin
d. Redifferentiation	iv. Indole

(1) a–iii, b–i, c–ii, d–iv

(2) a–iii, b–ii, c–i, d–iv

(3) a–iii, b–i, c–iv, d–ii

(4) a–iv, b–iii, c–ii, d–I

BPD085

NEET-UG 2022

24. Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants:

(1) Gibberellin (2) Ethylene

(3) Cytokinin (4) ABA

BPD086

25. Which one of the following plants does not show plasticity ?

(1) Coriander (2) Buttercup

(3) Maize (4) Cotton

BPD087

26. The gaseous plant growth regulator is used in plants to :

(1) promote root growth and root hair formation to increase the absorption surface

(2) help overcome apical dominance

(3) kill dicotyledonous weeds in the fields

(4) speed up the malting process

BPD088

Re-NEET-UG 2022

27. Which of the following growth regulators is an adenine derivative ?

(1) Auxin (2) Cytokinin

(3) Ethylene (4) Abscisic acid

BPD089

28. The phenomenon by which the undividing parenchyma cells start to divide mitotically during plant tissue culture is called as :

(1) Differentiation (2) Dedifferentiation

(3) Redifferentiation (4) Secondary growth

BPD090

29. The ability of plants to follow different pathways in response to environment leading to formation of different kinds of structures is called :

(1) Redifferentiation (2) Development

(3) Plasticity (4) Differentiation

BPD091

NEET-UG 2023

30. Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production ?

(1) Gibberellic Acid (2) Zeatin

(3) Abscisic Acid (4) Indole-3-butyric Acid

BPD147

31. Which hormone promotes internode/ petiole elongation in deep water rice?

(1) Kinetin (2) Ethylene

(3) 2, 4–D (4) GA_3

BPD148

32. In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as:

(1) Dedifferentiation (2) Development

(3) Senescence (4) Differentiation

BPD149

NEET-UG 2023 (Manipur)

33. The phenomenon which is influenced by auxin and also played a major role in its discovery :

(1) Phototropism (2) Root initiation

(3) Gravitropism (4) Apical Dominance

BPD150

34. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Auxin	(I)	Promotes female flower formation in cucumber
(B)	Gibberellin	(II)	Overcoming apical dominance
(C)	Cytokinin	(III)	Increase in the length of grape stalks
(D)	Ethylene	(IV)	Promotes flowering in pineapple

Choose the **correct** answer from the options given below :

(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(3) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

BPD151

NEET-UG 2024

35. Auxin is used by gardeners to prepare weed-free lawns. But no damage is caused to grass as auxin

(1) promotes apical dominance.

(2) promotes abscission of mature leaves only.

(3) does not affect mature monocotyledons plants.

(4) can help in cell division in grasses, to produce growth.

BPD165

36. Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield ?

(1) Auxin (2) Gibberellin

(3) Cytokinin (4) Abscisic acid

BPD166

RE-NEET-UG 2024

37. Match **List-I** with **List-II**.

List-I	List-II
A. Abscisic acid	I. Promotes female flowers in cucumber
B. Ethylene	II. Helps seeds to Withstand desiccation
C. Gibberellin	III. Helps in nutrient mobilisation
D. Cytokinin	IV. Promotes bolting in beet, cabbage etc.

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-IV, D-I

(2) A-III, B-II, C-I, D-IV

(3) A-II, B-I, C-IV, D-III

(4) A-II, B-I, C-III, D-IV

BPD167

38. F. Skoog observed that callus proliferated from the internodal segments of tobacco stem when auxin was supplied with one of the following except :-

(1) Extract of Vascular tissue

(2) Coconut milk

(3) Abscisic acid

(4) Yeast Extract

BPD168

39. Given below are some statements about plant growth regulators.

(A) All GAs are acidic in nature.

(B) Auxins are antagonists to GAs.

(C) Zeatin was isolated from coconut milk.

(D) Ethylene induces flowering in Mango.

(E) Abscisic acid induces parthenocarpy.

Choose the **correct** set of statements from the option given below :-

(1) A, C, D (2) B, E

(3) A, B, C (4) B, D, E

BPD169

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	1	4	3	2	3	4	2	2	4	1	3	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	3	4	3	3	3	2	3	1	2	2	3	1
Question	31	32	33	34	35	36	37	38	39						
Answer	2	1	1	2	3	2	3	3	1						

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which of the following is a characteristic feature of meristematic cells ?
(1) Cells are rich in protoplasm
(2) Cell walls are primary in nature
(3) Presence of large conspicuous nuclei
(4) All of the above

BPD092

2. Who isolated Auxin from tip of coleoptiles of oat seedlings?
(1) Chales Darwin (2) Kogl
(3) F.W. Went (4) Skoog

BPD093

3. 'Bakanae disease of rice seedlings' was caused by *Gibberella fujikuroi'* Which is -
(1) a Bacteria (2) an Algae
(3) a Protista (4) a Fungus

BPD094

4. Hormone, which is sprayed on young conifers to hasten the maturity period and early seed production, is -
(1) Ethylene (2) ABA
(3) Gibberellin (4) Cytokinin

BPD095

5. Ethylene is used to initiate flowering and for synchronising fruit set in -
(1) Rice (2) Potato
(3) Pineapple (4) Cherry

BPD096

6. Growth is measured by a variety of parameters, among following which is not included in these?
(1) Fresh weight (2) Dry weight
(3) Surface area (4) Cell type

BPD097

7. Spraying sugarcane crop with gibberellins increases the length of the stem, thus increasing the yield by as much as :-
(1) 30 tonnes per acre
(2) 40 tonnes per acre
(3) 20 tonnes per acre
(4) 50 tonnes per acre

BPD098

8. Among the following which is not present naturally in plants?
(1) Auxin (2) Gibberellin
(3) Kinetin (4) Zeatin

BPD099

9. Among the following which is not a function of ethylene?
(1) Breaks bud and seed dormancy
(2) Promotes root growth
(3) Delay senescence
(4) Initiate flowering

BPD100

10. Plants follow different pathways in response to environment to form different kinds of structures, that is called :-
(1) Plasticity (2) Senescence
(3) Photoperiodism (4) Vernalisation

BPD101

11. Which causes fruits like apple to elongate & improve its shape ?
(1) Ethylene (2) Gibberellins
(3) Auxin (4) ABA

BPD102

12. 2, 4-D is a :-
(1) Synthetic auxin
(2) Natural auxin
(3) Synthetic cytokinin
(4) Natural cytokinin

BPD103

13. Which of the following increases the tolerance of plants to various kind of stresses?

(1) ABA (2) Ethylene

(3) Auxin (4) Cytokinin

BPD104

14. Which of following is used to speed up the malting process in brewing industry ?

(1) Auxin (2) GA_3

(3) Kinetin (4) ABA

BPD105

15. Cells/tissues arising out of the same meristem have different structures at maturity, this phenomenon in plants is called:

(1) Open growth

(2) Open differentiation

(3) Indeterminate growth

(4) Determinate differentiation

BPD106

16. The gaseous plant growth regulator ethylene could fit in the group of :-

(1) Plant growth inhibitors

(2) Plant growth promoters

(3) Both (1) and (2)

(4) Neither inhibitors nor promoters

BPD107

17. Which of the following physiological effects, is common to both Auxin and Ethylene?

(1) Sprouting of potato tuber

(2) Petiole elongation in deep water rice plants

(3) Flowering in pineapple

(4) Weed killing

BPD108

18. Which of the following is not correctly matched?

(1) Delay leaf senescence $\Rightarrow$ Cytokinin

(2) Induce rooting in a twig $\Rightarrow$ Auxin

(3) Induce growth in axillary bud $\Rightarrow$ Auxin

(4) Quickly ripen a fruit $\Rightarrow$ Ethylene

BPD109

19. Regarding factors affecting the development, PGR is the :-

(1) Extrinsic control

(2) Intrinsic intra cellular control

(3) Neither intrinsic nor extrinsic control

(4) Intrinsic inter cellular control

BPD110

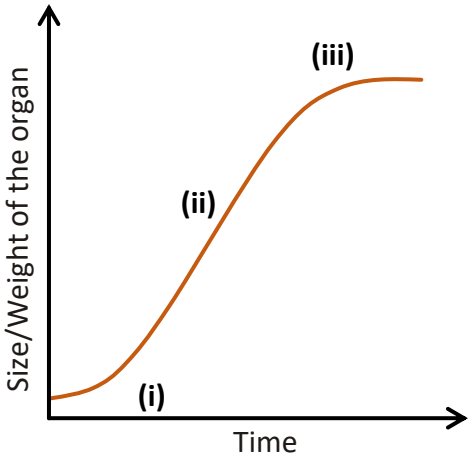
20. Arithmetic growth is related to :-

(1) $L_t = L_0e^{rt}$ (2) $W_1 = W_0e^{rt}$

(3) $L_t = L_0 + rt$ (4) $L_0 = L_te^{rt}$

BPD111

21. Go through the following figure :-



Choose the correct labelling.

	(i)	(ii)	(iii)
(1)	Log phase	Exponential phase	Stationary phase
(2)	Lag phase	Exponential phase	Stationary phase

(3)	Exponential phase	Log phase	Stationary phase
(4)	Stationary phase	Exponential phase	Log phase

BPD173

22. Which statement is not correct about ethylene ?

(1) Synthesized in large amount by tissues undergoing senescence.

(2) Enhances respiration rate during ripening of the fruits

(3) Breaks dormancy in peanut seeds

(4) Promotes male flowers in cucumber

BPD113

23. Which of the following plant growth regulators is responsible for 'respiratory climactic'?

(1) Auxin (2) Ethylene

(3) ABA (4) Cytokinin

BPD114

24. (i) Promote nutrient mobilisation

(ii) Overcome the apical dominance

(iii) Lateral shoot growth

All the above physiological effects are related to which of the following plant growth regulators?

(1) Auxins (2) Gibberellins

(3) Abscisic acid (4) Cytokinins

BPD115

25. Which of the following statements are correct regarding growth?

(A) In plants, the form of growth is open and localised

(B) Swelling of piece of wood in water is considered as growth since it involve the increase in size

(C) Growth is accompanied by metabolic processes

(D) Growth, at a cellular level, is a result of increase in the amount of protoplasm

(1) A, B, C and D (2) A and B

(3) B, C and D (4) A, C and D

BPD116

26. Meristematic phase of growth is characterised by :-

(1) Increased vacuolation

(2) Maximal size in terms of protoplasmic modifications

(3) Cells those are rich in protoplasm and having thin cell walls with abundant plasmodesmata

(4) Cell enlargement

BPD117

27. Which of the following is/are related to the type of growth in which both the progeny cells, arise from mother cell, retain the ability to divide?

(A) Sigmoid curve

(B) expressed as $W_1 = W_0 + rt$

(C) Linear curve

(D) Three phases - Lag, exponential and stationary

(1) A and D (2) A, B and D

(3) Only C (4) B and C

BPD118

28. Match the following :

(A) Auxin	(i) Derivatives of carotenoids
(B) Gibberellin	(ii) Gas
(C) Cytokinin	(iii) Adenine derivatives
(D) Ethylene	(iv) Terpenes
(E) Abscisic acid	(v) Indole compounds

(1) A - i, B - ii, C - iii, D - iv, E - v
(2) A - ii, B - i, C - v, D - iii, E - iv
(3) A - v, B - iv, C - iii, D - ii, E - i
(4) A - iv, B - iii, C - i, D - ii, E - v

BPD119

29. Math the following :

(A) Human urine	(i) Ethylene
(B) Coconut milk	(ii) GA_3
(C) Ripened oranges	(iii) Auxin
(D) *Gibberella fujikuroi*	(iv) Cytokinin

(1) A - iii, B - iv, C - i, D - ii
(2) A - i, B - ii, C - iii, D - iv
(3) A - iv, B - iii, C - ii, D - i
(4) A - ii, B - i, C - iv, D - iii

BPD120

30. Match the following :

(A) Cytokinin	(i) Weed free lawns
(B) Auxin	(ii) Brewing industry
(C) Abscisic acid	(iii) Root hair formation
(D) Ethylene	(iv) Overcome apical dominance
(E) Gibberellin	(v) Stress hormone

(1) A- iv, B- i, C- v, D- ii, E- iii
(2) A- i, B- ii, C- v, D- iii, E- iv
(3) A- iii, B- iv, C- v, D- i, E- ii
(4) A- iv, B- i, C- v, D- iii, E- ii

BPD121

31. Which of the following occur naturally in plants?
(1) 2, 4-dichlorophenoxyacetic acid
(2) Indole butyric acid
(3) Naphthalene acetic acid
(4) Kinetin

BPD122

32. Consider the following statements.
(I) The basis to the phenomenon of growth is formation of new protoplasm.
(II) Growth is maximum is zone of cell elongation.
Choose the correct option.
(1) Statement I is true, but II is false.
(2) Statement I is false, but II is true.
(3) Statement I and II are true.
(4) Statement I and II are false.

BPD174

33. Swelling of piece of wood when placed in water is not considered as growth because:
(1) It does not occur at expense of energy
(2) It is not a metabolic change
(3) It is reversible process
(4) All of the above

BPD125

34. Plant growth is unique because :
(1) It is intrinsic
(2) It occurs at the expense of energy
(3) Plant retains the capacity for unlimited growth throughout their life
(4) Its accompanied by metabolic processes

BPD126

35. Growth at a cellular level, is principally a consequence of increase in amount of :
(1) Cell wall material (2) Water
(3) Protoplasm (4) Cell sap

BPD127

36. Cells with increased vacuolation, cell enlargement and new cell wall deposition are the characteristic features of which phase of growth ?
(1) Meristematic phase
(2) Elongation phase
(3) Maturation phase
(4) Differentiation phase

BPD128

37. Trees showing seasonal activities, represent what kind of growth curve ?
(1) Sigmoid (2) Linear
(3) J-shaped (4) Hyperbola
BPD129

38. Quantitative comparisons between the growth of living systems can be made by :
(1) Absolute growth rate
(2) Relative growth rate
(3) Exponential growth rate
(4) Both (1) and (2)
BPD130

39. Nutrients both macro and micro essential elements are required by plant during growth for:
(1) Synthesis of protoplasm
(2) As source of energy
(3) Enzyme of activation
(4) All of the above
BPD131

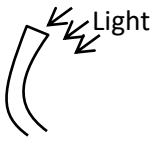
40. Parenchyma cells that are made to divide under controlled laboratory conditions during plant tissue culture, represents :
(1) Differentiation
(2) Dedifferentiation
(3) Redifferentiation
(4) Undifferentiated mass of cells
BPD132

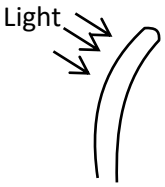
41. Select out the incorrect match :
(1) GA - speed up malting process
(2) Auxin - Xylem differentiation
(3) Cytokinin - Adventitious shoot formation
(4) Ethylene - Lateral shoot growth
BPD133

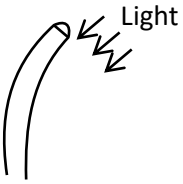
42. Removal of shoot tip (decapitation) is useful in:
(1) Promotion of Apical dominance
(2) Bolting
(3) Tea plantation (4) Senescence
BPD134

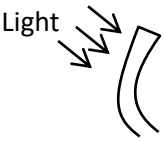
43. Which of the following phenomenon in plants requires interaction of more than one plant growth regulator :-
(1) Dormancy in seeds/buds
(2) Apical dominance
(3) Senescence
(4) All of the above
BPD135

44. Which of diagram is **correct** about the curvature of coleoptile ?

(1) 

(2) 

(3) 

(4) 

BPD136

45. Which of the following is/are correct about 'kinetin' ?
(1) Kinetin occurs naturally in plants
(2) It is a modified form of a purine
(3) Isolated from herring egg DNA
(4) All of the above
BPD137

46. Which of the following plant growth regulators (PGRs) belongs to that group of PGRs which play an important role in plant response to wounds?
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Abscisic acid
BPD138

47. Which of the following plant growth regulators is synthesised in large amounts by tissues undergoing senescence and ripening fruits ?
(1) Gibberellin (2) Zeatin
(3) IBA (4) Ethylene
BPD139

(B) ANALYTICAL QUESTIONS

48. Select out the incorrect match :-
(1) Water → Cell enlargement
(2) Oxygen → Enzyme activation
(3) Minerals → Osmotic regulation
(4) Light → Flowering

BPD140

49. Which of the following activities does not involve the role of GA?
(1) Seed germination
(2) Malting
(3) Root growth
(4) Flowering in LDPs

BPD141

50. Removal of shoot tips (decapitation) usually results in the growth of lateral buds. It is related to the removal of effect of which plant hormone?
(1) Ethylene (2) Auxin
(3) Cytokinin (4) Gibberellic acid

BPD142

51. Which plant hormone promotes the growth of lateral buds ?
(1) Auxin (2) Cytokinin
(3) Gibberellin (4) Abscisic acid

BPD143

52. Hormone involved in induction of α-amylase in Barley endosperm :-
(1) GA (2) NAA
(3) Ethylene (4) ABA

BPD146

53. How does pruning help in making the hedge dense
(1) The apical shoot grows faster after pruning
(2) It releases wound hormones
(3) It induces the differentiation of new shoots from the root stock
(4) It frees axillary buds from apical dominance

BPD046

54. Parthenocarpic tomato fruits can be produced by?
(1) raising the plants from vernalized seeds
(2) treating the plants with phenylmercuric acetate
(3) removing androecium of flowers before pollen grains are released
(4) treating the plants with low concentrations of gibberellic acid and auxins

BPD047

55. An enzyme that can stimulate germination of barley seeds is
(1) Protease (2) Invertase
(3) α-amylase (4) Lipase

BPD048

56. Which one of the following pairs, is not correctly matched?
(1) IAA - Cell wall elongation
(2) Abscisic acid - Stomatal closure
(3) Gibberellic acid - Leaf fall
(4) Cytokinin - Cell division

BPD049

57. "Foolish seeding" disease of rice led to the discovery of: -
(1) IAA (2) GA
(3) ABA (4) 2, 4-D

BPD051

58. Senescence as an active developmental cellular process in the growth and functioning of a flowering plant, is indicated in: -
(1) Annual plants
(2) Floral parts
(3) Vessels and tracheid differentiation
(4) Leaf abscission

BPD052

59. One of the synthetic auxin is: -

(1) IAA (2) GA (3) IBA (4) NAA

BPD054

60. Which one of the following acids is a derivative of carotenoids?

(1) Indole-3-acetic acid

(2) Gibberellic acid

(3) Abscisic acid

(4) Indole butyric acid

BPD055

61. The annular and spirally thickened conducting elements generally develop in the protoxylem when the root or stem is: -

(1) Differentiating (2) Maturing

(3) Elongating (4) Widening

BPD056

62. Phototropic curvature is the result of uneven distribution of:

(1) Gibberellin (2) Phytochrome

(3) Cytokinins (4) Auxin

BPD057

63. Root development is promoted by:

(1) Auxin (2) Gibberellin

(3) Ethylene (4) Abscisic acid

BPD059

64. Which one of the following generally acts as an antagonist to gibberellins?

(1) ABA (2) IAA

(3) Zeatin (4) Ethylene

BPD060

65. Through their effect on plant growth regulators, what do the temperature and light control in the plants?

(1) Closure of stomata (2) Fruit elongation

(3) Apical dominance (4) Flowering

BPD062

66. **Assertion :** Gibberellins promote bolting in beet, cabbages and many plants with rosette habit.

Reason : It promotes internode elongation just prior to flowering.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BPD155

67. **Assertion :** Plants retain the capacity for unlimited growth throughout their life.

Reason : Plants have meristems in their body.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BPD156

68. **Assertion:** Cytokinin helps in the delay of leaf senescence.

Reason: Cytokinin promote nutrient mobilisation.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BPD157

69. Match the following :

(A)	Auxin	(i)	Derivatives of carotenoids
(B)	Gibberellin	(ii)	Gas
(C)	Cytokinin	(iii)	Adenine derivatives
(D)	Ethylene	(iv)	Terpenes
(E)	Abscisic acid	(v)	Indole compounds

(1) A-i, B-ii, C-iii, D-iv, E-v

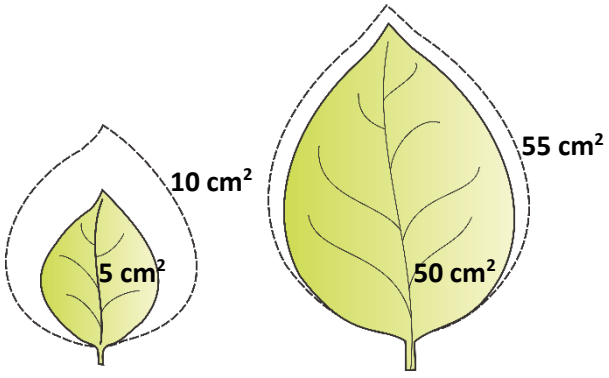
(2) A-ii, B-i, C-v, D-iii, E-iv

(3) A-v, B-iv, C-iii, D-ii, E-i

(4) A-iv, B-iii, C-i, D-ii, E-v

BPD158

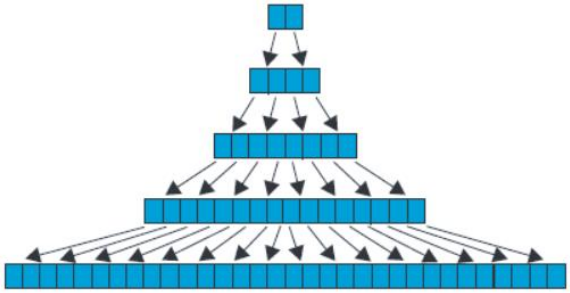
70. The given figures shows the growth of two leaves in a given time. If, AGR = absolute growth and RGR = relative growth rate, then select the correct option :-



	AGR for leaf A	RGR for leaf A	AGR for leaf B	RGR for leaf B
(1)	5	5%	5	5%
(2)	10	10%	55	10%
(3)	5	100 %	5	10 %
(4)	5	100 %	5	100 %

BPD159

71. What type of growth is represented in the following diagram ?



(1) Arithmetic

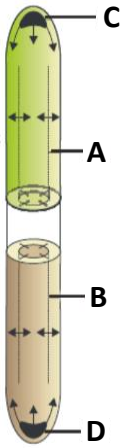
(2) Geometric

(3) Root elongation at a constant rate

(4) Both (1) & (2)

BPD160

72. Choose the correct option for A,B,C or D :-



(1) A - Primary growth

(2) B - Secondary growth

(3) C - Lateral growth

(4) D - Terminal

BPD161

73. One single maize root apical meristem can give rise to more than

(1) 17,500 new cells per minute

(2) 17,500 new cells per hour

(3) 17,500 new cells per day

(4) 17,500 new cells per week

BPD162

74. Match the columns :

	Column-I		Column-II
(1)	Human urine	(a)	Cytokinin
(2)	*Gibberella fujikuroi*	(b)	Auxin
(3)	Herring fish DNA	(c)	Ethylene
(4)	Ripening fruits	(d)	Abscisic acid
(5)	Aged leaves of plants	(e)	Gibberellins

(1) 1-e, 2-d, 3-c, 4-b, 5-a

(2) 1-b, 2-c, 3-d, 4-e, 5-a

(3) 1-a, 2-e, 3-b, 4-d, 5-c

(4) 1-b, 2-e, 3-a, 4-c, 5-d

BPD163

75. **Assertion:** Development is a term that includes all changes that an organism goes through during its life cycle.

Reason: Plants follows different pathways in response to environment or phases of life to form different kinds of structures.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

BPD164

EXERCISE-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	4	3	3	4	3	3	3	1	2	1	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	4	3	2	4	2	4	4	3	1	3	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	4	3	3	2	1	4	4	2	4	3	4	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	2	3	2	2	1	4	4	3	3	2	4	4	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	4	1	1	4	1	1	1	3	3	2	2	2	4	2

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION TO AEROBIC RESPIRATION (KREBS CYCLE)

1. Energy obtained by a cell from catabolic reaction is stored immediately in the form of
(1) Pyruvic acid (2) Glucose
(3) ATP (4) DNA
BRP001

2. Number of ATP produced from one pyruvic acid during conversion to acetyl CoA –
(1) 6 (2) 3
(3) 12 (4) 15
BRP002

3. The formation of Acetyl Co-A from pyruvic acid is the result of its :-
(1) Reduction
(2) Dehydration
(3) Phosphorylation
(4) Oxidative decarboxylation
BRP003

4. Pyruvate dehydrogenase complex is used in converting -
(1) Pyruvate to glucose
(2) Glucose to pyruvate
(3) Pyruvic acid to lactic acid
(4) Pyruvate to acetyl Co-A
BRP004

5. The first member of TCA cycle is :-
(1) Oxalosuccinic acid
(2) Oxaloacetic acid
(3) Citric acid
(4) Cis aconitic acid
BRP005

6. Excess of ATP inhibits the enzyme :-
(1) Phosphofructokinase
(2) Hexokinase
(3) Aldolase (Lyases)
(4) Pyruvate decarboxylase
BRP006

7. End product of glycolysis is :-
(1) Citric acid
(2) Glyceraldehyde
(3) Phosphoglyceraldehyde
(4) Pyruvic acid
BRP007

8. Respiration in plants :-
(1) Occurs only during day
(2) Results in the formation of vitamins
(3) Occurs both during day and night
(4) Often requires CO_2
BRP008

9. The end products of respiration in plants are:
(1) CO_2, H_2O and energy
(2) Starch and O_2
(3) Sugar and O_2
(4) H_2O and energy
BRP009

10. Common immediate source of energy in cellular activity is :-
(1) Glucose (2) Aldohexose
(3) ATP (4) NAD
BRP010

11. The net gain of ATP molecules by glycolysis in aerobic condition is
(1) Zero (2) Two (3) Four (4) Eight
BRP011

12. Respiratory enzymes are localised in :-
(1) Ribosomes (2) Chloroplast
(3) Mitochondria (4) Chromoplast
BRP012

13. The organism in which Kreb's cycle does not occur in mitochondria is :-
(1) Yeast (2) *E.coli*
(3) *Ulothrix* (4) Plants
BRP013

14. The first preferred respiratory substrate is :-
(1) Glucose (2) Fats
(3) Proteins (4) Polypeptides

BRP014

15. Respiration occurs in :-
(1) All living cells both in day and night
(2) Non green cells only in day
(3) Non green cells in both day and night
(4) All living cells in day only

BRP015

16. Respiration may take place -
(1) In the presence of O_2
(2) In the absence of O_2
(3) In the presence or absence of O_2
(4) Only in the presence of CO_2

BRP016

17. The common phase between aerobic & anaerobic respiration is :-
(1) TCA cycle
(2) Kreb's cycle
(3) Glycolysis
(4) Photo respiration

BRP017

18. The enzyme which converts glucose to glucose-6-phosphate -
(1) Phosphorylase
(2) Glucophosphorylase
(3) Hexokinase
(4) Phosphoglucomutase

BRP018

19. Product formed by the activity of malic dehydrogenase is :-
(1) Fumaric acid (2) Malic acid
(3) Oxaloacetic acid (4) Succinic acid

BRP019

20. Which of the following is 5-carbon compound of Krebs cycle ?
(1) Citric acid
(2) Fumaric acid
(3) Oxalosuccinic acid
(4) α-Ketoglutaric acid

BRP020

21. SLP net gain of ATP from one molecule of glucose during glycolysis or EMP pathway -
(1) 2 ATP (2) 6 ATP
(3) 36 ATP (4) 38 ATP

BRP021

22. Which enzyme breaks the fructose-1, 6-Bisphosphate ?
(1) Hexokinase
(2) Phosphatase
(3) Aldolase
(4) Phosphofructokinase

BRP022

23. Acceptor of acetyl Co-A in Krebs-cycle is :-
(1) Malic acid
(2) Fumaric acid
(3) α-ketoglutaric acid
(4) Oxaloacetic acid

BRP023

24. In which one of the following do the two names refer to one and the same thing ?
(1) Krebs cycle and Calvin cycle
(2) Tricarboxylic acid cycle and citric acid cycle
(3) Citric acid cycle and Calvin cycle
(4) Tricarboxylic acid cycle and urea cycle

BRP024

AEROBIC RESPIRATION (ETS) TO REPIRATORY QUOTIENT

25. Which component of ETS is mobile carrier ?
(1) UQ (CO-Q) (2) Cyt - a
(3) Cyto - b (4) Cyt - f
BRP025

26. R.Q. is less than one at the time of respiration of :-
(1) Starch (2) Organic acid
(3) Glucose (4) Protein
BRP026

27. Conversion of pyruvic acid into ethyl alcohol is mediated by –
(1) Phosphatase
(2) Dehydrogenase
(3) Decarboxylase & dehydrogenase
(4) Catalase
BRP027

28. Which of the following is link between carbohydrate and fat metabolism ?
(1) CO_2 (2) Acetyl Co-A
(3) Pyruvic acid (4) Citric acid
BRP028

29. In the electron transport chain during terminal oxidation, the cytochrome, which donates electrons to O_2 is ?
(1) Cytochrome-b (2) Cytochrome-C
(3) Cytochrome-a_3 (4) Cytochrome-f
BRP029

30. The incomplete breakdown of sugars in anaerobic respiration results in the formation of :-
(1) Fructose and water (2) Glucose and CO_2
(3) Alcohol and CO_2 (4) Water and CO_2
BRP030

31. Cytochromes are concerned with :-
(1) Protein synthesis (2) Cellular digestion
(3) Cell division (4) Cell-respiration
BRP031

32. In respiration pyruvic acid is :-
(1) Formed only when oxygen is available
(2) One of the products of Krebs cycle
(3) Broken down into Acetyl Co-A and CO_2
(4) Oxidised into Alcohol
BRP032

33. Number of ATP molecules formed during aerobic respiration in break down of one glucose molecule via malate aspartate shuttle, is :-
(1) 38 (2) 18 (3) 28 (4) 4
BRP033

34. What causes R.Q. to vary ?
(1) Respiratory Substrate
(2) Light & O_2
(3) Respiratory Product
(4) Temperature
BRP034

35. When the evolution of CO_2 is more than the intake of O_2, the respired substrate should be:
(1) Fatty acid (2) organic acid
(3) Glucose (4) Polysaccharides
BRP035

36. **Assertion :** Breaking down processes within the living organism is anabolism and synthesis is catabolism.
Reason : The proteins would be degraded by proteases and the individual amino acids (after deamination) depending on their structure would enter the pathway at some stage within the Krebs cycle or even as pyruvate or acetyl CoA.
(1) A and R both are correct.
(2) A and R both are incorrect.
(3) A is correct, R is incorrect.
(4) A is incorrect, R is correct.
BRP130

37. How many molecules of ATP are produced per molecule of $FADH_2$ oxidised ?
(1) One (2) Two (3) Three (4) four
BRP037

38. How many ATP molecules produced from the complete oxidation of a molecule of acetyl Co-A ?
(1) 38 ATP (2) 15 ATP
(3) 12 ATP (4) 4 ATP
BRP038

39. How many ATP equivalents are produced by the oxidation of succinate into fumarate?
(1) 1 ATP (2) 2 ATP
(3) 3 ATP (4) 4 ATP
BRP039

40. How many net ATP generates in aerobic respiration via glycerol phosphate shuttle in eukaryotes ?
(1) 38 ATP (2) 36 ATP
(3) 40 ATP (4) 80 ATP
BRP040

41. How many ATP are produced from one molecule of Acetyl CoA ?
(1) 12 ATP (2) 10 ATP
(3) 5 ATP (4) 14 ATP
BRP131

42. Anaerobic respiration takes place in :-
(1) Ribosome (2) Mitochondria
(3) Cytoplasm (4) Vacuole
BRP042

43. During the formation of bread, it becomes porous due to release of CO_2 by the action of:-
(1) Yeast (2) Bacteria
(3) Virus (4) Protozoans
BRP043

44. Chemiosmotic theory of ATP synthesis in the chloroplast and mitochondria is based on :-
(1) Proton gradient
(2) Accumulation of K ions
(3) Accumulation of Na ions
(4) Accumulation of Cl ions
BRP044

45. Curd formation is the result of :-
(1) Aerobic respiration
(2) Glycolysis only
(3) Lactic acid fermentation
(4) Alcoholic fermentation
BRP045

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	4	2	1	4	3	1	3	4	3	2	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	3	4	1	3	4	2	1	4	3	2	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	1	1	2	4	2	3	2	2	1	3	1	1	3

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. In which one of the following processes CO_2 is **not released**?

(1) Aerobic respiration in plants

(2) Aerobic respiration in animals

(3) Alcoholic fermentation

(4) Lactate fermentation

BRP058

NEET-II 2016

2. Which of the following biomolecules is common to respiration-mediated breakdown of fats, carbohydrates and proteins?

(1) Pyruvic acid

(2) Acetyl CoA

(3) Glucose-6-phosphate

(4) Fructose 1,6-bisphosphate

BRP059

NEET-UG 2017

3. Which statement is wrong for Krebs' cycle?

(1) There is one point in the cycle where FAD^+ is reduced to $FADH_2$

(2) During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised

(3) The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid

(4) There are three points in the cycle where NAD^+ is reduced to $NADH + H^+$

BRP060

NEET-UG 2018

4. What is the role of NAD^+ in cellular respiration?

(1) It functions as an enzyme

(2) It functions as an electron carrier

(3) It is a nucleotide source for ATP synthesis

(4) It is the final electron acceptor for anaerobic respiration

BRP061

5. Which of these statements is ***incorrect***?

(1) Enzymes of TCA cycle are present in mitochondrial matrix.

(2) Glycolysis occurs in cytosol.

(3) Glycolysis operates as long as it is supplied with NAD that can pick up hydrogen atoms.

(4) Oxidative phosphorylation takes place in outer mitochondrial membrane.

BRP062

NEET-UG 2019

6. Respiratory Quotient (RQ) value of tripalmitin is:

(1) 0.9 (2) 0.7

(3) 0.07 (4) 0.09

BRP063

7. Conversion of glucose to glucose-6-phosphate, the first irreversible reaction of glycolysis, is catalyzed by:

(1) Aldolase

(2) Hexokinase

(3) Enolase

(4) Phosphofructokinase

BRP064

NEET-UG 2019 (Odisha)

8. Where is respiratory electron transport system (ETS) located in plants?

(1) Mitochondrial matrix

(2) Outer mitochondrial membrane

(3) Inner mitochondrial membrane

(4) Intermembrane space

BRP065

NEET-UG 2020

9. The number of substrate level phosphorylations in one turn of citric acid cycle is :

(1) Three (2) Zero

(3) One (4) Two

BRP066

NEET-UG [COVID-19] 2020

10. Pyruvate dehydrogenase activity during aerobic respiration requires :-

(1) Calcium (2) Iron

(3) Cobalt (4) Magnesium

BRP067

NEET-UG 2021

11. Which of the following statements is incorrect?

(1) During aerobic respiration, role of oxygen is limited to the terminal stage.

(2) In ETC (Electron Transport Chain), one molecule of NADH + H^+ gives rise to 2 ATP molecules, and one $FADH_2$ gives rise to 3 ATP molecules.

(3) ATP is synthesized through complex V.

(4) Oxidation-reduction reactions produce proton gradient in respiration.

BRP068

NEET-UG 2022

12. What is the net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid ?

(1) Six (2) Two (3) Eight (4) Four

BRP069

13. What amount of energy is released from glucose during lactic acid fermentation?

(1) More than 18%

(2) About 10%

(3) Less than 7%

(4) Approximately 15%

BRP070

Re-NEET-UG 2022

14. The number of time(s) decarboxylation of isocitrate occurs during single TCA cycle is :

(1) One (2) Two (3) Three (4) Four

BRP071

NEET-UG 2023

15. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R:**

Assertion (A) : ATP is used at two steps in glycolysis.

Reason (R) : First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1-6-diphosphate.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

(2) **A** is true but **R** is false.

(3) **A** is false but **R** is true.

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

BRP122

16. Match **List I** with **List II**.

	List-I		List-II
A.	Oxidative decarboxylation	I.	Citrate synthase
B.	Glycolysis	II.	Pyruvate dehydrogenase
C.	Oxidative phosphorylation	III.	Electron transport system
D.	Tricarboxylic acid cycle	IV.	EMP pathway

Choose the correct answer from the options given below :-

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-I, C-II, D-IV

(3) A-II, B-IV, C-III, D-I

(4) A-III, B-IV, C-II, D-I

BRP123

NEET-UG 2023 (Manipur)

17. How many times decarboxylation occurs during each TCA cycle ?

(1) Thrice (2) Many

(3) Once (4) Twice

BRP124

18. Fatty acids are connected with the respiratory pathway through :-

(1) Acetyl CoA

(2) α-Ketoglutaric acid

(3) Dihydroxy acetone phosphate

(4) Pyruvic acid

BRP125

NEET-UG 2024

19. Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.

(1) Malic acid $\rightarrow$ Oxaloacetic acid

(2) Succinic acid $\rightarrow$ Malic acid

(3) Succinyl-CoA $\rightarrow$ Succinic acid

(4) Isocitrate $\rightarrow$ α-ketoglutaric acid

BRP126

20. Match **List I** with **List II**.

	List-I		List-II
A.	Citric acid cycle	I.	Cytoplasm
B.	Glycolysis	II.	Mitochondrial matrix
C.	Electron transport system	III.	Intermembrane space of mitochondria
D.	Proton gradient	IV.	Inner mitochondrial membrane

Choose the **correct** answer from the options given below :-

(1) A–I, B–II, C–III, D–IV

(2) A–II, B–I, C–IV, D–III

(3) A–III, B–IV, C–I, D–II

(4) A–IV, B–III, C–II, D–I

BRP127

RE-NEET-UG 2024

21. Match **List-I** with **List-II**.

	List-I		List-II
A.	ETS Complex I	I.	NADH Dehydrogenase
B.	ETS Complex II	II.	Cytochrome bC_1
C.	ETS Complex III	III.	Cytochrome C oxidase
D.	ETS Complex IV	IV.	Succinate Dehydrogenase

Choose the **correct** answer from the options given below :-

(1) A-IV, B-I, C-III, D-II

(2) A-I, B-IV, C-II, D-III

(3) A-III, B-I, C-IV, D-II

(4) A-I, B-II, C-IV, D-III

BRP128

22. Which of the following are **correct** about cellular respiration ?

(A) Cellular respiration is the breaking of C-C bonds of complex organic molecules by oxidation.

(B) The entire cellular respiration takes place in Mitochondria.

(C) Fermentation takes place under anaerobic condition in germinating seeds.

(D) The fate of pyruvate formed during glycolysis depends on the type of organism also.

(E) Water is formed during respiration as a result of O_2 accepting electrons and getting reduced

Choose the **correct** answer from the options given below :-

(1) A, C, D, E only (2) A, B, E only

(3) A, B, C, E only (4) B, C, D, E only

BRP129

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	2	4	2	2	3	3	4	2	2/3	3	2	4
Question	16	17	18	19	20	21	22								
Answer	3	4	1	3	2	2	1								

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Glycolysis is present in :-
(1) Plants and animals only.
(2) Plants, animals and fungi only.
(3) Animals only.
(4) All living organisms.
BRP072

2. The RQ for proteins is :
(1) 1 (2) 0.7 (3) 0.9 (4) 0
BRP073

3. Glycerol would enter the respiratory pathway only after being converted to –
(1) Fatty acid (2) Acetyl CoA
(3) PGAL (4) Pyruvic acid
BRP074

4. Final Decarboxylation in Krebs cycle occur between-
(1) Citric acid and α-ketoglutaric acid
(2) α-ketoglutaric acid and succinic acid
(3) Succinic acid and malic acid
(4) Malic acid and oxaloacetic acid
BRP075

5. Pyruvate, which is formed by the glycolytic catabolism of carbohydrates in the cytosol, after it enters mitochondrial matrix, undergoes :-
(1) Oxidative carboxylation
(2) Oxidative decarboxylation
(3) Reductive carboxylation
(4) Carboxylation
BRP076

6. The products formed from glycolysis are :-
(1) Pyruvate, CO_2, ATP
(2) Pyruvate, ATP, $NADH+H^+$
(3) Acetyl CoA, ATP, $NADH+H^+$
(4) Acetyl CoA, ATP, H_2O
BRP077

7. During conversion of succinyl Co-A to succinic acid a molecule of GTP is formed, this process is known as :-
(1) Oxidative phosphorylation
(2) Substrate level phosphorylation
(3) Photophosphorylation
(4) Terminal oxidation
BRP078

8. In Glycolysis, glucose form two molecules of pyruvic acid, this is called :-
(1) Complete oxidation
(2) Partial oxidation
(3) Photooxidation
(4) Terminal oxidation
BRP079

9. How many ATP molecules are directly synthesised in glycolysis?
(1) 6ATP (2) 4ATP
(3) 36ATP (4) 8ATP
BRP080

10. When carbohydrates are used as substrate & anaerobically oxidised; the value of RQ will be:
(1) 1 (2) 0
(3) ∞ (4) less than 1
BRP081

11. Which act as final hydrogen acceptor in E.T.S.?
(1) Oxygen (2) NAD^+
(3) FAD (4) $NADP^+$
BRP082

12. Number of ATP produced when 1 molecule of pyruvic acid is used as respiratory substrate?
(1) 15 (2) 12 (3) 3 (4) 36
BRP083

13. Which of the following is the energy yielding step of glycolysis?
(1) Glucose → Glucose-6-phosphate
(2) BPGA → PGA
(3) Fructose 1, 6 biphosphate → PGAL
(4) Phosphoglycerate→ phosphoenolpyruvate

BRP084

14. The passing on of the electrons removed as part of the hydrogen atoms to molecular oxygen with synthesis of ATP, the site of this process located–
(1) in the matrix of the mitochondria
(2) in the cytoplasm
(3) in the intermembrane space of mitochondria
(4) on the inner membrane of the mitochondria

BRP085

15. In TCA cycle, isomerisation of citrate is followed by how many decarboxylation(s) to form succinic acid?
(1) Two (2) Three
(3) One (4) Four

BRP086

16. In glycolysis there is no :-
(1) Oxidation
(2) Decarboxylation
(3) Dehydrogenation
(4) Splitting of C-C bonds

BRP087

17. Which of the following is not an intermediate of Krebs cycle ?
(1) OAA (2) α-ketoglutarate
(3) Malate (4) PGAL

BRP088

18. What is the product of two decarboxylations and two oxidation of isocitric acid ?
(1) α-ketoglutaric acid (2) Succinyl Co A
(3) Succinic acid (4) Fumaric acid

BRP089

19. The respiratory quotient depends upon the:-
(1) Type of respiratory substrate used during respiration
(2) Amount of respiratory substrate used during respiration
(3) Duration of respiration
(4) Type of coenzymes used during respiration

BRP090

20. In mitochondrial ETS, cytochrome 'C' acts as a mobile carrier for transfer of electrons between :-
(1) Complex III and IV
(2) Complex I and II
(3) Complex II and III
(4) Complex IV and V

BRP091

21. The energy released by oxidation of respiratory substrates :
(A) Comes out in a single step to increase the possibility of maximum ATP production
(B) Is not used directly
(C) Is used directly in the energy requiring processes of the organisms
(D) Is trapped as chemical energy in the energy currency of the cell
(1) C and D are incorrect
(2) B and D are correct
(3) A and B are correct
(4) A and D are incorrect

BRP092

22. How many ATP molecules and during which steps, are directly synthesised in EMP pathway from one glucose molecule?

(1) 4 ATP, 2 in each PEP to pyruvic acid and BPGA to PGA

(2) 8 ATP, 4 in each PEP to pyruvic acid and BPGA to PGA

(3) 2 ATP, 1 in each Glucose to Glucose-6-P and Fructose-6-P to Fructose 1, 6 bisphosphate

(4) 4 ATP, 2 in each Glucose to Glucose-6-P and Fructose-6-P to Fructose 1, 6 bisphosphate

BRP093

23. Which of the following enzyme(s) is/are involved in the conversion of pyruvic acid into CO_2 and ethanol?

(1) Pyruvic acid dehydrogenase

(2) Alcohol decarboxylase

(3) Both (1) and (2)

(4) Pyruvic acid decarboxylase

BRP094

24. The complete oxidation of one molecule of pyruvate by the stepwise removal of all the hydrogen atoms:

(1) Leaving six molecules of CO_2

(2) Leaving two molecules of CO_2

(3) Leaving four molecules of CO_2

(4) Leaving three molecules of CO_2

BRP095

25. In aerobic respiration, the ultimate or final electron acceptor is :

(1) Atomic oxygen (2) Molecular oxygen

(3) Cytochrome a_3 (4) Water

BRP096

26. Fermentation differs from aerobic respiration :-

(1) In having partial breakdown of glucose

(2) In producing less ATP per glucose

(3) In having slow oxidation of $NADH_2$ to NAD^+

(4) All of the above

BRP097

27. Complete oxidation of which of the following respiratory substrate evolve less volume of CO_2 as compare to volume of O_2 consumed ?

(1) Fats (2) Proteins

(3) Carbohydrates (4) Both (1) and (2)

BRP098

28. What is the significance of respiration ?

(1) Production of cellular energy currency

(2) Provides carbon skeleton as precursor for synthesis of various chemicals

(3) loss of weight

(4) Both (1) and (2)

BRP099

29. Plants do not present great demands for gaseous exchange because :

(1) They are autotrophic

(2) Photosynthesis and respiration work mutually

(3) In plants there is less need of energy

(4) Plants are regulators

BRP100

30. Select out the correct sequence of glycolytic steps:

(1) PGAL $\rightarrow$ 3-PGA $\rightarrow$ 1,3-BPGA $\rightarrow$ PEP

(2) PGAL $\rightarrow$ 1,3-BPGA $\rightarrow$ PEP $\rightarrow$ 3-PGA

(3) PGAL $\rightarrow$ 1,3-BPGA $\rightarrow$ 3-PGA $\rightarrow$ PEP

(4) PGAL $\rightarrow$ PEP $\rightarrow$ 1,3-BPGA $\rightarrow$ 2-PGA

BRP101

31. During respiration of *Yeast* which of the following enzyme is not used in oxygen stressed conditions ?
(1) Enolase
(2) Pyruvic acid decarboxylase
(3) Alcohol dehydrogenase
(4) Aconitase

BRP102

32. How much amount of energy present in glucose, get released during lactic acid and alcohol fermentation?
(1) 7 percent
(2) less than seven percent
(3) more than seven percent
(4) always 2 percent

BRP103

33. TCA cycle starts with condensation of acetyl group with :
(1) OAA (2) Water
(3) NAD (4) both (1) and (2)

BRP104

34. During TCA which of the following intermediate is a result of two successive decarboxylations ?
(1) Oxalosuccinic acid
(2) α-ketoglutaric acid
(3) Succinyl Co-A
(4) Cis aconitic acid

BRP105

35. Which of the following ETC complex is directly involved in reduction of oxygen ?
(1) Complex-I (2) Complex-II
(3) Complex-III (4) Complex-IV

BRP106

36. When proteins are respiratory substrates the ratio of CO_2/O_2 would be about :
(1) 1.0 (2) 0.7 (3) 0.9 (4) 1.3

BRP107

(B) ANALYTICAL QUESTIONS

37. At how many steps CO_2 is released in aerobic respiration?
(1) One or two (2) Three
(3) Five (4) Twelve

BRP108

38. Green plants kept in light produce ATP from the glucose. This process is :
(1) Photophosphorylation
(2) Hill reaction
(3) Oxidative phosphorylation
(4) β-oxidation

BRP109

39. $FADH_2$ produced in Kreb's-cycle from -
(1) Isocitrate (2) α-ketoglutarate
(3) Succinate (4) Malate

BRP110

40. Aerobic respiration is how many times useful than anaerobic respiration ?
(1) 2 (2) 8
(3) 19 (4) 38

BRP111

41. RQ is less than one for :-
(1) Proteins and organic acids
(2) Fatty acids and organic acids
(3) Fatty acids and proteins
(4) Organic acids and carbohydrates

BRP112

42. The step in which $NADH+H^+$ is not produced is:-
(1) Succinyl-CoA $\rightarrow$ Succinate
(2) Pyruvate $\rightarrow$ Acetyl-CoA
(3) α-ketoglutarate $\rightarrow$ Succinyl-CoA
(4) Malate $\rightarrow$ OAA

BRP113

43. Which intermediate of Krebs cycle is used as a substrate for amino acid biosynthesis ?

(1) Citric acid (2) α-ketoglutarate

(3) Succinyl-CoA (4) Isocitric acid

BRP114

44. Which of the following is a common feature between glycolysis and Krebs cycles?

(1) Release of CO_2.

(2) Formation of $FADH_2$.

(3) Site of the processes.

(4) Dehydrogenation.

BRP115

45. In lactic acid fermentation, the final electron acceptor is -

(1) Acetaldehyde (2) Ethyl alcohol

(3) NADH + H^+ (4) Pyruvic acid

BRP116

46. How many molecules of CO_2 are released in Krebs cycle per glucose molecule ?

(1) Two (2) Three

(3) Four (4) Six

BRP117

47. After entry of pyruvic acid in mitochondrial matrix, which of the following does not take place?

(1) Oxidation

(2) Decarboxylation

(3) Oxidative - decarboxylation

(4) ATP mediated phosphorylation

BRP118

48. What is incorrect about anaerobic respiration?

(1) Partial breakdown of glucose

(2) Net gain of 2ATP

(3) Rapid oxidation of NADH + H^+ to NAD^+

(4) Reduction of pyruvic acid

BRP119

49. How many NADH+H^+ are formed in per turn of Krebs cycle ?

(1) Two (2) Three

(3) Four (4) Five

BRP120

50. During alcoholic fermentation by yeast two molecules of glucose produce -

(1) 3 molecules of ethanol+3 molecules of CO_2

(2) 6 molecules of ethanol+6 molecules of CO_2

(3) 2 molecules of ethanol+2 molecules of CO_2

(4) 4 molecules of ethanol+4 molecules of CO_2

BRP121

51. Curing of tea leaves is brought about by the activity of -

(1) Viruses (2) Fungi

(3) Bacteria (4) Mycorrhiza

BRP046

52. How many ATP molecules could maximally be generated from one molecule of glucose, if the complete oxidation of one mole of glucose to CO_2 and H_2O yields 686 kcal and the useful chemical energy available in the high energy phosphate bond of one mole of ATP is 12 Kcal?

(1) 57 (2) 1

(3) 2 (4) 30

BRP047

53. The overall goal of glycolysis, Krebs cycle and the electron transport system is the formation of:

(1) Nucleic acids

(2) ATP in small stepwise units

(3) ATP in one large oxidation reaction

(4) Sugars

BRP048

54. All enzymes of TCA cycle are located in the mitochondrial matrix except one which is located in inner mitochondrial membrane in eukaryotes and in cytosol in prokaryotes. This enzyme is: -

(1) Succinate dehydrogenase

(2) Lactate dehydrogenase

(3) Isocitrate dehydrogenase

(4) Malate dehydrogenase

BRP049

55. Which one of the following mammalian cells are not capable of metabolising glucose to carbon-dioxide aerobically?

(1) Red blood cells

(2) White blood cells

(3) Unstriated muscle cells

(4) Liver cells

BRP050

56. The chemiosmotic coupling hypothesis of oxidative phosphorylation proposes that adenosine triphosphate (ATP) is form because: -

(1) A proton gradient forms across the inner membrane

(2) There is a change in the permeability of the inner mitochondrial membrane toward adenosine diphosphate (ADP)

(3) High energy bonds are formed in mitochondrial proteins

(4) ADP is pumped out of the matrix into the intermembrane space

BRP051

57. Read the following statements with respect to Krebs cycle and select the correct option:

(a) First stable product is 3C compound.

(b) At three steps NAD^+ is reduced.

(c) GTP is produced when succinyl CoA is converted to succinic acid.

(d) Citric acid is the first stable product.

(1) a, b & c are correct

(2) b, c & d are correct

(3) Only a is correct

(4) a, b & d are correct

BRP132

58. Aerobic respiratory pathway is appropriately termed: -

(1) Parabolic (2) Amphibolic

(3) Anabolic (4) Catabolic

BRP053

59. The energy-releasing metabolic process in which substrate is oxidised without an external electron acceptor is called:

(1) Glycolysis

(2) Fermentation

(3) Aerobic respiration

(4) Photorespiration

BRP054

60. In mitochondria, protons accumulate in the:

(1) Matrix

(2) Outer membrane

(3) Inner membrane

(4) Intermembrane space

BRP055

61. Which of the metabolites is common to respiration- mediated breakdown of fats, carbohydrates and proteins?

(1) Acetyl CoA

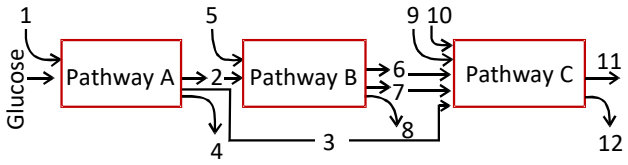
(2) Glucose-6-phosphate

(3) Fructose-1,6-bisphosphate

(4) Pyruvic acid

BRP056

62. The three boxes in this diagram represent the three major biosynthetic pathways in aerobic respiration. Arrows represent net reactants or products.



Arrow numbered 4, 8 and 12 can all be:

(1) FAD^+ or $FADH_2$ (2) NADH

(3) ATP (4) H_2O

BRP057

EXERCISE-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	2	2	2	2	2	2	3	1	1	2	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	2	1	1	2	1	4	4	2	4	4	4	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	4	3	4	3	2	3	3	3	3	1	2	4	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	4	3	2	4	3	1	2	1	1	1	2	2	2	4
Question	61	62													
Answer	1	3													

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION TO HOW MANY PIGMENTS ARE INVOLVED IN PHOTOSYNTHESIS

1. Oxygen which is liberated during photosynthesis comes from -
(1) Carbon dioxide
(2) Water
(3) Chlorophyll
(4) Phosphoglyceric acid
BPP001

2. In which of the following process, the light energy is converted into chemical energy?
(1) Digestive action (2) Respiration
(3) Photosynthesis (4) Fermentation
BPP002

3. During photosynthesis the oxygen in glucose comes from :-
(1) Water
(2) Carbon dioxide
(3) O_2 in air
(4) Both from water and CO_2 **BPP003**

4. Name the scientist, who first pointed out by bell jar experiment that plants purify foul air:
(1) Willstatter (2) Robert Hooke
(3) Priestley (4) Jean Senebier
BPP004

5. Moll's half leaf experiment explains that :-
(1) Carbon dioxide is essential for photosynthesis
(2) Chlorophyll and water are necessary for photosynthesis
(3) Light and water are essential for photosynthesis
(4) All the above are correct
BPP005

6. Oxygen during photosynthesis comes from water was proved with the help of O^{18} experiment by:-
(1) Ruben and Kamen (2) Hill
(3) Warburg (4) Blackman
BPP006

7. Name the scientist who pointed out the importance of different wavelengths of light using a green algae and aerobic bacteria :-
(1) Priestley (2) Ingenhousz
(3) K.V. Thimann (4) Englemann
BPP007

8. Photosynthesis is :-
(1) Oxidative, exergonic, catabolic
(2) Redox-reaction, endergonic, anabolic
(3) Reductive, exergonic, anabolic
(4) Reductive, endergonic, catabolic
BPP008

9. The significance of light and green part in photosynthesis was discovered by :-
(1) Priestley (2) Ingenhousz
(3) Englemann (4) Blackman
BPP009

10. Which one of the following pigment does not occur in the chloroplast?
(1) Carotene (2) Xanthophyll
(3) Chlorophyll 'b' (4) Anthocyanin
BPP010

11. Chlorophyll contains :
(1) Fe (2) Mg
(3) K (4) Mn
BPP011

12. Which pigment is water soluble?
(1) Chlorophyll (2) Carotene
(3) Anthocyanin (4) Xanthophyll
BPP012

13. Chlorophyll is present :
(1) In the grana of chloroplasts
(2) On the surface of chloroplasts
(3) Dispersed through out the chloroplasts
(4) In the stroma of chloroplasts
BPP013

14. Which colour of light gives maximum absorption peak by chlorophyll 'a' ?
(1) Blue light (2) Green light
(3) Violet light (4) Red light
BPP014

15. The number of pigment molecules in photosystem is:
(1) 250 - 400 (2) 300 - 900
(3) 500 - 600 (4) 50 -100
BPP016

16. The main difference between chlorophyll 'a' and 'b' is:
(1) Chlorophyll 'a' is a linear chain compound and 'b' is branched chain
(2) Chlorophyll 'a' has no Mg^{+} ion in center of molecule
(3) In chlorophyll 'a' there is $-CH_3$ group whereas in 'b' it is –CHO group
(4) All of the above
BPP017

17. Basic common structure of all chlorophyll comprises of :-
(1) Cytochrome system
(2) Flavoproteins
(3) Porphyrin system
(4) Plastocyanin
BPP018

18. What is the by product of bacterial photosynthesis?
(1) O_2 (2) CO_2 (3) S (4) H_2S
BPP019

19. Universal chlorophyll is :-
(1) Chl - 'a' (2) Chl - 'b'
(3) Chl - 'c' (4) Chl - 'e'
BPP021

20. Which one of the following is precursor of protochlorophyll?
(1) Acetyl COA
(2) Succinyl COA
(3) Oxaloacetic acid
(4) α-ketoglutarate
BPP022

21. Photosynthesis is an oxidation reduction process, the materials that is oxidised is :-
(1) CO_2 (2) NADP
(3) H_2O (4) PGA
BPP023

22. During photosynthesis :-
(1) Water is reduced & CO_2 is oxidized
(2) CO_2 is reduced & water is oxidized
(3) Both CO_2 & water get reduced
(4) Both CO_2 & water get oxidized
BPP024

23. Which photosynthetic pigment converts nascent oxygen to molecular oxygen ?
(1) Chlorophyll-a (2) Carotenoids
(3) Phycobilins (4) Chlorophyll-b
BPP025

24. Hill reaction occurs in :-
(1) High altitude plants only
(2) Total darkness
(3) Presence of ferricyanide
(4) Absence of water
BPP026

25. Photosynthetically active radiation (PAR) represents the following range of wavelength.
(1) 400 - 700 nm (2) 500 - 600 nm
(3) 450 - 950 nm (4) 340 - 450 nm
BPP027

26. Which one of the following categories of organisms do not evolve oxygen during photosynthesis ?
(1) Red algae
(2) Photosynthetic bacteria
(3) C_4-plants with Kranz anatomy
(4) Blue green algae
BPP028

27. The site for dark reaction of photosynthesis is:
(1) Stroma
(2) Grana thylakoid
(3) Intergranal thylakoid
(4) Mitochondria
BPP029

WHAT IS LIGHT REACTION AND THE ELECTRON TRANSPORT

28. Discovery of Emerson effect has shown the existence of :-
(1) Two distinct photosystems
(2) Light and dark reactions of photosynthesis
(3) Photophosphorylation
(4) Photorespiration
BPP030

29. Wavelength of light responsible for Emerson's enhancement effect :-
(1) only 680 nm
(2) only 680 nm↓
(3) infra red wavelength
(4) Both 680 nm and more than 680 nm
BPP031

30. The process of photo-phosphorylation take place in :-
(1) Chloroplast (2) Ribosomes
(3) Mitochondria (4) Cell-wall
BPP032

31. In pigment system-I, reaction centre is :-
(1) P-600 (2) P-680
(3) P-700 (4) P-720
BPP033

32. Which of the following is the site of photolysis of water ?
(1) Stroma of chloroplast
(2) Cristae of chloroplast
(3) Ribosome of chloroplast
(4) Lumen surface of thylakoid membrane
BPP034

33. The first step in photosynthesis is :-
(1) Joining of three carbon atoms to from glucose
(2) Formation of ATP
(3) Ionization of water
(4) Excitement of an electron of chlorophyll by a photon of light.
BPP035

34. The product of light reaction is :-
(1) ATP & NADPH(H^+)
(2) NADPH(H^+) & glucose
(3) Only ATP
(4) O_2 & glucose
BPP036

35. Which one of the following concerns with photophosphorylation ?
(1) ADP + AMP $\xrightarrow{\text{Light}}$ ATP
(2) ADP + organic PO_4 $\xrightarrow{\text{Light}}$ ATP
(3) ADP + Inorganic PO_4 $\xrightarrow{\text{Light}}$ ATP
(4) AMP + Inorganic PO_4 $\xrightarrow{\text{Light}}$ ATP
BPP037

36. During photochemical reaction of photosynthesis.
(1) liberation of O_2 takes place
(2) Formation of ATP and $NADPH_2$ take place
(3) Liberation of O_2, formation of ATP, and $NADPH_2$ takes place
(4) Assimilation of CO_2 takes place

BPP038

37. Which of the following is excited molecule during photosynthesis ?
(1) Chlorophyll (2) Oxygen
(3) Carbon dioxide (4) Water

BPP039

38. During splitting of H_2O, H^+ is ultimately captured by :-
(1) Chlorophyll (2) NADP
(3) O_2 (4) Cytochrome

BPP040

39. At the time of splitting of H_2O, which initially captures the electron :-
(1) Chlorophyll (2) NADP
(3) OH^- (4) Cytochrome

BPP041

40. In cyclic photophosphorylation which one of the following is formed ?
(1) NADP & ATP
(2) ATP
(3) NADH + H^+ and O_2
(4) NADPH + H^+, ATP and O_2

BPP042

41. Photooxidation of water in photosynthesis is associated with :-
(1) Cytochrome b_6
(2) Pigment system–I
(3) Pigment system–II
(4) Plastocyanin

BPP043

42. During ATP synthesis electron pass through:-
(1) Water (2) Cytochromes
(3) O_2 (4) CO_2

BPP044

43. Which pigment system immediately donates e^- for the reduction of NADP ?
(1) PS-II (2) PS-I
(3) CO_2 (4) Plastoquinone

BPP045

44. Which element are presents in OEC (Oxygen evolving complex) ?
(1) Mn^{++} (2) Cl^-
(3) Ca^{++} (4) All

BPP046

WHERE ARE THE ATP AND NADPH USED

45. The path of CO_2 in the dark reactions of photosynthesis was successfully traced by the use of the :-
(1) O_2^{18} (2) $C^{14}O_2$
(3) P^{32} (4) X - rays

BPP047

46. Which of the following protein is most abundant on the earth?
(1) Catalase (2) RuBisCO
(3) Amylase (4) Pepsin

BPP048

47. The function of ATP in photosynthesis is the transfer of energy from the :-
(1) Dark reaction to the light reaction
(2) Light reaction to the dark reaction
(3) Chloroplasts to mitochondria
(4) Mitochondria to chloroplasts

BPP049

48. In photosynthesis, hydrogen is transferred from the light reactions to dark reactions by:–
(1) NAD (2) DNA (3) ATP (4) NADP
BPP050

49. NADPH(H^+) is also called :-
(1) Real power
(2) Oxidising agent
(3) Power house of energy
(4) Reducing power
BPP051

50. Fixation of 1 CO_2 requires :-
(1) 6NADPH(H^+) & 3ATP
(2) 2NADPH(H^+) & 3ATP
(3) 4NADPH(H^+) & 3ATP
(4) 5NADPH(H^+) & 3ATP
BPP052

51. Connecting link between light phase and dark phase of photosynthesis, is :-
(1) Only ATP (2) Only NADPH(H^+)
(3) Only NADH + H^+ (4) ATP & NADPH(H^+)
BPP053

52. In photosynthesis CO_2 combines with :-
(1) RUBP (2) ATP (3) ADP (4) PGA
BPP054

53. During the dark reactions of photosynthesis:–
(1) Water splits
(2) CO_2 is reduced to organic compounds
(3) Chlorophyll is activated
(4) Stable C_6-sugar is broken into three carbon sugars
BPP055

54. The enzyme that fixes atmospheric CO_2 in C_4 plants is :-
(1) PEP carboxylase (2) Hexokinase
(3) RUBP oxygenase (4) Hydrogenase
BPP056

55. During photosynthesis when PGA is changed into phosphoglyceraldehyde which of the following reaction occur ?
(1) Oxidation (2) Reduction
(3) Electrolysis (4) Hydrolysis
BPP057

56. Carbon refixation in C_4 plants occurs in chloroplasts of :-
(1) Palisade tissue
(2) Spongy Mesophyll
(3) Bundle sheath cells
(4) Guard cells
BPP058

57. Tropical plants like sugarcane show high efficiency of CO_2 fixation because of :-
(1) Calvin cycle
(2) Hatch - Slack cycle
(3) Cyclic photophosphorylation
(4) TCA Cycle
BPP059

58. "Kranz" type of anatomy is found in :-
(1) C_4 plant (2) C_3 plant
(3) Succulent plants (4) All of the above
BPP060

59. Which of the following is C_4 plants ?
(1) Maize (2) *Atriplex*
(3) Sugarcane (4) All of the above
BPP061

60. Which pair is wrong ?
(1) C_3 plant - maize
(2) Calvin cycle - PGA
(3) Hatch and Slack cycle - Maize
(4) C_4-plant - Kranz Anatomy
BPP062

61. C_4 plants are found among :-
(1) Only Gramineae family
(2) Only monocots
(3) Only dicots
(4) Monocots as well as dicots
BPP063

62. In case of C_4 pathway, the first step is :-
(1) CO_2 combines with RUBP
(2) CO_2 combines with PGA
(3) CO_2 combines with PEP
(4) CO_2 combines with PGAL

BPP064

63. In dark reaction, first reaction is the :-
(1) Carboxylation (2) Decarboxylation
(3) Dehydrogenation (4) Deamination

BPP065

64. Number of ATP molecules required for regeneration phase of RUBP during synthesis of 1 glucose molecule :-
(1) 6 (2) 12 (3) 18 (4) 30

BPP066

65. Isotopes employed to study the process of photosynthesis reaction.
(1) S^{35} and P^{32} (2) C^{14} and O^{18}
(3) N^{14} and Co^{60} (4) N^{14} and O^{18}

BPP067

66. Chloroplast is present in bundle sheath cells of :-
(1) C_3 plants
(2) C_4 plants
(3) CAM plants
(4) Photorespiring plants

BPP068

67. CO_2 is accepted by RUBP in C_4 plants in :-
(1) Mesophyll cells
(2) Bundle sheath cells
(3) Stomatal guard cells
(4) Epidermal cells

BPP069

68. Bundle sheath chloroplasts of C_4 plant are :
(1) Large & agranal (2) Large & granal
(3) Small & agranal (4) Small & granal

BPP070

69. In addition to the 12 molecules of NADPH(H^+), the energy required for the synthesis of one mole of hexose by C_3 and C_4 pathway is -
(1) 18 & 18 molecules of ATP respectively
(2) 30 & 30 molecules of ATP respectively
(3) 18 & 30 molecules of ATP respectively
(4) 30 & 18 molecules of ATP respectively

BPP071

70. How many Calvin cycles would generate one molecule of glucose/hexose ?
(1) One cycle
(2) Three cycles
(3) Six cycles
(4) Twelve cycles

BPP072

71. CAM - plants are :-
(1) Succulent xerophyte
(2) Hydrophytes
(3) Epiphytes
(4) Mesophytes

BPP073

72. The first stable product of Calvin cycle and Hatch stack pathway are :-
(1) 4-C and 3-C compounds respectively
(2) 4-C and 6-C compounds respectively
(3) 3-C and 4-C compounds respectively
(4) 5-C and 4-C compounds respectively

BPP074

73. Which of the following was used during discovery of Calvin cycle ?
(1) *Spirogyra*
(2) *Volvox*
(3) *Chlamydomonas*
(4) *Chlorella*

BPP075

PHOTORESPIRATION & FACTOR AFFECTING PHOTOSYNTHESIS

74. Which one of the following have high CO_2 compensation point ?
(1) C_2 plants (2) C_3 plants
(3) C_4 plants (4) CAM plants
BPP076

75. First stable product of Photorespiration is :-
(1) PGAL
(2) Glycerate
(3) Glycine
(4) Phosphoglycolate
BPP077

76. Photorespiration is favoured by :-
(1) Low light intensity
(2) Low O_2 & high CO_2
(3) Low temperature
(4) High O_2 & Low CO_2
BPP078

77. What is C_2 - Cycle ?
(1) Glycolate cycle (2) Calvin cycle
(3) Kreb's cycle (4) TCA - cycle
BPP079

78. Compensation point means :-
(1) When the rate of photosynthesis is equal to rate of respiration
(2) When there is neither photosynthesis nor respiration
(3) When the entire food manufactured in photosynthesis remains unutilized
(4) When availability of water equalise with necessity of water.
BPP080

79. What does not occur in photorespiration
(1) Utilization of O_2
(2) Production of CO_2
(3) Synthesis of ATP
(4) Use of ATP
BPP081

80. DCMU is an inhibitor of :-
(1) PS-I
(2) PS-II
(3) Calvin cycle
(4) Kreb's cycle
BPP082

81. Main factor which limits the rate of photosynthesis on a clear day is :-
(1) Chlorophyll (2) Light
(3) CO_2 (4) Water
BPP083

82. Which one of the following is wrong in relation to photorespiration ?
(1) It occurs in chloroplast
(2) It occurs in day time only
(3) It is a characteristic of C_4 plants
(4) It is a characteristic of C_3 plants
BPP084

83. The law of limiting factor for photosynthesis was given by :-
(1) R. Hill (2) Krebs
(3) Calvin (4) Blackman
BPP085

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	3	1	1	4	2	2	4	2	3	1	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	1	2	3	2	2	3	1	2	1	1	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	4	1	3	3	1	2	1	2	3	2	2	4	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	4	4	2	4	1	2	1	2	3	2	1	4	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	3	1	1	2	2	2	1	3	3	1	3	4	2	4
Question	76	77	78	79	80	81	82	83							
Answer	4	1	1	3	2	3	3	4							

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Anoxygenic photosynthesis is characteristic of-

(1) *Rhodospirillum* (2) *Spirogyra*

(3) *Chlamydomonas* (4) *Ulva*

BPP103

Re-AIPMT 2015

2. Chromatophores take part in: -

(1) Respiration (2) Photosynthesis

(3) Growth (4) Movement

BPP104

3. In photosynthesis, the light-independent reactions take place at :-

(1) Stromal matrix (2) Thylakoid lumen

(3) Photosystem-I (4) Photosystem-II

BPP105

NEET-I 2016

4. Emerson's enhancement effect and red drop have been instrumental in the discovery of: -

(1) Photophosphorylation and non-cyclic electron transport

(2) Two photosystems operating simultaneously

(3) Photophosphorylation and cyclic electron transport

(4) Oxidative phosphorylation

BPP106

5. In a chloroplast the highest number of protons are found in :-

(1) Stroma

(2) Lumen of thylakoids

(3) Inter membrane space

(4) Antennae complex

BPP107

6. A plant in your garden avoids photorespiratory losses, has improved water use efficiency shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant?

(1) C_3 (2) C_4

(3) CAM (4) Nitrogen fixer

BPP108

NEET-II 2016

7. The process which makes major difference between C_3 and C_4 plants is: -

(1) Photorespiration (2) Respiration

(3) Glycolysis (4) Calvin cycle

BPP109

NEET-UG 2017

8. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?

(1) Increasing atmospheric CO_2 concentration up to 0.05% can enhance CO_2 fixation rate

(2) C_3 plants respond to higher temperatures with enhanced photosynthesis while C_4 plants have much lower temperature optimum

(3) Tomato is a greenhouse crop which can be grown in CO_2 - enriched atmosphere for higher yield

(4) Light saturation for CO_2 fixation occurs at 10% of full sunlight

BPP110

9. Phosphonyl pyruvate (PEP) is the primary CO_2 acceptor in :

(1) C_4 plants (2) C_2 plants

(3) C_3 and C_4 plants (4) C_3 plants

BPP111

NEET-UG 2018

10. Oxygen is not produced during photosynthesis by: -

(1) Green Sulphur bacteria

(2) *Nostoc*

(3) *Cycas*

(4) *Chara*

BPP112

11. Which of the following is not a product of light reaction of photosynthesis?

(1) ATP (2) NADH

(3) NADPH (4) Oxygen

BPP113

NEET-UG 2019 (Odisha)

12. In Hatch and Slack pathway, the primary CO_2 acceptor is: -

(1) Oxaloacetic acid

(2) Phosphoglyceric acid

(3) Phosphoenol pyruvate

(4) RuBisCO

BPP114

13. One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the culture by splitting light through a prism. He observed that bacteria accumulated mainly in the region of:

(1) Violet and green light

(2) Indigo and green light

(3) Orange and yellow light

(4) Blue and red light

BPP115

NEET-UG 2020

14. In light reaction, plastoquinone facilitates the transfer of electrons from :

(1) PS-I to ATP synthase

(2) PS-II to $Cytb_6f$ complex

(3) $Cytb_6f$ complex to PS-I

(4) PS-I to $NADP^+$

BPP116

15. The oxygenation activity of RuBisCO enzyme in photorespiration leads to the formation of :-

(1) 1 molecule of 4 C compound and 1 molecule of 2 C compound.

(2) 2 molecules of 3 C compound

(3) 1 molecule of 3 C compound

(4) 1 molecule of 6 C compound

BPP117

NEET-UG (COVID-19) 2020

16. During non-cyclic photophosphorylation, when electrons are lost from the reaction centre at PS-II, what is the source which replaces these electrons?

(1) Oxygen (2) Water

(3) Carbon dioxide (4) Light

BPP118

17. Which of the following statements is incorrect?

(1) RuBisCO is a bifunctional enzyme

(2) In C_4 plants, the site of RuBisCO activity is mesophyll cell

(3) The substrate molecule for RuBisCO activity is a 5-carbon compound

(4) RuBisCO action requires ATP and NADPH

BPP119

NEET-UG 2021

18. The first stable product of CO_2 fixation in sorghum is :

(1) Pyruvic acid

(2) Oxaloacetic acid

(3) Succinic acid

(4) Phosphoglyceric acid

BPP120

19. Which of the following statements is incorrect?

(1) Both ATP and NADPH + H^+ are synthesized during non-cyclic photophosphorylation.

(2) Stroma lamellae have PS-I only and lack NADP reductase.

(3) Grana lamellae have both PS-I and PS-II.

(4) Cyclic photophosphorylation involves both PS-I and PS-II.

BPP121

NEET-UG [Paper-2] 2021

20. In the plastid, the site of dark and light reaction is :-

(1) Grana and stroma respectively

(2) Grana and matrix respectively

(3) Matrix and grana respectively

(4) Stroma and grana respectively

BPP122

21. The maximum absorption of light by chlorophyll occurs in which region of the absorption spectrum.

(1) Red and green

(2) Yellow and green

(3) Blue and red

(4) Brown and red

BPP123

22. Non-cyclic photophosphorylation results in the formation of

(1) ATP and NADH + H^+

(2) ATP only

(3) ATP and NADPH + H^+

(4) ATP, ADP and NADH + H^+

BPP124

NEET-UG 2022

23. Given below are two statements :

Statement I: The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells.

Statement II: Mesophyll cells of C_4 plants lack RuBisCO enzyme.

In the light of the above statements, choose the correct answer from the options given below :-

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

BPP125

24. Which one of the following is not true regarding the release of energy during ATP synthesis through chemiosmosis?

It involves :-

(1) Breakdown of electron gradient.

(2) Movement of protons across the membrane to the stroma.

(3) Reduction of NADP to $NADPH_2$ on the stroma side of the membrane.

(4) Breakdown of proton gradient.

BPP126

25. What is the role of large bundle sheath cells found around the vascular bundles in C_4 plants?

(1) To increase the number of chloroplasts for the operation of Calvin cycle.

(2) To enable the plant to tolerate high temperature.

(3) To protect the vascular tissue from high light intensity.

(4) To provide the site for photorespiratory pathway.

BPP127

Re-NEET-UG 2022

26. When one CO_2 molecule is fixed as one molecule of triose phosphate, which of the following photochemically made, high energy chemical intermediates are used in the reduction phase ?

(1) 1 ATP + 1 NADPH (2) 1 ATP + 2 NADPH

(3) 2 ATP + 1 NADPH (4) 2 ATP + 2 NADPH

BPP128

27. Match **List-I** with **List-II**.

List-I	**List-II**
(a) Porins	(i) Pink coloured nodules
(b) leg haemoglobin	(ii) Lumen of thylakoid
(c) H^+ accumulation	(iii) Amphibolic pathway
(d) Respiration	(iv) Huge pores in outer membrane of mitochondria

Choose the **correct** answer from the options given below :-

(1) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

(2) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

(3) (a)–(iii), (b)–(iv), (c)–(ii), (d)–(i)

(4) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)

BPP129

28. Identify the correct statements regarding chemiosmotic hypothesis :

(a) Splitting of the water molecule takes place on the inner side of the membrane.

(b) Protons accumulate within the lumen of the thylakoids.

(c) Primary acceptor of electron transfers the electrons to an electron carrier.

(d) NADP reductase enzyme is located on the stroma side of the membrane.

(e) Protons increase in number in stroma.

Choose the correct answer from the options given below:

(1) (a), (b) and (e) (2) (a), (b) and (d)

(3) (b), (c) and (d) (4) (b), (c) and (e)

BPP130

NEET-UG 2023

29. How many ATP and $NADPH_2$ are required for the synthesis of one molecule of Glucose during Calvin cycle ?

(1) 18 ATP and 12 $NADPH_2$

(2) 12 ATP and 16 $NADPH_2$

(3) 18 ATP and 16 $NADPH_2$

(4) 12 ATP and 12 $NADPH_2$

BPP201

30. Which micronutrient is required for splitting of water molecule during photosynthesis ?

(1) Molybdenum (2) Magnesium

(3) Copper (4) Manganese

BPP202

31. The reaction centre in PS-II has an absorption maxima at :-

(1) 700 nm (2) 660 nm

(3) 780 nm (4) 680 nm

BPP203

32. Which of the following combinations is required for chemiosmosis?

(1) membrane, proton pump, proton gradient, NADP synthase

(2) proton pump, electron gradient, ATP synthase

(3) proton pump, electron gradient, NADP synthase

(4) membrane, proton pump, proton gradient, ATP synthase

BPP204

NEET-UG 2023 (Manipur)

33. Match List-I with List-II.

	List-I		List-II
(A)	Chlorophyll a	(I)	Yellow to yellow orange
(B)	Chlorophyll b	(II)	Yellow green
(C)	Xanthophyll	(III)	Blue green
(D)	Carotenoid	(IV)	Yellow

Choose the **correct** answer from the options given below :-

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

BPP205

34. Given below are two statements.

Statement I : RuBisCO is the most abundant enzyme in the world.

Statement II : Photorespiration does not occur in C_4 plants.

In the light of the above statements, choose the **most appropriate** answer from the options given below :-

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

BPP206

35. Which out of the following statements is **incorrect** ?

(1) Grana lamellae have both PS-I and PS-II

(2) Cyclic photophosphorylation involves both PS-I and PS-II

(3) Both ATP and NADPH + H^+ are synthesised during non-cyclic photophosphorylation.

(4) Stroma lamellae lack PS-II and NADP reductase

BPP207

NEET-UG 2024

36. Which of the following are required for the dark reaction of photosynthesis?

A. Light B. Chlorophyll

C. CO_2 D. ATP

E. NADPH

Choose the correct answers from the options given below :-

(1) A, B and C only (2) B, C and D only

(3) C, D and E only (4) D and E only

BPP208

37. How many molecules of ATP and NADPH are required for every molecule of CO_2 fixed in the Calvin cycle?

(1) 2 molecules of ATP and 3 molecules of NADPH.

(2) 2 molecules of ATP and 2 molecules of NADPH.

(3) 3 molecules of ATP and 3 molecules of NADPH.

(4) 3 molecules of ATP and 2 molecules of NADPH.

BPP209

38. **Statement-I :** In C_3 plants, some O_2 binds to RuBisCO, hence CO_2 fixation is decreased.

Statement-II : In C_4 plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

BPP210

RE-NEET-UG 2024

39. Which of the following are required for the light reaction of Photosynthesis?

A. CO_2 B. O_2

C. H_2O D. Chlorophyll

E. Light

Choose the **correct** answer from the options given below :-

(1) A, C, D and E only

(2) C, D and E only

(3) A and B only

(4) A, C and E only

BPP211

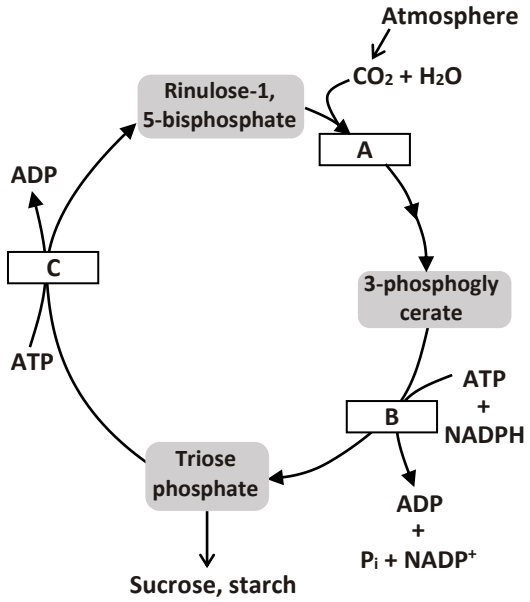
40. Which one of the following products diffuses out of chloroplast during photosynthesis?

(1) ADP (2) NADPH

(3) O_2 (4) ATP

BPP212

41. Observe the given figure. Identify the different stages labelled with alphabets by selecting the **correct** option.



(1) A-Carboxylation, B-Regeneration, C-Reduction

(2) A- Reduction, B-Decarboxylation C-Regeneration

(3) A-Carboxylation, B-Reduction, C-Regeneration

(4) A-Reduction, B-Carboxylation, C-Regeneration

BPP213

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	1	2	2	2	1	2	1	1	2	3	4	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	2	4	4	3	3	4	1	1	1	2	2	1	4
Question	31	32	33	34	35	36	37	38	39	40	41				
Answer	4	4	1	3	2	3	4	3	2	3	3				

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. "Only green part of the plants could release oxygen" this is shown by -
(1) Priestley (2) Moll
(3) Ingenhousz (4) Julius Van Sachs

BPP131

2. In Calvin cycle which step involve utilisation of 2 molecules of ATP for phosphorylation and two of $NADPH_2$ per CO_2 molecule fixed?
(1) Regeneration
(2) Carboxylation
(3) Reduction
(4) All of above

BPP132

3. Which character make C_4 plants special ?
(1) Special type of leaf anatomy
(2) They can tolerate high temperature
(3) Show response to high light intensities
(4) All of the above

BPP133

4. Most abundant enzyme in the world is -
(1) Collagen
(2) RuBisCO
(3) PEPcase
(4) Pyruvate dikinase

BPP134

5. For every CO_2 molecule entering the Calvin cycle, required energy is :-
(1) 3 ATP and 2 NADPH
(2) 2 ATP and 3 NADPH
(3) 3 ATP and 3 NADPH
(4) 2 ATP and 2 NADPH

BPP135

6. In C_4 plants, bundle sheath cells are characterised by :-
(1) Large number of chloroplasts
(2) Thick walls impervious to gaseous exchange
(3) No intercellular spaces
(4) All the above

BPP136

7. In C_4 plants photorespiration does not occur, due to:-
(1) Presence of RuBisCO enzyme
(2) Presence of high light intensity
(3) Increased concentration of CO_2 at RuBisCO enzyme site
(4) Presence of bundle sheath cells.

BPP137

8. In photosynthesis during ATP formation accumulation of protons occurs :-
(1) in lumen of thylakoid
(2) in stroma of chloroplast
(3) in periplastidial space
(4) at outer surface of thylakoid

BPP138

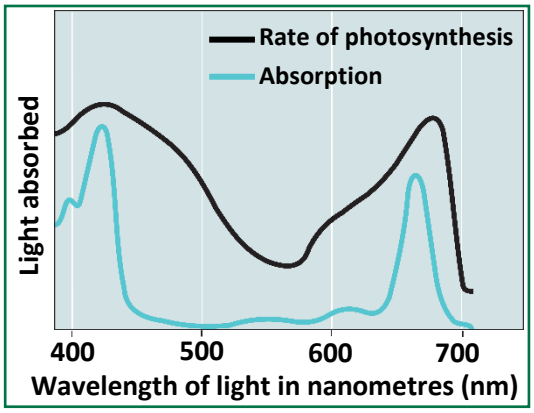
9. By using the bell jar, which of the following showed that sunlight is essential to plant process that purifies the air?
(1) Joseph priestley (2) Julius Von Sachs
(3) T.W. Engelmann (4) Jan Ingenhousz

BPP139

10. Radioisotopic technique was used to prove some facts about :-
(1) Light reaction of photosynthesis
(2) Biosynthetic phase of photosynthesis
(3) Both (1) and (2)
(4) Different shapes of chloroplasts in plants

BPP140

11. Represented below is the graph showing action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll 'a'. Which one of the following is **correct** for this ?



(1) There is a complete one to one overlap between the absorption and action spectrum

(2) By this overlapping it can be concluded that chlorophyll 'a' is the chief pigment of photosynthesis

(3) This overlapping shows that only chlorophyll 'a' absorb the entire light energy

(4) These graphs show that photosynthesis occurs only in the wavelengths of blue and red light

BPP141

12. The plants with Hatch and Slack pathway are adapted to :-

(1) Humid temperate regions

(2) Dry temperate regions

(3) Humid tropical regions

(4) Dry tropical regions

BPP142

13. Which of the following is correct?

(1) The C_3 and C_4 plants respond similarly to CO_2 concentrations.

(2) At low light conditions C_4 plants show increase in the rates of photosynthesis.

(3) C_4 plants not respond to high temperature.

(4) The C_3 and C_4 plants respond differently to CO_2 concentrations.

BPP143

14. Word dark reaction for biosynthetic phase of photosynthesis is misnomer word because:

(1) Dark reaction occur during dark

(2) Dark reaction occur during day

(3) Dark reaction is completely free from light

(4) Dark reaction occur during short dark intervals

BPP144

15. Photosynthetic photosystem is consist of :-

(1) Reaction centre only

(2) LHC only

(3) Both LHC & reaction centre

(4) Proteins only

BPP145

16. In grana lamellae, presence of which of the following specify that 'Z' scheme found only in grana lamellae?

(1) PS-I, PS-II

(2) PS-II & NADP reductase

(3) PS-I & NADP reductase

(4) Mg^{++}

BPP146

17. An elegant experiment with an aquatic plant showed that in bright sunlight, small bubbles were formed around the green parts while in the dark they did not. It was done by :-

(1) Jan Ingenhousz (2) Van Sachs

(3) Priestley (4) Van Niel

BPP147

18. How many ATP are required for regeneration of RUBP in one Calvin cycle?

(1) 6 ATP (2) 1 ATP

(3) 3 ATP (4) 12 ATP

BPP148

19. Based on his studies of purple & green bacteria who demonstrated that photosynthesis is essentially a light dependent reaction in which hydrogen from a suitable oxidisable compound reduces CO_2 to carbohydrates?

(1) Van Niel (2) Von Sachs

(3) Calvin (4) Priestley

BPP149

20. Which of following is correct about dark reaction?

(1) It is not directly light driven but are dependent on products of light reaction (ATP & NADPH)

(2) It occurs in stroma of chloroplast by enzymatic reactions

(3) Both (1) & (2)

(4) It occurs in stroma lamellae

BPP150

21. Which of the following is most crucial step of Calvin cycle?

(1) Carboxylation

(2) Reduction

(3) Regeneration

(4) Glycolytic reversal

BPP151

22. Find out the **correct match** from the following table :-

	Column-I	Column-II	Column-III
(i)	Photochemical phase	Photosystem-II	Oxygen release
(ii)	Paper Chromatography	Chlorophyll 'b'	Yellow orange
(iii)	Carotenoids	Chief pigment	Red light absorption

(1) (i) only (2) (i) and (ii)

(3) (iii) only (4) (ii) and (iii)

BPP152

23. During photosynthesis, plants mainly utilise the red and blue regions of visible spectrum, for the first time it was concluded by :

(1) Jan Ingenhousz

(2) Joseph Priestley

(3) T.W. Engelmann

(4) Cornelius Van Niel

BPP153

24. Which of the following conclusions regarding photosynthesis was proved by using radioisotopic techniques ?

(1) Light is essential

(2) O_2 comes from H_2O and not from CO_2

(3) Glucose is stored as starch

(4) Exchange of gases with environment

BPP154

25. The dark reactions of the photosynthesis :

(1) occur in darkness

(2) are not light dependent

(3) are not directly light driven

(4) occur in membrane system of chloroplast

BPP155

26. Electrons from which of following reduces $NADP^+$ to NADPH + H^+ during Z-scheme of photosynthesis?
(1) Photosystem-I (2) Water
(3) Carbon dioxide (4) Photosystem-II

BPP156

27. During photosynthesis the stroma lamellae of chloroplast could perform :
(1) the process of dark reaction in which ATP utilised
(2) the process of light reaction which produce NADPH + H^+
(3) the process of dark reaction which utilise NADPH + H^+
(4) the process of light reaction which produce ATP

BPP157

28. Which of the following statements are correct regarding synthesis of ATP in chloroplast during photosynthesis ?
(A) Splitting of water in stroma helps in creation of proton gradient
(B) Cytochrome complex helps in the release of protons in the lumen of thylakoid by accepting electrons from hydrogen carrier.
(C) Movement of protons across the membrane to the stroma through the F_0 of the ATPase is coupled with ATP synthesis.
(D) Reduction of $NADP^+$ to NADPH + H^+ is also a cause for creation of proton gradient.
(1) All statements are correct
(2) C and D
(3) A and B
(4) B, C and D

BPP158

29. What is the correct ratio of ATP utilisation in steps of Calvin cycle?
(1) Reduction : Regeneration :: 1 : 1
(2) Reduction : Regeneration :: 2 : 1
(3) Reduction : Regeneration :: 2 : 2
(4) Reduction : Regeneration :: 1 : 2

BPP159

30. The cells of C_4 plants those are rich in RuBisCO enzyme, also have which of the following characteristic (s)?
(1) Intercellular spaces absent
(2) Thick walls impervious to gaseous exchange
(3) Large number of chloroplast
(4) All of the above

BPP160

31. The productivity is better in C_4 plants because:
(1) They increase the intracellular concentration of CO_2 in mesophyll cells.
(2) In these plants RuBisCO has much greater affinity for O_2 than for CO_2.
(3) These plants can prevent competitive binding phenomena related to RuBisCO.
(4) These plants minimise the carboxylase activity of RuBisCO.

BPP161

32. C_3 plants respond to higher CO_2 concentration by showing increased rates of photosynthesis because:
(1) Current availability of CO_2 levels is limiting to the C_3 plants.
(2) C_3 plants show saturation at about 360 $\mu\ell L^{-1}$ concentration of CO_2.
(3) These plants responds to high CO_2 concentration even in low light conditions.
(4) In these plants RuBisCO shows only carboxylation.

BPP162

33. $2H_2A + CO_2 \longrightarrow 2A + CH_2O + H_2$
in this given equation H_2A represents to :
(1) Suitable reducible compounds
(2) Suitable oxidisable compound
(3) Suitable buffer
(4) Both (1) and (2)

BPP163

34. The membrane system of chloroplast is responsible for :
(1) Trapping the light energy
(2) Synthesis of ATP & NADPH
(3) Enzymatic reactions for CO_2 incorporation
(4) Both (1) and (2)

BPP164

35. How does PS-II supply electrons continuously?
(1) By removing electrons from photon
(2) By removing electrons from H_2O
(3) By removing electrons from CO_2
(4) By removing electrons from constituent carotenoids

BPP165

36. Which of the following is not always required for chemiosmosis ?
(1) Membrane (2) Proton pump
(3) OEC (4) ATP synthase

BPP166

37. Classification of biosynthetic phase of dark reaction as C_3 & C_4 is primarily based on.
(1) Initial CO_2 fixation
(2) Final CO_2 assimilation
(3) First CO_2 receptor
(4) Number of ATP get consumed

BPP167

38. Which of the following is not special about C_4 plants?
(1) Responsiveness to high light intensities
(2) Lack of photorespiration
(3) Greater productivity
(4) Scotoactive stomata

BPP168

39. Which of the following is not a plant factor regulating photosynthesis ?
(1) Age of leaf
(2) Number of mesophyll cells
(3) Atmospheric CO_2 concentration
(4) Amount of chlorophyll

BPP169

40. CO_2 is the major limiting factor for photosynthesis because the concentration of CO_2 is very low in the atmosphere which is :-
(1) between 0.01 and 0.02 per cent
(2) between 0.02 and 0.03 per cent
(3) between 0.03 and 0.04 per cent
(4) between 0.06 and 0.07 per cent

BPP170

41. (a) Formation of ATP and NADPH
(b) Use of ATP
(c) Reduction of PGA
(d) Water splitting
(e) Oxygen release
Which of the above processes are included in the dark reaction of photosynthesis ?
(1) a, b and c
(2) c and e
(3) b and c
(4) b, c and e

BPP171

42. Which of the following statements is correct in relation to the photosynthesis ?

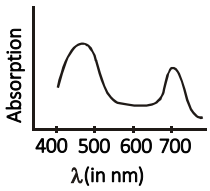
(1) It is the redox reaction during which reduction of H_2O and oxidation of CO_2 occurs

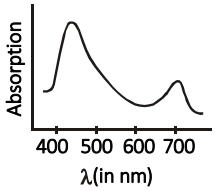
(2) First of all it started in *Chlorella*

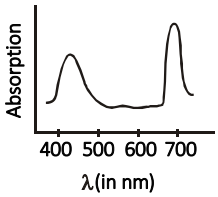
(3) It converts light energy into chemical energy in night

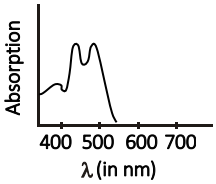
(4) Photosynthesis results in synthesis of sugar

BPP172

43. (A) 

(B) 

(C) 

(D) 

Regarding to given curves select the incorrect identification :-

(1) A is the absorption spectrum of chl 'a'

(2) B is the absorption spectrum of the chl 'b'

(3) C is the absorption spectrum of chl 'b'

(4) D is the absorption spectrum of carotenoids.

BPP173

44. Which of the following step(s) of Calvin cycle require(s) both ATP and NADPH(H^+) ?

(1) Carboxylation and regeneration

(2) Carboxylation

(3) Reduction

(4) Regeneration

BPP174

45. During photosynthesis, fixation of one CO_2 molecule in maize (A) and tomato (B) plant require :-

(1) [A] - 5 ATP & 2 NADPH + $2H^+$
[B] - 3 ATP & 2 NADPH + $2H^+$

(2) [A] - 3 ATP & 2 NADPH + $2H^+$
[B] - 3 ATP & 2 NADPH + $2H^+$

(3) [A] - 2 ATP & 2 NADPH + $2H^+$
[B] - 3 ATP & 2 NADPH + $2H^+$

(4) [A] - 30 ATP & 12 NADPH + $12H^+$
[B] - 18 ATP & 12 NADPH + $12H^+$

BPP175

46. How many carbons are present in first CO_2 acceptor in *sorghum* ?

(1) 5 (2) 4

(3) 3 (4) 2

BPP176

47. How many electrons are required to produce sufficient reducing power [NADPH + H^+] to fix 6 CO_2 molecules to form glucose?

(1) 6 electrons (2) 12 electrons

(3) 18 electrons (4) 24 electrons

BPP177

48. Which of the following is correct for photosynthesis?

(1) Use light energy to drive the synthesis of organic compounds

(2) Primary source of all food on earth

(3) Responsible for release of oxygen into the atmosphere

(4) All of the above

BPP178

49. Studies by which of the following scientists showed that the green substance in plants is located in special bodies within plant cells?

(1) T.W. Engelmann

(2) Julius Von Sachs

(3) Cornelius Van Niel

(4) Joseph Priestley

BPP179

50. Action spectrum of photosynthesis roughly resembles the :-

(1) Absorption spectrum of chlorophyll a

(2) Absorption spectrum of chlorophyll b

(3) Both (1) and (2)

(4) Absorption spectrum of carotenoids

BPP180

51. Which of the following is correct for the carbon reactions of photosynthesis ?

(1) They occur in the membranous system of chloroplast

(2) They are not light dependent

(3) They synthesize sugar, ATP and NADPH

(4) They are not directly light driven

BPP181

52. $2H_2O \longrightarrow 4H^+ + O_2 + 4e^-$

Above reaction is a part of :-

(1) Biosynthetic phase of photosynthesis in C_3 plants

(2) Z-Scheme of light reaction of photosynthesis

(3) Hatch and slack pathway

(4) Cyclic photophosphorylation

BPP182

53. Respectively during the creation of proton gradient and breakdown of proton gradient in chloroplast, protons move :-

(1) from the stroma to cytoplasm and from the cytoplasm to stroma

(2) from the cytoplasm to stroma and from stroma to cytoplasm

(3) from the thylakoid lumen to stroma and from the stroma to thylakoid lumen

(4) from the stroma to thylakoid lumen and from the thylakoid lumen to stroma

BPP183

54. In photosynthesis (Choose the correct one):

(1) Phosphorylation can occur in light reaction only

(2) Phosphorylation can occur in biosynthetic phase only

(3) Phosphorylation can occur in both light reaction and biosynthetic phase

(4) Phosphorylation not occurs at all

BPP184

55. Which of the following is a probable reason that why cyclic photophosphorylation takes place in photosynthesis?

(1) To supply extra carbon dioxide to Calvin cycle

(2) To utilise the light of wavelength of 680 nm which is not utilised in noncyclic photophosphorylation

(3) To produce extra NADPH for biosynthetic phase

(4) To meet the difference in number of ATP and NADPH

BPP185

56. C_4 plants are special plants having Kranz anatomy in their leaves, here term 'Kranz' is a reflection of :-
(1) Large number of chloroplasts in mesophyll cells.
(2) Capacity to tolerate high temperatures.
(3) Absence of chloroplasts in bundle sheath cells.
(4) Ring like arrangement of cells.

BPP186

57. During the dark reaction of photosynthesis in C_4 plants (Choose the correct one) :-
(1) Regeneration step occurs in mesophyll cell only.
(2) Regeneration step occurs in bundle sheath cell only.
(3) Regeneration steps occur in both mesophyll cell and bundle sheath cell.
(4) There is no step of regeneration.

BPP187

58. $RuBP + A \xrightarrow{RuBisCO} 2 \times$ Phosphoglycerate
$RuBP + B \xrightarrow{RuBisCO}$ Phosphoglycerate + Phosphoglycolate
Choose correct match for A and B from given options :-
(1) A = NADPH, B = CO_2
(2) A = O_2, B = CO_2
(3) A = CO_2, B = O_2
(4) A = CO_2, B = NADPH

BPP188

59. Before the discovery of RuBP as primary acceptor of CO_2 in Calvin cycle, it was believed that the primary acceptor would be:
(1) 6 carbon compound
(2) 2 carbon compound
(3) 3 carbon compound
(4) 4 carbon compound

BPP189

(B) ANALYTICAL QUESTIONS

60. Glycolate accumulates in chloroplast at :-
(1) Low temp
(2) Low CO_2
(3) Visible light illumination
(4) High CO_2

BPP190

61. For the synthesis of one molecule of sucrose, how may ATP molecules are required in reduction step of Calvin cycle?
(1) 24 ATP (2) 18 ATP
(3) 2 ATP (4) 12 ATP

BPP191

62. Which of the following is the product of oxidation during photosynthesis ?
(1) Carbon dioxide (2) Glucose
(3) Water (4) Oxygen

BPP192

63. In Calvin cycle, CO_2 is fused with primary acceptor molecule to form 3C compound phosphoglycerate. If in this reaction CO_2 provides one carbon then how many carbons are present in primary acceptor molecule ?
(1) Five carbons
(2) Two carbons
(3) Six carbons
(4) Three carbons

BPP193

64. In photosynthesis, $NADPH + H^+$ are oxidised in :-
(1) Calvin cycle
(2) Non-cyclic photophosphorylation
(3) Cyclic photophosphorylation
(4) Z-scheme

BPP194

65. Enzyme, which can catalyse both carboxylation and oxygenation of RuBP in the chloroplast, is **not** found in which of the following cells ?
(1) Mesophyll cells of C_4 plants.
(2) Mesophyll cells of C_3 plants.
(3) Mesophyll cells of CAM plants.
(4) Bundle sheath cells of C_4 plants.

BPP195

66. The product of light reaction of photosynthesis, which is not common between cyclic and non-cyclic photophosphorylation and is not utilized in Calvin cycle, is :-
(1) O_2 (2) ATP
(3) $NADPH_2$ (4) O_2 and ATP

BPP196

67. Which phase of Calvin cycle does/do not use any energy rich molecule ?
(1) Carboxylation only
(2) Carboxylation and reduction
(3) Reduction and regeneration
(4) Reduction only

BPP197

68. The C_4 plants differ from C_3 plants with reference to the :-
(1) Primary acceptor of CO_2 in dark reaction
(2) Type of end product of photosynthesis.
(3) Number of $NADPH_2$ that are consumed in synthesis of hexose.
(4) Type of pigments involved in photosynthesis.

BPP198

69. The common feature between non-cyclic and cyclic photophosphorylation is :-
(1) Proton gradient formation
(2) Release of O_2
(3) Formation of $NADPH + H^+$
(4) Photolysis of water

BPP199

70. The second acceptor of CO_2 in C_4 plants is :-
(1) PEP (2) Malate
(3) Aspartate (4) RuBP

BPP200

71. In photosystem-I, the first electron acceptor is :-
(1) Plastocyanin
(2) An iron-sulphur protein
(3) Ferredoxin
(4) Cytochrome

BPP086

72. During photorespiration, the oxygen consuming reaction(s) occur in :-
(1) Grana of chloroplasts and peroxisomes.
(2) Stroma of chloroplasts.
(3) Stroma of chloroplasts and mitochondria
(4) Stroma of chloroplasts and peroxisomes

BPP087

73. The first acceptor of electrons from an excited chlorophyll molecule of photosystem II is :-
(1) Quinone
(2) Cytochrome
(3) Iron-sulphur protein
(4) Ferredoxin

BPP088

74. In the leaves of C_4 plants, malic acid formation during CO_2 fixation occurs in the cells of :-
(1) Epidermis (2) Mesophyll
(3) Bundle Sheath (4) Phloem

BPP089

75. In leaves of C_4 plants malic acid synthesis during CO_2 fixation occurs in :-
(1) Bundle sheath
(2) Guard cells
(3) Epidermal cells
(4) Mesophyll cells

BPP090

76. The C_4 plants are photosynthetically more efficient than C_3 plants because :-

(1) The CO_2 efflux is not prevented.

(2) They have more chloroplasts.

(3) The CO_2 compensation point is more.

(4) CO_2 generated during photorespiration is trapped and recycled through PEP carboxylase.

BPP091

77. Electrons from excited chlorophyll molecule of photosystem II are accepted first by :-

(1) Quinone (2) Ferredoxin

(3) Cytochrome-b (4) Cytochrome-f

BPP092

78. Oxygenic photosynthesis occurs in :-

(1) *Oscillatoria* (2) *Rhodospirillum*

(3) *Chlorobium* (4) *Chromatium*

BPP093

79. Cyclic photophosphorylation results in the formation of :-

(1) ATP and NADPH

(2) ATP, NADPH and O_2

(3) ATP

(4) NADPH

BPP094

80. PGA as the first CO_2 fixation product was discovered in photosynthesis of :-

(1) Bryophyte (2) Gymnosperm

(3) Angiosperm (4) Alga

BPP095

81. C_4 plants are more efficient in photosynthesis than C_3 plants due to :-

(1) Higher leaf area

(2) Presence of larger number of chloroplasts in the leaf cells.

(3) Presence of thin cuticle.

(4) Lower rate of photorespiration.

BPP096

82. Read the following four statements, A, B, C and D and select the right option having both correct statements.

Statements:

(A) Z scheme of light reaction takes place in present of PSI only.

(B) Only PSI is functional in cyclic photophosphorylation.

(C) Cyclic photophosphorylation results into synthesis of ATP and NADPH(H^+) .

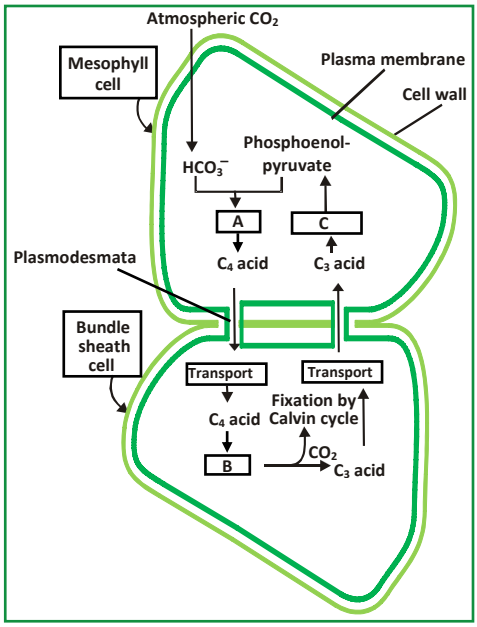
(D) Stroma lamellae lack PSII as well as NADP reductase.

(1) A and B (2) B and C

(3) C and D (4) B and D

BPP097

83. Study the pathway given below: -



In which of the following options correct words for all the three blanks A, B and C are indicated?

	A	B	C
(1)	Fixation	Transamination	Regeneration
(2)	Fixation	Decarboxylation	Regeneration
(3)	Carboxylation	Decarboxylation	Reduction
(4)	Decarboxylation	Reduction	Regeneration

BPP098

84. In Kranz anatomy, the bundle sheath cells have :-

(1) thick walls, many intercellular spaces and few chloroplasts.

(2) thin walls, many intercellular spaces and no chloroplasts.

(3) thick walls, no intercellular spaces and large number of chloroplasts.

(4) thin walls, no intercellular spaces and several chloroplasts.

BPP100

85. A process that makes important difference between C_3 and C_4 plants is :-

(1) Photosynthesis

(2) Photorespiration

(3) Transpiration

(4) Glycolysis

BPP101

86. The correct sequence of cell organelles during photorespiration is :-

(1) Chloroplast-peroxisome-mitochondria

(2) Chloroplast-vacuole-peroxisome.

(3) Chloroplast-Golgi bodies-mitochondria

(4) Chloroplast-Rough endoplasmic reticulum-Dictyosomes.

BPP102

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	4	2	1	4	3	1	4	3	2	4	4	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	2	1	3	1	1	3	2	3	1	4	4	2	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	2	4	2	3	1	4	3	3	3	4	3	3	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	4	4	2	3	4	2	4	1	4	4	3	3	2	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	4	1	1	1	1	1	1	1	4	2	4	1	2	4
Question	76	77	78	79	80	81	82	83	84	85	86				
Answer	2	1	1	3	4	2	4	2	3	2	1				

Exercise - I (Conceptual Questions) Build Up Your Understanding

1. The population of an insect species shows an explosive increase in numbers during rainy season followed by its disappearance at the end of the season. What does this show?

(1) The population of its predators increases enormously

(2) S-shaped or sigmoid growth of this insect

(3) The food plants mature and die at the end of the rainy season.

(4) Its population growth curve is of J-type

BOP001

2. The formula for exponential population growth is -

(1) $\frac{dt}{dN} = rN$ (2) $\frac{dN}{rN} = dt$

(3) $\frac{rN}{dN} = dt$ (4) $\frac{dN}{dt} = rN$

BOP002

3. Regarding life history variations. Which among the following is incorrect?

(1) Breeding once in life time — Bamboo

(2) Breeding many times in life time — Birds

(3) Production in large number of small size offspring — Mammals

(4) Production in small number of large size offspring — Birds

BOP003

4. $\frac{dN}{dt} = rN\left(\frac{K-N}{N}\right)$

In above equation $\left(\frac{K-N}{N}\right)$ represent: -

(1) Carrying capacity

(2) Environmental resistance

(3) Rate of change in population density

(4) Instrinsic rate of growth

BOP004

5. In a pond there are 400 lotus plant last year, through reproduction 16 new lotus plants are added, taking the current population to 416, then calculate the birth rate.

(1) 0.4 offspring per year

(2) 0.6 offspring per year

(3) 0.08 offspring per year

(4) 0.04 offspring per year

BOP005

6. A country has a high number of reproductive individual than pre-reproductive individual, what is correct about the population?

(1) Population is expanding

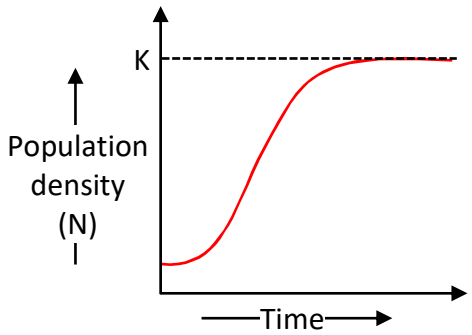
(2) Population is declining

(3) Population is stable

(4) Cannot be predicted

BOP006

7. In a logistic growth curve of population, environmental resistance is maximum when



(1) value of 'r' is high (2) value of 'r' is low

(3) N approaches K (4) K approaches N

BOP007

8. A group of individuals of same species constitute:

(1) Organism (2) Community

(3) Biome (4) Population

BOP008

9. Which of the following sequences is correct according to ecological hierarchy :-

(1) Population, variety, community, ecosystem

(2) Population, community, ecosystem, Biome

(3) Population, community, species, ecosystem

(4) Species, variety, ecosystem, population

BOP009

10. Properties of populations are not present in the "A" but arise as a result of interactions among the "B" comprising the population. "A" and "B" are :-

(1) Organisms and ecosystems

(2) Biomes and organisms

(3) Organism and organisms

(4) Ecosystems and communities.

BOP010

11. Fill in the blank.

A ..... is a unit of land with a natural boundary having a mosaic of patches. There patches generally represent different .....

(1) Biome, Landscape

(2) Biome, Ecosystem

(3) Landscape, Ecosystem

(4) Landscape, Biome

BOP011

12. In ecological hierarchy the abiotic factors are first time included at which hierarchical level

(1) Community Level

(2) Population Level

(3) Ecosystem Level

(4) Landscape Level

BOP014

13. Biosphere is :

(1) a component in the ecosystem

(2) composed of the plants present in the soil

(3) life in the outer space

(4) composed of all living organisms present on earth which interact with the physical environment

BOP015

14. The species which establishes an essential link with other species to help the later in some vital activity, is called :-

(1) Critical link species

(2) Key stone species

(3) Endemic species

(4) Ecophene

BOP016

15. Species which play, a vital role in controlling the relative abundance of other species, and control functioning of the community is called

(1) Dominant species

(2) Key stone species

(3) Link species

(4) Network species

BOP017

16. Kangaroo rat in desert is an example of ?

(1) Endemic species

(2) Key stone species

(3) Critical link species

(4) None of these

BOP018

17. Which of the following is not an example of endemic species ?

(1) Fig tree

(2) Kangaroo

(3) Kiwi

(4) *Metasequoia*

BOP019

18. Mycorrhizal association is
(1) Mutualism (2) Commensalism
(3) Parasitism (4) Exploitation
BOP020

19. In general, species diversity _____ as we move away from the equator towards the poles
(1) Increases
(2) Decrease
(3) Constant
(4) Can not say anything
BOP021

20. Which of the following is not a characteristics of a community?
(1) Dominance (2) Species diversity
(3) Natality (4) Stratification
BOP022

21. The group of organisms of different species forms a
(1) Community (2) Population
(3) Ecosystem (4) Biome
BOP026

22. Organisms which breed only once in their life is :-
(1) Pacific Salmon fish (2) Pacific Tuna fish
(3) Pacific Shark (4) Pacific Oysters
BOP027

23. Sexual deceit is exhibited by :
(1) Yucca plant (2) *Ophrys* plant
(3) Wasp species (4) Fig Tree
BOP028

24. Which of the following is an example of co–evolution?
(1) Barnacle & Whale
(2) Clown fish & Sea Anemone
(3) Hermit crab & Sea Anemone
(4) Bees & orchid flower
BOP029

25. A relationship between algae & fungi is an example of:
(1) Commensalism (2) Protocooperation
(3) Mutualism (4) Parasitism
BOP030

26. Lichens are the associations of
(1) Protist and fungus
(2) Algae and bacterium
(3) Fungus and algae
(4) Fungus and virus
BOP031

27. Which one of the following population interaction is (+, +) type :
(1) Commensalism (2) Mutualism
(3) Parasitism (4) Amensalism
BOP032

28. A wasp pollinating a fig flower is an example of:
(1) Commensalism
(2) Protocooperation
(3) Mutualism
(4) Parasitism
BOP033

29. Which one of the following interactions share a characteristic where none of the organism can complete its life cycle without the other participating organism?
(1) *E.coli* and Human
(2) Cattle egret bird and Cattle
(3) Hermit Crab and Sea Anemone
(4) *Yucca* and Pronuba
BOP034

30. Which of the following species relation is not an example of detrimental relation?
(1) Viscum – Oak
(2) Sparrow – Seed
(3) Loranthus-Mango
(4) Fig tree – Wasp
BOP035

31. They may reduce the survival, growth and reproduction of the other organism and reduce its population density, they might render the organism more vulnerable to predation by making it physically weak. They are:

(1) Predator (2) Parasites

(3) Ammensals (4) Commensal

BOP036

32. Animals eating plants are categorised separately as herbivores, they are in a broad ecological context not very different from:

(1) Parasite (2) Predator

(3) Commensal (4) Amensal

BOP037

33. Which statement is true for Connell's elegant field experiment?

(1) Performed on prickly pear cactus

(2) Shows resource partitioning

(3) Was perform in rocky intertidal area of Scotland

(4) Was done on Pisaster

BOP038

34. The prickly pear cactus in the early 1920 introduced into :-

(1) American pacific cost

(2) Rocky sea coasts of Scotland

(3) Galapagos islands

(4) Australia

BOP039

35. Which of the following concept is correct about amensalism?

(1) Both species are benefitted

(2) One species is benefitted and the other is neither benefitted nor harmed

(3) One species is harmed whereas the other is unaffected

(4) One species is benefitted and the other is harmed

BOP040

36. Endemic plants -

(1) Cosmopolitan

(2) Occur in a particular area

(3) Occur at high altitudes

(4) Occur on north pole

BOP041

37. Increase of population under optimum condition is termed.

(1) Reproductive ability

(2) Secondary production

(3) Biotic potential

(4) Biomass

BOP042

38. Occurrence of endemic species in South America and Australia due to :-

(1) These species has been extinct from other regions

(2) Continental separation

(3) There is no terrestrial route to these places

(4) Retrogressive evolution

BOP043

39. In a population unrestricted reproductive capacity is called as:

(1) Biotic potential

(2) Fertility

(3) Carrying capacity

(4) Birth rate

BOP044

40. What is true for individuals of same species

(1) Live in same niche

(2) Live in same habitat

(3) Interbreeding

(4) Live in different habitat

BOP045

41. When the two ecosystems overlap each other the area is called.

(1) Ecotone (2) Niche

(3) Edge effect (4) Ecotypes

BOP046

42. Group of two or more than two plant species is called as :-
(1) Plant community
(2) Animal ecosystem
(3) Plant ecosystem
(4) Ecological niche

BOP050

43. Competition for food, light and space is most severe in -
(1) Closely related species growing in the same area (in the same niche)
(2) Closely related species growing in different habitat
(3) Distantly related species growing in the same habitat
(4) Distantly related species growing in different habitat

BOP053

44. Most successful parasites are those which do not
(1) Grow free (2) Kill their host
(3) Reproduce sexually (4) Survive in soil

BOP054

45. The basic unit of ecological study is :-
(1) species (2) organism
(3) community (4) biosphere

BOP055

46. Mycorrhizae relationship between fungi and roots of higher plants is ?
(1) Parasitic relationship
(2) Saprophytic relationship
(3) Symbiotic relationship
(4) Epiphytic relationship

BOP056

47. Parasites adversely affect :-
(1) Survival of host.
(2) Growth of host.
(3) Reproduction potential of host.
(4) All of the above

BOP057

48. Identify the correct match :-

Column-I		Column-II	
(i)	Species diversity	(a)	Great influence on community stability
(ii)	Species dominance	(b)	Zonation according to the need of light
(iii)	Stratification	(c)	Different types of species in a community
(iv)	Keystone species	(d)	Highest number of one type of species

(1) i-c, ii-b, iii-a, iv-d
(2) i-c, ii-d, iii-b, iv-a
(3) i-b, ii-a, iii-d, iv-c
(4) i-b, ii-d, iii-a, iv-c

BOP058

49. The group of organisms of different species form a :-
(1) Community (2) Population
(3) Ecosystem (4) Biome

BOP060

50. Which of the following is an epiphyte ?
(1) Orchid (2) Lianas
(3) *Santalum* (4) Mango

BOP062

51. The correct statement for parasites is/are :-
(a) Host specific parasites & hosts tend to co-evolve
(b) Parasites have highly developed sense organs
(c) Parasites may reduce population density of host
(d) Parasites have highly developed digestive system
(1) a and b (2) b and c
(3) a and c (4) a and d

BOP063

52. A predator can be -

(1) Carnivore only (2) Herbivore only

(3) Mammal only (4) Both (1) and (2)

BOP169

53. Niche overlap indicates

(1) sharing of one or more resources between the two species

(2) mutualism between two species

(3) active cooperation between two species

(4) two different parasites on the same host

BOP065

54. A high density of elephant population in an area can result in :-

(1) Predation on one another

(2) Mutualism

(3) Intra specific competition

(4) Inter specific competition

BOP067

55. Geometric representation of age structure is a characteristic of :-

(1) Ecosystem (2) Biotic community

(3) Population (4) Landscape

BOP068

56. Which one of the following is one of the characteristics of a biological community?

(1) Sex-ratio (2) Stratification

(3) Natality (4) Mortality

BOP073

57. Which one of the following is categorised as a parasite in true sense?

(1) The female Anopheles bites and sucks blood from humans.

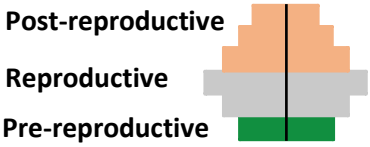
(2) Human foetus developing inside the uterus draws nourishment from the mother.

(3) Head louse living on the human scalp as well as laying eggs on human hair.

(4) The cuckoo (koel) lays its eggs in crow's nest.

BOP080

58. What type of human population is represented by the following age pyramid?



(1) Vanishing population

(2) Stable population

(3) Declining population

(4) Expanding population

BOP082

59. The logistic population growth is expressed by the equation :

(1) $\frac{dN}{dt} = rN\left(\frac{N-K}{N}\right)$

(2) $\frac{dt}{dN} = rN\left(\frac{K-N}{K}\right)$

(3) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

(4) $\frac{dN}{dt} = rN$

BOP083

60. *Cuscuta* is an example of: -

(1) Predation

(2) Endoparasitism

(3) Ectoparasitism

(4) Brood parasitism

BOP085

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	2	4	2	3	4	2	3	3	3	4	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	1	2	3	1	1	2	4	3	3	2	3	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	3	4	3	2	3	2	1	3	1	1	1	2	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	4	2	1	1	3	4	1	3	3	2	3	3	3	3

SOLUTION

5. Given : No. of lotus plants in last year = 400

No. of lotus plants in current year = 416

No. of new lotus plants = 16

$$\text{Birth rate} = \frac{\text{No. of birth in a year}}{\text{No. of individuals initial Population}}$$

$$= \frac{16}{400} = 0.04 \text{ Lotus plants / Lotus plant / year}$$

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. Most animals are tree dwellers in a:-

(1) Thorn woodland

(2) Temperate deciduous forest

(3) Tropical rain forest

(4) Coniferous forest

BOP087

2. Vertical distribution of different species occupying different levels in a biotic community is known as:

(1) Stratification

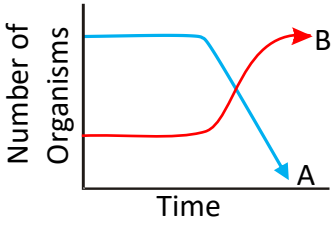
(2) Zonation

(3) Pyramid

(4) Divergence

BOP088

3. The following graph depicts changes in two populations (A and B) of herbivores in a grassy field A possible reason for these changes is that:



(1) Population B competed more successfully for food than population A

(2) Population A produced more offspring than population B

(3) Population A consumed the members of population B

(4) Both plant populations in this habitat decreased

BOP089

4. In which of the following interactions both partners are adversely affected?

(1) Mutualism

(2) Competition

(3) Predation

(4) Parasitism

BOP090

NEET-I 2016

5. When does the growth rate of a population following the logistic model equal zero? The logistic model is given as $\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$:-

(1) when N/K is exactly one.

(2) when N nears the carrying capacity of the habitat.

(3) when N/K equals zero.

(4) when death rate is greater than birth rate.

BOP091

6. Gause's principle of competitive exclusion states that:

(1) More abundant species will exclude the less abundant species through competition.

(2) Competition for the same resources excludes species having different food preferences.

(3) No two species can occupy the same niche indefinitely for the same limiting resources.

(4) Larger organisms exclude smaller ones through competition.

BOP092

7. Which of the following would appear as the pioneer organisms on bare rocks?
(1) Lichens (2) Liverworts
(3) Mosses (4) Green algae

BOP093

NEET-II 2016

8. The principle of competitive exclusion was stated by :-
(1) MacArthur
(2) Verhulst and Pearl
(3) C. Darwin
(4) G.F. Gause

BOP094

NEET-UG 2017

9. Asymptote in a logistic growth curve is obtained when:
(1) K = N
(2) K > N
(3) K < N
(4) The value of 'r' approaches zero

BOP095

10. Presence of plants arranged into well defined vertical layers depending on their height can be seen best in:
(1) Tropical Rain Forest
(2) Grassland
(3) Temperate Forest
(4) Tropical Savannah

BOP096

11. Mycorrhiza are the example of :
(1) Amensalism
(2) Antibiosis
(3) Mutualism
(4) Fungistasis

BOP097

12. In a hypothetical population of 100 individual having 'r' = 0.5/female/year, what will be the population size in 6 years (with e = 2.72) showing exponential rate of growth ?
(1) 1218 (2) 739 (3) 2012 (4) 448

BOP098

NEET-UG 2018

13. Niche is -
(1) all the biological factors in the organism environment
(2) the physical space where an organism live
(3) the range of temperature that the organism needs to live
(4) the functional role played by the organism where it lives

BOP099

14. Which one of the following plants shows a very close relationship with a species of moth, where none of the two can complete its life cycle without the other?
(1) Hydrilla (2) Yucca
(3) Banana (4) Viola

BOP100

15. Which of the following flowers only once in its life-time?
(1) Bamboo species (2) Jack fruit
(3) Mango (4) Papaya

BOP101

16. Natality refers to-
(1) Death rate
(2) Birth rate
(3) Number of individuals leaving the habitat
(4) Number of individuals entering a habitat

BOP102

17. In a growing population of a country
(1) pre-reproductive individuals are more than the reproductive individuals.
(2) reproductive individuals are less than the post-reproductive individuals.
(3) reproductive and pre-reproductive individuals are equal in number.
(4) pre-reproductive individuals are less than the reproductive individuals.

BOP103

NEET-UG 2019 (Odisha)

18. Carnivorous animals - lions and leopards, occupy the same niche but lions predate mostly larger animals and leopards take smaller ones. This mechanism of competition is referred to as :-
(1) Character displacement
(2) Altruism
(3) Resource partitioning
(4) Competitive exclusion

BOP104

19. Match Column-I with Column-II.

Column-I		Column-II	
(a)	Saprophyte	(i)	Symbiotic association of fungi with plant roots
(b)	Parasite	(ii)	Decomposition of dead organic materials
(C)	Lichens	(iii)	Living on living plants or animals
(d)	Mycorrhiza	(iv)	Symbiotic association of algae and fungi

Choose the **correct** answer from the options given below :-

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(ii)	(i)	(iii)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

BOP105

20. Between which among the following, the relationship is not an example of commensalism?
(1) Orchid and the tree on which it grows
(2) Cattle Egret and grazing cattle
(3) Sea Anemone and Clown fish
(4) Female wasp and fig species

BOP106

NEET-UG 2020

21. Which of the following is not an attribute of a population?
(1) Species interaction
(2) Sex ratio
(3) Natality
(4) Mortality

BOP107

22. Match the items in Column-I with those in Column-II :

Column I	Column II
(a) Herbivores-Plants	(i) Commensalism
(b) Mycorrhiza-Plants	(ii) Mutualism
(c) Sheep-Cattle	(iii) Predation
(d) Orchid-Tree	(iv) Competition

Select the **correct** option from following:
(1) (a)–(iv), (b)–(ii), (c)–(i), (d)–(iii)
(2) (a)–(iii), (b)–(ii), (c)–(iv), (d)–(i)
(3) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)
(4) (a)–(i), (b)–(iii), (c)–(iv), (d)–(ii)

BOP108

23. The impact of immigration on population density is: -
(1) Negative
(2) Both positive and negative
(3) Neutralized by natality
(4) Positive

BOP109

NEET 2021

24. Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival?
(1) Resource partitioning
(2) Competitive release
(3) Mutualism
(4) Predation

BOP110

25. Amensalism can be represented as:
(1) Species A (–); Species B (0)
(2) Species A (+); Species B (+)
(3) Species A (–); Species B (–)
(4) Species A (+); Species B (0)

BOP111

26. In the exponential growth equation $N_t = N_0e^{rt}$, e represents:
(1) The base of number logarithms
(2) The base of exponential logarithms
(3) The base of natural logarithms
(4) The base of geometric logarithms

BOP112

27. Match **List-I** with **List-II**.

	List-I		List-II
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical Adaptation	(iv)	Polar seal

Choose the **correct** answer from the options given below.
(1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
(2) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

BOP113

28. If a pond has 40 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 48, what will be the birth rate?
(1) 0.6 offspring per lotus per year
(2) 0.5 offspring per lotus per year
(3) 0.2 offspring per lotus per year
(4) 0.1 offspring per lotus per year

BOP115

NEET-UG 2022

29. Which one of the following statements cannot be connected to Predation?
(1) It might lead to extinction of a species
(2) Both the interacting species are negatively impacted
(3) It is necessitated by nature to maintain the ecological balance
(4) It helps in maintaining species diversity in a community

BOP116

30. While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (–) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (–) for another species involved in the interaction?
(1) Amensalism (2) Commensalism
(3) Competition (4) Predation

BOP117

31. Two butterfly species are competing for the same nectar of a flower in a garden. To survive and coexist together, they may avoid competition in the same garden by:
(1) feeding at the same time
(2) choosing different foraging patterns
(3) increasing time spent on attacking each other
(4) predating on each other

BOP118

32. The pioneer species in a hydrarch succession are:
(1) Free-floating angiosperms
(2) Submerged rooted plants
(3) Phytoplanktons
(4) Filamentous algae

BOP119

33. The species that come to appear in bare area are called:
(1) Pioneer species
(2) Invasive species
(3) Competitive species
(4) Species of seral community

BOP120

34. All successions irrespective of the habitat proceed to which type of climax community?
(1) Xeric
(2) Mesic
(3) Hydrophytic
(4) Edaphic

BOP121

NEET-UG 2023

35. Given below are two statements :-
Statement-I : Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.
Statement II : In general, carnivores are more adversely affected by competition than herbivores.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is correct but **Statement II** is false.
(3) **Statement I** is incorrect but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

BOP149

36. Match **List I** with **List II** :

List-I (Interaction)		**List-II (Species A and B)**	
A.	Mutualism	I.	+(A), 0(B)
B.	Commensalism	II.	–(A), 0(B)
C.	Amensalism	III.	+(A), –(B)
D.	Parasitism	IV.	+(A), +(B)

Choose the **correct** answer from the options given below:-
(1) A–IV, B–I, C–II, D–III
(2) A–IV, B–III, C–I, D–II
(3) A–III, B–I, C–IV, D–II
(4) A–IV, B–II, C–I, D–III

BOP150

37. Match **List I** with **List II.**

List-I (Interacting species)		**List-II (Name of Interaction)**	
A.	A Leopard and a Lion in a forest/ grassland	I.	Competition
B.	A Cuckoo laying egg in a Crow's nest	II.	Brood parasitism
C.	Fungi and root of a higher plant in Mycorrizae	III.	Mutualism
D.	A cattle egret and a Cattle in a field	IV.	Commensalism

Choose the **correct** answer from the options given below :-
(1) A-I, B-II, C-IV, D-III
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-III, C-I, D-IV
(4) A-I, B-II, C-III, D-IV

BOP151

38. Match **List I** with **List II.**

	List-I		List-II
A.	Logistic growth	I.	Unlimited resource availability condition
B.	Exponential growth	II.	Limited resource availability condition
C.	Expanding age pyramid	III.	The percent individuals of pre-reproductive age is largest followed by reproductive and post reproductive age groups
D.	Stable age pyramid	IV.	The percent individuals of pre reproductive and reproductive age group are same

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-IV, C-III, D-I

(4) A-II, B-I, C-III, D-IV

BOP152

NEET-UG 2023 (Manipur)

39. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Hydrarch succession	(I)	Gradual change in the species composition
(B)	Xerarch succession	(II)	Faster and climax reached quickly
(C)	Ecological succession	(III)	Lichens to mesic conditions
(D)	Secondary succession	(IV)	Phytoplankton to mesic conditions

Choose the **correct** answer from the options given below :-

(1) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

BOP153

40. Plants offer rewards to animals in the form of pollen and nectar and the animals facilitate the pollination process. This is an example of :

(1) Amensalism (2) Competition

(3) Commensalism (4) Mutualism

BOP154

41. The species of plants that plays a vital role in controlling the relative abundance of other species in a community is called _____ .

(1) alien species

(2) endemic species

(3) exotic species

(4) keystone species

BOP155

42. For chemical defence against herbivores, *Calotropis* has :

(1) cardiac glycosides

(2) strychnine

(3) toxic ricin

(4) distasteful quinine

BOP156

NEET-UG 2024

43. The equation of Verhulst-Pearl logistic growth is $\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$

From this equation, K indicates :

(1) Intrinsic rate of natural increase

(2) Biotic potential

(3) Carrying capacity

(4) Population density

BOP170

44. **Statement-I:** Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.

Statement-II: According to Gause's principle, during competition, the inferior will be eliminated. This may be true if resources are limiting.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

BOP171

RE-NEET-UG 2024

45. When will the population density increase, under special conditions ?

(1) deaths exceeds number of births and also number of emigrants equals number of immigrants.

(2) births plus number of immigrants equals number of deaths plus number of emigrants.

(3) births plus number of emigrants is more than the number of deaths plus number of immigrants.

(4) births plus number of immigrants is more than the sum of number of deaths and number of emigrants.

BOP172

46. Match **List-I** with **List-II** :

	List-I		**List-II**
A.	Migratory flamingos and resident fish in South American lakes	I.	Interference Competition
B.	Abingdon tortoise became extinct after introduction of goats in their habitat	II.	Competitive release
C.	*Chathamalus* expands its distributional range in the absence of *Balanus*	III.	Resource Partitioning
D.	Five closely related species of Warblers feeding in different locations on same tree	IV.	Interspecific competition

(1) A-I, B-IV, C-III, D-II

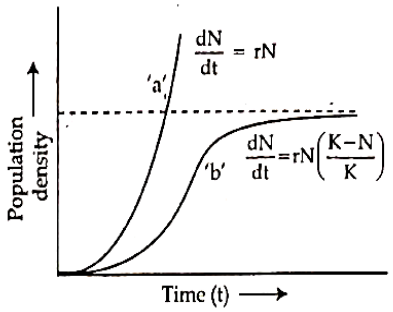
(2) A-IV, B-I, C-II, D -III

(3) A-III, B-I, C-II, D-IV

(4) A-II, B-IV, C-III, D-I

BOP173

47. What do 'a' and 'b' represent in the following population growth curve ?



(1) 'a' represents exponential growth when responses are not limiting the growth; and 'b' represents logistic growth when responses
are limiting the growth

(2) 'a' represents logistic growth when responses are not limiting the growth; 'b' represents exponential growth when responses are limiting the growth.

(3) 'a' represents carrying capacity and 'b' shows logistic growth when responses are limiting the growth.

(4) 'a' represents exponential growth when responses are not limiting the growth and 'b' shows carrying capacity.

BOP174

48. Match **List-I** with **List-II**.

	List-I		List-II
A.	Predator	I.	*Ophrys*
B.	Mutualism	II.	*Pisaster*
C.	Parasitism	III.	Female wasp and fig
D.	Sexual deceit	IV.	*Plasmodium*

Choose the **correct** answer from the options given below :-

(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-I, C-II, D -III

(3) A-II, B-III, C-I, D-IV

(4) A-II, B-III, C-IV, D-I

BOP175

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	1	2	1	3	1	4	1	1	3	3	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	3	4	4	1	2	4	1	1	3	3	3	2	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	1	2	2	1	4	4	4	4	4	1	3	4	4
Question	46	47	48												
Answer	2	1	4												

SOLUTION

12. Given :

$N_0 = 100$, $N_t = ?$, $r = 0.5$, $t = 6$ yrs

$N_0 = N_t e^{rt}$

$= 100 \times 2.72^{(0.5 \times 6)}$

$= 100 \times 2.72^3$

$= 100 \times 20.12$

$N_t = 2012$

27. (a) Polar seal (Cold climate) has shorter limbs to reduce heat loss (Allen's Rule).

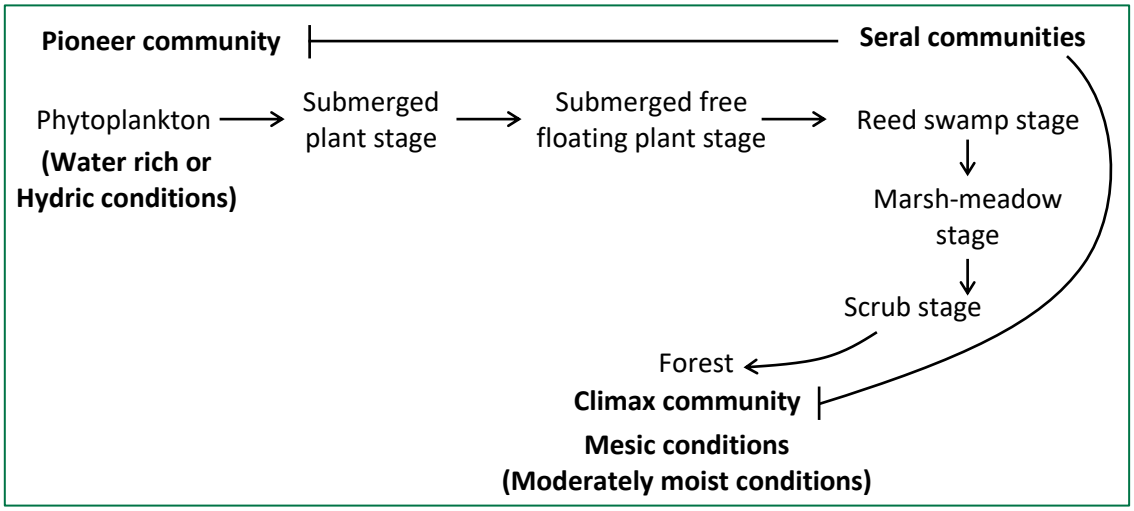
(b) Kangaroo rat use metabolic water for its requirement. It is a physiological adaptation

(c) Desert lizard comes out of burrow when its body temperature goes down and goes back to the burrow after sun-bath. It is a behavioural adaptation.

(d) Marine fishes at depth has special proteins that help in their survival at greater depths and it is a physiological adaptation.

32. **Ecological succession is the process by which plant and animal communities gradually change over time.**

Succession that takes place in water body is called hydrarch succession, during hydrarch / hydrosere occurring in a newly formed water body following gradual and sequential changes takes place.



33. Pioneer species are the first organisms to colonize a new area, such as a barren environment or a disturbed ecosystem.

34. The succession leads to mesic (medium) conditions to attain more biodiversity and stability.

39. (A) Hydrarch succession — Succession in water body.

(B) Xerarch succession — Succession dry environments, such as deserts and sand dunes.

(C) Ecological succession — Gradual change in the species composition.

(D) Secondary succession — Process by which plants and animals recolonize a disturbed area, using the remaining soil and nutrients to rebuild the ecosystem.

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Select the statement which explains best parastism :-
(1) One organism is benefited
(2) Both the organism are benefited
(3) One organism is benefited, other is not affected
(4) One organism is benefited, other is affected

BOP122

2. The role of an organism in the ecological system is known as
(1) Habitat (2) Herbivory
(3) Niche (4) Interaction

BOP123

3. Select the statement which best explains commensalism: -
(1) One organism is benefited
(2) Both the organisms are benefited
(3) One organism is benefitted, other is not affected
(4) One organism is benefitted, other is affected.

BOP124

4. Community is defined as aggregation of: -
(1) Individuals of the same kind
(2) Individuals of same species
(3) Individuals of a population
(4) Populations of different species.

BOP125

5. Parasite can be explained as an organism which depends on others: -
(1) For food
(2) For shelter
(3) For both food and shelter
(4) For Reproduction

BOP126

6. An orchid plant growing on the branch of mango tree, what is the interaction between orchid & mango ?
(1) Parasitism
(2) Commensalism
(3) Protocoperation
(4) Mutualism

BOP127

7. Mark the important defence mechanism in plants against herbivory :-
(1) Spines (2) Toxic Chemical
(3) Both (1) & (2) (4) None of these

BOP128

8. Species A (–) & Species B(o) shows the following interaction: -
(1) Amensalism (2) Predation
(3) Mutualism (4) Competition

BOP129

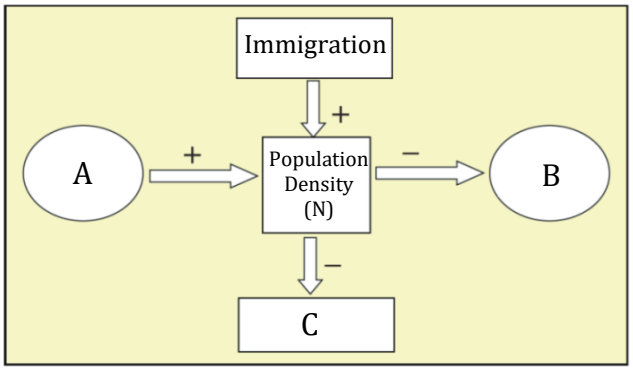
9. The age pyramid with broad base indicates:
(1) High percentage of young individuals
(2) Low percentage of young individuals
(3) High percentage of old individuals
(4) Low percentage of old individuals

BOP130

10. In a decline population of a country:
(1) Number of pre reproductive is more than reproductive.
(2) Number of pre reproductive is less than reproductive.
(3) Number of pre reproductive is equal to reproductive.
(4) Reproductive are less than post reproductive.

BOP131

11. In the diagram given above, which of the following option correctly represents A, B and C.



(1) A = Death rate, B = Birth rate, C = Emigration

(2) A = Birth rate, B = Death rate, C = Emigration

(3) A = Emigration, B = Death rate, C = Birth rate

(4) A = Death rate, B = Emigration, C = Birth rate

BOP132

12. Which of the following equation is/are correct for the population density (N) at time t + 1 ?

N = Density at time 't'

B = Natality

D = Mortality

I = Immigration

E = Emigration

(A) $N_{(t+1)} = N_t + [(B + I) – (D + E)]$

(B) $N_{(t+1)} = N_t + [(B – D) + (I – E)]$

(C) $N_{(t+1)} = N_t + [(B + I) + (D + E)]$

(D) $N_{(t+1)} = N_t – [(B – D) + (I – E)]$

(1) Only A

(2) Only A and B

(3) Only C

(4) A, B, C and D

BOP133

13. In a population there are higher number of pre-reproductive individuals, moderate number of reproductive individuals and less post reproductive individuals are present. This type of population represents: -

(1) Population of developed countries

(2) Population of developing country

(3) Stable growth

(4) Declining population

BOP134

14. In a new habitat which is just being colonised which will play significant role in population growth: -

(1) Birth rate (2) Emigration

(3) Migration (4) Immigration

BOP135

(B) ANALYTICAL QUESTIONS

15. What will happen if the number of organism increased at a place: -

(1) Inter species competition

(2) Intra species competition

(3) Both

(4) None

BOP137

16. Which of the following is a correct pair:

(1) *Cuscuta* - parasite

(2) *Dischidia* - insectivorous

(3) *Opuntia* - predator

(4) *Capsella* - hydrophyte

BOP139

17. Which one of the following correctly represents an organism and its ecological niche?

(1) ***Vallisneria*** and pond

(2) Desert locust ***(Schistocerca)*** and desert

(3) Plant lice (aphids) and leaf

(4) Vultures and dense forest

BOP141

18. In a population birth rate is 0.15 and death rate is 0.08 during a unit time period. What is the value of r (intrinsic rate of natural increase) for given population?

(1) 0.23 (2) 0.07

(3) 0.05 (4) 0.25

BOP142

19. A population has more young individuals, compared to older individuals. What would be the status of the population after some years:

(1) It will decline

(2) It will stabilize

(3) It will first decline and then stabilize

(4) It will increase

BOP143

20. In a month of January Siberian cranes migrate from Russia to India for breeding, a survey was done till December total population of Siberian cranes = 1200

Birth rate = 400

Mortality rate = 200

Number of cranes immigrated = 600

Number of cranes emigrated = 300

Calculate the total population

(1) 1500 (2) 1000

(3) 2000 (4) 1700

BOP144

21. If a population of 50 *Paramoecium* present in a pool increase to 150 after an hour, what would be the growth rate of population ?

(1) 50 per hours

(2) 200 per hour

(3) 5 per hour

(4) 100 per hour

BOP145

22. Lichens are well known combination of an alga and a fungus where fungus has :

(1) An epiphytic relationship with the alga

(2) A parasitic relationship with the alga

(3) A symbiotic relationship with the alga

(4) A saprophytic relationship with the alga

BOP146

23. Which one of the following is a matching pair of certain organisms(s) and the kind of association

(1) Shark and sucker fish- amensalism

(2) Algae and fungi in lichens-mutualism

(3) Orchids growing on trees-parasitism

(4) ***Cuscuta*** (dodder) growing on other flowering plants-predation

BOP147

24. The table below gives the populations (in thousands) of ten species (A – J) in four areas (a – d) consisting of the number of habitats given within brackets against each. Study the table and answer the question which follows: -

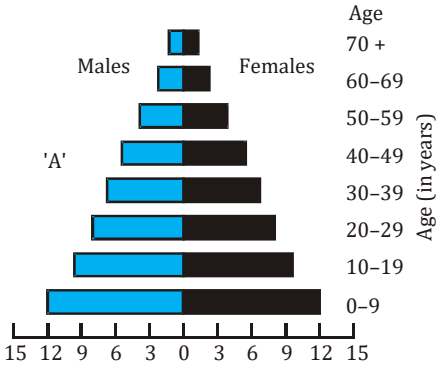
Area and Number of habitats	Species and their populations (in thousands) in the areas									
	A	B	C	D	E	F	G	H	I	J
a (11)	2.3	1.2	0.52	6	-	3.1	1.1	9	-	10.3
b (11)	10.2	-	0.62	-	1.5	3	-	8.2	1.1	11.2
c (13)	11.3	0.9	0.48	2.4	1.4	4.2	0.8	8.4	2.2	4.1
d (12)	3.2	10.2	11.1	4.8	0.4	3.3	0.8	7.3	11.3	2.1

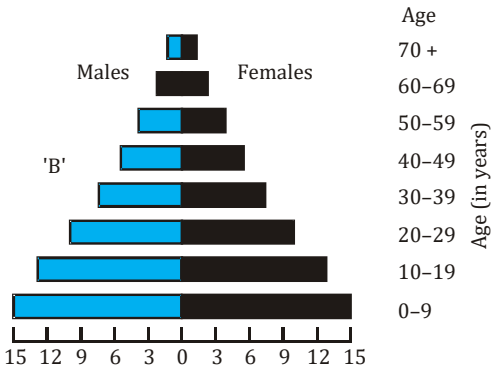
Which area out of a to d shows maximum species diversity?

(1) d (2) a (3) b (4) c

BOP070

25. A country with a high rate of population growth took measures to reduce it. The figure below shows age-sex pyramids of populations A and B twenty years apart. Select the correct interpretation about them:





(1) "A" is the earlier pyramid and no change has occurred in the growth rate

(2) "A" is more recent and shows slight reduction in the growth rate

(3) "B" is the earlier pyramid and shows stabilised growth rate

(4) "B" is more recent showing that population is very young

BOP072

26. Study the four statements (a-d) given below and select the two correct ones out of them

(a) A lion eating a deer and a sparrow feeding on grain are ecologically similar in being consumers

(b) Predator star fish Pisaster helps in maintaining species diversity of some invertebrates

(c) Predators ultimately lead to the extinction of prey species

(d) Production of chemicals such as nicotine, strychnine by the plants are metabolic disorders

The two correct statements are :

(1) (a) and (b) (2) (b) and (c)

(3) (c) and (d) (4) (a) and (d)

BOP074

27. Which one of the following is most appropriately defined?

(1) Amensalism is a relationship in which one species is benefited whereas the other is unaffected.

(2) Predator is an organism that catches and kills other organism for food.

(3) Parasite is an organism which always lives inside the body of other organism and may kill it.

(4) Host is an organism which provides food to another organism.

BOP077

28. A sedentary sea anemone gets attached to the shell lining of hermit crab. The association is :

(1) Amensalism (2) Ectoparasitism

(3) Symboisis (4) Commensalism

BOP086

29. Animals eating plants are categorised separately as herbivores, they are in a broad ecological context not very different from :-

(1) Parasite (2) Predator

(3) Commensal (4) Amensal

BOP157

30. Which statements are true about inter specific interactions ?

(i) One is benefitted and other is uneffected in mutualism.

(ii) Both partners are benefitted in commensalism.

(iii) One kills and eats another in predation.

(iv) Both partners can be benefitted in symbiosis.

(1) (i) and (ii) only (2) (i) and (iii) only

(3) (ii) and (iii) only (4) (iii) and (iv) only

BOP158

31. They may reduce the survival, growth and reproduction of the other organism and reduce its population density, they might render the organism more vulnerable to predation by making it physically weak, they are :-

(1) Predator (2) Parasite

(3) Amensal (4) Commensal

BOP159

32. Choose the correct match :-

(a) Co-evolution – Endoparasite & Host

(b) Mutualism – Relation of algae & fungi

(c) Sexual deceit – Shown by *Opuntia plant*

(d) Marine fishes and Copepods – Parasitism

(1) Only a & b (2) Only b & c

(3) Only c & d (4) a, b & d

BOP160

33. Select the incorrect statement :-

(1) Carnivores appear to be more adversely affected by competition than Herbivores

(2) Species diversity in a community depends upon size of area

(3) The cattle egret and grazing cattle is an example of commensalism

(4) Clear stratification is found in tropical rain forest

BOP161

34. How many on the given population interections are positive type of interaction ?

Orchids and bees, fig and wasp, barnacles and whale, goat and abingdon tortoise, flamingoes and fish.

(1) One (2) Two

(3) Three (4) Four

BOP162

35. You can never see any cattle or goats browsing on *Calotropis* because :-

(1) They produce highly poisonous cardiac glycosides

(2) They have spines instead of leaves

(3) They are distasteful to their herbivores

(4) They are too long in heights to reach by herbivores

BOP163

36. **Assertion:** Some organism breed only once in their lifetime while others breed many times during their lifetime.

Reason: Population evolve to minimise their reproductive fitness.

(1) Both assertion and reason are true and the reason is the correct explanation of the assertion

(2) Both assertion and reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

BOP164

37. Why predator are prudent in nature ?

(1) They keep prey population under control

(2) They maintain species diversity

(3) They do not over exploit its prey

(4) They are too efficient to kill their pray.

BOP165

38. Competition for food, light and space is most severe between two

(1) Closely related species growing in the same area.

(2) Closely related species growing in different habitat.

(3) Distantly related species growing in the same habitat.

(4) Distantly related species growing in different habitat.

BOP166

39. Read the following statement for population interaction and find out correct option :

(A) Some species of insects and frogs are cryptically coloured to avoid being detected easily by the predator.

(B) Lichens represent a mutualistic relationship between a fungus and photosynthesising algae.

(C) Amensalism is 0, –type of interspecific interaction.

(1) Only A, C is correct

(2) Only B, C is correct

(3) Only C is incorrect

(4) All are correct

BOP167

40. Consider the following four statements and select the option which includes all the correct ones:-

(a) Sparrow eating any seed is not less than a predator.

(b) Commensalism is +/– type of relationship.

(c) Nearly 25% of all insects are known to be phytophagous.

(d) Mutualism is +/+ type of relationship.

(1) Statements- (a), (c), (d)

(2) Statements- (a), (b)

(3) Statements- (a), (b), (c)

(4) Statements- (b), (d)

BOP168

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	4	3	2	3	1	1	2	2	2	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	2	4	4	4	3	2	1	2	1	2	4	2	4
Question	31	32	33	34	35	36	37	38	39	40					
Answer	2	4	1	3	1	3	3	1	4	1					

SOLUTION

18. b = 0.15 ,

d = 0.08

r = (b – d)

r = 0.15 – 0.08

r = 0.07

20. N_t = 1200 , I = 600

B = 400 , E = 300

D = 200 , $N_{(t+1)}$ = ?

$N_{(t+1)} = N_t = [(B+I) – (D+E)]$

= 1200 + [(400 + 600) – (200 + 300)]

= 1200 + [1000 – 500]

$N_{(t+1)}$ = 1200 + 500 = 1700

21. Initial population = 50

Population after an hour = 150

Increase in population in an hour

= 150 – 50 = 100

So, growth rate = 100 per hour

24. Area ‘d’ contains members of all 10 species and has less number of habitats (12) than area ‘c’.

25. Pyramid ‘A’ shows less number of pre-reproductive population than pyramid ‘B’.

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

1. Ecosystem term coined by :-
(1) Odum (2) Mishra
(3) Reiter (4) Tansley
BES001

2. Which one is not a functional aspect of ecosystem?
(1) Energy flow (2) Productivity
(3) Decomposition (4) Stratification
BES002

3. The minimum energy is stored at which of the following trophic level in any ecosystem
(1) Carnivores (2) Herbivores
(3) Producers (4) Decomposers
BES003

4. The source of energy in an ecosystem is :-
(1) Sunlight (2) DNA
(3) ATP (4) RNA
BES004

5. Ecosystem may be defined as :-
(1) A localized association of several plants and animals
(2) Different communities of plants, animals and microbes together with their physico-chemical environment.
(3) Different communities of plants and microbes along with their physico-chemical environment
(4) None of the above
BES005

6. The importance of ecosystem lies in: -
(1) Flow of energy
(2) Cycling of materials
(3) Both the above
(4) None of the above
BES006

7. Ecosystem is:-
(1) Any functional unit that includes the whole community in a given area interacting with the abiotic factors
(2) A group of green plants
(3) A group of animals interacting with environment
(4) Man and pets living together
BES007

8. Who proposed that ecosystem is symbol of structure & function of nature :-
(1) Gardner (2) Odum
(3) Tansley (4) Reiter
BES008

9. Largest ecosystem of the world are: -
(1) Forests (2) Grass lands
(3) Great lakes (4) Oceans
BES009

10. Which of the following is a manmade artificial ecosystem?
(1) Grassland ecosystem
(2) Forest ecosystem
(3) Ecosystem of artificial lakes & dams
(4) None of these
BES010

11. Nepenthes (Insectivorous pitcher plant) is:
(1) Producer (2) Consumer
(3) Both (1) & (2) (4) None of these
BES011

12. Which one is omnivorous?
(1) Cow (2) Lion
(3) Deer (4) Man
BES012

13. Which biotic components mainly help in recycling of minerals?
(1) Producers (2) Consumers
(3) Decomposers (4) All the above
BES013

14. Trophic levels are formed by :-
(1) Only plants
(2) Only carnivores
(3) Only animals
(4) Organisms linked in food chain
BES014

15. In a forest ecosystem green plants are :-
(1) Primary producers
(2) Consumers
(3) Primary consumers
(4) Decomposers
BES015

16. In an ecosystem the function of the producers is to :-
(1) Convert organic compounds into inorganic compounds
(2) Trap solar energy and convert it into chemical energy
(3) Utilize chemical energy
(4) Release energy
BES016

17. With regard to ecological food chain, man is a :-
(1) Consumer
(2) Producer
(3) Both consumer & producer
(4) Decomposer
BES017

18. A plant, being eaten by a herbivore which in turn is eaten by a carnivore makes :-
(1) Food chain (2) Web of Food
(3) Omnivores (4)Interdependence
BES018

19. When peacock, eats snake which eats insects depends on green plants, the peacock is?
(1) A primary consumer
(2) A primary decomposer
(3) A final decomposer of plants
(4) The apex of the food pyramid
BES019

20. If we completely remove decomposers from an ecosystem, the ecosystem functioning will be adversely affected because: -
(1) Mineral movement will be blocked
(2) Herbivores will not receive solar energy
(3) Energy flow will be blocked
(4) Rate of decomposition of other components will be very high
BES020

21. Path of energy flow in an ecosystem is :-
(1) Herbivorous→ producer → carnivorous → decomposer
(2) Herbivorous→ carnivorous → producer → decomposer
(3) Producer→ carnivorous → herbivorous → decomposer
(4) Producer→ herbivorous → carnivorous → decomposer
BES021

22. Pyramids of energy are :-
(1) Always upright (2) Always Inverted
(3) Mostly upright (4) Mostly inverted
BES022

23. The ecological pyramid of numbers in pond ecosystem is :-
(1) Upright
(2) Inverted
(3) May upright or Inverted
(4) First upright then inverted
BES023

24. An ecosystem resists change because it is in a state of :-
(1) Homoeostasis
(2) Regular Illumination
(3) Static Imbalance
(4) Food accumulation
BES024

25. What is true about any ecosystem?
(1) It is self-regulatory
(2) It is self-sustained
(3) Top carnivores have climax trophic level position
(4) All of these above

BES025

26. The Pyramid of numbers in grassland ecosystem will be :-
(1) Upright (2) Inverted
(3) Irregular (4) Linear

BES026

27. Which ecosystem has maximum number of producers in an unit area?
(1) Pond (2) Grassland
(3) Forest (4) Tundra

BES027

28. The storage of energy at consumer level is known as :-
(1) Gross primary production
(2) Secondary productivity
(3) Net primary productivity
(4) Net productivity

BES028

29. Gross primary productivity is :-
(1) Rate at which organic molecules are formed in an autotroph
(2) Rate at which organic molecules are used up by an autotroph
(3) Storage of organic molecules in the body of an autotroph
(4) Rate at which organic molecules are transferred to next higher trophic level

BES029

30. Carbon cycle includes (the following is a logical sequence) :-
(1) Producer – consumer – decomposer
(2) Decomposer – consumer – producer
(3) Producer – decomposer – consumer
(4) Consumer – producer – decomposer

BES030

31. The flow of materials from non-living components to living components and back to the non-living components in a more or less cyclic manner is called a :-
(1) Gaseous cycle
(2) Sedimentary cycle
(3) Biogeochemical cycle
(4) Hydrologic cycle

BES031

32. A group of interconnected food chains is called:
(1) Pyramid of energy
(2) Food web
(3) Food cycle
(4) Complex food chain

BES032

33. Which of these is the food chain which runs from larger to smaller organisms?
(1) Grazing food chain
(2) Saprotrophic food chain
(3) Predatory food chain
(4) Parasitic food chain

BES033

34. A food chain starts from: -
(1) Nitrogen fixing organisms
(2) Photosynthetic organisms
(3) Respiration
(4) Decomposers

BES034

35. The functional unit of nature is called: -
(1) Community (2) Ecosystem
(3) Population (4) Succession

BES035

36. Primary productivity of plants in an ecosystem depends on :-
(1) A variety of environmental factors
(2) Availability of nutrients
(3) Photosynthetic capacity of plants
(4) All of the above

BES036

37. The annual net primary productivity of land is about :-
(1) 170 billion tons (2) 70 billion tons
(3) 55 billion tons (4) 115 billion tons
BES037

38. Which of the following is included in decomposition?
(1) Fragmentation (2) Leaching
(3) Catabolism (4) All of the above
BES038

39. Decomposition rate is slow if detritus is rich in :-
(1) Nitrogen (2) Humus
(3) Sugars (4) Lignin and Chitin
BES039

40. Match the columns I, II and III, and choose the correct combination from the options given :-

column I		Column II		Column III	
a.	1^0 Producer	1.	T_1	K.	Grasshopper
b.	1^0 consumer	2.	T_2	L.	Wolf
c.	2^0 Consumer	3.	T_3	M.	Lion
d.	3^0 consumer	4.	T_4	N.	Phytoplankton

(1) a - 1 - K, b - 2 - L, c - 3 - M, d - 4 - N
(2) a - 1 - N, b - 2 - M, c - 3 - K, d - 4 - L
(3) a - 1 - N, b - 2 - K, c - 3 - L, d - 4 - M
(4) a - 2 - N, b - 3 - K, c - 4 - L, d - 1 - M
BES040

41. A plant receives 10^6 J of sunlight energy. The energy reached at the tertiary consumer will be: -
(1) 1000 J (2) 100 J
(3) 10 J (4) 1 J
BES041

42. An inverted pyramid of biomass can be found in which ecosystem: -
(1) Forest (2) Pond
(3) Grass land (4) Tundra
BES042

43. In an ecosystem:
(1) Primary producers are more than primary consumers
(2) Primary consumers are larger than primary producers
(3) Secondary consumers are larger than primary producers
(4) Primary consumers are least depend on primary producers
BES043

44. Large ecosystems are called -
(1) Biomes (2) Ecotone
(3) Ecads (4) Biocoenosis
BES045

45. Which one is not a functional aspect of ecosytem ?
(1) Energy flow (2) Productivity
(3) Decomposition (4) Stratification
BES046

46. Vultures in an ecosystem are :
(1) Predators (2) Scavangers
(3) Consumers (4) Top carnivores
BES047

47. The maximum energy is stored at which of the following trophic level in any ecosystem
(1) Producers (2) Herbivores
(3) Carnivores (4) Top carnivores
BES048

48. Limiting factor for productivity in lake is :-
(1) Phosphorous (2) Nitrogen
(3) Carbon (4) Sulphur
BES163

49. In an ecosystem, energy flow is always :-
(1) Bidirectional
(2) Unidirectional
(3) In all directions
(4) More than one option is correct
BES164

50. What is formed when a biological community is integrated with its physical environment?
(1) Biosphere (2) Biome
(3) Ecosystem (4) Population
BES165

51. Based on ecology which is not essential for an ecosystem to self-sustain-
(1) Consumers (2) Producers
(3) Decomposers (1) Both (1) and (2)
BES166

52. **Statement-1 :** Death of organism is the beginning of the detritus food chain/web.
Statement 2 : Amount of energy decreases at successive trophic levels.
(1) Both statement 1 and 2 are incorrect
(2) Statement 1 is correct but statement 2 is incorrect
(3) Statement 1 is incorrect but statement 2 is correct
(4) Both statement 1 and 2 are correct
BES167

53. A pond is a :-
(1) Biome
(2) Natural ecosystem
(3) Artificial ecosystem
(4) Community of plants & animals
BES056

54. Which of the following are not given any place in ecological pyramids –
(1) Herbivores (2) Carnivores
(3) Producers (4) Saprophytes
BES168

55. The bulk of nitrogen in nature is fixed by -
(1) Lighting
(2) Chemical industries
(3) Denitrifying bacteria
(4) Symbiotic bacteria
BES063

56. A good soil is that which -
(1) holds whole of the water entering into it
(2) Allows limited amount of water into it
(3) Allows the water to percolate slowly into it
(4) Allows the water to pass very quickly from it
BES068

57. The soil near the surface is usually darker then the soil about one meter down. This is because the top soil is
(1) Young & wet
(2) Richer in organic matter
(3) Richer in Ca & Mg
(4) Dry
BES069

58. A soil is said to be fertile when -
(1) It is rich in organic matter
(2) It has capacity to hold water
(3) It has a capacity to hold nutrients
(4) It holds water & all essential nutrients in a definite proportion
BES070

59. Forests near equator region are called -
(1) Deciduous
(2) Tropical rain forests
(3) Coniferous forests
(4) Temperate forests
BES072

60. All the living organisms and non-living factors of the earth constitute -
(1) Biosphere
(2) Community
(3) Biome
(4) Association
BES073

61. The term biosphere is used for the zone of the earth where life exists -
(1) On the lithospere
(2) In the hydrosphere
(3) In the lithosphere and hydrosphere
(4) In the lithosphere, hydrosphere and atmosphere

BES074

62. A biosphere is composed of
(1) Living organisms
(2) Living organisms + Lithosphere
(3) Living organisms + lithosphere + atmosphere
(4) Living organisms + lithosphere + atmosphere + hydrosphere

BES082

63. Which one of the following is *not* used for construction of ecological pyramids?
(1) Rate of energy flow
(2) Fresh weight
(3) Dry weight
(4) Number of individuals

BES083

64. Which one of the following ecosystem types has the highest annual net primary productivity?
(1) Temperate deciduous forest
(2) Tropical rain forest
(3) Tropical deciduous forest
(4) Temperate evergreen forest

BES084

65. Which one of the following types of organisms occupy more than one trophic level in a pond ecosystem?
(1) Frog (2) Phytoplankton
(3) Fish (4) Zooplankton

BES086

66. The biomass available for consumption by the herbivores and the decomposers is called :-
(1) Gross primary productivity
(2) Net primary productivity
(3) Secondary productivity
(4) Standing crop

BES087

67. Mass of living matter at a trophic level in an area at any time is called: -
(1) Standing crop (2) Detritus
(3) Humus (4) Standing state

BES089

68. Of the total incident solar radiation, the proportion of PAR is: -
(1) About 70% (2) About 60%
(3) Less than 50% (4) More than 80%

BES090

69. Which one of the following animals may occupy more than one trophic level in the same ecosystem at the same time?
(1) Frog (2) Sparrow
(3) Lion (4) Goat

BES093

70. The breakdown of detritus into smaller particles by earthworm is a process called
(1) Catabolism (2) Humification
(3) Fragmentation (4) Mineralisation

BES094

71. Which one of the following is not a functional unit of an ecosystem: -
(1) Productivity
(2) Species diversity
(3) Energy flow
(4) Decomposition

BES095

72. Identify the possible link "A" in the following food chain: -

Plant → Insect → Frog → "A" → Eagle

(1) Cobra (2) Parrot

(3) Rabbit (4) Wolf

BES099

73. Secondary productivity is rate of formation of new organic matter by :-

(1) Decomposer (2) Producer

(3) Parasite (4) Consumer

BES103

74. Which one of the following is functional characteristics of ecosystem ?

(1) Stratification

(2) Productivity

(3) Species dominancy

(4) Species diversity

BES187

75. In an ecosystem :-

(1) Primary producers are more than primary consumers.

(2) Primary consumers are always larger than primary producers.

(3) Secondary consumers are always larger than primary producers.

(4) Primary consumers are least depend on primary producers.

BES188

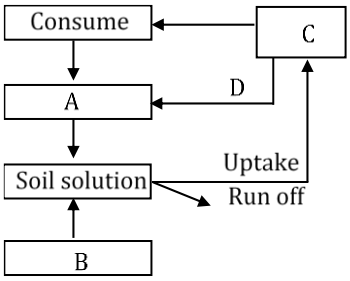
EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	1	1	2	3	1	3	4	3	3	4	3	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	1	4	1	4	1	1	1	4	1	1	2	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	4	2	2	4	4	4	4	3	3	2	1	1	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	1	2	3	1	4	2	4	4	3	2	4	2	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	4	2	2	3	2	1	3	2	3	2	1	4	2	1

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Given below is a simplified model of phosphorus cycling in a terrestrial ecosystem with four blanks (A-D). Identify the blanks :-



	A	B	C	D
(1)	Rock minerals	Detritus	Litter fall	Producers
(2)	Litter fall	Producers	Rock minerals	Detritus
(3)	Detritus	Rock minerals	Producer	Litter fall
(4)	Producers	Litter fall	Rock minerals	Detritus

BES105

2. If 20 J of energy is trapped at producer level, then how much energy will be available to peacock as food in the following chain?

plant → mice → snake → peacock

(1) 0.02 J (2) 0.002 J

(3) 0.2 J (4) 0.0002 J

BES106

AIPMT 2015

3. The mass of living material at a trophic level at a particular time is called :-

(1) Standing state

(2) Net primary productivity

(3) Standing crop

(4) Gross primary productivity

BES107

4. In an ecosystem the rate of production of organic matter during photosynthesis is termed as :-

(1) Gross primary productivity

(2) Secondary productivity

(3) Net productivity

(4) Net primary productivity

BES108

5. Secondary Succession takes place on/in :-

(1) Degraded forest

(2) Newly created pond

(3) Newly cooled lava

(4) Bare rock

BES109

6. In which of the following both pairs have corrected combination?

(1)	Gaseous nutrient cycle	Sulphur and Phosphorus
	Sedimentary nutrient cycle	Carbon and Nitrogen
(2)	Gaseous nutrient cycle	Carbon and Nitrogen
	Sedimentary nutrient cycle	Sulphur and Phosphorus
(3)	Gaseous nutrient cycle	Carbon and Sulphur
	Sedimentary nutrient cycle	Nitrogen and Phosphorus
(4)	Gaseous nutrient cycle	Nitrogen and Sulphur
	Sedimentary nutrient cycle	Carbon and Phosphorus

BES110

7. During ecological succession :-

(1) The changes lead to a community that is in near equilibrium with the environment and is called pioneer community

(2) The gradual and predictable change in species composition occurs in a given area

(3) The establishment of a new biotic community is very fast in its primary phase

(4) The numbers and types of animals remain constant

BES111

NEET-I 2016

8. The term ecosystem was coined by :-

(1) E. P. Odum (2) A. G. Tansley

(3) E. Haeckel (4) E. Warming

BES112

NEET-II 2016

9. The primary producers of the deep-sea hydrothermal vent ecosystem are :-

(1) Blue-green algae

(2) Coral reefs

(3) Green algae

(4) Chemosynthetic bacteria

BES113

NEET-UG 2017

10. Which ecosystem has the maximum biomass?

(1) Grassland ecosystem

(2) Pond ecosystem

(3) Lake ecosystem

(4) Forest ecosystem

BES114

11. The sequential events from initial stage till climax stage in a succession are called :-

(1) Ecesis

(2) Sere

(3) Nudation

(4) Migration

BES115

NEET-UG 2019

12. Which of the following ecological pyramids is generally inverted?

(1) Pyramid of numbers in grassland

(2) Pyramid of energy

(3) Pyramid of biomass in a forest

(4) Pyramid of biomass in a sea

BES116

13. Which of the following statements is incorrect?

(1) Biomass decreases from first to fourth trophic level

(2) Energy content gradually increases from first to fourth trophic level

(3) Number of individuals decreases from first trophic level to fourth trophic level

(4) Energy content gradually decreases from first to fourth trophic level

BES117

14. The rate decomposition is faster in the ecosystem due to following factors EXCEPT

(1) Detritus rich in sugars

(2) Warm and moist environment

(3) Presence of aerobic soil microbes

(4) Detritus richer in lignin and chitin

BES118

NEET-UG 2020

15. In relation of Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is correct?

(1) There is no relationship between Gross primary productivity and Net primary productivity.

(2) Gross primary productivity is always less than net primary productivity.

(3) Gross primary productivity is always more than net primary productivity.

(4) Gross primary productivity and Net primary productivity are one and same.

BES119

NEET 2021

16. **Assertion (A) :** A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) : Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

BES189

NEET-UG 2021

17. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as

(1) Climax

(2) Climax community

(3) Standing state

(4) Standing crop

BES120

18. Which of the following statements is not correct?

(1) Pyramid of biomass in sea is generally inverted.

(2) Pyramid of biomass in sea is generally upright.

(3) Pyramid of energy is always upright.

(4) Pyramid of numbers in a grassland ecosystem is upright.

BES121

19. In the equation GPP - R = NPP

R represents :-

(1) Radiant energy

(2) Retardation factor

(3) Environment factor

(4) Respiration losses

BES122

NEET-UG 2022

20. The amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis is called:

(1) Secondary production

(2) Primary production

(3) Gross primary production

(4) Net primary production

BES123

21. Which one of the following will accelerate phosphorus cycle?

(1) Volcanic activity

(2) Weathering of rocks

(3) Rain fall and storms

(4) Burning of fossil fuels

BES124

22. Match the List-I with List-II

List-I		List-II	
(a)	Carbon dissolved in oceans	(i)	55 billion tons
(b)	Annual fixation of carbon through photosynthesis	(ii)	71%
(c)	PAR captured by plants	(iii)	4×10^{13} kg
(d)	Productivity of oceans	(iv)	2 to 10%

Choose the correct answer from the options given below:

(1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(3) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(4) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

BES125

NEET-UG 2023

23. Identify the **correct** statements :

A. Detrivores perform fragmentation.

B. The humus is further degraded by some microbes during mineralization.

C. Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.

D. The detritus food chain begins with living organisms.

E. Earthworms break down detritus into smaller particles by a process called catabolism.

Choose the **correct** answer from the option given below:

(1) B, C, D only

(2) C, D, E only

(3) D, E, A only

(4) A, B, C only

BES159

24. In the equation

GPP – R = NPP

GPP is Gross Primary Productivity

NPP is Net Primary Productivity

R here is ____.

(1) Respiratory quotient

(2) Respiratory loss

(3) Reproductive allocation

(4) Photosynthetically active radiation

BES160

NEET-UG 2023 (Manipur)

25. The amount of nutrients such as carbon, nitrogen, potassium and calcium present in the soil at any given time is referred to as :

(1) Standing state

(2) Standing crop

(3) Humus

(4) Detritus

BES161

26. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A person goes to high altitude and experiences "Altitude Sickness" with symptoms like breathing difficulty and heart palpitations.

Reason (R) : Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **(A)** is true but **(R)** is false

(2) **(A)** is false but **(R)** is true

(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(4) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

BES162

NEET-UG 2024

27. In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is 100 *x (kcal* m^{-2}*)* yr^{-1}, what would be the GPP (Gross Primary Productivity) of the third trophic level of the same ecosystem ?

(1) $\frac{x}{10}(kcal\,m^{-2})yr^{-1}$

(2) *x(kcal* m^{-2}*)*yr^{-1}

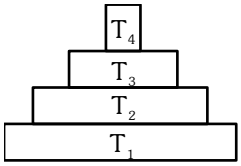
(3) 10*x(kcal* m^{-2}*)*yr^{-1}

(4) $\frac{100x}{3x}(kcal\,m^{-2})yr^{-1}$

BES169

RE-NEET-UG 2024

28. Consider the pyramid of energy of an ecosystem given below :



If T_4 is equivalent to 1000J, what is the value at T_1.

(1) $\frac{10000}{10}$ J (2) $\frac{10000}{10} \times 4$ J

(3) 10,000 J (4) 10,00,000 J

BES170

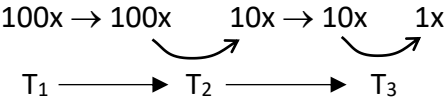
EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	3	1	1	2	2	2	4	4	2	4	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	1	3	2	4	2	2	3	4	2	1	3	3	4		

SOLUTION

16. If we go high altitude place we experience 'Altitude sickness' it's symptoms include nausea, fatigue & heart palpitation. This is because of low atmospheric pressure of high altitudes the body does not get enough oxygen. The body compensates low oxygen availability by increasing red blood cell production, decreasing the binding capacity of haemoglobin by increasing breathing rate. This is gradual physiological adjustment (Acclimatisation) of organism (human being) to the slowly changing new environmental conditions.

27. NPP = 100x Kcal/m^2/year



So 10x Kcal/m^2/year is available to third trophic level as gross productivity at third tropic level.

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Adaptation to low temperature and freezing in animals occurs due to the production of :-
(1) Antifreeze proteins (2) Chaperonins
(3) Proline (4) Analine
BES126

2. Which one of the following has the largest population in a food chain ?
(1) Producers
(2) Primary consumers
(3) Secondary consumers
(4) Decomposers
BES127

3. The second trophic level in a lake is :-
(1) Phytoplankons (2) Zooplanktons
(3) Benthos (4) Fishes
BES128

4. Secondary producers are :-
(1) Herbivores
(2) Producers
(3) Carnivores
(4) None of the above
BES129

5. What is the percentage of photosynthetically active radiation (PAR) in the incident solar radiations ?
(1) 100 % (2) 50 %
(3) 1–5 % (4) 2 – 10 %
BES130

6. Which of the following is an abiotic components of the ecosystem ?
(1) Bacteria (2) Humus
(3) Plants (4) Fungi
BES131

7. Which of the following processes helps in Nutrient conservation ?
(1) Mineralisation (2) Immobilisation
(3) Leaching (4) Nitrification
BES132

8. Which of the following represents the sedimentary type of nutrient cycle ?
(1) Sulphur (2) Phosphorus
(3) Nitrogen (4) Oxygen
BES133

9. Photosynthetically active radiation (PAR) represents the following range of wave length.
(1) 400 – 700 nm (2) 500 – 600 nm
(3) 450 – 950 nm (4) 340 – 450 nm
BES134

10. The detritus food chain (DFC) begins with :-
(1) Producers (2) Grasses
(3) Dead organic matter (4) Humus
BES171

11. Rate of capture of solar energy or total production of organic matter is called as-
(1) Gross primary productivity
(2) Gross secondary productivity
(3) Net primary productivity
(4) Secondary productivity
BES172

12. A sparrow is a _____ consumer when it eats seeds, fruits, peas, and a _____ consumer when it eats insects and worms.
(1) Secondary, Primary
(2) Primary, Secondary
(3) Tertiary, Secondar
(4) Primary, Tertiary
BES173

(B) ANALYTICAL QUESTIONS

13. Total amount of energy trapped by green plants in food is called :-
(1) Gross primary production
(2) Net primary production
(3) Standing crop
(4) Standing state

BES140

14. Bacteria are essential in carbon cycle as :-
(1) Decomposer (2) Synthesizer
(3) Consumer (4) Pri. Producer

BES141

15. In which biome a new plant may adapt soon:-
(1) Tropical rain forest (2) Desert
(3) Mangrove (4) Sea island

BES142

16. Concentration of DDT is highest in :-
(1) Primary consumer
(2) Producers
(3) Top consumer
(4) Decomposers

BES143

17. Percentage energy transferred to higher trophic level in food chain is :-
(1) 1% (2) 10% (3) 90% (4) 100%

BES144

18. Biotic and abiotic components form :-
(1) Community (2) Ecosystem
(3) Population (4) Species

BES145

19. Species diversity is maximum in :-
(1) Tropical rain forest (2) Temperate forest
(3) Deserts (4) Hill slops

BES146

20. Insectivorous plants grow in the soil which is deficient in :-
(1) Mg (2) Ca (3) P (4) N

BES147

21. Which is the reason for highest biomass in aquatic ecosystem-
(1) Nano plankton, blue green algae, green algae
(2) Sea grass, and slime molds
(3) Benthonic and brown algae
(4) Diatoms

BES149

22. Pneumatophores are found in-
(1) The vegetation which is found in marshy and saline lake
(2) The vegetation which found in saline soil
(3) Xerophytes
(4) Epiphytes

BES150

23. In which of the following plant sunken stomata are found :-
(1) *Nerium* (2) Hydrilla
(3) Mango (4) Guava

BES151

24. An ecosystem which can be easily damaged but can recover after some time if damaging effect stops will having :-
(1) High stability and low resilience
(2) Low stability and low resilience
(3) High stability and high resilience
(4) Low stability and high resilience

BES153

25. In which one of the following habitats does the diurnal temperature of soil surface vary most ?
(1) Forest (2) Desert
(3) Grassland (4) Shrub land

BES154

26. Which of the following is **not** true for a species?

(1) Members of a species can interbreed

(2) Gene flow does not occur between the populations of a species

(3) Each species is reproductively isolated from every other species

(4) Variations occur among members of a species

BES157

27. Given below is one of the types of ecological pyramids. This type represents



(1) Pyramid of numbers in a grassland

(2) Pyramid of biomass in a fallow land

(3) Pyramid of biomass in a lake

(4) Energy pyramid in a spring

BES158

28. Consider the following statements concerning food chains :-

(a) Removal of 80% tigers from an area resulted in greatly increased growth of vegetation.

(b) Removal of most of the carnivores resulted in an increased population of deers.

(c) The length of food chains is generally limited to 3-4 trophic levels due to energy loss.

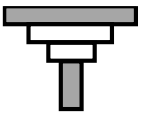
(d) The length of food chains may very from 2 to 8 trophic levels.

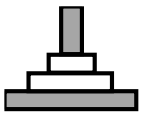
Which two of the above statements are **correct**?

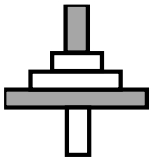
(1) a, d (2) a, b (3) b, c (4) c, d

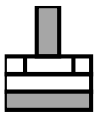
BES085

29. Which of the following representations shows the pyramid of numbers in a forest ecosystem: -

(A) 

(B) 

(C) 

(D) 

(1) A (2) B (3) C (4) D

BES088

30. Which one of the following statements for pyramid of energy is incorrect, whereas the remaining three are correct?

(1) Its base is broad

(2) It shows energy content of different trophic level organisms

(3) It is inverted in shape

(4) It is upright in shape

BES092

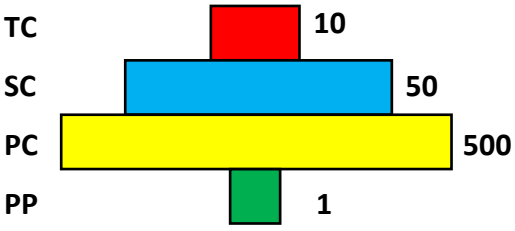
31. The upright pyramid of number is absent in

(1) Lake (2) Grassland

(3) Pond (4) Forest

BES096

32. Given below is an imaginary pyramid of numbers. What could be one of the possibilities about certain organisms at some of the different levels?



(1) Level one PP is "Pipal trees" and the level SC is "sheep"

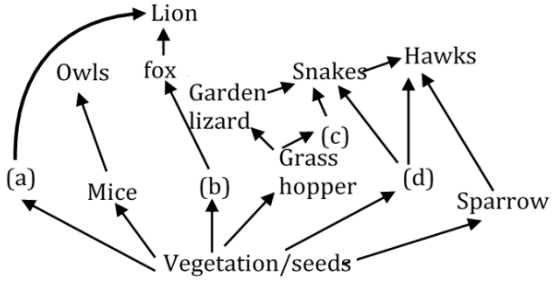
(2) Level PC is "rats" and level SC is "cats"

(3) Level PC is "insects" and level SC is "small insectivorous birds

(4) Level PP is "Phytoplanktons" in sea and "Whale" on top level TC.

BES098

33. Identify the likely organisms (a), (b), (c) and (d) in the food web shown below :-



	(a)	(b)	(c)	(d)
(1)	rat	dog	tortoise	crow
(2)	squirrel	cat	rat	pigeon
(3)	deer	rabbit	frog	rat
(4)	dog	squirrel	bat	deer

BES101

34. The rate of formation of new organic matter by rabbit in a grassland, is called :-

(1) Net primary productivity

(2) Gross primary productivity

(3) Net productivity

(4) Secondary productivity

BES102

35. Which one of the following processes during decomposition is correctly described?

(1) Leaching — Water soluble inorganic nutrients rise to the top layers of soil

(2) Fragmentation — Carried out by organisms such as earthworm

(3) Humification — Leads to the accumulation of a dark coloured substance humus which undergoes microbial action at a very fast rate

(4) Catabolism — Last step in the decomposition under fully anaerobic condition

BES104

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	2	1	2	2	2	2	1	3	1	2	1	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	2	1	4	3	1	1	4	2	2	3	3	2	3
Question	31	32	33	34	35										
Answer	4	3	3	4	2										

Exercise - I (Conceptual Questions) Build Up Your Understanding

1. The world summit on sustainable development held in 2002 in :
(1) Rio de Janeiro
(2) Johannesburg (in South Africa)
(3) Kyoto
(4) Canada

BAC001

2. The Earth summit, 1992, was held at:
(1) Rio de janeiro (2) Kyoto
(3) St. Petersburg (4) Washington

BAC002

3. Find the wrongly matched pair:
(1) Endemism – Species confined to a region and not found anywhere else
(2) Hotspots – Western ghats
(3) Sacred groves – Jaintia hills of Rajasthan
(4) Ex situ conservation – Zoological parks

BAC003

4. Which one the following is false about habitat loss and fragmentation?
(1) This is the most important cause driving animals and plants to extinction
(2) The most dramatic examples of habitat loss come from temperate rain forests
(3) Once covering more than 14% of the earths now cover not more than 6%.
(4) The amazon rain forest harbouring probably millions of species.

BAC004

5. Which of the following is an example of alien species invading a new ecosystem resulting in biodiversity losses?
(1) Introduction of Nile perch into lake Victoria in east Africa
(2) Introduction of water hyacinth in India
(3) Introduction of African catfish in our rivers
(4) All of the above.

BAC005

6. Biodiversity loss occurs due to:
A. Habital loss and fragmentation
B. Co-extinction
C. over-exploitation
D. Alien species invasion.
(1) A, B
(2) A, B, C
(3) B, C, D
(4) A, B, C, D

BAC006

7. A traditional strategy for protection of biodiversity in India is :
(1) Ex-situ conservation
(2) In-situ conservation
(3) Biosphere reserve
(4) Sacred groves

BAC007

8. India is a mega diversity country. Its share of the global species diversity is :
(1) 2.4% (2) 8.1%
(3) 30% (4) 16%

BAC008

9. The species-area relationship studied by Alexander Von Humboldt is a straight line on a logarithmic scale and is represented by equation :-

Log S = Log C + Z Log A

In this equation 'A' stands for: -

(1) Area

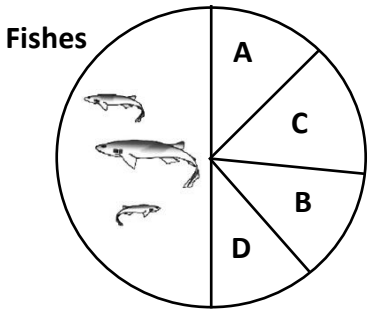
(2) Species richness

(3) Slope of the line

(4) Regression coefficient

BAC009

10. The given figure shows global biodiversity of proportionate number of species of major taxa of vertebrates. Identify A,B, C and D correctly :-



(1) A-Mammals; B – Birds; C – Reptiles; D – Amphibians

(2) A-Amphibians; B – Reptiles; C – Birds; D – Mammals

(3) A-Mammals; B – Reptiles; C – Birds; D – Amphibians

(4) A-Fishes; B – Mammals; C – Birds; D – Reptiles

BAC010

11. Find the odd one out w.r.t. exotic species:

(1) *Parthenium* (2) *Rauwolfia*

(3) *Eichhornia* (4) *Lantana*

BAC011

12. Which of the following is an ex-situ conservation strategy?

(1) Sacred groves

(2) Biodiversity hotspot

(3) Botanical garden

(4) National Park

BAC012

13. Which of the following will not result in loss of biodiversity?

(1) Over-exploitation of natural resources

(2) Introduction of alien species

(3) Keeping animal in a zoological garden

(4) Commercial exploitation of hot spots

BAC013

14. Match some examples of recent extinctions of animals in the column-A with their locations (Countries) in column - B :-

Column-A		Column-B	
(a)	Steller's Sea cow	(i)	Africa
(b)	Thylacine	(ii)	Mauritius
(c)	Dodo	(iii)	India
(d)	Quagga	(iv)	Russia
		(v)	Australia

(1) a–(iv), b–(v), c–(ii), d–(i)

(2) a–(iv), b–(v), c–(iii), d–(i)

(3) a–(iv), b–(v), c–(i), d–(ii)

(4) a–(iii), b–(iv), c–(ii), d–(i)

BAC014

15. The extinction of passenger pigeon was due to-

(1) Increased number of predatory birds

(2) Over exploitation

(3) Non- availability of the food

(4) Disease in pigeon

BAC015

16. The one horned rhinoceros is specific to which of the following sanctuary :

(1) Bandipur (2) Kaziranga

(3) Corbett Park (4) Manas

BAC016

17. The convention on biological diversity held in Rio de janeiro in 1992 is known as

(1) CITES Convention

(2) The Earth Summit

(3) G-16 Summit

(4) MAB Programme

BAC017

18. Introduction of Nile Perch in Lake Victoria of South Africa resulted in :

(1) Excessive growth of water weeds

(2) Elimination of water weeds

(3) Elimination of many species of Cichlid fish

(4) Excessive growth of cichlid fish

BAC018

19. Most diverse among vertebrates

(1) Insects (2) Fishes

(3) Birds (4) Reptiles

BAC019

20. According to Robert May number of species present on Earth are: -

(1) 1.5 Million (2) 1.7 Million

(3) 7 Million (4) 70 Million

BAC020

21. A table of species diversity in different area is given below.

Area	Taxonomic group				
	A	B	C	D	E
I	3	3	–	8	9
II	10	4	3	2	–
III	5	4	4	5	6
IV	4	–	3	2	2

Which area is seemingly more diverse ?

(1) III (2) I (3) II (4) IV

BAC021

22. Which is example of extinct species ?

(1) Red panda (2) Hangul

(3) Dodo (4) Black buck

BAC022

23. How many hot spots of biodiversity in the world have been identified till date by Norman myers?

(1) 34 (2) 17

(3) 51 (4) 3

BAC023

24. A renewable natural resource is :

(1) Forest (2) Coal

(3) Petroleum (4) Minerals

BAC024

25. Which of the following forest is known as the "lungs of the Planet" (earth) and their total contribution of oxygen to earth?

(1) Rain forest of Kerela, 25%

(2) Amazon rain forest, 20%

(3) Taiga forest, 10%

(4) Amazon rain forest, 45%

BAC025

26. The main cause for the extinction of more than 200 species of cichlid fish in the Lake Victoria in East Africa was :-

(1) Over-exploitation

(2) Co-extinction

(3) Alien-species invasion

(4) Habitat fragmentation

BAC026

27. Endangered or threatened animals are taken out from their natural habitat and can be given special care and protection by establishing: -
(1) Zoological park
(2) National park
(3) Biosphere reserve
(4) Sanctuary

BAC027

28. All of the following are included in *'ex-situ'* conservation except:
(1) Wildlife safari parks
(2) Sacred groves
(3) Botanical garden
(4) Seed banks

BAC028

29. Alexander Von Humboldt described for the first time :
(1) Species Area relationship
(2) Population growth equation
(3) Ecological biodiversity
(4) Laws of limiting factor

BAC029

30. Rivet popper hypothesis was proposed by
(1) David Tilman's (2) Paul Ehrlich
(3) Robert May (4) Edward Wilson

BAC030

31. One of the important function of botanical gardens is that :-
(1) In situ conservation of biodiversity
(2) One can observe only dried plants (non-living) there
(3) They allow ex situ conservation of Biodiversity
(4) They provide the natural habitat for wildlife

BAC031

32. The 'Evil Quartet' is the sobriquet used to describe causes of biodiversity loss. This 'Evil Quartet' involve:
(1) Habitat loss & fragmentation
(2) Alien species invasions
(3) Overexploitation & coextinction
(4) All of the above

BAC032

33. How many of the following are broadly utiltarian reasons for conserving biodiversity?
(a) Firewood
(b) Fibres
(c) Pollination
(d) Product of medicinal importance
(1) One (2) Three
(3) Four (4) Two

BAC033

34. Which of the following is wrong about hotspots?
(1) They bear less than 2% area of earth
(2) They show high degree of endemism
(3) They are rich in biodiversity
(4) They represent ex-situ conservation

BAC034

35. Find the correct option w.r.t. defore-station:
(1) Decrease a major greenhouse gas
(2) Promote better hydrological cycle
(3) Promote biodiversity conservation
(4) Loss of biodiversity

BAC035

36. Which of the following is not an invasive alien species in the Indian context?
(1) *Lantana* (2) *Mangifera*
(3) *Parthenium* (4) *Eichhornia*

BAC036

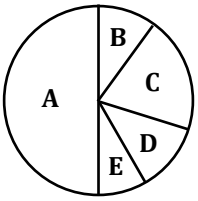
37. The concept of Joint Forest Management (JFM) involves
(1) Conservation of forest and agricultural land by the government only
(2) Conservation of forests and wildlife by the local communities only
(3) Working of government with close association of the local communities for protecting and managing forests
(4) Over exploitation of beneficial forest products only which case biodiversity

BAC037

38. Endangered or threatened animals and plants are taken out from their natural habitat and can be given special care and protection by establishing:-
(1) Zoological park and Botanical Garden
(2) National park
(3) Biosphere reserve
(4) Sanctuary

BAC038

39. The following pie chart is showing the global biodiversity of vertebrates:



Which sectors represent fishes and birds species respectively?
(1) E and B (2) B and D
(3) B and C (4) A and C

BAC039

40. Amongst the global species diversity, more than 70 percent are of
(1) Animals (2) Insects
(3) Fungi (4) Plants

BAC040

41. According to "Rivet popper hypothesis" related to biodiversity and ecosystem stability. Which of the following is referred as aeroplane?
(1) Species (2)Key-stone species
(3) Ecosystem (4) Human

BAC132

42. Which is not a renewable source -
(1) Forest (2) Coal
(3) Water (4) Forest organism

BAC042

43. Biosphere refers to
(1) Plants of the world
(2) Special plants
(3) Area occupied by living beings
(4) Plants of a particular area

BAC043

44. Which of the following is the non-conventional source of energy
(1) Coal
(2) Petroleum
(3) Electricity from nuclear power plants
(4) Solar radiations

BAC044

45. Petroleum resource is
(1) Renewable
(2) Non-renewable
(3) Synthetic
(4) Infinite & non-conventional

BAC045

46. Red data book is famous for -
(1) Extinct plants and animals only
(2) Extinct plants only
(3) Endangered plants and animals
(4) Extinct animals only

BAC046

47. Green book contains: -
(1) The list of endangered plants
(2) The list of extinct plants
(3) The list of rare plants grown in botanical gardens
(4) Flora of certain area

BAC047

48. The method by which endangered plant species are conserved in a botanical garden or in some controlled circumstances -
(1) Afforestation
(2) In situ conservation
(3) Ex situ conservation
(4) None of the above

BAC048

49. Which one of the following may be the reason for extinction of plant species due to human activities -
(1) Earthquakes (2) Pollution
(3) Diseases (4) Evolution

BAC049

50. The main aim of plant conservation is -
(1) To conserve the necessary ecological activities and life supporting systems
(2) To conserve species diversity and range of genetic meterial
(3) Both the above
(4) None of the above

BAC050

51. Which of the following species is in an endangered state:
(1) Indian bustard & rhino
(2) Asiatic donkey
(3) Hangul-Kashmir Stag
(4) All the above

BAC051

52. Wild life protection act was enacted in India in
(1) 1947 (2) 1962 (3) 1972 (4) 1992

BAC052

53. Number of wild life is continuously decreasing. What is the main reason of this: -
(1) Predation
(2) Cutting down of forest
(3) Destruction of habitat
(4) Hunting

BAC053

54. One of the following is associated with the conservation of forests
(1) Kaziranga (2) Ghana
(3) Silent valley (4) Gir

BAC054

55. Which one of the following is not included under in-situ conservation?
(1) Biosphere reserve
(2) National park
(3) Sanctuary
(4) Botanical garden

BAC056

56. Which of the following is considered a hot-spot of biodiversity in India ?
(1) Eastern Ghats
(2) Aravalli Hills
(3) Western Ghats
(4) Indo-Gangetic Plain

BAC058

57. Which one of the following is not observed in biodiversity hotspots?
(1) Lesser inter–specific competition
(2) Species richness
(3) Endemism
(4) Accelerated species loss

BAC061

58. World summit on sustainable Development (2002) was held in:-
(1) Argentina
(2) South Africa
(3) Brazil
(4) Sweden

BAC062

59. Which one of the following is an example of ex-situ conservation?
(1) National park
(2) Wildlife sanctuary
(3) Seed bank
(4) Sacred groves

BAC063

60. A renewable exhaustible natural resource is
(1) Forest (2) Coal
(3) Petroleum (4) Minerals

BAC064

61. Which one of the following have the highest number of species in nature?
(1) Fungi (2) Insects
(3) Birds (4) Angiosperms

BAC066

62. Which one of the following shows maximum genetic diversity in India?
(1) Groundnut (2) Rice
(3) Maize (4) Mango

BAC067

63. Biodiversity of a geographical region represents:
(1) Species endemic to the region
(2) Endangered species found in the region
(3) The diversity in the organisms living in the region.
(4) Genetic diversity present in the dominant species of the region.

BAC068

64. Which one of the following areas in India, is a hot spot of biodiversity?
(1) Sunderbans
(2) Western Ghats
(3) Eastern Ghats
(4) Gangetic plain

BAC069

65. The highest number of species in the world is represented by :-
(1) Algae
(2) Lichens
(3) Fungi
(4) Mosses

BAC070

66. Select the correct statement about biodiversity:-
(1) Western Ghats have a very high degree of species richness and endemism.
(2) Conservation of biodiversity is just a fad pursued by the developed countries
(3) The desert areas of Rajasthan and Gujarat have a very high level of desert animal species as well as numerous rare animals
(4) Large scale planting of Bt cotton has no adverse effect on biodiversity

BAC071

67. Sacred groves are specially useful in :-
(1) year round flow of water in rivers
(2) conserving rare and threatened species
(3) generating environmental awareness
(4) preventing soil erosion

BAC072

68. Which one of the following is not used for ex situ plant conservation ?

(1) Botanical Gardens

(2) Field gene banks

(3) Seed banks

(4) Shifting cultivation

BAC073

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	3	2	4	4	4	2	1	3	2	3	3	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	3	2	3	1	3	1	1	2	3	1	2	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	1	4	4	2	3	1	4	1	3	2	3	4	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	3	2	3	4	3	3	3	4	3	1	2	3	1
Question	61	62	63	64	65	66	67	68							
Answer	2	2	3	2	3	1	2	4							

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

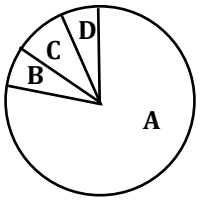
1. An example of ex situ conservation is :-
(1) National Park
(2) Seed Bank
(3) Wildlife Sanctuary
(4) Sacred Grove

BAC074

2. A species facing extremely high risk of extinction in the immediate future is called
(1) Vulnerable
(2) Endemic
(3) Critically Endangered
(4) Extinct

BAC075

3. Given below is the representation of the extent of global diversity of invertebrates. What groups the four portions (A-D) represent respectively :-



	A	B	C	D
(1)	Insects	Crustaceans	Other animal groups	Molluscs
(2)	Crustaceans	Insects	Molluscs	Other animal groups
(3)	Molluscs	Other animal groups	Crustaceans Insects	Insects
(4)	Insects	Molluscs	Crustaceans	Other animal groups

BAC076

4. Just as a person moving from Delhi to Shimla to escape the heat for the duration of hot summer, thousands of migratory birds from Siberia and other extremely cold northern regions move to :-
(1) Western Ghat
(2) Meghalaya
(3) Corbett National Park
(4) Keolado National Park

BAC077

AIPMT 2015

5. Cryopreservation of gametes of threatened species in viable and fertile condition can be referred to as:-
(1) Advanced ex-situ conservation of biodiversity
(2) In situ conservation by sacred groves
(3) In situ cryo-conservation of biodiversity
(4) In situ conservation of biodiversity

BAC078

6. The UN Conference of Parties on climate change in the year 2011 was held in :-
(1) South Africa (2) Peru
(3) Qatar (4) Poland

BAC079

7. In which of the following both pairs have correct combination :-
(1) In situ conservation: Cryopreservation
Ex situ conservation: Wildlife Sanctuary
(2) In situ conservation: Seed Bank
Ex situ conservation: National Park
(3) In situ conservation: Tissue culture
Ex situ conservation: Sacred groves
(4) In situ conservation: National Park
Ex situ conservation: Botanical Garden

BAC080

8. The species confined to a particular region and not found elsewhere is termed as :

(1) Rare (2) Keystone

(3) Alien (4) Endemic

BAC081

AIPMT 2016

9. A system of rotating crops with legume or grass pasture to improve soil structure and fertility is called:-

(1) Ley farming

(2) Contour farming

(3) Strip farming

(4) Shifting agriculture

BAC082

10. Joint Forest Management Concept was introduced in India during :

(1) 1960 s (2) 1970 s

(3) 1980 s (4) 1990 s

BAC083

11. How many hot spots of biodiversity in the world have been identified till date by Norman Myers?

(1) 34 (2) 43

(3) 17 (4) 25

BAC084

12. Which of the following National Parks is home to the famous musk deer or hangul?

(1) Eaglenest Wildlife Sanctuary, Arunachal Pradesh

(2) Dachigam National Park, Jammu & Kashmir

(3) Keibul Lamjao National Park, Manipur

(4) Bandhavgarh National Park, Madhya Pradesh

BAC085

13. Which of the following is the most important cause of animals and plants being driven to extinction ?

(1) Over - exploitation

(2) Alien species invasion

(3) Habitat loss and fragmentation

(4) Co-extinctions

BAC086

AIPMT 2017

14. Which one of the following is related to Ex-situ conservation of threatened animals and plants ?

(1) Biodiversity hot spots

(2) Amazon rainforest

(3) Himalayan region

(4) Wildlife safari parks

BAC087

15. Alexander Von Humbolt described for the first time:

(1) Laws of limiting factor

(2) Species area relationships

(3) Population Growth equation

(4) Ecological Biodiversity

BAC088

16. The region of Biosphere Reserve which is legally protected and where no human activity is allowed is known as:

(1) Buffer zone (2) Transition zone

(3) Restoration zone (4) Core zone

BAC089

17. Van Mahotsava is a festival of:

(1) Planting trees in open areas

(2) Taking oath to protect trees

(3) Worshipping trees

(4) Conservation of sacred groves

BAC091

18. Which scientist proposed 'Rivet popper hypothesis' related to biodiversity and Ecosystems?
(1) Alexander von Humboldt
(2) Paul Ehrlich
(3) David Tilman
(4) Tansley

BAC092

AIPMT 2018

19. All of the following are included in 'Ex-situ conservation' except
(1) Wildlife safari parks
(2) Sacred groves
(3) Botanical gardens
(4) Seed banks

BAC093

AIPMT 2019

20. Which one of the following is not a method of in situ conservation of biodiversity ?
(1) Biosphere Reserve
(2) Wildlife Sanctuary
(3) Botanical Garden
(4) Sacred Grove

BAC094

21. The Earth Summit held in Rio de Janeiro in 1992 was called :
(1) to reduce CO_2 emissions and global warming.
(2) for conservation of biodiversity and sustainable utilization of its benefits.
(3) to assess threat posed to native species by invasive weed species.
(4) for immediate steps to discontinue use of CFCs that were damaging the ozone layer.

BAC095

22. Which of the following is the most important causes for animals and plants being driven to extinction?
(1) Habitat loss and fragmentation
(2) Drought and floods
(3) Economic exploitation
(4) Alien species invasion

BAC096

23. Decline in the population of Indian native fishes due to introduction of Clarias gariepinus in river Yamuna can be categoriesd as :-
(1) Co-extinction
(2) Habitat fragmentation
(3) Over exploitation
(4) Alien species invasion

BAC097

24. Western Ghats have a large number of plant and animal species that are not found anywhere else. Which of the following terms will you use to notify such species?
(1) Endemic (2) Vulnerable
(3) Threatened (4) Keystone

BAC098

AIPMT 2020

25. According to Robert May, the global species diversity is about :
(1) 7 million (2) 1.5 million
(3) 20 million (4) 50 million

BAC099

26. Which of the following regions of the globe exhibits highest species diversity?
(1) Amazon forests
(2) Western Ghats of India
(3) Madagascar
(4) Himalayas

BAC100

27. According to Alexander von Humboldt :
(1) Species richness decreases with increasing area of exploration
(2) Species richness increases with increasing area, but only up to limit
(3) There is no relationship between species richness and area explored.
(4) Species richness goes on increasing with increasing area of exploration

BAC101

28. In the following in each set a conservation approach and an example of method of conservation are given
(a) In situ conservation - Biosphere Reserve
(b) Ex situ conservation - Sacred groves
(c) In situ conservation - Seed bank
(d) Ex situ conservation - Cryopreservation
Select the option with correct match of approach and method :
(1) (a) and (c) (2) (a) and (d)
(3) (b) and (d) (4) (a) and (b)

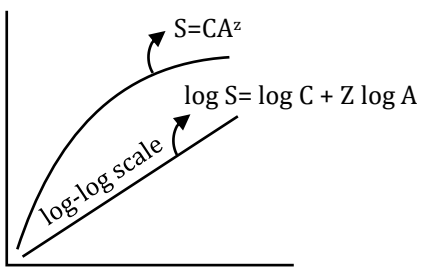
BAC102

AIPMT 2021

29. Find the wrongly matched pair.
(1) *Clarias gariepinus* – African cat fish
(2) Himalaya – Hotspot
(3) Aravalli hills – Sacred groves
(4) Earth Summit – South Africa

BAC103

30. The following graph shows



(1) Plant–animal relationship
(2) Algae–fungi relationship
(3) Species–volume relationship
(4) Species-area relationship

BAC104

31. Loss of biodiversity in a region may lead to
(1) Decline in plant production
(2) Lowered resistance to drought
(3) Increased variability in disease cycles
(4) All of the above

BAC105

AIPMT 2022

32. Habitat loss and fragmentation, over exploitation, alien species invasion and co-extinction are causes for:
(1) Competition
(2) Biodiversity loss
(3) Natality
(4) Population explosion

BAC106

33. Which of the following is not a method of ex situ conservation?
(1) National Parks
(2) Micropropagation
(3) Cryopreservation
(4) In vitro fertilization

BAC107

34. Give the correct descending order of organisms with reference to their estimated number found in Amazon forest.
(a) Plants (b) Invertebrates
(c) Fishes (d) Mammals
(e) Birds
Choose the **correct** answer from the options given below :-
(1) (a) > (b) > (e) > (d) > (c)
(2) (a) > (c) > (d) > (b) > (e)
(3) (b) > (a) > (e) > (d) > (c)
(4) (b) > (a) > (c) > (e) > (d)

BAC108

35. The World Summit on sustainable development held in 2002 in Johannesburg, South Africa pledged for:

(1) A significant reduction in the current rate of biodiversity loss.

(2) Declaration of more biodiversity hotspots.

(3) Increase in agricultural production

(4) Collection and preservation of seeds of different genetic strains of commercially important plants.

BAC109

36. Frugivorous birds are found in large numbers in tropical forests mainly because of :

(1) lack of niche specialisation

(2) higher annual rainfall

(3) availability of fruits throughout the year

(4) temperature conducive for their breeding

BAC110

37. Match **List-I** with **List-II**.

List-I		List-II	
(a)	Sacred groves	(i)	Alien species
(b)	Zoological park	(ii)	Release of large quantity of oxygen
(c)	Nile perch	(iii)	Ex-situ conservation
(d)	Amazon forest	(iv)	Khasi Hills in Meghalaya

Choose the **correct** answer from the options given below :-

(1) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)

(2) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)

(3) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

(4) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

BAC111

38. Western Ghats have a large number of plants and animal species that are not found any where else. Which of the following term is used to notify such species?

(1) Threatened species

(2) Keystone species

(3) Endemic species

(4) Vulnerable species

BAC112

NEET-UG 2023

39. Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species ?

(1) Over exploitation for economic gain

(2) Alien species invasions

(3) Co-extinctions

(4) Habitat loss and fragmentation

BAC128

40. The historic Convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year:

(1) 1992 (2) 1986 (3) 2002 (4) 1985

BAC129

NEET-UG 2023 (Manipur)

41. In 'rivet popper hypothesis', Paul Ehrlich compared the rivets in an airplane to :

(1) species within a genus

(2) genetic diversity

(3) ecosystem

(4) genera within a family

BAC130

42. Which of the following is/are cause(s) of biodiversity losses ?

(1) Over-exploitation, habitat loss and fragmentation.

(2) Climate change only

(3) Over-Exploitation only

(4) Habitat loss and fragmentation only

BAC131

NEET-UG 2024

43. List of endangered species was released by-

(1) GEAC (2) WWF

(3) FOAM (4) IUCN

BAC136

44. Tropical regions show greatest level of species richness because

A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.

B. Tropical environments are more seasonal.

C. More solar energy is available in tropics.

D. Constant environments promote niche specialization.

E. Tropical environments are constant and predictable.

Choose the **correct** answer from the options given below :-

(1) A, C, D and E only

(2) A and B only

(3) A, B and E only

(4) A, B and D only

BAC137

45. The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called ;-

(1) *in-situ* conservation

(2) Biodiversity conservation

(3) Semi-conservative method

(4) Sustainable development

BAC138

46. These are regarded as major causes of biodiversity loss :-

A. Over exploitation

B. Co-extinction

C. Mutation

D. Habitat loss and fragmentation

E. Migration

Choose the **correct** option :-

(1) A, C and D only

(2) A, B, C and D only

(3) A, B and E only

(4) A, B and D only

BAC139

47. Match **List I** with **List II.**

	List I		List II
A.	Robert May	I.	Species-Area relationship
B.	Alexander von Humboldt	II.	Long term ecosystem experiment using out door plots
C.	Paul Ehrlich	III.	Global species diversity at about 7 million
D.	David Tilman	IV.	Rivet popper hypothesis

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-I, C-IV, D-II

(3) A-I, B-III, C-II, D-IV

(4) A-III, B-IV, C-II, D-I

BAC140

RE-NEET-UG 2024

48. The regions with high level of species richess, high degree of endemism and a loss of 70% of the species and habitat are identified as :-

(1) Natural Reserves

(2) Sacred Groves

(3) Biodiversity Hotspots

(4) Biogeographical Regions

BAC141

49. Which one of the following is **not** included under *in-situ* conservation ?

(1) Wild-life sanctuary

(2) Botanical garden

(3) Biosphere reserve

(4) National park

BAC142

50. Cryo preservation technique is used for :

(1) Protection of environment

(2) Proteciton of Biodiversity hotspots.

(3) Preservation of gametes in viable and fertile condition for a long period

(4) *In-situ* conservation

BAC143

51. Match **List-I** with **List-II** :

	List-I		List-II
A.	Biodiversity hotspot	I.	Khasi and Jantia hills in Meghalaya
B.	Sacred groves	II.	World Summit on Sustainable Development 2002
C.	Johannesburg South Africa	III.	*Parthenium*
D.	Alien species invasion	IV.	Western Ghats

Choose the **correct** answer from the options given below :-

(1) A-IV, B-I, C-II, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-I, B-IV, C-III, D-II

(4) A-III, B-I, C-II, D-IV

BAC144

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	4	4	1	1	4	4	1	3	1	2	3	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	2	2	3	2	1	4	1	1	1	2	2	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	1	4	1	3	1	3	4	1	1	1	4	1	2
Question	46	47	48	49	50	51									
Answer	4	2	3	2	3	1									

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Direct economic benefit from nature are included in-
(1) Broadly utilitarian
(2) Ethical
(3) Social forestry
(4) Narrowly utilitarian

BAC145

2. The earth summit held at Rio de janeiro in 1992 resulted in to :-
(1) Compilation of Red list
(2) Establishment of Biosphere Reserves
(3) Conservation of Biodiversity
(4) IUCN

BAC114

(B) ANALYTICAL QUESTIONS

3. Reason for elimination of wild life is :-
(1) Deforestation (2) Forest fire
(3) Floods (4) Less Rain fall

BAC115

4. Best economic method to harvest the solar energy :-
(1) Solar cell
(2) Energy plantation
(3) Cultivation of sugar cane then energy obtain by burning it.
(4) Solar cooker.

BAC116

5. Main reason of disturbance of biological diversity:-
(1) Green house effect
(2) Hunting
(3) Soil erosion
(4) Destruction of natural habitats

BAC117

6. Best method to preserve the wild relatives of plants :-
(1) By growing them in natural habitats
(2) Gene library
(3) By storing seeds
(4) Cryopreservation

BAC118

7. What shall be the effect of destruction of wild life :–
(1) Wild gene of disease resistance will not be obtained
(2) Soil erosion
(3) Floods
(4) Green house effect

BAC119

8. Which of the following are generally considered to be the major reasons why biodiversity should be conserved-
(i) Economic benefits
(ii) Ethical reason
(iii) Political advantages
(iv) For scientific study
(v) Ecological benefits
(1) (i), (iii), (iv) (2) (i), (ii), (iv)
(3) (iii), (v) (4) (ii), (iii)

BAC134

9. Presently India has _____ Biosphere reserve, _____ National parks and _____ wildlife sanctuaries.
(1) 20, 90, 500
(2) 14, 85, 348
(3) 11, 90, 448
(4) 14, 90, 448

BAC135

10. Which one of the following is a pair of endangered species

(1) Garden lizard and Mexican poppy

(2) Rhesus monkey and Sal tree

(3) Indian peacock and carrot grass

(4) Hornbill and Indian Aconite

BAC123

11. If the Bengal Tiger becomes extinct

(1) Hyenas and wolves will become scarce

(2) The wild areas will be safe for man and domestic animals

(3) Its gene pool will be lost for ever

(4) The populations of beautiful animals like deers will get stabilized.

BAC124

12. Which one of the following pairs of geographical areas show maximum biodiversity in our country:-

(1) Sunderbans and Rann of Kutch

(2) Eastern Ghats and West Bengal

(3) Eastern Himalaya and Western Ghats

(4) Kerala and Punjab

BAC125

13. One of the **ex situ** conservation methods for endangered species is :-

(1) Wildlife Sanctuaries

(2) Biosphere Reserves

(3) Cryopreservation

(4) National Parks

BAC126

14. Exploring molecular, genetic and species level diversity for products of economic importance is called

(1) Biodiversity

(2) Biopiracy

(3) Bioprospecting

(4) Bioremediation

BAC127

15. Which of the following pairs of an animal and a plant represents endangered organisms in India?

(1) Cinchona and Leopard

(2) Banyan and Black buck

(3) Bentinckia nicobarica and Red Panda

(4) Tamarind and Rhesus monkey

BAC055

16. Which one of the following is the correctly matched pair of an endangered animal and a National Park?

(1) Wild ass — Dudhwa National Park

(2) Great Indian Bustard—Keoladeo National Park

(3) Lion— Corbett National Park

(4) Rhinoceros—Kaziranga National Park

BAC057

17. Identify the odd combination of the habitat and the particular animal concerned :

(1) Rann of Kutch — Wild Ass

(2) Dachigam National Park—Snow Leopard

(3) Sunderbans — Bengal Tiger

(4) Periyar — Elephant

BAC059

18. Which one of the following pairs of organisms are exotic species introduced in India ?

(1) Nile perch, *Ficus religiosa*

(2) *Ficus religiosa*, *Lantana camara*

(3) *Lantana camara*, Water hyacinth

(4) Water hyacinth, *Prosopis cineraria*

BAC060

19. The Indian Rhinoceros is a natural inhabitant of which one of the Indian states?

(1) Uttar Pradesh

(2) Himachal Pradesh

(3) Assam

(4) Uttarakhand

BAC065

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	1	2	4	1	1	2	4	4	3	3	3	3	3
Question	16	17	18	19											
Answer	4	2	3	3											